Notes on Algebra

MTH 619/620

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Contents

1 Monoids and groups

1.1 Definition. A *monoid* is a set M together with a map

$$M \times M \to M, \quad (x, y) \mapsto x \cdot y$$

such that

(i) $(x \cdot y) \cdot z = x \cdot (y \cdot z) \;\; \forall x, y, z \in M$ (associativity);

(ii) $\exists e \in M$ such that
$$x \cdot e = e \cdot x = x$$
for all $x \in M$ ($e =$ the identity element of M).

1.2 Examples.

1) $\mathbb{Z}$ with addition of integers ($e = 0$)

2) $\mathbb{Z}$ with multiplication of integers ($e = 1$)

3) $M_n(\mathbb{R}) = \{$the set of all $n \times n$ matrices with coefficients in $\mathbb{R}\}$ with matrix multiplication ($e = I =$ the identity matrix)

4) $U =$ any set
$$P(U) := \{\text{the set of all subsets of } U\}$$
$P(U)$ is a monoid with $A \cdot B := A \cup B$ and $e = \varnothing$.

5) Let $U =$ any set
$$F(U) := \{\text{the set of all functions } f\colon U \to U\}$$
$F(U)$ is a monoid with multiplication given by composition of functions ($e = \mathrm{id}_U =$ the identity function).

1.3 Definition. A monoid is *commutative* if $x \cdot y = y \cdot x$ for all $x, y \in M$.

1.4 Example. Monoids 1), 2), 4) in 1.2 are commutative; 3), 5) are not.

1.5 Note. Associativity implies that for $x_1, \dots, x_k \in M$ the expression

$$x_1 \cdot x_2 \cdot \dots \cdot x_k$$

has the same value regardless how we place parentheses within it; e.g.:

$$(x_1 \cdot x_2) \cdot (x_3 \cdot x_4) = ((x_1 \cdot x_2) \cdot x_3) \cdot x_4 = x_1 \cdot ((x_2 \cdot x_3) \cdot x_4)$$

etc.

1.6 Note. A monoid has only one identity element: if $e, e' \in M$ are identity elements then

$$e = e \cdot e' = e'$$

1.7 Definition. A *group* is a monoid G such that for any $x \in G$ there is $y \in G$ satistying $x \cdot y = e = y \cdot x$.

The element y is called the *inverse* of x and it is denoted by x^{-1} (or by $-x$ in the additive notation).

A group G is *commutative* (or abelian) if $x \cdot y = y \cdot x$ for all $x, y \in G$.

1.8 Examples.

1) $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ with addition
2) $\mathbb{Q}^* = \mathbb{Q} - \{0\}$, $\mathbb{R}^* = \mathbb{R} - \{0\}$, $\mathbb{C}^* = \mathbb{C} - \{0\}$ with multiplication
3) $GL_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\}$ with matrix multiplication
   (the $n \times n$ general linear group)
4) $SL_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid \det(A) = 1\}$ with matrix multiplication
   (the $n \times n$ special linear group)
5) Let $U =$ be any set and let

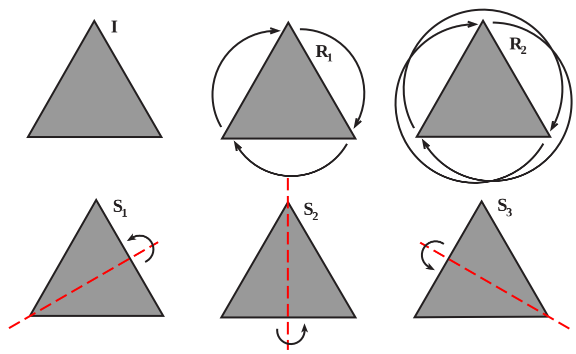
   $$\mathrm{Perm}(U) := \{f \colon U \to U \mid f \text{ is a bijection}\}$$

   $\mathrm{Perm}(U)$ with composition of functions is a group (the group of permutations of U)

   Note. If $U = \{1, 2, \dots, n\}$ then $\mathrm{Perm}(U)$ is called the *symmetric group on n letters* and it is denoted by S_n.

7) Let $T =$ an equilateral triangle

$$G_T = \{I, R_1, R_2, S_1, S_2, S_3\}$$



$G_T =$ the group of symmetries of T.

1.9 Proposition (Cancellation Law)**.** *If G is a group, $x, y, x \in G$ and*

$$xy = xz$$

then $y = z$.

Proof.

$$\begin{aligned} xy &= xz \\ x^{-1}xy &= x^{-1}xz \\ y &= z \end{aligned}$$

□

1.10 Note. The cancellation law does not hold for monoids. E.g. in $M_2(\mathbb{R})$ take

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \ B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \ C = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Then $AB = AC$ but $A \neq C$.

2 Subgroups

2.1 Definition. If G is a group then a *subgroup* of G is a subset $H \subseteq G$ such that

(i) $e \in H$;

(ii) if $x, y \in H$ then $xy \in H$;

(iii) if $x \in H$ then $x^{-1} \in H$.

2.2 Note. A subgroup of a group is by itself a group.

2.3 Examples.

1) If G is a group then G, $\{e\}$ are subgroups of G
2) $\mathbb{Z}$ is a subgroup of $\mathbb{Q}$, which is a subgroup of $\mathbb{R}$, which is a subgroup of $\mathbb{C}$.
3) $SL_n(\mathbb{R})$ is a subgroup of $GL_n(\mathbb{R})$
4) $H = \{I, R_1, R_2\}$ is a subgroup of G_T

2.4 Note. If $\{H_i\}_{i \in I}$ is a family of subgroups of G then $\bigcap_{i \in I} H_i$ is also a subgroup of G.

2.5 Definition. If G is a group and S is a subset of G then denote

$$\langle S \rangle = \text{the smallest subgroup of } G \text{ that contains } S$$

$\langle S \rangle$ is the *subgroup of G generated by the set S.*

2.6 Proposition. *If $S \subseteq G$ then $\langle S \rangle$ consists of all elements of the form*

$$x_1^{\pm 1} x_2^{\pm 1} \cdot \dots \cdot x_k^{\pm 1}$$

where $x_1, \dots, x_k \in S$.

Proof. Exercise. $\square$

2.7 Definition. A set $S \subseteq G$ *generates* G if $\langle S \rangle = G$.

2.8 Example. $S = \{S_1, S_2\}$ generates G_T.

2.9 Definition. A group G is *finitely generated* if it is generated by some finite subset $S \subseteq G$.

2.10 Note.

- Every finite group is finitely generated.
- Some infinite groups are finitely generated; e.g. $\mathbb{Z} = \langle 1 \rangle$.

2.11 Definition. A group G is *cyclic* if $G = \langle a \rangle$ for some $a \in G$

2.12 Note. If G is cyclic, $G = \langle a \rangle$ then every element $g \in G$ is of the form

$$g = a^n$$

for some $n \in \mathbb{Z}$ (where $a^{-n} := (a^{-1})^n$, $a^0 = e$).

2.13 Examples.

1) $\mathbb{Z} = \langle 1 \rangle$ is cyclic.

2) $H := \{I, R_1, R_2\} \subseteq G_T$ is cyclic:

$$H = \langle R_1 \rangle \quad \text{and} \quad H = \langle R_2 \rangle$$

3 Homomorphisms of groups

3.1 Definition. Let G, H be groups. A function $f\colon G \to H$ is a *group homomorphism* if for any $a, b \in G$ we have

$$f(ab) = f(a)f(b)$$

3.2 Proposition. *If $f\colon G \to H$ is a homomorphism of groups and e_G, e_H denote identity elements in, respectively, G and H then*

(i) $f(e_G) = e_H$

(ii) $f(a^{-1}) = f(a)^{-1}$ *for any $a \in G$.*

Proof. (i) We have

$$f(e_G) = f(e_G \cdot e_G) = f(e_G) \cdot f(e_G)$$

Multiplying this equation by $f(e_G)^{-1}$ we obtain $e_H = f(e_G)$.

(ii) Since by (i) we have $f(e_G) = e_H$ therefore

$$f(a) \cdot f(a^{-1}) = f(a \cdot a^{-1}) = f(e_G) = e_H$$

It is now enough to multiply this equation from the left by $f(a)^{-1}$. $\square$

3.3 Definition. A homomorphism $f\colon G \to H$ is an *isomorphism* if there is a homomorphism $g\colon H \to G$ such that $g \circ f = \mathrm{id}_G$ and $f \circ g = \mathrm{id}_H$.

3.4 Proposition. *A map $f\colon G \to H$ is an isomorphism of groups iff f is a homomorphism and a bijection.*

Proof. Exercise. $\square$

3.5 Definition. If there exists an isomorphism $f\colon G \to H$ then we say that the groups G and H are *isomorphic* and we write $G \cong H$.

3.6 Definition. A homomorphism $f\colon G \to G$ is called an *endomorphism* of G. An isomorphism $f\colon G \to G$ is called an *automorphism* of G.

3.7 Examples.

1) $\mathrm{id}_G\colon G \to G$ is an automorphism of G.

2) $f\colon G \to G,\ \ f(g) = e\ \ \forall_{g\in G}$ is an endomorphism of G.

3) If $f\colon G \to H$, $g\colon H \to K$ are homomorphisms then so is $g \circ f\colon G \to K$.

4) For $g \in G$ define
$$c_g\colon G \to G, \quad c_g(a) := gag^{-1}$$
Check: c_g is an automorphism of G. Automorphisms of this form are called *inner automorphisms* of G.

   Note. If G is an abelian group then $c_g = \mathrm{id}_G$ for all $g \in G$.

5) Recall: $GL_n(\mathbb{R}) = \{A \in M_n \mid \det(A) \neq 0\}$, $\mathbb{R}^* = \mathbb{R} - \{0\}$

   We have the determinant function:
$$\det\colon GL_n(\mathbb{R}) \to \mathbb{R}^*$$
Since $\det(AB) = \det(A) \cdot \det(B)$ this function is a homomorphism.

6) Let $G \subseteq GL_2(\mathbb{R})$
$$G := \left\{ \begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix} \;\middle|\; r \in \mathbb{R} \right\}$$
G is a subgroup of $GL_2(\mathbb{R})$:
$$\begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & r+s \\ 0 & 1 \end{pmatrix}$$
$$\begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -r \\ 0 & 1 \end{pmatrix}$$
We have homomorphisms:
$$f\colon \mathbb{R} \to G \ \text{ and } \ g\colon \mathbb{R} \to G$$

where

$$f(r) = \begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix}, \quad g\left(\begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix}\right) = r$$

Since $g \circ f = \mathrm{id}_G$, $f \circ g = \mathbb{R}$ we get $G \cong \mathbb{R}$.

3.8 Definition. If G is a group then

$$|G| := \text{the number of elements of G}$$

$|G|$ is called the *order* of G.

3.9 Example. $|G_T| = 6$, $|\mathbb{Z}| = \infty$.

3.10 Note. If $G \cong H$ then $|G| = |H|$.

4 The kernel and the image of a homomorphism

4.1 Proposition. *Let $f\colon G \to H$ be a homomorphism.*

1) *If G' is a subgroup of G then $f(G')$ is a subgroup of H.*
2) *If H' is a subgroup of H then $f^{-1}(H')$ is a subgroup of G.*

Proof. Exercise. □

4.2 Definition. If $f\colon G \to H$ is a homomorphism then

- the *image* of f is the subgroup

$$\mathrm{Im}(f) := f(G) \subseteq H$$

- the *kernel* of f is the subgroup

$$\mathrm{Ker}(f) := f^{-1}(e_H) \subseteq G$$

4.3 Note. $f\colon G \to H$ is an epimorphism (onto) iff $\mathrm{Im}(f) = H$.

4.4 Proposition. *$f\colon G \to H$ is a monomorphism (1-1) iff $\mathrm{Ker}(f) = \{e_G\}$*

Proof. $(\Rightarrow)$ We have $f(e_G) = e_H$. Thus if f is 1-1 then $f(g) = e_H$ only if $g = e_H$. In other words we have then $\mathrm{Ker}(f) = \{e_H\}$.

$(\Leftarrow)$ Assume that $\mathrm{Ker}(f) = \{e_G\}$ and let $f(a) = f(b)$ for some $a, b \in G$. We have:

$$f(ab^{-1}) = f(a)f(b)^{-1} = e_H$$

so $ab^{-1} \in \mathrm{Ker}(f)$. Therefore $ab^{-1} = e_G$, and so $a = b$.

□

4.5 Problem. Let G be a group, and let H be a subgroup of G. Is there a homomorphism

$$f\colon G \to K$$

such that $\mathrm{Ker}(f) = H$?

4.6 Note. The dual problem is trivial: if H is a subgroup of G then we have the inclusion homomorphism

$$i\colon H \hookrightarrow G$$

and $\mathrm{Im}(i) = H$. It follows that any subgroup of G is an image of some homomorphism.

4.7 Definition. A subgroup $H \subseteq G$ is a *normal subgroup* if for every $h \in H$ we have

$$aha^{-1} \in H \quad \forall a \in G$$

4.8 Notation. If H is a normal subgroup of G then we write $H \lhd G$

4.9 Proposition. *If $f\colon G \to H$ is a homomorphism then $\mathrm{Ker}(f)$ is a normal subgroup of G.*

Proof. If $a \in G$, $h \in \mathrm{Ker}(f)$ then

$$f(aha^{-1}) = f(a)f(h)f(a)^{-1} = f(a) \cdot e \cdot f(a)^{-1} = f(a)f(a)^{-1} = e$$

so $aha^{-1} \in \mathrm{Ker}(f)$. $\square$

4.10 Examples.

1) Any subgroup of an abelian group is normal.
2) $H := \{I, R_1, R_2\}$ is a normal subgroup of G_T (check!).
3) $K := \{I, S_1\}$ is not a normal subgroup of G_T (check!). As a consequence K cannot be the kernel of any homomorphism $G_T \to G$.

5 Normal subgroups, cosets and quotient groups

Recall. If $f\colon G \to K$ is a homomorphism then $\mathrm{Ker}(f)$ is a normal subgroup of G.

Next goal: If H is a normal subgroup of G then there is a homomorphism $f\colon G \to K$ such that $H = \mathrm{Ker}(f)$.

5.1 Definition. If H is a subgroup of G then a *left coset* of H in G is a subset of G of the form

$$aH := \{ah \mid h \in H\}$$

for some $a \in G$.

A *right coset* of H in G is a subset of G of the form

$$Ha := \{ha \mid h \in H\}$$

for some $a \in G$.

5.2 Example.

Recall: $G_T = \{I, R_1, R_2, S_1, S_2, S_3\}$. Take $H := \{I, S_1\}$. We have:

$IH = \{I \cdot I, I \cdot S_1\} = \{I, S_1\} = H$
$S_1H = \{S_1 \cdot I, S_1 \cdot S_1\} = \{S_1, I\}$
$S_2H = \{S_2 \cdot I, S_2 \cdot S_1\} = \{S_2, R_2\}$
$S_3H = \{S_3 \cdot I, S_3 \cdot S_1\} = \{S_3, R_1\}$
$R_1H = \{R_1 \cdot I, R_1 \cdot S_1\} = \{R_1, S_3\}$
$R_2H = \{R_2 \cdot I, R_2 \cdot S_1\} = \{R_2, S_2\}$

Note: $IH = S_1H$, $S_2H = R_2H$, $S_3H = R_1H$

5.3 Lemma. *If G is a group and H is a subgroup of G then*

$$aH = bH \quad \textit{iff} \quad a^{-1}b \in H$$

Proof. $(\Rightarrow)$ Let $aH = bH$. Since $e \in H$ thus

$$b = be \in bH = aH$$

so $b = ah$ for some $h \in H$. Therefore $a^{-1}b = h \in H$.

$(\Leftarrow)$ Assume that $a^{-1}b \in H$. For any $h \in H$ we have

$$ah = a(a^{-1}b)(a^{-1}b)^{-1}h = b((a^{-1}b)^{-1})h \in bH$$

This gives: $aH \subseteq bH$. Also for any $h \in H$ we have:

$$bh = (aa^{-1})bh = a(a^{-1}b)h \in aH$$

so $bH \subseteq aH$. Therefore $aH = bH$. $\square$

5.4 Proposition. *If H is a subgroup of G then for any $a, b \in G$ either*

$$aH = bH \quad \textit{or} \quad aH \cap bH = \varnothing$$

Proof. Let $aH \cap bH \neq \varnothing$ and let $c \in aH \cap bH$. Then

$$ah_1 = c = bh_2$$

for some $h_1, h_2 \in H$. This gives

$$a^{-1}b = h_1h_2^{-1} \in H$$

and so $aH = bH$ by (5.3). $\square$

5.5 Corollary. *If H is a subgroup of G then every element of G belongs to one and only left coset of H.*

5.6 Note. In general $aH \neq Ha$. For example, If $H \subseteq G_T$, $H = \{I, S_1\}$ then

$$S_2H = \{S_2, R_2\}, \quad HS_2 = \{S_2, R_1\}$$

5.7 Proposition. *A subgroup H of G is normal iff*

$$aH = Ha \quad \forall a \in G$$

Proof. Exercise. $\square$

5.8 Notation. If H is a subgroup of G then

$$G/H := \text{the set of all left cosets of } H \text{ in } G$$

5.9. Multiplication of cosets. Let $H \subseteq G$, $aH, bH \in G/H$. Define

$$aH \cdot bH := (ab)H$$

5.10 Note. In general this is not well defined, i.e. we may have $aH = a'H$, $bH = b'H$ but $(ab)H \neq (a'b')H$.

For example, take $H = \{I, S_1\} \subseteq G_T$. Recall:

$$S_2H = R_2H = \{S_2, R_2\}, \; S_3H = R_1H = \{S_3, R_1\}$$

However:

$$(S_2S_3)H = R_1H = \{R_1, S_2\}$$
$$(R_2R_1)H = IH = \{I, S_1\}$$

5.11 Proposition. *If H is a normal subgroup of G then the multiplication of cosets given in (5.9) is well defined.*

Proof. If $H \lhd G$ then by (5.7) we have $aH = Ha \;\; \forall_{a \in G}$.

Let $aH = a'H$, $bH = b'H$. Then

$$(ab)H = a(bH) = a(b'H) = a(Hb') = (aH)b' = (a'H)b' = a'(b'H) = (a'b')H$$

$\square$

5.12 Corollary/Definition. If $H \lhd G$ then G/H is a group with multiplication defined by (5.9). The identity elements in G/H is the coset $eH = H \in G/H$. The inverse of a coset aH is the coset $a^{-1}H$.

The group G/H is called the *quotient group* (or the *factor group*) of G by H.

5.13 Example.

Take $\mathbb{Z}$, the additive group of integers. Since $\mathbb{Z}$ is abelian every its subgroup is normal. For $n \in \mathbb{Z}$, $n \geq 2$ define

$$n\mathbb{Z} = \{na \mid a \in \mathbb{Z}\}$$

e.g. $2\mathbb{Z} = \{\cdots - 4, -2, 0, 2, 4, \dots\}$, $5\mathbb{Z} = \{\dots, -10, -5, 0, 5, 10, \dots\}$

Note: $n\mathbb{Z}$ is a subgroup of $\mathbb{Z}$.

Cosets of $n\mathbb{Z}$ in $\mathbb{Z}$:

$$k + n\mathbb{Z} = \{k + na \mid a \in \mathbb{Z}\}$$

e.g. $1 + 5\mathbb{Z} = \{\cdots - 9, -4, 1, 6, 11, \dots\}$, $3 + 5\mathbb{Z} = \{\dots, -7, -2, 3, 8, 13, \dots\}$

Note: $k + n\mathbb{Z} = l + n\mathbb{Z}$ iff $(k - l) \in n\mathbb{Z}$ i.e. iff $k = l + na$ for some $a \in \mathbb{Z}$.

E.g.:

$$1 + 5\mathbb{Z} = 6 + 5\mathbb{Z} = 11 + 5\mathbb{Z} = -4 + 5\mathbb{Z}$$

Recall: if $n, k \in \mathbb{Z}$ then there is a unique number $l \in \{0, 1, \dots, n-1\}$ such that

$$k = l + na$$

for some $a \in \mathbb{Z}$. Thus every coset of $n\mathbb{Z}$ can be uniquely written as $l + n\mathbb{Z}$ where $l \in \{0, 1, \dots, n-1\}$.

Denote $\bar{l} := l + n\mathbb{Z}$. Then $\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$

The addition table in $\mathbb{Z}/5\mathbb{Z}$:

$+$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$
$\bar{0}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{0}$
$\bar{2}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{0}$	$\bar{1}$
$\bar{3}$	$\bar{3}$	$\bar{4}$	$\bar{0}$	$\bar{1}$	$\bar{2}$
$\bar{4}$	$\bar{4}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$

Recall: A group G is cyclic if it is generated by a single element: $G = \langle a \rangle$ for some $a \in G$.

Note: For every n the group $\mathbb{Z}/n\mathbb{Z}$ is cyclic: $\mathbb{Z}/n\mathbb{Z} = \langle \bar{1} \rangle$.

5.14 Note. If $H \lhd G$ then we have a homomorphism

$$\pi\colon G \to G/H, \quad \pi(a) := aH$$

This is the *canonical epimorphism* of G onto G/H.

We have:

$$\begin{aligned}
\operatorname{Ker}(\pi) &= \{a \in G \mid \pi(a) = eH\} \\
&= \{a \in G \mid aH = eH\} \\
&= \{a \in G \mid e^{-1}a \in H\} \\
&= \{a \in G \mid a \in H\} \\
&= H
\end{aligned}$$

5.15 Corollary. *A subgroup $H \subseteq G$ is the kernel of some homomorphism $f\colon G \to K$ iff H is a normal subgroup*

6 Isomorphism theorems

6.1 Theorem. *If $f\colon G \to H$ is a homomorphism then there is a unique homomorphism*

$$\bar{f}\colon G/\operatorname{Ker}(f) \to H$$

such that the following diagram commutes:

$$\begin{tikzcd}
G \arrow[r, "f"] \arrow[d, "\pi"'] & H \\
G/\operatorname{Ker}(f) \arrow[ur, dashed, "\bar{f}"'] &
\end{tikzcd}$$

Moreover, $\bar{f}$ is a monomorphism and $\operatorname{Im}(\bar{f}) = \operatorname{Im}(f)$.

Proof. Denote: $K := \operatorname{Ker}(f)$. Define

$$\bar{f}\colon G/K \to H, \quad \bar{f}(aK) := f(a)$$

We have:

1) $\bar{f}$ is well defined:

If $aK = bK$ then $a^{-1}b \in K$, so $f(a^{-1}b) = e$. Thus

$$f(b) = f(aa^{-1}b) = f(a)f(a^{-1}b) = f(a)$$

2) $\bar{f}$ is a homomorphism (check).

3) $\bar{f}$ is a unique homomorphism satisfying $\bar{f} \circ \pi = f$

Indeed, if $g\colon G/K \to H$ is some other homomorphism and $g \circ \pi = f$ then

$$f(a) = g \circ \pi(a) = g(aK)$$

and so $g(aK) = \bar{f}(aK)$ for all $aK \in G/K$.

4) $\bar{f}$ is 1-1:

We need: $\operatorname{Ker}(\bar{f}) = \{eK\}$.
We have: if $\bar{f}(aK) = e$ then $f(a) = e$, so $a \in K$ and so $aK = eK$.

5) $\operatorname{Im}(f) = \operatorname{Im}(\bar{f})$ (obvious).

$\square$

6.2 First Isomorphism Theorem. *If $f\colon G \to H$ is an epimorphism then*

$$G/\operatorname{Ker}(f) \cong H$$

Proof. Take the map $\bar{f}\colon G/\operatorname{Ker}(f) \to H$. Then $\operatorname{Im}(\bar{f}) = \operatorname{Im}(f) = H$, so $\bar{f}$ is an epimorphism. Also, $\bar{f}$ is 1-1. Therefore $\bar{f}$ is a bijective homomorphism and thus it is an isomorphism. $\square$

6.3 Example.

Recall:

$$GL_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\}$$
$$SL_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid \det(A) = 1\}$$

$SL_n(\mathbb{R})$ is a normal subgroup of $GL_n(\mathbb{R})$.

We have the homomorphism

$$\det\colon GL_n(\mathbb{R}) \to \mathbb{R}^*$$

Since this is an epimorphism and $\operatorname{Ker}(\det) = SL_n(\mathbb{R})$ we get

$$GL_n(\mathbb{R})/SL_n(\mathbb{R}) \cong \mathbb{R}^*$$

6.4 Theorem. *If G is a cyclic group then $G \cong \{e\}$ or $G \cong \mathbb{Z}$ or $G \cong \mathbb{Z}/n\mathbb{Z}$ for some $n \geq 2$.*

Proof. Let $G = \langle a \rangle$ for some $a \in G$. Define

$$f\colon \mathbb{Z} \to G, \quad f(n) := a^n$$

Notice that

1) f is a homomorphism

2) f is onto.

Thus by the First Isomorphism Theorem $G \cong \mathbb{Z}/\operatorname{Ker}(f)$.

Check: all subgroups $H \subseteq \mathbb{Z}$ are of the form $H = n\mathbb{Z}$ for some $n \geq 0$.

It follows that $G \cong \mathbb{Z}/n\mathbb{Z}$ for some $n \geq 0$. Also:

- if $n = 0$ then $n\mathbb{Z} = 0\mathbb{Z} = \{0\}$ and $G \cong \mathbb{Z}/\{0\} \cong \mathbb{Z}$
- if $n = 1$ then $n\mathbb{Z} = 1\mathbb{Z} = \mathbb{Z}$ and $G \cong \mathbb{Z}/\mathbb{Z} \cong \{e\}$
- if $n \geq 2$ then $G \cong \mathbb{Z}/n\mathbb{Z}$.

□

6.5 Notation. If H, K are subgroups of G then

$$HK := \{hk \in G \mid h \in H, k \in K\}$$

6.6 Lemma. *If H, K are subgroups of G then HK is a subgroup of G iff*

$$HK = KH$$

Proof. Exercise. □

6.7 Second Isomorphism Theorem. *If H, K are subgroups of G and $H \lhd G$ then KH is a subgroup of G, $(H \cap K) \lhd K$ and*

$$K/(H \cap K) \cong KH/H$$

Proof. Exercise. □

6.8 Third Isomorphism Theorem. *Let* $K \subseteq H \subseteq G$. *If* K, H *are normal subgroups of* G *then* $K \lhd H$, $H/K \lhd G/K$ *and*

$$(G/K)/(H/K) \cong G/H$$

Proof. Exercise. $\square$

7 Index of a subgroup and order of an element

7.1 Definition. Let H be a subgroup of G. Then

$$[G\colon H] := \text{ the number of distinct left cosets of } H \text{ in } G$$

This number is called the *index* of H in G.

7.2 Note.

1) The number of left cosets of H in G is the same as the number of right cosets (check!), so also

$$[G\colon H] = \text{ the number of distinct right cosets of } H \text{ in } G$$

2) If H is a normal subgroup of G then $[G : H] = |G/H|$.

7.3 Lemma. *If H is a subgroup of G then every left (and right) coset of H in G has the same number of elements as H.*

Proof. If $a \in G$ then the map of sets

$$f : H \to aH, \quad f(h) = ah$$

is a bijection (check!). □

7.4 Theorem (Lagrange)**.** *If H is a subgroup of G then*

$$|G| = [G : H]|H|$$

Proof. Recall:

1) G is a disjoint union of left cosets of H (5.5)
2) each left coset has as many elements as H (7.3)

This gives:

$|G| = $ (number of left cosets of H) $\cdot$ (number of elements of H) $= [G : H] \cdot |H|$

$\square$

7.5 Definition. If $a \in G$ then the order $|a|$ of a is the order of the subgroup $\langle a \rangle \subseteq G$ generated by a.

7.6 Proposition. *$|a| = n$ if n is the smallest positive integer such that $a^n = e$, and $|a| = \infty$ if such n does not exist.*

Proof. Take the homomorphism

$$f\colon \mathbb{Z} \to \langle a \rangle, \quad f(k) = a^k$$

Note that f is onto. If $a^n \neq e$ for all $n > 0$ then $\mathrm{Ker}(f) = \{0\}$. Then f is an isomorphism and so $|\langle a \rangle| = |\mathbb{Z}| = \infty$.

If $a^n = e$ for some $n > 0$ and n is the smallest positive number with this property then $\mathrm{Ker}(f) = n\mathbb{Z}$ (check!). Therefore $\langle a \rangle \cong \mathbb{Z}/n\mathbb{Z}$ and so $|\langle a \rangle| = |\mathbb{Z}/n\mathbb{Z}| = n$.

$\square$

7.7 Proposition. *If G is a finite group and $a \in G$ then $|a|$ divides $|G|$.*

Proof. Follows from Lagrange's theorem (7.4). $\square$

7.8 Note. If a number n divides $|G|$ then G need not contain an element of order n. E.g. $|G_T| = 6$ but G_T does not have an element of order 6:

$$|S_1| = |S_2| = |S_3| = 2, \quad |R_1| = |R_2| = 3, \quad |I| = 1$$

We will see later that if p is a prime number that divides $|G|$ then G contains an element of order p.

7.9 Definition. An element $a \in G$ is a *torsion element* if $|a| < \infty$. A group G is a *torsion group* if all its elements are torsion elements. A group is *torsion free* if it has not torsion elements.

7.10 Examples.

1) Every finite group is a torsion group.
2) $\mathbb{Q}/\mathbb{Z}$ is also a torsion group.
3) $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ are torsion free.
4) $\mathbb{Q}^*$ is neither torsion nor torsion free: $2 \in \mathbb{Q}^*$ has infinite order, $-1 \in \mathbb{Q}^*$ has order 2.

8 Free groups and presentations of groups

8.1. Let S be a set. A *word* in S is a finite sequence of the form

$$w = x_1^{\lambda_1} x_2^{\lambda_2} \cdot \dots \cdot x_k^{\lambda_k}$$

where $x_i \in S$ and $\lambda_i = \pm 1$ for $i = 1, 2, \dots, k$. Also, take

$$e := \text{“the empty word”}$$

(i.e. the word corresponding to the sequence of length 0).

We identify two words if one can be obtained from the other by a series of “cancellations” and “insertions” of subwords of the form xx^{-1} and $x^{-1}x$, e.g.:

$$x_1x_2x_3x_3^{-1} \sim x_1x_2 \sim x_1x_4^{-1}x_4x_2 \sim x_1x_4^{-1}x_1x_1^{-1}x_4x_2$$

$$x_1x_1^{-1} \sim e$$

Let $F(S)$ be the set of equivalence classes of words under this equivalence relation.

Note. A word is *reduced* if it does not contain any subwords of the form xx^{-1} or $x^{-1}x$. Every equivalence class in $F(S)$ is represented by a unique reduced word.

Define multiplication in $F(S)$ by concatenation of words, e.g.:

$$(x_1x_2x_1^{-1}) \cdot (x_2x_3^{-1}) = x_1x_2x_1^{-1}x_2x_3^{-1}$$

F(S) with this multiplication becomes a group:

- the identity element in $F(S)$: e
- inverses in $F(S)$: $(x_1x_2x_3^{-1}x_2)^{-1} = x_2^{-1}x_3x_2^{-1}x_1^{-1}$

8.2 Definition. $F(S)$ is called the *free group* generated by the set S.

In general, a group G is *free* if $G \cong F(S)$ for some set S.

8.3 Note.

- If $S = \varnothing$ then $F(S) = \{e\}$ is the trivial group.
- If S consists of a single element, $S = \{x\}$ then $F(S)$ is an infinite cyclic group, so $F(S) \cong \mathbb{Z}$.

8.4 Note. We have a map of sets

$$i\colon S \to F(S), \quad i(x) = x$$

8.5 Theorem (The universal property of free groups)**.**
Let S be a set and G be a group. For any map of sets $f\colon S \to G$ there exists a unique homomorphism $\bar{f}\colon F(S) \to G$ such that the following diagram commutes:

$$\begin{tikzcd} S \arrow[r, "f"] \arrow[d, "i"'] & G \\ F(S) \arrow[ur, dashed, "\bar{f}"'] & \end{tikzcd}$$

Proof. $\bar{f}$ is defined by

$$\bar{f}(x_1^{\lambda_1} x_2^{\lambda_2} \cdot \dots \cdot x_k^{\lambda_k}) := f(x_1)^{\lambda_1} f(x_2)^{\lambda_2} \cdot \dots \cdot f(x_k)^{\lambda_k}$$

$\square$

8.6 Corollary. *Every group is the homeomorphic image of a free group.*

Proof. Let G be a group. Take the set

$$S := \{x_g \mid g \in G\}$$

We have a map of sets

$$f\colon S \to G, \quad f(x_g) := g$$

This gives a homomorphism $\bar{f}\colon F(S) \to G$. Since f is onto thus also $\bar{f}$ is onto, i.e. $G = \bar{f}(F(S))$. □

8.7 Note. By Corollary 8.6 and the First Isomorphism Theorem (6.2) we have

$$G \cong F(S)/\operatorname{Ker}(\bar{f})$$

One can show that any subgroup of a free group is free. In particular $\operatorname{Ker}(\bar{f})$ is free. This shows that any group is isomorphic to a quotient of two free groups.

8.8 Definition. Let S be a set and let R be a subset of $F(S)$. Then

$$\langle S \mid R\rangle := F(S)/H$$

where H is the smallest normal subgroup of $F(S)$ such that $R \subseteq H$. We say then that

- elements of S are *generators* of $\langle S \mid R\rangle$
- elements of R are *relations* (or *relators*) in $\langle S \mid R\rangle$

8.9 Definition. If G is a group and $G \cong \langle S \mid R\rangle$ for some set S and some subset $R \subseteq F(S)$ then we say that $\langle S \mid R\rangle$ is a *presentation* of G.

We say that a group G is *finitely presentable* if it has a presentation such that both S and R are finite sets.

8.10 Examples.

1) $F(S) \cong \langle S \mid \varnothing\rangle$

2) $\mathbb{Z}/n\mathbb{Z} \cong \langle x \mid x^n \rangle$

Note: in particular $\mathbb{Z}/6\mathbb{Z} \cong \langle x \mid x^6 \rangle$. Here is a different presentation of $\mathbb{Z}/6\mathbb{Z}$:

$$\mathbb{Z}/6\mathbb{Z} \cong \langle x, y \mid x^2,\ y^3,\ xyxy^2 \rangle$$

3) $G_T \cong \langle x, y \mid x^2, y^3, xyxy \rangle$ (isomorphism: $x \mapsto S_1$, $y \mapsto R_1$)

4) Recall: $S_n =$ the symmetric group on n letters (1.8)

$$S_n \cong \langle x_1, \ldots x_{n-1} \mid x_i^2,\ (x_i x_{i+1})^3,\ (x_i x_j)^2 \text{ for } |i-j| > 1 \rangle$$

(isomorphism: $x_i \mapsto \sigma_i$ where $\sigma_i \colon \{1, \ldots, n\} \to \{1, \ldots, n\}$, $\sigma_i(i) = i+1$, $\sigma_i(i+1) = i$ and $\sigma_i(j) = j$ for $j \neq i, i+1$).

9 Direct products, direct sums, and free abelian groups

9.1 Definition. A *direct product* of a family of groups $\{G_i\}_{i\in I}$ is a group $\prod_{i\in I} G_i$ defined as follows. As a set $\prod_{i\in I} G_i$ is the cartesian product of the groups G_i. Given elements $(a_i)_{i\in I}, (b_i)_{i\in I} \in \prod_{i\in I} G_i$ we set

$$(a_i)_{i\in I} \cdot (b_i)_{i\in I} := (a_i b_i)_{i\in I}$$

9.2 Definition. A *weak direct product* of a family of groups $\{G_i\}_{i\in I}$ is the subgroup of $\prod_{i\in I} G_i$ given by

$$\prod_{i\in I}^{\mathrm{w}} G_i := \{(a_i)_{i\in I} \mid a_i \neq e_i \in G_i \text{ for finitely many } i \text{ only}\}$$

If all groups G_i are abelian then $\prod^{\mathrm{w}}_{i\in I} G_i$ is denoted $\bigoplus_{i\in I} G_i$ and it is called the *direct sum* of $\{G_i\}_{i\in I}$.

9.3 Note. If I is a finite set then $\prod_{i\in I} G_i = \prod^{\mathrm{w}}_{i\in I} G_i$.

9.4 Example.

$$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} = \{(0,0),(0,1),(1,0),(1,1)\}$$

Note. $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ is a the smallest non-cyclic group. It is called the Klein four group.

9.5 Example.

$$\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} = \{(0,0),(0,1),(0,2),(1,0),(1,1),(1,2)\}$$

Note. $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$ is a cyclic group ($(1,1)$ is a generator), thus

$$\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \cong \mathbb{Z}/6\mathbb{Z}$$

9.6. Let S be a set. Denote by $F_{\mathsf{ab}}(S)$ the set of all expressions of the form

$$\sum_{x\in S} k_x x$$

where $k_x \in \mathbb{Z}$ and $k_x \neq 0$ for finitely many $x \in X$ only.

$F_{\mathsf{ab}}(S)$ is an abelian group with addition defined by

$$\sum_{x\in S} k_x x + \sum_{x\in S} l_x x := \sum_{x\in S}(k_x + l_x)x$$

9.7 Definition. The group $F_{\mathsf{ab}}(S)$ is called the *free abelian group* generated by the set S.

In general a group G is *free abelian* if $G \cong F_{\mathsf{ab}}(S)$ for some set S.

9.8 Proposition. *If S is a set then*

$$F_{\mathsf{ab}}(S) \cong \bigoplus_{x\in S} \mathbb{Z}$$

Proof. The isomorphism is given by

$$f\colon F_{\mathsf{ab}}(S) \to \bigoplus_{x\in S}\mathbb{Z}, \qquad f(\sum_{x\in S} k_x x) = (k_x)_{x\in S}$$

$\square$

9.9 Note. We have a map of sets

$$i\colon S \to F_{\mathsf{ab}}(S), \quad i(x) = 1\cdot x$$

9.10 Theorem (The universal property of free abelian groups)**.**
Let S be a set and G be an abelian group. For any map of sets $f\colon S \to G$ there exists a unique homomorphism $\bar{f}\colon F(S) \to G$ such that the following diagram commutes:

$$\begin{tikzcd}
S \arrow[r, "f"] \arrow[d, "i"'] & G \\
F_{\textsf{ab}}(S) \arrow[ur, dashed, "\bar{f}"'] &
\end{tikzcd}$$

Proof. Define $\bar{f}$ by

$$\bar{f}\left(\sum_{x\in S} k_x x\right) := \sum_{x\in S} k_x f(x)$$

Note: this is well defined since $k_x = 0$ for almost all $x \in S$. $\square$

10 Categories and functors

10.1 Definition. A *category* $\mathcal{C}$ consists of

1) a collection of *objects* $\mathrm{Ob}(\mathcal{C})$
2) for any $a, b \in \mathrm{Ob}(\mathcal{C})$ a set $\mathrm{Hom}_{\mathcal{C}}(a, b)$ of *morphisms* from a to b
3) for any $a, b, c \in \mathrm{Ob}(\mathcal{C})$ a function ("composition law")

$$\begin{array}{ccccc} \mathrm{Hom}_{\mathcal{C}}(a,b) & \times & \mathrm{Hom}_{\mathcal{C}}(b,c) & \to & \mathrm{Hom}_{\mathcal{C}}(a,c) \\ (f & , & g) & \mapsto & g \circ f \end{array}$$

such that the following conditions are satisfied:

- Associativity.
$$f \circ (g \circ h) = (f \circ g) \circ h$$
for any morphisms f, g, h for which these compositions are defined.

- Identity. For any $c \in \mathrm{Ob}(\mathcal{C})$ there is a morphism $\mathrm{id}_c \in \mathrm{Hom}_{\mathcal{C}}(c, c)$ such that
$$f \circ \mathrm{id}_c = f, \quad \mathrm{id}_c \circ g = g$$
for any $f \in \mathrm{Hom}_{\mathcal{C}}(c, d), g \in \mathrm{Hom}_{\mathcal{C}}(b, c)$.

10.2 Examples.

1) $\mathcal{S}et$ = the category of all sets.
   - $\mathrm{Ob}(\mathcal{S}et)$ = the collection of all sets
   - $\mathrm{Hom}_{\mathcal{S}et}(A, B) = \{$ all maps of sets $f \colon A \to B$ $\}$

2) $\mathcal{G}r$ = the category of all groups
   - $\mathrm{Ob}(\mathcal{G}r)$ = the collection of all groups
   - $\mathrm{Hom}_{\mathcal{G}r}(G, H) = \{$ all homomorphisms $f \colon G \to H$ $\}$

3) $\mathcal{A}b$ = the category of all abelian groups

- $\mathrm{Ob}(\mathcal{A}b)$ = the collection of all abelian groups
- $\mathrm{Hom}_{\mathcal{A}b}(G, H) = \{$ all homomorphisms $f\colon G \to H$ $\}$

4) $\mathcal{T}op$ = the category of all topological spaces

- $\mathrm{Ob}(\mathcal{T}op)$ = the collection of all topological spaces
- $\mathrm{Hom}_{\mathcal{T}op}(X, Y) = \{$ all continuous maps $f\colon X \to Y$ $\}$

5) Let G be a group. Define a category $\mathcal{C}_G$ as follows:

- $\mathrm{Ob}(\mathcal{C}_G) = \{*\}$
- $\mathrm{Hom}_{\mathcal{C}_G}(*, *) = \{$ elements of G $\}$
- composition of morphisms = multiplication in G

6) A very small category $\mathcal{C}$:

$$c \xrightarrow{f} d$$

- $\mathrm{Ob}(\mathcal{C}) = \{c, d\}$
- $\mathrm{Hom}_{\mathcal{C}}(c, d) = \{f\}$
  $\mathrm{Hom}_{\mathcal{C}}(d, c) = \varnothing$
  $\mathrm{Hom}_{\mathcal{C}}(c, c) = \mathrm{id}_c$
  $\mathrm{Hom}_{\mathcal{C}}(d, d) = \mathrm{id}_d$

10.3 Definition. A morphism $f\colon c \to d$ in a category $\mathcal{C}$ is an *isomorphism* if there exists a morphism $g\colon d \to c$ such that $gf = \mathrm{id}_c$ and $fg = \mathrm{id}_d$.

If for some $c, d \in \mathcal{C}$ there exist an isomorphism $f\colon c \to d$ then we say that the objects c and d are *isomorphic* and we write $c \cong d$.

10.4 Note. For an object $c \in \mathcal{C}$ define

$$\mathrm{Aut}(c) := \{ \text{ all isomorphisms } f\colon c \to c \ \}$$

$\mathrm{Aut}(c)$ with composition of morphisms is a group.

10.5 Definition. Let $\mathcal{C}, \mathcal{D}$ be categories. A *(covariant) functor* $F\colon \mathcal{C} \to \mathcal{D}$ consists of

1) an assignment
$$\mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D}), \quad c \mapsto F(c)$$

2) for every $c, c' \in \mathcal{C}$ a function
$$\mathrm{Hom}_{\mathcal{C}}(c, c') \to \mathrm{Hom}_{\mathcal{D}}(F(c), F(c')), \quad f \mapsto F(f)$$
such that $F(gf) = F(g)F(f)$ and $F(id_c) = \mathrm{id}_{F(c)}$.

10.6 Note. If $F\colon \mathcal{C} \to \mathcal{D}$ is a functor and $f\colon c \to c'$ is an isomorphism in $\mathcal{C}$ then $F(f)\colon F(c) \to F(c')$ is an isomorphism in $\mathcal{D}$.

In particular if $c \cong c'$ in $\mathcal{C}$ then $F(c) \cong F(c')$ in $\mathcal{D}$.

10.7 Examples.

1) $U\colon \mathcal{G}r \to \mathcal{S}et$

   If $G \in \mathcal{G}r$ then $U(G) = \{$ the set of elements of G $\}$

   If $f\colon G \to H$ is a homomorphism then $U(f)\colon U(G) \to U(H)$ is the map of sets underlying this homomorphism.

2) $U\colon \mathcal{A}b \to \mathcal{S}et$,

   defined the same way as in 1).

Note. The functors U in 1), 2) are called *forgetful functors*.

3) Let G be a group. The *commutator* of $a, b \in G$ is the element

$$[a, b] := aba^{-1}b^{-1}$$

Note: $[a, b] = e$ iff $ab = ba$.

The *commutator subgroup* of G is the subgroup $[G, G] \subseteq G$ generated by the set $S = \{[a, b] \mid a, b \in G\}$.

Note.

(a) $[G, G] = \{e\}$ iff G is an abelian group.

(b) $[G, G]$ is a normal subgroup of G (check!).

(c) $G/[G, G]$ is an abelian group (check!).

(d) If $f\colon G \to H$ is a homomorphism then $f([G, G]) \subseteq [H, H]$.

(e) If $f\colon G \to H$ is a homomorphism then f induces a homomorphism

$$f_{\mathsf{ab}}\colon G/[G, G] \to H/[H, H]$$

given by $f_{\mathsf{ab}}(a[G, G]) = f(a)[H, H]$.

The *abelianization functor* $\mathrm{Ab}\colon \mathcal{G}r \to \mathcal{A}b$ is given by

$$\mathrm{Ab}(G) := G/[G, G], \qquad \mathrm{Ab}(f) := f_{\mathsf{ab}}$$

4) Recall: if S is a set then $F(S)$ is the free group generated by S.

A map of sets $f\colon S \to T$ defines a homomorphism

$$\tilde{f}\colon F(S) \to F(T)$$

given by $\tilde{f}(x_1^{\lambda_1} x_2^{\lambda_2} \cdot \dots \cdot x_k^{\lambda_k}) = f(x_1)^{\lambda_1} f(x_2)^{\lambda_2} \cdot \dots \cdot f(x_k)^{\lambda_k}$.

Check: the assignment

$$S \mapsto F(S), \quad (f\colon S \to T) \mapsto (\tilde{f}\colon F(S) \to F(T))$$

Defines a functor $F\colon \mathcal{S}et \to \mathcal{G}r$. This is the *free group functor*.

5) Similarly we have the *free abelian group functor*

$$F_{\mathsf{ab}} \colon \mathcal{S}et \to \mathcal{A}b$$

where

- $F_{\mathsf{ab}}(S) =$ the free abelian group generated by the set S
- if $f \colon S \to T$ then $F_{\mathsf{ab}}(f) \colon F_{\mathsf{ab}}(S) \to F_{\mathsf{ab}}(T)$ is given by

$$F_{\mathsf{ab}}(f)\left(\sum_{x \in S} k_x x\right) = \sum_{x \in S} k_x f(x)$$

11 Adjoint functors

11.1 Definition. Given two functors

$$L\colon \mathcal{C} \to \mathcal{D} \quad \text{and} \quad R\colon \mathcal{D} \to \mathcal{C}$$

we say that L is the *left adjoint functor* of R and that R is the *right adjoint functor* of L if for any object $c \in \mathcal{C}$ we have a morphism $\eta_c\colon c \to RL(c)$ such that:

1) for any morphism $f\colon c \to c'$ in $\mathcal{C}$ the following diagram commutes:

$$\begin{tikzcd} c \arrow[r, "f"] \arrow[d, "\eta_c"'] & c' \arrow[d, "\eta_{c'}"] \\ RL(c) \arrow[r, "RL(f)"'] & RL(c') \end{tikzcd}$$

2) for any $c \in \mathcal{C}$ and $d \in \mathcal{D}$ the map of sets

$$\begin{aligned} \mathrm{Hom}_{\mathcal{D}}(L(c), d) &\longrightarrow \mathrm{Hom}_{\mathcal{C}}(c, R(d)) \\ (L(c) \xrightarrow{f} d) \quad &\longmapsto \quad (c \xrightarrow{\eta_c} RL(c) \xrightarrow{R(f)} R(d)) \end{aligned}$$

is a bijection.

In such situation we say that (L, R) is an *adjoint pair of functors*.

11.2 Note.

1) The collection of morphisms $\{\eta_c\}_{c\in\mathcal{C}}$ is called the *unit of adjunction* of (L, R).

2) For any adjoint pair (L, R) we also have morphisms $\{\varepsilon_d\colon LR(d) \to d\}_{d\in\mathcal{D}}$ satisfying analogous conditions as $\{\eta_c\}_{c\in\mathcal{C}}$. This collection of morphisms is called the *counit of the adjunction*.

11.3 Note. The morphism η_c is universal in the following sense. For any $d \in \mathcal{D}$ and any morphism $f\colon c \to R(d)$ in $\mathcal{C}$ there is a unique morphism $\bar{f}\colon L(c) \to d$

in $\mathcal{D}$ such that the following diagram commutes:

$$\begin{tikzcd} c \arrow[r, "f"] \arrow[d, "\eta_c"'] & R(d) \\ RL(c) \arrow[ur, dashed, "R(\bar{f})"'] & \end{tikzcd}$$

This property is equivalent to part 2) of Definition 11.1.

11.4 Examples.

1) Recall that we have functors:

$$F\colon \mathcal{S}et \to \mathcal{G}r, \qquad \mathcal{G}r \leftarrow \mathcal{S}et\colon U$$

where $F =$ free group functor, $U =$ forgetful functor.

The pair (F, U) is an adjoint pair. For $S \in \mathcal{S}et$ the unit of adjunction is given by the function

$$i_S\colon S \to UF(S), \quad i_S(x) = x$$

The universal property of free groups (9.10) says that for any $G \in \mathcal{G}r$ and any map of sets $f\colon S \to U(G)$ there is a unique homomorphism $\bar{f}\colon F(S) \to G$ such that we have a commutative diagram

$$\begin{tikzcd} S \arrow[r, "f"] \arrow[d, "i_S"'] & U(G) \\ UF(S) \arrow[ur, dashed, "U(\bar{f})"'] & \end{tikzcd}$$

2) We have functors

$$F_{\mathsf{ab}}\colon \mathcal{S}et \to \mathcal{A}b, \qquad \mathcal{S}et \leftarrow \mathcal{A}b\colon U$$

where $F_{\mathsf{ab}} =$ free abelian group functor, $U =$ forgetful functor. Similarly as in 1) one can check that (F_{ab}, U) is an adjoint pair.

3) Recall that we have the abelianization functor

$$\mathrm{Ab}\colon \mathcal{G}r \to \mathcal{A}b, \qquad \mathrm{Ab}(G) = G/[G,G]$$

This functor is left adjoint to the inclusion functor

$$J\colon \mathcal{A}b \to \mathcal{G}r, \qquad J(G) = G$$

(check!).

11.5 Note. It is not true that every functor has a left or right adjoint.

12 Categorical products and coproducts

12.1 Definition. Let $\{c_i\}_{i\in I}$ be a family of objects in a category $\mathcal{C}$. A *(categorical) product* of the family $\{c_i\}_{i\in I}$ is an object $p \in \mathcal{C}$ equipped with morphisms $\pi_i \colon p \to c_i$ for all $i \in I$ that satisfies the following universal property. For any object $d \in \mathcal{C}$ and a family of morphisms $\{f_i \colon d \to c_i\}_{i\in I}$ there exists a unique morphism $f \colon d \to p$ such that $\pi_i f = f_i$ for all $i \in I$.

$$\begin{tikzcd}
d \arrow[dr, dashed, "f"] \arrow[drr, bend left, "f_1"] \arrow[ddr, bend right, "f_2"'] & & \\
& p \arrow[r, "\pi_1"'] \arrow[d, "\pi_2"] & c_1 \\
& c_2 &
\end{tikzcd}$$

12.2 Note. If a categorical product of $\{c_i\}_{i\in I}$ exists then it is defined uniquely up to isomorphism. We then write:

$$p = \prod_{i\in I} c_i$$

12.3 Examples.

1) In the category of groups $\mathcal{G}r$ the categorical product of a family $\{G_i\}_{i\in I}$ is the direct product of groups $\prod_{i\in I} G_i$.

   Indeed, we have projection homomorphisms:

   $$\pi_{i_0} \colon \prod_{i\in I} G_i \to G_{i_0}, \quad \pi_{i_0}((g_i)_{i\in I}) = g_{i_0}$$

   Also, if for some group H we have homomorphisms $f_i \colon H \to G_i$ then this defines a homomorphism

   $$f \colon H \to \prod_{i\in I} G_i, \quad f(h) = (f_i(h))_{i\in I}$$

   Moreover, f is the unique homomorphism such that we have $\pi_i f = f_i$.

2) By a similar argument if $\{G_i\}_{i\in I}$ is a family of abelian groups then the direct product $\prod_{i\in I} G_i$ is the categorical product of the family $\{G_i\}_{i\in I}$ in the category $\mathcal{A}b$.

3) In the category $\mathcal{S}et$ the categorical product of a family of sets $\{A_i\}_{i\in I}$ is the cartesian product of sets $\prod_{i\in I} A_i$.

12.4 Definition. Let $\{c_i\}_{i\in I}$ be a family of objects in a category $\mathcal{C}$. A *(categorical) coproduct* of the family $\{c_i\}_{i\in I}$ is an object $d \in \mathcal{C}$ equipped with morphisms $\varepsilon_i\colon c_i \to d$ for all $i \in I$ that satisfies the following universal property. For any object $b \in \mathcal{C}$ and a family of morphisms $\{f_i\colon c_i \to b\}_{i\in I}$ there exists a unique morphism $f\colon d \to b$ such that $f\varepsilon_i = f_i$ for all $i \in I$.

$$\begin{tikzcd}
 & c_1 \arrow[d, "\varepsilon_1"'] \arrow[ddr, bend left, "f_1"] & \\
c_2 \arrow[r, "\varepsilon_2"] \arrow[drr, bend right, "f_2"'] & d \arrow[dr, dashed, "f"] & \\
 & & b
\end{tikzcd}$$

12.5 Note. If a categorical coproduct of $\{c_i\}_{i\in I}$ exists then it is defined uniquely up to isomorphism. We then write:

$$d = \coprod_{i\in I} c_i$$

12.6 Examples.

1) In the category of sets $\mathcal{S}et$ the categorical coproduct of a family of sets $\{A_i\}_{i\in I}$ is the disjoint union of sets $\coprod_{i\in I} A_i$.

2) In the category of abelian groups $\mathcal{A}b$ the categorical coproduct of a family of abelian groups $\{G_i\}_{i\in I}$ is the direct sum $\bigoplus_{i\in I} G_i$.

The homomorphisms $\varepsilon_{i_0}\colon G_{i_0} \to \bigoplus_{i\in I} G_i$ are given by $g \mapsto (g_i)_{i\in I}$ where

$$g_i = \begin{cases} g & \text{if } \ i = i_0 \\ e_{G_i} & \text{otherwise} \end{cases}$$

Given an abelian group H and homomorphisms $f_i\colon G_i \to H$ we have a homomorphism

$$f\colon \bigoplus_{i\in I} G_i \to H, \qquad f((g_i)_{i\in I}) = \sum_{i\in I} f_i(g_i)$$

This is the unique homomorphism satisfying $f\varepsilon_i = f_i$ for all $i \in I$.

3) If $\{G_i\}_{i\in I}$ is a family of groups then $\prod^{\mathrm{w}}_{i\in I} G_i$ is not, in general, a coproduct of $\{G_i\}_{i\in I}$. Take e.g. $G_1 = \mathbb{Z}/2\mathbb{Z}$, $G_2 = \mathbb{Z}/3\mathbb{Z}$. We have homomorphisms

$$\begin{aligned} f_1\colon \mathbb{Z}/2\mathbb{Z} \to G_T, &\qquad f(1) = S_1 \\ f_2\colon \mathbb{Z}/3\mathbb{Z} \to G_T, &\qquad f(1) = R_1 \end{aligned}$$

However, there is no homomorphism $f\colon \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \to G_T$ such that $f\varepsilon_i = f$ for $i = 1, 2$.

12.7. Construction of coproducts in $\mathcal{G}r$.

Let $\{G_i\}_{i\in I}$ be a family of groups, and let $S = \coprod_{i\in I} G_i$ be the disjoint union of sets of elements of these groups. A *word* in S is a sequence

$$a_1 a_2 \dots a_k$$

where $k \geq 0$ and $a_1, a_2, \dots, a_k \in S$. Consider the equivalence relation of words generated by the following conditions:

1) if e_{G_i} is the trivial element in G_i for some $i \in I$ then

$$a_1 \dots a_j a_{j+1} \dots a_k \ \sim \ a_i \dots a_j e_{G_i} a_{j+1} \dots a_k$$

2) if a_j, a_{j+1} belong to the same group G_i for some $i \in I$ then

$$a_1 \dots a_j a_{j+1} \dots a_k \sim a_1 \dots \underbrace{(a_j a_{j+1})}_{\text{product in } G_i} \dots a_k$$

Denote

$$\prod_{i\in I}^{*} G_i := \{ \text{ equivalence classes of words } \}$$

This set is a group with multiplication defined by concatenation of words.

12.8 Definition. The group $\prod^{*}_{i\in I} G_i$ is called the *free product* of the family $\{G_i\}_{i\in I}$

12.9 Proposition. *If $\{G_i\}_{i\in I}$ is a family of groups then $\prod^{*}_{i\in I} G_i$ is the coproduct of the family $\{G_i\}_{i\in I}$ in the category of groups.*

12.10 Note. The free product $\prod^{*}_{i\in I} \mathbb{Z}$ is isomorphic to the free group generated by the set I.

13 More on free abelian groups

Recall. G is a free abelian group if

$$G \cong \bigoplus_{i \in I} \mathbb{Z}$$

for some set I.

13.1 Definition. Let G be an abelian group. A set $B \subseteq G$ is a *basis* of G if

- B generates G
- if for some $x_1, \dots x_k \in B$ and $n_1, \dots, n_k \in \mathbb{Z}$ we have

  $$n_1 x_1 + \cdots + n_k x_k = 0$$

  then $n_1 = \cdots = n_k = 0$.

13.2 Theorem. *An abelian group G has a basis iff G is a free abelian group.*

Proof. If B is a basis of G then the map

$$f \colon \bigoplus_{x \in B} \mathbb{Z} \to G, \quad f((n_x)_{x \in B}) = \sum_{x \in B} n_x x$$

is an isomorphism (check!).

Conversely, if we have an isomorphism

$$f \colon \bigoplus_{i \in I} \mathbb{Z} \to G$$

then for $j \in I$ take $\delta_j \in \bigoplus_{i \in I} \mathbb{Z}$ where $\delta_j = (n_i)_{i \in I}$ such that

$$n_i = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

Then $B = \{f(\delta_j)\}_{i \in I}$ is a basis of G (check!).

$\square$

13.3 Proposition. *If G is a free abelian group then any two bases of G have the same cardinality.*

Notation. If X is a set then $|X| =$ cardinality of X.

13.4 Lemma. *If $\{G_i\}_{i\in I}$ is a family of abelian groups and H_i is a subgroup of G_i for $i \in I$ then*

$$\bigoplus_{i\in I} G_1 \;/\; \bigoplus_{i\in I} H_i \;\;\cong\;\; \bigoplus_{i\in I}(G_i/H_i)$$

Proof. Exercise. □

Proof of Proposition 13.3.

Let B, B' be two bases of G. We want to show: $|B| = |B'|$.

We have isomorphisms

$$\bigoplus_{x\in B}\mathbb{Z} \cong G \cong \bigoplus_{y\in B'}\mathbb{Z}$$

Case 1. B, B' - finite sets , $|B| = m$, $|B'| = n$.

Take $2G := \{2a \mid a \in G\}$. This is a subgroup G. Since $G \cong \bigoplus_{x\in B}\mathbb{Z}$, using Lemma 13.4 we obtain

$$G/2G \cong \bigoplus_{x\in B}\mathbb{Z}/2\mathbb{Z}$$

Similarly, since $G \cong \bigoplus_{x\in B'}\mathbb{Z}$ we have

$$G/2G \cong \bigoplus_{y\in B'}\mathbb{Z}/2\mathbb{Z}$$

This gives:

$$2^m = |\bigoplus_{x\in B}\mathbb{Z}/2\mathbb{Z}| = |G/2G| = |\bigoplus_{y\in B'}\mathbb{Z}/2\mathbb{Z}| = 2^n$$

so $m = n$.

Case 2. B - finite set, B' - infinite set.

As before this would give:

$$\bigoplus_{x\in B} \mathbb{Z}/2\mathbb{Z} \cong \ G/2G \ \cong \bigoplus_{x\in B'} \mathbb{Z}/2\mathbb{Z}$$

This is however impossible since $\bigoplus_{x\in B} \mathbb{Z}/2\mathbb{Z}$ is a finite group and $\bigoplus_{y\in B'} \mathbb{Z}/2\mathbb{Z}$ is an infinite group.

Case 3. B, B' - infinite sets.

Check: if B is an infinite set then

$$|B| = |\bigoplus_{x\in B} \mathbb{Z}|$$

It follows that we have

$$|B| = |\bigoplus_{x\in B} \mathbb{Z}| = |G| = |\bigoplus_{y\in B'} \mathbb{Z}| = |B'|$$

□

13.5 Definition. If G is a free abelian group then the *rank* of G is the cardinality of a basis of G.

Note. By (13.3) rank of G does not depend on the choice of basis of G.

13.6 Theorem. *Let G be a free abelian group of a finite rank n and let H be a subgroup of G. Then H is a free abelian group and*

$$\operatorname{rank} H \leq \operatorname{rank} G$$

Note. This theorem is true also if G is a free abelian group of an infinite rank.

13.7 Lemma. *If $f\colon G \to H$ is an epimorphism of abelian groups and H is free abelian group then*

$$G \cong H \oplus \operatorname{Ker}(f)$$

Proof. Recall (from homework): if we have homomorphisms

$$f\colon G \to H, \quad g\colon H \to G$$

such that $fg = \mathrm{id}_H$ then $G \cong H \oplus \mathrm{Ker}(f)$. If follows that we only need to construct the homomorphism g.

Take a basis B of H. Since f is onto, for every $x \in B$ there is $a_x \in G$ such that $f(a_x) = x$. Let $g\colon H \to G$ be the unique homomorphism satisfying

$$g(x) = a_x$$

for all $x \in B$. Then $fg(x) = x$ for all $x \in B$ and so $fg = \mathrm{id}_H$. □

Proof of Theorem 13.6.

We can assume that $G = \mathbb{Z}^n$.

We want to show: if $H \subseteq \mathbb{Z}^n$ then H is a free abelian group and $\mathrm{rank}\, H \leq n$.

Induction with respect to n:

If $n = 1$ then $H = k\mathbb{Z}$ for some $k \geq 0$ so $H = \{0\}$ or $H \cong \mathbb{Z}$.

Next, assume that for some n every subgroup of $\mathbb{Z}^n$ is a free abelian group of rank $\leq n$, and let $H \subseteq \mathbb{Z}^{n+1}$. Take the homomorphism

$$f\colon \mathbb{Z}^{n+1} \to \mathbb{Z}, \quad f(m_1, \ldots, m_{n+1}) = m_{n+1}$$

We have:

$$\mathrm{Ker}(f) = \{(m_1, \ldots, m_n, 0) \mid m_i \in \mathbb{Z}\} \cong \mathbb{Z}^n$$

We have an epimorphism:

$$f|_H\colon H \to \mathrm{Im}(f)$$

Since $\mathrm{Im}(f|_H) \subseteq \mathbb{Z}$, thus $\mathrm{Im}(f|_H)$ is a free abelian group and so by Lemma 13.7 we have

$$G \cong \mathrm{Im}(f|_H) \oplus \mathrm{Ker}(f|_H)$$

We also have:

$$\mathrm{Ker}(f|_H) = \mathrm{Ker}(f) \cap H$$

It follows that that $\mathrm{Ker}(f|_H)$ is a subgroup of $\mathrm{Ker}(f)$, and since $\mathrm{Ker}(f)$ is a free abelian group of rank n by the inductive assumption we get that $\mathrm{Ker}(f|_H)$ is a free abelian group of rank $\leq n$. Therefore

$$H \cong \underbrace{\mathrm{Im}(f|_H)}_{\substack{\text{free abelian} \\ \text{rank } \leq 1}} \oplus \underbrace{\mathrm{Ker}(f|_H)}_{\substack{\text{free abelian} \\ \text{rank } \leq n}}$$

and so H is a free abelian group of rank $\leq n+1$. $\square$

14 Finitely generated abelian groups

Goal. Describe all isomorphism types of finitely generated abelian groups.

14.1 Proposition. *If G is an abelian group generated by n elements then*

$$G \cong F/H$$

where F is a free abelian group of rank n and H is some subgroup of F.

Proof. We have $G = \langle a_1, \dots a_n \rangle$ for some $a_1, \dots, a_n \in G$. Let $\{x_1, \dots, x_n\}$ be a basis of F. We have a homomorphism

$$f\colon F \to G, \qquad f(x_i) = a_i$$

Take $H = \mathrm{Ker}(f)$. Since f is an epimorphism by the First Isomorphism Theorem (6.2) we have

$$G \cong F/H$$

$\square$

Recall. If $\operatorname{rank} F = n$ and $H \subseteq F$ then H is a free abelian group of rank $\leq n$.

14.2 Theorem. *Let F be a free abelian group of rank n and let H be a subgroup of F There exists a basis $\{x_1, \dots, x_n\}$ of F and integers $d_1, \dots, d_r > 0$ such that*

- $d_i | d_{i+1}$ *for* $i = 1, \dots, r$
- $\{d_1 x_1, \dots, d_r x_r\}$ *is a basis of* H.

14.3 Theorem. *If G is a finitely generated abelian group then*

$$G \ \cong \ (\mathbb{Z}/d_1\mathbb{Z}) \oplus \ldots \oplus (\mathbb{Z}/d_r\mathbb{Z}) \oplus \mathbb{Z}^k$$

for some $k \geq 0$ and $d_1, \dots, d_r > 0$ such that $d_i | d_{i+1}$ for $i = 1, \dots, r$.

Proof. By (14.1) we have

$$G \cong F/H$$

for some free abelian group F of finite rank and $H \subseteq F$. Let $\operatorname{rank} F = n$. By Theorem 14.2 there is a basis $\{x_1, \ldots, x_n\}$ of F such that $\{d_1 x_1, \ldots, d_r x_r\}$ is a basis of H for some $d_1, \ldots, d_r > 0$, $d_i | d_{i+1}$.

We have an isomorphism

$$f\colon F \to \underbrace{\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}}_{n \text{ times}}$$

where $f(x_1) = (1, 0, \ldots, 0), \quad f(x_2) = (0, 1, \ldots, 0), \ \ldots \ , f(x_n) = (0, \ldots, 0, 1)$
Notice that

$$f(H) = d_1\mathbb{Z} \oplus \ldots \oplus d_r\mathbb{Z} \ \oplus \{0\} \oplus \ldots \oplus \{0\}$$

. This gives

$$G \cong (\mathbb{Z} \oplus \cdots \oplus \mathbb{Z})/(d_1\mathbb{Z} \oplus \ldots \oplus d_r\mathbb{Z} \ \oplus \{0\} \oplus \ldots \oplus \{0\})$$

Using (13.4) we obtain

$$G \cong \ (\mathbb{Z}/d_1\mathbb{Z}) \oplus \ldots \oplus (\mathbb{Z}/d_r\mathbb{Z}) \oplus \mathbb{Z}^k$$

where $k = n - r$. □

Proof of Theorem 14.2.

Let $\{y_1, \ldots, y_n\}$ be any basis of F and let $\{h_1, \ldots, h_m\} \subseteq H$ be any set generating H. We have

$$h_i = a_{i1}y_1 + a_{i2}y_2 + a_{in}y_n$$

for some $a_{ij} \in \mathbb{Z}$. Consider the matrix

$$A = \begin{pmatrix} a_{11} & \ldots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \ldots & a_{mn} \end{pmatrix}$$

Note: columns of A correspond to basis elements of F and rows of A correspond to generators of H.

Consider the following operations on matrices:

1) interchange of two rows
2) multiplication of a row by (-1)
3) addition of a multiple of one row to another row.

These operations are called *elementary row operations* for matrices of integers. *Elementary column operations* are defined analogously.

Notice that:

- application of an elementary row operation to the matrix A corresponds to replacing of the set of generators of H by another set of generators of H;
- application of an elementary column operation to the matrix A corresponds to passing to a new basis of F and rewriting the generators of H in terms of this new basis.

Key step. Starting with any matrix of integers A and applying a sequence of elementary row and column operations we can obtain a matrix of the form

$$B = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}$$

where 0's denote zero matrices (of appropriate dimensions) and D is a square diagonal matrix

$$D = \begin{pmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_r \end{pmatrix}$$

for some $d_1, \dots, d_r > 0$ such that $d_i | d_{i+1}$ for all i.

Note. The matrix B is called the *Smith normal form* of the matrix A.

Let $\{x_1, \dots, x_n\}$ be the basis of F corresponding to columns of the matrix B. In terms of this basis a set of generators of H is given by $\{d_1 x_1, \dots, d_r x_r, 0, \dots, 0\}$. It follows that $\{d_1 x_1, \dots d_r x_r\}$ is a basis of H.

$\square$

Key step. How to compute the Smith normal form of a matrix A.

Step 1. Produce a matrix of the form

$$A_1 = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & A_1' & \\ 0 & & & \end{pmatrix}$$

This can be done as follows.

(1a) By interchanging rows and columns if necessary we can make sure that $a_{11} \neq 0$. Also, by multiplying the first row by (-1) we can get $a_{11} > 0$.

(1b) By adding multiples of the first row to the other rows (and multiples of the first column to the other columns) we can make all other entries in the first row and the first column positive and smaller than a_{11}.

(1c) If all these other entries of the first row and column are 0 we are done. If some entry is non-zero then by replacing rows (or columns) we can move that entry to the $(1,1)$-position. Then we go back to (1b).

(1d) After a finite number of iterations we get a matrix of the form of A_1.

Step 2. Given a matrix A_1 as above make sure that the entry a_{11} divides all entries of A_1'.

This can be done as follows.

(2a) If all entries of A_1' are already divisible by a_{11} we are done.

(2b) If some entry a_{ij} is not divisible by a_{11} add the i-th row to the first row. Then go back to (1b).

(2c) After a finite number of iterations we get a matrix of the form of A_1 with a_{11} dividing all entries of A'.

Step 3. We are done with the first row and and the first column. Next we apply the same steps recursively to reduce A_1'.

14.4 Lemma. *Let G be a group and let $a_1, \dots, a_k \in G$ be elements such that $|a_i| = n_i$. Assume that for $i = 1, \dots, k$ we have*

$$\gcd(n_i, \prod_{j \neq i} n_i) = 1$$

Then $|a_1 \cdot \ldots \cdot a_k| = n_1 \cdot \ldots \cdot n_k$.

Proof. Exercise. □

14.5 Corollary. *If $m > 0$ is an integer and $m = p_1^{n_1} \cdot \ldots \cdot p_k^{n_k}$ where $p_1, \dots, p_k$ are distinct primes then*

$$\mathbb{Z}/m\mathbb{Z} \cong (\mathbb{Z}/p_1^{n_1}\mathbb{Z}) \oplus \ldots \oplus (\mathbb{Z}/p_k^{n_k}\mathbb{Z})$$

Proof. It is enough to show that the group $(\mathbb{Z}/p_1^{n_1}\mathbb{Z}) \oplus \ldots \oplus (\mathbb{Z}/p_k^{n_k}\mathbb{Z})$ contains an element of order n. This follows however from Lemma 14.4. □

14.6 Theorem. *If G is a finitely generated abelian group then G is isomorphic to a finite direct sum groups $\mathbb{Z}$ and $\mathbb{Z}/p^n\mathbb{Z}$ where p is a prime.*

Proof. This follows directly from (14.3) and (14.5). □

14.7 Theorem. *For any finitely generated abelian group G there are unique (up to order) integers $p_1^{n_1}, \dots, p_s^{n_s} > 1$ where $p_1, \dots, p_s$ are primes, and a unique integer $k \geq 0$ such that*

$$G \;\cong\; (\mathbb{Z}/p_1^{n_1}\mathbb{Z}) \oplus \ldots \oplus (\mathbb{Z}/p_k^{n_s}\mathbb{Z}) \oplus \mathbb{Z}^k$$

14.8 Note. The integers $p_1^{n_1}, \dots, p_s^{n_s}$ are called the *elementary divisors* of the group G.

14.9 Lemma. *If p is a prime number and*

$$0 < n_1 \leq \cdots \leq n_s \quad 0 < m_1 \leq \cdots \leq m_r$$

be integers such that

$$(\mathbb{Z}/p^{n_1}\mathbb{Z}) \oplus \cdots \oplus (\mathbb{Z}/p^{n_s}\mathbb{Z}) \ \cong \ (\mathbb{Z}/p^{m_1}\mathbb{Z}) \oplus \cdots \oplus (\mathbb{Z}/p^{m_r}\mathbb{Z})$$

Then $s = r$ and $n_i = m_i$ for all i.

Proof. Exercise. $\square$

Proof of Theorem 14.7.
Assume that

$$G \ \cong \ (\mathbb{Z}/p_1^{n_1}\mathbb{Z}) \oplus \ldots \oplus (\mathbb{Z}/p_s^{n_s}\mathbb{Z}) \oplus \mathbb{Z}^k \tag{1}$$

and

$$G \ \cong \ (\mathbb{Z}/q_1^{m_1}\mathbb{Z}) \oplus \ldots \oplus (\mathbb{Z}/q_r^{m_r}\mathbb{Z}) \oplus \mathbb{Z}^l \tag{2}$$

where $k, l \geq 0$, p_i, q_j are primes and and $n_1, m_j > 0$. We want to show that $k = l$, $r = s$ and (after reordering) $p_i^{n_i} = q_i^{m_i}$ for all i.

Define

$$G_t := \{a \in G \mid \ |a| < \infty\}$$

This is the *torsion subgroup* of G. From the isomorphism (1) we obtain:

$$G_t \cong (\mathbb{Z}/p_1^{n_1}\mathbb{Z}) \oplus \ldots \oplus (\mathbb{Z}/p_s^{n_s}\mathbb{Z})$$

while from the isomorphism (2) we get

$$G_t \cong \ (\mathbb{Z}/q_1^{m_1}\mathbb{Z}) \oplus \ldots \oplus (\mathbb{Z}/q_r^{m_r}\mathbb{Z})$$

If follows that

$$G/G_t \cong \mathbb{Z}^k \quad \text{and} \quad G/G_t \cong \mathbb{Z}^l$$

Using Proposition 13.3 we obtain from here that $k = l$.

Next, we want to show that the family of integers $\{p_i^{n_i} \mid i = 1, \ldots, s\}$ is the same as the family $\{q_j^{m_j} \mid j = 1, \ldots, r\}$.

For a prime p define

$$G_p := \{a \in G \mid |a| = p^l \text{ for some } l > 0\}$$

From the isomorphisms (1) and (2) we get

$$\bigoplus_{p_i = p} (\mathbb{Z}/p_i^{n_i}\mathbb{Z}) \;\cong\; G_p \cong \bigoplus_{q_j = p} (\mathbb{Z}/q_j^{m_j}\mathbb{Z})$$

Using Lemma 14.9 we obtain that the family $\{p_i^{n_i} \mid p_i = p\}$ is the same as $\{q_j^{n_j} \mid q_j = p\}$. Since this holds for all primes p we are done.

$\square$

15 Permutation representations and G-sets

Recall. If $\mathcal{C}$ is a category and $c \in \mathcal{C}$ then

$$\mathrm{Aut}(c) = \text{the group of automorphisms of } c$$

15.1 Definition. A *representation* of a group G in a category $\mathcal{C}$ is a homomorphism

$$\varrho\colon G \to \mathrm{Aut}(c)$$

Special types of representations:

- linear representations = representations in the category of vector spaces
- permutation representations = representations in the category of sets.

15.2 Note. Let S be a set and let

$$\varrho\colon G \to \mathrm{Aut}(S)$$

be a permutation representation of G. The homomorphism ϱ defines a map

$$G \times S \to S, \qquad (a, x) \mapsto \varrho(a)(x)$$

Denote $a \cdot x := \varrho(a)(x)$. We have:

1) $(ab) \cdot x = a \cdot (b \cdot x)$ for $a, b \in G$, $x \in S$
2) $e \cdot x = x$ for all $x \in S$

15.3 Definition. An *action of a group G on a set S* is a map

$$G \times S \to S, \qquad (a, x) \mapsto a \cdot x$$

satisfying conditions 1) - 2) above.

A *G-set* is a set S equipped with an action of the group G.

15.4 Note. A permutation representation $G \to \mathrm{Aut}(S)$ determines a G-set structure on the set S. Conversely, the structure of a G-set on S determines a permutation representation $G \to \mathrm{Aut}(S)$.

15.5 Examples.

1) Take $S = G$. The map

$$G \times G \to G, \qquad (a, b) \mapsto ab$$

defines a G-set structure on G. We say that *the group G acts on itself by left translations*.

2) Let H be a subgroup of G. Take $S = G/H$. The map

$$G \times G/H \to G/H, \qquad (a, bH) \mapsto (ab)H$$

defines a G-set structure on G/H.

3) Take again $S = G$. The map

$$G \times G \to G, \qquad (a, b) \mapsto aba^{-1}$$

defines another G-set structure on G. We say that *the group G acts on itself by conjugations*.

4) $\mathbb{R}$ is a $\mathbb{Z}/2\mathbb{Z}$-set with the action

$$\mathbb{Z}/2\mathbb{Z} \times \mathbb{R} \to \mathbb{R}$$

given by $0 \cdot x = x$, $1 \cdot x = -x$.

15.6 Definition. If S, T are G-sets then a *G-equivariant map* is a function

$$f \colon S \to T$$

such that $f(a \cdot x) = a \cdot f(x)$ for all $a \in G$, $x \in S$.

15.7 Note. For a given group G the collection of G-sets and G-equivariant map forms a category $\mathcal{S}et^G$.

15.8 Proposition. *A G-equivariant map*

$$f\colon S \to T$$

is an isomorphism of G-sets iff f is a bijection.

15.9. Proposition/Definition. Let S be a G-set and let $x \in G$. The set

$$G_x := \{a \in G \mid a \cdot x = x\}$$

is a subgroup of G. It is called the *stabilizer* of x (or the *isotropy group* of x).

15.10 Definition. Let S be a G-set. The action of the group G on S is *transitive* if for any $x, y \in S$ there is $a \in G$ such that $a \cdot x = y$.

15.11 Proposition. *For any transitive G-set S we have an isomorphism of G-sets:*

$$S \cong G/G_x$$

where x is any element of S.

Proof. We have a map

$$f\colon G/G_x \to S, \qquad f(aG_x) = a \cdot x$$

Check that

- f is well defined
- f is G-equivariant
- f is a bijection.

It follows that f an isomorphism of G-sets. $\square$

15.12 Proposition. *If S is a transitive G-set and $x, y \in S$ then*

$$G_x = a^{-1} G_y a$$

where $y = a \cdot x$

Proof. For $b \in G_y$ we have

$$(a^{-1}ba) \cdot x = (a^{-1}b) \cdot y = a^{-1} \cdot y = x$$

It follows that $a^{-1}G_y a \subseteq G_x$. On the other hand, if $c \in G_x$ then

$$(aca^{-1}) \cdot y = (ac) \cdot x = a \cdot x = y$$

Thus $aca^{-1} \in G_y$ and so $c = a^{-1}(aca^{-1})a \in a^{-1}G_y a$. This shows that $G_x \subseteq a^{-1}G_y a$. $\square$

15.13 Definition. If S is a G-set then the *orbit* of an element $x \in S$ is the set

$$\mathrm{Orb}(x) := \{y \in S \mid y = a \cdot x \text{ for some } a \in G\}$$

15.14 Proposition. *Let S be a G-set.*

1) *If $x \in S$ then $\mathrm{Orb}(x)$ is the unique subset of S such that*
    - $x \in \mathrm{Orb}(x)$
    - *G acts transitively on $\mathrm{Orb}(x)$.*
2) *If $x, y \in S$ then either $\mathrm{Orb}(x) \cap \mathrm{Orb}(y) = \varnothing$ or $\mathrm{Orb}(x) = \mathrm{Orb}(y)$.*
3) *If $x, y \in S$ then $\mathrm{Orb}(x) = \mathrm{Orb}(y)$ iff $x = a \cdot y$ for some $a \in G$.*

Proof. Exercise. $\square$

15.15 Corollary. *Every G-set S is a disjoint union of its orbits. For an element $x \in S$ we have an isomorphism of G-sets*

$$\mathrm{Orb}(x) \cong G/G_x$$

Proof. This follows directly from Proposition 15.14 and Proposition 15.11. $\square$

15.16 Corollary. *Let G be a finite group and let S be a finite G-set .*

1) *For any $x \in S$ we have*

$$|\mathrm{Orb}(x)| \cdot |G_x| = |G|$$

2) *If $\mathrm{Orb}(x_1), \ldots \mathrm{Orb}(x_n)$ are all distinct orbits of S then*

$$|S| = \sum_{i=1}^{n} |\mathrm{Orb}(x_i)|$$

Proof.

1) By Corollary 15.15 we have $|\mathrm{Orb}(x)| = [G : G_x]$, so by Lagrange's Theorem (7.4)

$$|G| = [G : G_x] \cdot |G_x| = |\mathrm{Orb}(x)| \cdot |G_x|$$

. $\square$

16 Some applications of G-sets

Recall. If G is a finite group and $a \in G$ then $|a|$ divides $|G|$.

16.1 Theorem (Cauchy)**.** *Let G be a finite group. If $|G|$ is divisible by a prime number p then there exists $a \in G$ such that $|a| = p$.*

Proof. Let
$$S = \{(a_1, \dots, a_p) \mid a_i \in G,\ a_1 \cdot \dots \cdot a_p = e\}$$
Notice that $(a_1, \dots, a_p) \in S$ iff $a_1, \dots, a_{p-1}$ are arbitrary elements of G and $a_p = (a_1 \cdot \dots \cdot a_{p-1})^{-1}$. If follows that
$$|S| = |G|^{p-1}$$
In particular p divides S. Define an action of $\mathbb{Z}/p\mathbb{Z}$ on S by cyclic permutations:
$$k \cdot (a_1, \dots, a_p) := (a_{k+1}, \dots, a_p, a_1, \dots, a_k)$$
for $k \in \mathbb{Z}/p\mathbb{Z}$.

Notice that:

1) The number of elements of each orbit of this action divides $|\mathbb{Z}/p\mathbb{Z}| = p$. Therefore for any $(a_1, \dots, a_p) \in S$ we have
$$|\mathrm{Orb}((a_1, \dots, a_p))| = p \quad \text{or} \quad |\mathrm{Orb}((a_1, \dots, a_p))| = 1$$

2) $|\mathrm{Orb}((a_1, \dots, a_p))| = 1$ iff $a_1 = \dots = a_p$. We have then
$$a_1^p = a_1 \cdot \dots \cdot a_p = e$$

3) $|\mathrm{Orb}((e, \dots, e))| = 1$

Assume that $\mathrm{Orb}((e, \dots, e))$ is the only orbit consisting of only one element. Then every other orbit of S has p elements. From part 2) of Corollary 15.16 we obtain then
$$|S| = kp + 1$$
for some k. This is however impossible since p divides $|S|$. Therefore there is some element $(a_1, \dots, a_p) \in S$ such that $(a_1, \dots, a_p) \neq (e, \dots, e)$ and $|\mathrm{Orb}((a_1, \dots, a_p))| = 1$. It follows that $a_1 \neq e$ and $a_1^p = e$. Thus $|a_1| = p$. $\square$

16.2 Definition. Let p be a prime number. A group G is a *p-group* if order of every element of G is a power of p.

16.3 Proposition. *If G is a finite group then G is a p-group iff $|G| = p^n$ for some n.*

Proof. If $|G| = p^n$ and $a \in G$ then by (7.7) $|a|$ divides p^n, so $|a| = p^r$ for some $r \geq 0$.

Conversely, assume that $|G| = qm$ for some prime $q \neq p$. Then by Cauchy's Theorem (16.1) there is an element $a \in G$ such that $|a| = q$. Therefore G is not a p-group. $\square$

Recall. If G is a group then the center of G is the subgroup

$$Z(G) = \{a \in G \mid ab = ba \text{ for all } b \in G\}$$

Note. In general it may happen that $Z(G) = \{e\}$ (take e.g. $G = G_T$).

16.4 Theorem. *If G is a finite p-group then $Z(G) \neq \{e\}$.*

Proof. Consider the action of G on itself by conjugations:

$$G \times G \to G, \qquad a \cdot b := aba^{-1}$$

Let $|G| = p^n$. Notice that:

1) If $b \in G$ then $|\mathrm{Orb}(b)|$ divides p^n, so $|\mathrm{Orb}(b)| = p^r$ for some $r \geq 0$. In particular either p divides $|\mathrm{Orb}(b)|$ or $|\mathrm{Orb}(b)| = 1$.

2) $|\mathrm{Orb}(b)| = 1$ iff $b \in Z(G)$.

As a consequence if $Z(G) = \{e\}$ then $|\mathrm{Orb}(e)| = 1$ and p divides the number of elements in every other orbit. From part 2) of Corollary 15.16 we obtain then

$$p^n = |G| = kp + 1$$

which is impossible. Therefore $Z(G) \neq \{e\}$.

$\square$

16.5 Note. Not every p-group is abelian. For example take the group of quaternions Q_8 (see Hungerford p. 33) and the group of symmetries of a square D_4^* (see Hungerford p. 25). These group are non-abelian, but $|Q_8| = |D_4^*| = 8$, so they are 2-groups.

16.6 Definition. Let S be a G-set. A element $x \in S$ is a *fixed point* of the action of G on S if $a \cdot x = x$ for all $a \in G$.

Note. If S is a G-set then x is a fixed point of the action of G iff $|\mathrm{Orb}(x)| = 1$.

17 The Sylow theorems

Recall. If G is a finite group and $H \subseteq G$ is a subgroup then $|H|$ divides $|G|$.

Note.

1) If G is an abelian group and n divides $|G|$ then there is a subgroup $H \subseteq G$ such that $|H| = n$ (homework).
2) This is not true in general when G is a non-abelian group (homework).

17.1 Definition. Let G be a finite group such that $|G| = p^r m$ where p is a prime and $p \nmid m$. We say that a subgroup $P \subseteq G$ is a Sylow p-subgroup of G if $|P| = p^r$.

17.2 First Sylow Theorem. *If G is a finite group and p is a prime number dividing $|G|$ then G contains a Sylow p-subgroup.*

Proof.

Assume that $|G| = p^r m$ where $p \nmid m$.

Let S be the set of all subsets $A \subseteq G$ such that $|A| = p^r$. Define an action of G on S as follows. If $A \in S$, $A = \{a_1, \dots, a_{p^r}\}$ and $b \in G$ then

$$b \cdot A := \{ba_1, \dots, ba_{p^r}\}$$

Notice that

$$|S| = \binom{p^r m}{p^r} = \prod_{j=1}^{p^r} \frac{p^r(m-1)+j}{j}$$

so p does not divide $|S|$. A a consequence there is an orbit $\mathrm{Orb}(A_0)$ in S such that $p \nmid |\mathrm{Orb}(A_0)|$. Take the stabilizer G_{A_0}. We want to show that $|G_{A_0}| = p^r$. We have

$$p^r m = |G| = |\mathrm{Orb}(A_0)| \cdot |G_{A_0}|$$

so p^r divides $|G_{A_0}|$. On the other hand, notice that for any $a \in G$ there is some element $b \cdot A_0 \in \mathrm{Orb}(A_0)$ such that $a \in b \cdot A_0$. Ineed, if $A_0 = \{a_1, \dots, a_{p^r}\}$ then

$$(aa_1^{-1}) \cdot A_0 = \{a, \dots\}$$

Since G has $p^r m$ elements and each set $b \cdot A_0$ contains p^r elements, thus we must have $|\mathrm{Orb}(A_0)| \geq m$, and consequently $|G_{A_0}| \leq p^r$. It follows that $|G_{A_0}| = p^r$. $\square$

17.3 Second Sylow Theorem. *If P is a Sylow p-subgroup of a finite group G then for any p-subgroup $H \subseteq G$ we have*

$$aHa^{-1} \subseteq P$$

for some $a \in G$.

In particular if P, P' are two Sylow p-subgroups of G then $P = aP'a^{-1}$ for some $a \in G$.

Proof. Assume that $|G| = p^r m$ where $p \nmid m$ and let $|H| = p^s$.

Take G/P, the set of left cosets of P. The group H acts on G/P by left translations:

$$H \times G/P \to G/P, \qquad a \cdot (bP) := (ab)P$$

Since $|H| = p^s$, for every $bP \in G/P$ we have $|\mathrm{Orb}(bP)| = p^k$ for some $k \leq s$. So, either p divides $|\mathrm{Orb}(bP)|$ or $|\mathrm{Orb}(bP)| = 1$.

On the other hand $|G/P| = [G : P] = m$, and $p \nmid m$, so there must be an orbit whose number of elements is not divisible by p. It follows that there is $b_0P \in G/P$ such that

$$|\mathrm{Orb}(b_0P)| = 1$$

As a consequence $ab_0P = b_0P$ for all $a \in H$, i.e. $Hb_0P \subseteq b_0P$. Since $Hb_0 \subseteq Hb_0P$ we obtain

$$Hb_0 \subseteq b_0P$$

or equivalently: $b_0^{-1}Hb_0 \subseteq P$. $\square$

17.4 Example. Take the group G_T. We have: $|G_T| = 6 = 3 \cdot 2$.

- Sylow 3-subgroup of G_T: $\{I, R_1, R_2\}$
- Sylow 2-subgroups of G_T: $P = \{I, S_1\}$, $P' = \{I, S_2\}$, $P'' = \{I, S_3\}$.

We have: $P = R_1 P' R_1^{-1}$ and $P = R_2 P'' R_2^{-1}$.

17.5 Third Sylow Theorem. *Let G be a finite group such that $|G| = p^r m$ where $r > 0$ and $p \nmid m$. If s is the number of Sylow p-subgroups of G then*

1) $s \mid m$

2) $s \equiv 1 \pmod{p}$

17.6 Definition. Let G be a group and H be a subgroup of G. The *normalizer* of H in G is the subgroup $N_G(H) \subseteq G$ given by

$$N_G(H) = \{a \in G \mid aHa^{-1} = H\}$$

17.7 Note.

1) $H \lhd N_G(H)$. In fact, $N_G(H)$ is the biggest subgroup of G that contains H a its normal subgroup.

2) $H \lhd G$ iff $N_G(H) = G$.

3) Let S be the set of all subgroups of G. The group G acts on S by conjugations:

$$G \times S \to S, \qquad a \cdot H := aHa^{-1}$$

If H is a subgroup of G then

$$\mathrm{Orb}(H) = \{\text{ all subgroups of } G \text{ conjugate to } H\ \}$$
$$N_G(H) = \text{stabilizer of } H$$

By (15.16) this gives:

$$(\text{number of subgroups conjugate to } H) = |\mathrm{Orb}(H)| = [G : N_G(H)]$$

Proof of Theorem 17.5. Let P be a Sylow p-subgroup of G. By (17.3) all other Sylow p-subgroups of G are conjugate to P, so we get

$$s = (\text{number of subgroups conjugate to } P) = \ [G : N_G(P)]$$

Since $P \subseteq N_G$ by Lagrange's Theorem (7.4) we have

$$\underbrace{|G|}_{p^r m} = [G : N_G(P)] \cdot |N_G(P)| = \underbrace{[G : N_G(P)]}_{s} \cdot [N_G(P) : P] \cdot \underbrace{|P|}_{p^r}$$

It follows that $s \mid m$.

Next, let $S = \{P_1, \dots, P_s\}$ be the set of all Sylow p-subgroups of G. The group P_1 acts on S by conjugations:

$$P_1 \times S \to S, \qquad a \cdot P_i := aP_ia^{-1}$$

Notice that:

- $|\mathrm{Orb}(P_i)|$ divides $|P_1| = p^r$, so for $i = 1, \dots, s$ either p divides $|\mathrm{Orb}(P_i)|$ or $|\mathrm{Orb}(P_i)| = 1$.
- $|\mathrm{Orb}(P_1)| = 1$
- If $i > 1$ then $|\mathrm{Orb}(P_i)| > 1$.

  Indeed, if $|\mathrm{Orb}(P_i)| = 1$ for some $i > 1$ then $aP_ia^{-1} = P_i$ for all $a \in P_1$. Check: in this case P_1P_i is a p-subgroup of G. Since $P_1 \neq P_i$ we would also have $|P_1P_i| > p^r$ which is impossible.

As a consequence we get

$$s = |S| = \underbrace{|\mathrm{Orb}(P_1)|}_{1} + \sum \underbrace{|(\text{ all other orbits })|}_{\text{multiples of } p}$$

$\square$

Therefore $s \equiv 1 \pmod{p}$.

18 Application: groups of order pq

Recall. If G is a group, $|G| = p$ where p is a prime then $G \cong \mathbb{Z}/p\mathbb{Z}$.

Goal. Classify all groups of order pq where p, q are prime numbers.

18.1 Proposition. *If G is a group of order p^2 for some prime p then either $G \cong \mathbb{Z}/p^2\mathbb{Z}$ or $G \cong \mathbb{Z}/p\mathbb{Z} \oplus \mathbb{Z}/p\mathbb{Z}$.*

Proof. It is enough to show that G is abelian since then the statement follows from the classification of finitely generated abelian groups (14.7).

Since G is a p-group by Theorem 16.4 we have $Z(G) \neq \{e\}$. If $Z(G) = G$ then G is abelian. If $Z(G) \neq G$ then $G/Z(G)$ is a group of order p and thus it is a non-trivial cyclic group. This is however impossible by Problem 8 of HW 1. $\square$

18.2 Proposition. *If G is a group of order pq for some primes p, q such that $p > q$ and $q \nmid (p-1)$ then*

$$G \cong \mathbb{Z}/pq\mathbb{Z}$$

Proof. If is enough to show that G contains an element of order pq.

Let s_p denote the number of Sylow p-subgroups of G. By the Third Sylow Theorem (17.5) we have

$$s_p \mid q \quad \text{and} \quad s_p = 1 + kp$$

Since q is a prime the first condition gives $s_p = 1$ or $s_p = q$. Since $p > q$ the second condition implies then that $s_p = 1$.

Similarly, let s_q be the number of Sylow q-subgroups of G. We have

$$s_q \mid p \quad \text{and} \quad s_q = 1 + kq$$

The first condition gives $s_q = 1$ or $s_q = p$. If $s_q = p$ then the second condition gives $p = 1 + kq$, or $p - 1 = kq$. This is however impossible since $q \nmid (p-1)$. Therefore we have $s_q = 1$.

We obtain that G has exactly one Sylow p-subgroup P (of order p) and exactly one Sylow q-subgroup Q (of order q). By the Second Sylow Theorem (17.3) every element of G of order p belongs to the subgroup P and every element of order q belongs to the subgroup Q. It follows that G contains exactly $p - 1$ elements of order p, exactly $q - 1$ elements of order q, and one trivial element (of order 1). Since for all p, q we have

$$pq > (p-1) + (q-1) + 1$$

there are elements of G of order not equal to 1, p, or q. Any such element must have order pq.

□

Note. If $|G| = pq$, and $q \mid (p-1)$ then G need not be isomorphic to $\mathbb{Z}/pq\mathbb{Z}$ (take e.g. $G = G_T$).

18.3. Defintition. Let N, K be groups and let

$$\varphi\colon K \to \mathrm{Aut}(N)$$

be a homomorphism. The *semidirect product* of N and K with respect to φ is the group $N \rtimes_\varphi K$ such that

$$N \rtimes_\varphi K = N \times K$$

as sets. Multiplication in $N \rtimes_\varphi K$ is given by

$$(a_1, b_1) \cdot (a_2, b_2) := (a_1(\varphi(b_1)(a_2)), b_1 b_2)$$

18.4 Note.

1) If φ is the trivial homomorphism then $N \rtimes_\varphi K = N \times K$.

2) If $(a, b) \in N \rtimes_\varphi K$ then

$$(a, b)^{-1} = (\varphi(b^{-1})(a^{-1}), b^{-1})$$

3) $N \rtimes_\varphi K$ contains subgroups

$$N \cong \{(a, e) \mid a \in N\} \quad \text{and} \quad K \cong \{(e, b) \mid b \in K\}$$

We have $N \lhd N \rtimes_\varphi K$, and $N \cap K = \{(e, e)\}$.

18.5 Examples.

1) Notice that $\mathrm{Aut}(\mathbb{Z}/3\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$. Let

$$\varphi \colon \mathbb{Z}/2 \to \mathrm{Aut}(\mathbb{Z}/3\mathbb{Z})$$

be the isomorphism. Check:

$$\mathbb{Z}/3\mathbb{Z} \rtimes_\varphi \mathbb{Z}/2\mathbb{Z} \cong G_T$$

2) In general if p is a prime then we have

$$\mathrm{Aut}(\mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/(p-1)\mathbb{Z}$$

For any $n \mid (p-1)$ there is a unique cyclic subgroup $H \subseteq \mathrm{Aut}(\mathbb{Z}/p\mathbb{Z})$ of order n. Let

$$\varphi \colon \mathbb{Z}/n\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/p\mathbb{Z})$$

be any homomorphism such that $\mathrm{Im}(\varphi) = H$. Then $\mathbb{Z}/p\mathbb{Z} \rtimes_\varphi \mathbb{Z}/n\mathbb{Z}$. is a non-abelian group of order pn.

3) If N is any abelian group then the map

$$\mathrm{inv} \colon N \to N, \quad \mathrm{inv}(a) = a^{-1}$$

is an automorphism of N. This gives a homomorphism

$$\varphi\colon \mathbb{Z}/2\mathbb{Z} \to \mathrm{Aut}(N)$$

such that $\varphi(1) = \mathrm{inv}$. We obtain in this way a group $N \rtimes_\varphi \mathbb{Z}/2\mathbb{Z}$ of order $2|N|$.

Special cases:

• If $N = \mathbb{Z}/n\mathbb{Z}$ then this group is called the *dihedral group of order* $2n$ and it is denoted D_n. The group D_n is isomorphic to the group of all isometries of a regular polygon with n sides (exercise). In particular $D_3 \cong G_T$.

• If $N = \mathbb{Z}$ then this group is the *infinite dihedral group* and it is denoted by D_∞. The group D_∞ is isomorphic to the free product $\mathbb{Z}/2\mathbb{Z} * \mathbb{Z}/2\mathbb{Z}$ (exercise).

Recall. From HW 1: If G, H are abelian groups and $f\colon G \to H$, $g\colon H \to G$ are homomorphisms such that $fg = \mathrm{id}_H$ then $G \cong \mathrm{Ker}(f) \oplus H$.

18.6 Proposition. *If G, H are groups and $f\colon G \to H$, $g\colon H \to G$ are homomorphisms such that $fg = \mathrm{id}_H$ then*

$$G \cong \mathrm{Ker}(f) \rtimes_\varphi H$$

where $\varphi\colon H \to \mathrm{Aut}(\mathrm{Ker}(f))$ is given by

$$\varphi(b)(a) := g(b)ag(b)^{-1}$$

for $a \in \mathrm{Ker}(f)$, $b \in H$.

Proof. Exercise. □

18.7 Proposition. *Let p, q be prime numbers such that $p > q$ and $q \mid (p-1)$. Then, up to isomorphism, there are only two groups of order pq:*

- *the abelian group* $\mathbb{Z}/pq\mathbb{Z} \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$
- *the non-abelian group* $\mathbb{Z}/p\mathbb{Z} \rtimes_\varphi \mathbb{Z}/q\mathbb{Z}$ *where*

  $$\varphi\colon \mathbb{Z}/q\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/p\mathbb{Z})$$

  is any non-trivial homomorphism.

Proof. By the same argument as in the proof of Proposition 18.2 we get that G has only one Sylow p-subgroup. Call this subgroup P. We have $P \lhd G$.

Let Q be any Sylow q-subgroup. Consider the quotient map

$$f\colon G \to G/P$$

Take the restriction $f|_Q\colon Q \to G/P$. Notice that $\mathrm{Ker}(f|_Q) = \mathrm{Ker}(f) \cap Q = P \cap Q = \{e\}$, so $f|_Q$ is a monomorphism. In addition $|Q| = q = |G/P|$, so $f|_Q$ is an isomorphism. As a consequence we have a homomorphism

$$g\colon G/P \xrightarrow{(f|_Q)^{-1}} Q \hookrightarrow G$$

Since $fg = \mathrm{id}_{G/P}$ by Proposition 18.6 we obtain

$$G \cong P \rtimes_\varphi G/P$$

for some homomorphism $\varphi\colon G/P \to \mathrm{Aut}(P)$. Also, since $P \cong \mathbb{Z}/p\mathbb{Z}$ and $G/P \cong \mathbb{Z}/q\mathbb{Z}$ we get

$$G \cong \mathbb{Z}/p\mathbb{Z} \rtimes_\varphi \mathbb{Z}/q\mathbb{Z}$$

for some $\varphi\colon \mathbb{Z}/q\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/p\mathbb{Z})$.

If φ is the trivial homomorphism then $G \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z} \cong \mathbb{Z}/pq\mathbb{Z}$. If φ is non-trivial then G is a non-abelian group. Notice that since $q \mid (p-1)$ such non-trivial homomorphism exists by (18.5).

It remains to show that for any two non-trivial homomorphisms

$$\varphi, \psi\colon \mathbb{Z}/q\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/p\mathbb{Z})$$

we have an isomorphism

$$\mathbb{Z}/p\mathbb{Z} \rtimes_\varphi \mathbb{Z}/q\mathbb{Z} \cong \mathbb{Z}/p\mathbb{Z} \rtimes_\psi \mathbb{Z}/q\mathbb{Z}$$

(exercise). □

18.8 Example. For any odd prime p there are two non-isomorphic groups of order $2p$:

- the cyclic group $\mathbb{Z}/2p\mathbb{Z}$
- the dihedral group D_p.

H. U. Besche, B. Eick, E. A. O'Brien
A millennium project: constructing small groups
International Journal of Algebra and Computation 12(5) (2002) 623-644.

The number of groups of order at most 2000

	+0	+1	+2	+3	+4	+5	+6	+7	+8	+ 9
0		1	1	1	2	1	2	1	5	2
10	2	1	5	1	2	1	14	1	5	1
20	5	2	2	1	15	2	2	5	4	1
30	4	1	51	1	2	1	14	1	2	2
40	14	1	6	1	4	2	2	1	52	2
50	5	1	5	1	15	2	13	2	2	1
60	13	1	2	4	267	1	4	1	5	1
70	4	1	50	1	2	3	4	1	6	1
80	52	15	2	1	15	1	2	1	12	1
90	10	1	4	2	2	1	231	1	5	2
100	16	1	4	1	14	2	2	1	45	1
110	6	2	43	1	6	1	5	4	2	1
120	47	2	2	1	4	5	16	1	2328	2
130	4	1	10	1	2	5	15	1	4	1
140	11	1	2	1	197	1	2	6	5	1
150	13	1	12	2	4	2	18	1	2	1
160	238	1	55	1	5	2	2	1	57	2
170	4	5	4	1	4	2	42	1	2	1
180	37	1	4	2	12	1	6	1	4	13
190	4	1	1543	1	2	2	12	1	10	1
200	52	2	2	2	12	2	2	2	51	1
210	12	1	5	1	2	1	177	1	2	2
220	15	1	6	1	197	6	2	1	15	1
230	4	2	14	1	16	1	4	2	4	1
240	208	1	5	67	5	2	4	1	12	1
250	15	1	46	2	2	1	56092	1	6	1
260	15	2	2	1	39	1	4	1	4	1
270	30	1	54	5	2	4	10	1	2	4
280	40	1	4	1	4	2	4	1	1045	2
290	4	2	5	1	23	1	14	5	2	1
300	49	2	2	1	42	2	10	1	9	2
310	6	1	61	1	2	4	4	1	4	1
320	1640	1	4	1	176	2	2	2	15	1
330	12	1	4	5	2	1	228	1	5	1
340	15	1	18	5	12	1	2	1	12	1
350	10	14	195	1	4	2	5	2	2	1
360	162	2	2	3	11	1	6	1	42	2
370	4	1	15	1	4	7	12	1	60	1
380	11	2	2	1	20169	2	2	4	5	1
390	12	1	44	1	2	1	30	1	2	5
400	221	1	6	1	5	16	6	1	46	1
410	6	1	4	1	10	1	235	2	4	1
420	41	1	2	2	14	2	4	1	4	2
430	4	1	775	1	4	1	5	1	6	1
440	51	13	4	1	18	1	2	1	1396	1
450	34	1	5	2	2	1	54	1	2	5
460	11	1	12	1	51	4	2	1	55	1
470	4	2	12	1	6	2	11	2	2	1
480	1213	1	2	2	12	1	261	1	14	2
490	10	1	12	1	4	4	42	2	4	1

	+0	+1	+2	+3	+4	+5	+6	+7	+8	+ 9
500	56	1	2	1	202	2	6	6	4	1
510	8	1	10494213	15	2	1	15	1	4	1
520	49	1	10	1	4	6	2	1	170	2
530	4	2	9	1	4	1	12	1	2	2
540	119	1	2	2	246	1	24	1	5	4
550	16	1	39	1	2	2	4	1	16	1
560	180	1	2	1	10	1	2	49	12	1
570	12	1	11	1	4	2	8681	1	5	2
580	15	1	6	1	15	4	2	1	66	1
590	4	1	51	1	30	1	5	2	4	1
600	205	1	6	4	4	7	4	1	195	3
610	6	1	36	1	2	2	35	1	6	1
620	15	5	2	1	260	15	2	2	5	1
630	32	1	12	2	2	1	12	2	4	2
640	21541	1	4	1	9	2	4	1	757	1
650	10	5	4	1	6	2	53	5	4	1
660	40	1	2	2	12	1	18	1	4	2
670	4	1	1280	1	2	17	16	1	4	1
680	53	1	4	1	51	1	15	2	42	2
690	8	1	5	4	2	1	44	1	2	1
700	36	1	62	1	1387	1	2	1	10	1
710	6	4	15	1	12	2	4	1	2	1
720	840	1	5	2	5	2	13	1	40	504
730	4	1	18	1	2	6	195	2	10	1
740	15	5	4	1	54	1	2	2	11	1
750	39	1	42	1	4	2	189	1	2	2
760	39	1	6	1	4	2	2	1	1090235	1
770	12	1	5	1	16	4	15	5	2	1
780	53	1	4	5	172	1	4	1	5	1
790	4	2	137	1	2	1	4	1	24	1
800	1211	2	2	1	15	1	4	1	14	1
810	113	1	16	2	4	1	205	1	2	11
820	20	1	4	1	12	5	4	1	30	1
830	4	2	1630	2	6	1	9	13	2	1
840	186	2	2	1	4	2	10	2	51	2
850	10	1	10	1	4	5	12	1	12	1
860	11	2	2	1	4725	1	2	3	9	1
870	8	1	14	4	4	5	18	1	2	1
880	221	1	68	1	15	1	2	1	61	2
890	4	15	4	1	4	1	19349	2	2	1
900	150	1	4	7	15	2	6	1	4	2
910	8	1	222	1	2	4	5	1	30	1
920	39	2	2	1	34	2	2	4	235	1
930	18	2	5	1	2	2	222	1	4	2
940	11	1	6	1	42	13	4	1	15	1
950	10	1	42	1	10	2	4	1	2	1
960	11394	2	4	2	5	1	12	1	42	2
970	4	1	900	1	2	6	51	1	6	2
980	34	5	2	1	46	1	4	2	11	1
990	30	1	196	2	6	1	10	1	2	15

	+0	+1	+2	+3	+4	+5	+6	+7	+8	+ 9
1000	199	1	4	1	4	2	2	1	954	1
1010	6	2	13	1	23	2	12	2	2	1
1020	37	1	4	2	49487365422	4	66	2	5	19
1030	4	1	54	1	4	2	11	1	4	1
1040	231	1	2	1	36	2	2	2	12	1
1050	40	1	4	51	4	2	1028	1	5	1
1060	15	1	10	1	35	2	4	1	12	1
1070	4	4	42	1	4	2	5	1	10	1
1080	583	2	2	6	4	2	6	1	1681	6
1090	4	1	77	1	2	2	15	1	16	1
1100	51	2	4	1	170	1	4	5	5	1
1110	12	1	12	2	2	1	46	1	4	2
1120	1092	1	8	1	5	14	2	2	39	1
1130	4	2	4	1	254	1	42	2	2	1
1140	41	1	2	5	39	1	4	1	11	1
1150	10	1	157877	1	2	4	16	1	6	1
1160	49	13	4	1	18	1	4	1	53	1
1170	32	1	5	1	2	2	279	1	4	2
1180	11	1	4	3	235	2	2	1	99	1
1190	8	2	14	1	6	1	11	14	2	1
1200	1040	1	2	1	13	2	16	1	12	5
1210	27	1	12	1	2	69	1387	1	16	1
1220	20	2	4	1	164	4	2	2	4	1
1230	12	1	153	2	2	1	15	1	2	2
1240	51	1	30	1	4	1	4	1	1460	1
1250	55	4	5	1	12	2	14	1	4	1
1260	131	1	2	2	42	3	6	1	5	5
1270	4	1	44	1	10	3	11	1	10	1
1280	1116461	5	2	1	10	1	2	4	35	1
1290	12	1	11	1	2	1	3609	1	4	2
1300	50	1	24	1	12	2	2	1	18	1
1310	6	2	244	1	18	1	9	2	2	1
1320	181	1	2	51	4	2	12	1	42	1
1330	8	5	61	1	4	1	12	1	6	1
1340	11	2	4	1	11720	1	2	1	5	1
1350	112	1	52	1	2	2	12	1	4	4
1360	245	1	4	1	9	5	2	1	211	2
1370	4	2	38	1	6	15	195	15	6	2
1380	29	1	2	1	14	1	32	1	4	2
1390	4	1	198	1	4	8	5	1	4	1
1400	153	1	2	1	227	2	4	5	19324	1
1410	8	1	5	4	4	1	39	1	2	2
1420	15	4	16	1	53	6	4	1	40	1
1430	12	5	12	1	4	2	4	1	2	1
1440	5958	1	4	5	12	2	6	1	14	4
1450	10	1	40	1	2	2	179	1	1798	1
1460	15	2	4	1	61	1	2	5	4	1
1470	46	1	1387	1	6	2	36	2	2	1
1480	49	1	24	1	11	10	2	1	222	1
1490	4	3	5	1	10	1	41	2	4	1

	+0	+1	+2	+3	+4	+5	+6	+7	+8	+ 9
1500	174	1	2	2	195	2	4	1	15	1
1510	6	1	889	1	2	2	4	1	12	2
1520	178	13	2	1	15	4	4	1	12	1
1530	20	1	4	5	4	1	408641062	1	2	60
1540	36	1	4	1	15	2	2	1	46	1
1550	16	1	54	1	24	2	5	2	4	1
1560	221	1	4	1	11	1	30	1	928	2
1570	4	1	10	2	2	13	14	1	4	1
1580	11	2	6	1	697	1	4	3	5	1
1590	8	1	12	5	2	2	64	1	4	2
1600	10281	1	10	1	5	1	4	1	54	1
1610	8	2	11	1	4	1	51	6	2	1
1620	477	1	2	2	56	5	6	1	11	5
1630	4	1	1213	1	4	2	5	1	72	1
1640	68	2	2	1	12	1	2	13	42	1
1650	38	1	9	2	2	2	137	1	2	5
1660	11	1	6	1	21507	5	10	1	15	1
1670	4	1	34	2	60	2	4	5	2	1
1680	1005	2	5	2	5	1	4	1	12	1
1690	10	1	30	1	10	1	235	1	6	1
1700	50	309	4	2	39	7	2	1	11	1
1710	36	2	42	2	2	5	40	1	2	2
1720	39	1	12	1	4	3	2	1	47937	1
1730	4	2	5	1	13	1	35	4	4	1
1740	37	1	4	2	51	1	16	1	9	1
1750	30	2	64	1	2	14	4	1	4	1
1760	1285	1	2	1	228	1	2	5	53	1
1770	8	2	4	2	2	4	260	1	6	1
1780	15	1	110	1	12	2	4	1	12	1
1790	4	5	1083553	1	12	1	5	1	4	1
1800	749	1	4	2	11	3	30	1	54	13
1810	6	1	15	2	2	9	12	1	10	1
1820	35	2	2	1	1264	2	4	6	5	1
1830	18	1	14	2	4	1	117	1	2	2
1840	178	1	6	1	5	4	4	1	162	2
1850	10	1	4	1	16	1	1630	2	2	2
1860	56	1	10	15	15	1	4	1	4	2
1870	12	1	1096	1	2	21	9	1	6	1
1880	39	5	2	1	18	1	4	2	195	1
1890	120	1	9	2	2	1	54	1	4	4
1900	36	1	4	1	186	2	2	1	36	1
1910	6	15	12	1	8	1	4	5	4	1
1920	241004	1	5	1	15	4	10	1	15	2
1930	4	1	34	1	2	4	167	1	12	1
1940	15	1	2	1	3973	1	4	1	4	1
1950	40	1	235	11	2	1	15	1	6	1
1960	144	1	18	1	4	2	2	2	203	1
1970	4	15	15	1	12	2	39	1	4	1
1980	120	1	2	2	1388	1	6	1	13	4
1990	4	1	39	1	2	5	4	1	66	1
2000	963									

19 Group extensions and composition series

19.1 Definition. Let

$$\dots \longrightarrow G_i \xrightarrow{f_i} G_{i+1} \xrightarrow{f_{i+1}} G_{i+2} \longrightarrow \dots$$

be a sequence of groups and group homomorphisms. This sequence is *exact* if $\mathrm{Im}(f_i) = \mathrm{Ker}(f_{i+1})$ for all i.

19.2 Definition. A *short exact sequence* is an exact sequence of the form

$$1 \longrightarrow N \xrightarrow{f} G \xrightarrow{g} K \longrightarrow 1$$

(where 1 is the trivial group).

19.3 Note.

1) A sequence $1 \to N \xrightarrow{f} G \xrightarrow{g} K \to 1$ is a short exact sequence iff
   - f is a monomorphism
   - g is an epimorphism
   - $\mathrm{Im}(f) = \mathrm{Ker}(g)$.

2) If $H \lhd G$ then we have a short exact sequence

$$1 \longrightarrow H \longrightarrow G \longrightarrow G/H \longrightarrow 1$$

Morever, up to an isomorphism, every short exact sequence is of this form:

$$\begin{tikzcd}
1 \arrow[r] & N \arrow[r, "f"] \arrow[d, "\cong"'] & G \arrow[r, "g"] \arrow[d, "="'] & K \arrow[r] \arrow[d, "\cong"] & 1 \\
1 \arrow[r] & \mathrm{Ker}(g) \arrow[r] & G \arrow[r] & G/\mathrm{Ker}(g) \arrow[r] & 1
\end{tikzcd}$$

19.4 Definition. If a group G fits into a short exact sequence

$$1 \longrightarrow N \xrightarrow{f} G \xrightarrow{g} K \longrightarrow 1$$

then we say that G is an *extension of* K *by* N.

19.5 Example. For any $n > 1$ the dihedral group D_n and the cyclic group $\mathbb{Z}/2n\mathbb{Z}$ are non-isomorphic extensions of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}/2\mathbb{Z}$.

19.6 Definition. A group G is a *simple group* if $G \neq \{e\}$ and the only normal subgroups of G are G and $\{e\}$.

19.7 Example. $\mathbb{Z}/p\mathbb{Z}$ is a simple group for every prime p.

Note. A group G is simple iff it is not a non-trivial extension of any group.

19.8 Definition. If G is a group then a *normal series* of G is a sequence of subgroups

$$\{e\} = G_0 \subseteq G_1 \subseteq G_2 \subseteq \ldots \subseteq G_k = G$$

such that $G_{i-1} \lhd G_i$ for all i.

A *composition series* of G is a normal series such that all quotient groups G_i/G_{i-1} are simple.

19.9 Example. Take the dihedral group $D_4 = \mathbb{Z}/4\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$. We have a composition series

$$\{0\} \ \subseteq \ \mathbb{Z}/2\mathbb{Z} \ \subseteq \ \mathbb{Z}/4\mathbb{Z} \ \subseteq \ D_4$$

Another composition series of D_4:

$$\{0\} \ \subseteq \ \mathbb{Z}/2\mathbb{Z} \ \subseteq \ \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \ \subseteq \ D_4$$

19.10 Theorem (Jordan – Hölder)**.** *If $G \neq \{e\}$ is a finite group then*

1) *G has a composition series.*

2) *All composition series of G are equivalent in the following sense. If we have composition series*

$$\{e\} = G_0 \subseteq \ldots \subseteq G_k = G \quad \text{and} \quad \{e\} = H_0 \subseteq \ldots \subseteq H_l = G$$

then $k = l$ and there is a bijection $\sigma \colon \{1, \ldots, k\} \to \{1, \ldots, k\}$ such that for $i = 1, \ldots, k$ we have an isomorphism

$$G_i/G_{i-1} \cong H_{\sigma(i)}/H_{\sigma(i)-1}$$

Proof. Exercise (or see Hungerford p. 111). □

Upshot. If $G \neq \{e\}$ is a finite group then G can be obtained by taking successive extensions of simple groups as follows.

1) Take a composition series

$$\{e\} = G_0 \subseteq G_1 \subseteq G_2 \subseteq \ldots \subseteq G_k = G$$

2) For every $i = 1, \ldots, k$ we have a short exact sequence

$$1 \longrightarrow G_{i-1} \longrightarrow G_i \longrightarrow G_i/G_{i-1} \longrightarrow 1$$

where G_i/G_{i-1} is a simple group. Therefore

G_1 is an extension of G_1/G_0 by G_0
G_2 is an extension of G_2/G_1 by G_1
... ...
... ...
$G = G_k$ is an extension of G_k/G_{k-1} by G_{k-1}

The grand plan for classifying all finite groups (The Hölder Program)

1) Classify all finite simple groups.

2) For any two groups N, K describe all possible extensions of K by N.

Good news: part 1) is done. See

R. Solomon, *A brief history of the classification of the finite simple groups*, Bulletin AMS 38 (3) (2001), 315-352.

M. Aschbacher, *The status of the classification of the finite simple groups*, Notices AMS 51(7) (2004), 736-740.

20 Simple groups

Recall. A group $G \neq \{e\}$ is a simple group if the only normal subgroups of G are G and $\{e\}$.

Note. If G is a simple group then any non-trivial homomorphism $f\colon G \to H$ is a monomorphism.

20.1 Proposition. *If G is an abelian group then G is simple iff $G \cong \mathbb{Z}/p\mathbb{Z}$ for some prime p.*

Proof. Exercise. $\square$

20.2 Proposition. *If G is a simple p-group then $G \cong \mathbb{Z}/p\mathbb{Z}$.*

Proof. If G is a p-group then $Z(G) \neq \{e\}$ by (16.4). We have $Z(G) \lhd G$, so if G is simple we must have $G = Z(G)$. Therefore G is a simple abelian p-group, and so $G \cong \mathbb{Z}/p\mathbb{Z}$. $\square$

20.3 Lemma. *There are no non-abelian simple groups of order $p^r m$ where p is a prime, $r \geq 1$, $p \nmid m$ and $p^r m \nmid m!$.*

Proof. Assume that G is a simple, non-abelian group of such order. We must have $m > 1$ (since if $m = 1$ then G is a p-group). Let P be a Sylow p-subgroup of G. Consider the action of G on the left cosets G/P:

$$G \times G/P \to G/P, \qquad a \cdot bP = (ab)P$$

This action defines a homomorphism

$$\varrho\colon G \to \mathrm{Perm}(G/P)$$

where $\mathrm{Perm}(G/P)$ is the group of all permutations of the set G/P. Since $G \neq P$ this homomorphism is non-trivial, and so, since G is a simple group, ϱ is a monomorphism. Therefore G can be identified with a subgroup of $\mathrm{Perm}(G/P)$. By Lagrange's Theorem (7.4) we obtain that $|G|$ divides $|\mathrm{Perm}(G/P)|$. Since $|\mathrm{Perm}(G/P)| = m!$ this gives

$$p^r m \mid m!$$

which contradicts assumptions of the lemma. $\square$

20.4 Theorem. *There are no non-abelian simple groups of order* < 60.

Proof. Check: If $1 \leq n < 60$, and $n \neq 30, 40, 56$ then n is of the form $p^r m$ for some prime p, and $r, m \geq 1$ such that $p \nmid m$ and $p^r m \nmid m!$.

By Lemma 20.3 we obtain then that a non-abelian group G of order $n < 60$ may be simple only if $n = 30, 40$ and 56.

Assume that $|G| = 30 = 2 \cdot 3 \cdot 5$. We will show that G cannot be a simple group.

We argue by contradiction. Assume that G is simple and let s_3 be the number of Sylow 3-subgroups of G. We have

$$s_3 \mid 10 \quad \text{and} \quad s_3 \equiv 1 \pmod 3$$

It follows that either $s_3 = 1$ or $s_3 = 10$. Since G is simple $s_3 \neq 1$, so $s_3 = 10$.

Notice that if P, P' are two distinct Sylow 3-subgroups of G then $P \cap P' = \{e\}$. We obtain:

- G contains 10 Sylow 3-subgroups.
- each Sylow 3-subgroup contains 2 elements of order 3.

It follows that G contains 20 elements of order 3.

By a similar argument we obtain that

- G must contain 6 Sylow 5-subgroups.

– each Sylow 5-subgroup contains 4 elements of order 3.

so we have 24 elements of order 5 in G. This is however impossible, since $20 + 24 > 30 = |G|$.

In a similar way one can show that if $|G| = 40$ or $|G| = 56$ then G is not a simple group (exercise) $\square$

Next goal: there are infinitely many non-abelian simple finite groups. In particular there is a simple group of order 60.

21 Symmetric and alternating groups

Recall. The symmetric group on n letters is the group

$$S_n = \mathrm{Perm}(\{1, \ldots, n\})$$

21.1 Theorem (Cayley). *If G is a group of order n then G is isomorphic to a subgroup of S_n.*

Proof. Let S be the set of all elements of G. Consider the action of G on S

$$G \times S \to S, \quad a \cdot b := ab$$

This action defines a homomorphism $\varrho \colon G \to \mathrm{Perm}(S)$. Check: this homomorphism is 1-1. It follows that G is isomorphic to a subgroup of $\mathrm{Perm}(S)$. Finally, since $|S| = n$ we have $\mathrm{Perm}(S) \cong S_n$. $\square$

21.2 Notation. Denote

$$[n] := \{1, \ldots, n\}$$

If $\sigma \in S_n$, $\sigma \colon [n] \to [n]$ then we write

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & \ldots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \ldots & \sigma(n) \end{pmatrix}$$

21.3 Definition. A permutation $\sigma \in S_n$ is a *cycle of length* r (or r*-cycle*) if there are distinct integers $i_1, \ldots, i_r \in [n]$ such that

$$\sigma(i_1) = i_2,\ \sigma(i_2) = i_3,\ \ldots\ ,\ \sigma(i_r) = i_1$$

and $\sigma(j) = j$ for $j \neq i_1, \ldots, i_r$.

A cycle of length 2 is called a *transposition*.

Note. The only cycle of length 1 is the identity element in S_n.

21.4 Notation. If σ is a cycle as above then we write

$$\sigma = (i_1\ i_2\ \dots\ i_r)$$

21.5 Example. In S_5 we have

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 2 & 5 & 3 \end{pmatrix} = (2\ 4\ 5\ 3)$$

Note: $(2\ 4\ 5\ 3) = (4\ 5\ 3\ 2) = (5\ 3\ 2\ 4) = (3\ 2\ 4\ 5)$.

21.6 Definition. Permutations $\sigma, \tau \in S_n$ are *disjoint* if

$$\{i \in [n] \mid \sigma(i) \neq i\} \ \cap\ \{j \in [n] \mid \tau(j) \neq j\} = \varnothing$$

21.7 Proposition. *If σ, τ are disjoint permutations then $\sigma\tau = \tau\sigma$.*

Proof. Exercise. □

21.8 Proposition. *Every non-identity permutation $\sigma \in S_n$ is a product of disjoint cycles of length ≥ 2. Moreover, this decomposition into cycles is unique up to the order of factors.*

21.9 Example. Let $\sigma \in S_9$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 4 & 7 & 1 & 3 & 2 & 6 & 5 & 9 & 8 \end{pmatrix}$$

Then $\sigma = (1\ 4\ 3)(2\ 7\ 5)(8\ 9)$.

Proof of proposition 21.8. Consider the action of $\mathbb{Z}$ on the set $[n]$ given by

$$k \cdot i = \sigma^k(i)$$

for $k \in \mathbb{Z}$, $i \in [n]$. Notice that

$$\mathrm{Orb}(i) = \{\sigma^k(i) \mid k \in \mathbb{Z}\}$$

Define $\sigma_i \colon [n] \to [n]$

$$\sigma_i(j) = \begin{cases} \sigma(j) & \text{if } j \in \mathrm{Orb}(i) \\ j & \text{otherwise} \end{cases}$$

Notice that σ_i is a bijection since $\sigma(\mathrm{Orb}(i)) = \mathrm{Orb}(i)$. Thus $\sigma_i \in S_n$. Check:

1) σ_i is a cycle of length $|\mathrm{Orb}(i)|$.

2) if $\mathrm{Orb}(i_1), \dots, \mathrm{Orb}(i_r)$ are all distinct orbits of $[n]$ containing more than one element then $\sigma_{i_1}, \dots, \sigma_{i_r}$ are non-trivial, disjoint cycles and

$$\sigma = \sigma_{i_1} \cdot \ldots \cdot \sigma_{i_r}$$

Uniqueness of decomposition - easy. □

21.10 Proposition. *Every permutation $\sigma \in S_n$ is a product of (not necessarily disjoint) transpositions.*

Proof. By Proposition 21.8 it is enough to show that every cycle is a product of transpositions. We have:

$$(i_1\ i_2\ i_3\ \ldots\ i_r) = (i_1\ i_r)(i_1\ i_{r-1}) \cdot \ldots \cdot (i_1\ i_3)(i_1\ i_2)$$

□

Note. For $\sigma \in S_n$ we have a bijection

$$\sigma \times \sigma \colon [n] \times [n] \to [n] \times [n]$$

given by $\sigma \times \sigma(i, j) = (\sigma(i), \sigma(j))$. Define

$$S_\sigma := \{(i, j) \in [n] \times [n] \mid i > j \text{ and } \sigma(i) < \sigma(j)\}$$

21.11 Definition. A permutation $\sigma \in S_n$ is *even* (resp. *odd*) if the number of elements of S_σ is even (resp. odd).

21.12 Theorem. *1) The map* $\mathrm{sgn}\colon S_n \to \mathbb{Z}/2\mathbb{Z}$ *defined by*

$$\mathrm{sgn}(\sigma) = \begin{cases} 0 & \textit{if } \sigma \textit{ is even} \\ 1 & \textit{if } \sigma \textit{ is odd} \end{cases}$$

is a homomorphism.

2) If σ *is a transposition then* $\mathrm{sgn}(\sigma) = 1$*, so this homomorphism is non-trivial.*

Proof. 1) Let $\sigma, \tau \in S_n$. Denote $s_\sigma = |S_\sigma|$. We want to show

$$s_{\tau\sigma} \equiv s_\tau + s_\sigma \pmod 2$$

Let $[n]^+ := \{(i, j) \in [n] \times [n] \mid i > j\}$. Define subsets $P_\sigma, R_\sigma, P_\tau, R_\tau \subseteq [n]^+$ as follows:

$$\begin{aligned} P_\sigma :=& \{(i, j) \mid \sigma^{-1}(i) > \sigma^{-1}(j)\} \\ R_\sigma :=& \{(i, j) \mid \sigma^{-1}(i) < \sigma^{-1}(j)\} \\ P_\tau :=& \{(i, j) \mid \tau(i) > \tau(j)\} \\ R_\tau :=& \{(i, j) \mid \tau(i) < \tau(j)\} \end{aligned}$$

Notice that $s_\sigma = |R_\sigma|$ and $s_\tau = |R_\tau|$. Notice also that $(i, j) \in S_{\tau\sigma}$ iff either $(\tau(i), \tau(j)) \in P_\sigma \cap R_\tau$ or $(\tau(j), \tau(i)) \in R_\sigma \cap P_\tau$. This gives

$$s_{\tau\sigma} = |P_\sigma \cap R_\tau| + |R_\sigma \cap P_\tau|$$

On the other hand we have:

$$s_\sigma = |R_\sigma| = |R_\sigma \cap P_\tau| + |R_\sigma \cap R_\tau|$$

$$s_\tau = |R_\tau| = |P_\sigma \cap R_\tau| + |R_\sigma \cap R_\tau|$$

Therefore

$$s_\sigma + s_\tau = |R_\sigma \cap P_\tau| + |P_\sigma \cap R_\tau| + 2|R_\sigma \cap R_\tau| = s_{\tau\sigma} + 2|R_\sigma \cap R_\tau|$$

and so $s_\tau + s_\sigma \equiv s_{\tau\sigma} \pmod 2$.

2) Exercise. $\square$

21.13 Definition/Proposition. The set

$$A_n = \{\sigma \in S_n \mid \sigma \text{ is even}\}$$

is a normal subgroup of S_n. It is called the *alternating group on n letters*.

Proof. It is enough to notice that $A_n = \mathrm{Ker}(\mathrm{sgn})$. $\square$

Note. We have

$$S_n/A_n \cong \mathbb{Z}/2\mathbb{Z}$$

Since $|S_n| = n!$ thus $|A_n| = \frac{n!}{2}$.

21.14 Proposition. *If $\sigma \in S_n$ then σ is even (resp. odd) iff σ is a product of an even (resp. odd) number of transpositions.*

Proof. If $\sigma = \tau_1 \dots \tau_m$ where $\tau_1, \dots, \tau_m$ are transpositions then

$$\mathrm{sgn}(\sigma) = \mathrm{sgn}(\tau_1 \dots \tau_m) = \sum_{i=1}^{m} \mathrm{sgn}(\tau_i) = \sum_{i=1}^{m} 1$$

Thus $\mathrm{sgn}(\sigma) = 0$ iff m is even and $\mathrm{sgn}(\sigma) = 1$ iff m is odd.

□

Note. If follows that if a permutation $\sigma \in S_n$ is a product of an even number of transpositions then it cannot be written as a product of an odd number of transpositions (and vice versa).

21.15 Corollary. *A permutation $\sigma \in S_n$ is even iff*

$$\sigma = \sigma_1 \sigma_2 \dots \sigma_r$$

where σ_i is a cycle of length m_i and $\sum_{i=1}^{r}(m_i - 1)$ is even.

Proof. It is enough to notice that by the proof of Proposition 21.10 a cycle of length m is a product of $m - 1$ transpositions. □

Note. The usual notation for the sign of a permutation is

$$\mathrm{sgn}(\sigma) = \begin{cases} 1 & \text{if } \sigma \text{ is even} \\ -1 & \text{if } \sigma \text{ is odd} \end{cases}$$

where $\{-1, 1\} \cong \mathbb{Z}/2\mathbb{Z}$ is the multiplicative group of units in $\mathbb{Z}$.

22 Simplicity of alternating groups

22.1 Theorem. *The alternating group A_n is simple for $n \geq 5$.*

22.2 Lemma. *For $n \geq 3$ every element of A_n is a product of 3-cycles.*

Proof. It is enough to show that if $n \geq 3$ and τ, σ are transpositions in S_n then $\tau\sigma$ is a product of 3-cycles.

Case 1) τ, σ are disjoint transpositions: $\tau = (i\ j)$, $\sigma = (k\ l)$ for distinct elements $i, j, k, l \in [n]$. Then we have

$$\tau\sigma = (i\ j\ k)(j\ k\ l)$$

Case 2) τ, σ are not disjoint: $\tau = (i\ j)$, $\sigma = (j\ k)$. Then

$$\tau\sigma = (i\ j\ k)$$

□

22.3 Lemma. *If $n \geq 5$ and σ, σ' are 3-cycles in S_n then*

$$\sigma' = \tau\sigma\tau^{-1}$$

for some $\tau \in A_n$

Proof. Check: if $(i_1\ i_2\ \dots\ i_r)$ is a cycle in S_n then for any $\omega \in S_n$ we have

$$\omega(i_1\ i_2\ \dots\ i_r)\omega^{-1} = (\omega(i_1)\ \omega(i_2)\ \dots \omega(\ i_r))$$

If $\sigma = (i_1\ i_2\ i_3)$, $\sigma' = (j_1\ j_2\ j_3)$ then take $\omega \in S_n$ such that $\omega(i_k) = j_k$ for $k = 1, 2, 3$. We have

$$\sigma' = \omega\sigma\omega^{-1}$$

If $\omega \in A_n$ we can then take $\tau := \omega$.

Assume then that $\omega \notin A_n$. Since $n \geq 5$ there are $r, s \in [n]$ such that $(r\ s)$ and $\sigma = (i_1\ i_2\ i_3)$ are disjoint cycles. Take $\tau = \omega(r\ s)$. Then $\tau \in A_n$. Moreover, since $(r\ s)$ commutes with σ we have

$$\tau\sigma\tau^{-1} = \omega(r\ s)\sigma(r\ s)^{-1}\omega^{-1} = \omega\sigma\omega^{-1} = \sigma'$$

□

22.4 Corollary. *If $n \geq 5$ and H is a normal subgroup of A_n such that H contains some 3-cycle then $H = A_n$.*

Proof. By Lemma 22.3 H contains all 3-cycles, and so by Lemma 22.2 it contains all elements of A_n. □

Proof of Theorem 22.1. Let $n \geq 5$, $H \lhd A_n$ and $H \neq \{(1)\}$. We need to show that $H = A_n$. By Corollary 22.4 it will suffice to show that H contains some 3-cycle.

Let $(1) \neq \sigma$ be an element of H with the maximal number of fixed points in $[n]$. We will show that σ is 3-cycle. Take the decompositon of σ into dosjoint cycles:

$$\sigma = \sigma_1\sigma_2 \cdot \ldots \cdot \sigma_m$$

Case 1) $\sigma_1, \ldots, \sigma_m$ are transpositions.

Since $\sigma \in A_n$ we must then have $m \geq 2$. Say, $\sigma_1 = (i\ j)$, $\sigma_2 = (k\ l)$. Take $s \neq i, j, k, l$ and let $\tau = (k\ l\ s) \in A_n$. Since H is normal in A_n we have

$$\tau\sigma\tau^{-1}\sigma^{-1} \in H$$

Check:

1) $\tau\sigma\tau^{-1}\sigma^{-1} \neq (1)$ since $\tau\sigma\tau^{-1}\sigma^{-1}(k) \neq k$

2) $\tau\sigma\tau^{-1}\sigma^{-1}$ fixes every element of $[n]$ fixed by σ
3) $\tau\sigma\tau^{-1}\sigma^{-1}$ fixes i, j.

Thus $\tau\sigma\tau^{-1}\sigma^{-1}$ has more fixed points than σ which is impossible by the definition of σ.

Case 2) σ_r is a cycle of length ≥ 3 for some $1 \leq r \leq m$.

We can assume $r = 1$: $\sigma_1 = (i\ j\ k \dots)$. If $\sigma = \sigma_1$ and σ_1 is a 3-cycle we are done.

Otherwise σ must move at least two more elements, say p, q. In such case take $\tau = (k\ p\ q)$. We have

$$\tau\sigma\tau^{-1}\sigma^{-1} \in H$$

Check:

1) $\tau\sigma\tau^{-1}\sigma^{-1} \neq (1)$ since $\tau\sigma\tau^{-1}\sigma^{-1}(k) \neq k$
2) $\tau\sigma\tau^{-1}\sigma^{-1}$ fixes every element of $[n]$ fixed by σ
3) $\tau\sigma\tau^{-1}\sigma^{-1}$ fixes j.

Thus $\tau\sigma\tau^{-1}\sigma^{-1}$ has more fixed points than σ which is again impossible by the definition of σ.

As a consequence σ must be a 3-cycle.

$\square$

22.5. Classification of simple finite groups.

1) cyclic groups $\mathbb{Z}/p\mathbb{Z}$, p – prime
2) alternating groups A_n, $n \geq 5$
3) finite simple groups of Lie type, e.g. projective special linear groups

$$PSL_n(\mathbb{F}) := SL_n(\mathbb{F})/Z(SL_n(\mathbb{F}))$$

$\mathbb{F}$ -finite field, $n \geq 2$ (and $n > 2$ if $\mathbb{F} = \mathbb{F}_2$ or $\mathbb{F} = \mathbb{F}_3$).

4) 26 sporadic groups (the smallest: Mathieu group M_{11}, $|M_{11}| = 7920$, the biggest: the Monster M, $|M| \approx 8 \cdot 10^{53}$).

23 Solvable groups

Recall. Every finite group G has a composition series:

$$\{e\} = G_0 \subseteq \ldots \subseteq G_k = G$$

where $G_{i-1} \lhd G_i$ and G_i/G_{i-1} is a simple group.

23.1 Definition. A group G is *solvable* if it has a composition series

$$\{e\} = G_0 \subseteq \ldots \subseteq G_k = G$$

such that for every i the group G_i/G_{i-1} is a simple abelian group (i.e. $G_i/G_{i-1} \cong \mathbb{Z}/p_i\mathbb{Z}$ for some prime p_i).

23.2 Example.

1) Every finite abelian group is solvable.

2) For $n \geq 5$ the symmetric group S_n has a composition series

$$\{(1)\} \subseteq A_n \subseteq S_n$$

and so S_n is not solvable.

23.3 Proposition. *A finite group G is solvable iff it has a normal series*

$$\{e\} = H_0 \subseteq \ldots \subseteq H_l = G$$

such that H_j/H_{j-1} is an abelian group for all j.

Proof. Exercise. $\square$

Recall.

1) If G is a group the $[G, G]$ is the commutator subgroup of G

$$[G, G] = \langle \{aba^{-1}b^{-1} \ \mid \ a, b \in G\} \rangle$$

2) $[G, G]$ is the smallest normal subgroup of G such that $G/[G, G]$ is abelian: if G/H for some $H \lhd G$ then $[G, G] \subseteq H$.

23.4 Definition. For a group G the *derived series of* G is the normal series

$$\cdots \subseteq G^{(2)} \subseteq G^{(1)} \subseteq G^{(0)} = G$$

where $G^{i+1} = [G^{(i)}, G^{(i)}]$ for $i \geq 1$. The group $G^{(i)}$ is called the i*-th derived group of* G.

23.5 Theorem. *A group G is solvable iff $G^{(n)} = \{e\}$ for some $n \geq 0$.*

Proof. Exercise. □

23.6 Theorem.

1) Every subgroup of a solvable group is solvable.

2) Ever quotient group of a solvable group is solvable.

3) If $H \lhd G$, and both H and G/H are solvable groups then G is also solvable.

Proof.

1) If $H \subseteq G$ then $H^{(i)} \subseteq G^{(i)}$. Thus if $G^{(n)} = \{e\}$ then $H^{(n)} = \{e\}$.

2) For $H \lhd G$ take the canonical epimorphism $f \colon G \to G/H$. We have

$$f(G^{(i)}) = (G/H)^{(i)}$$

Therefore if $G^{(n)} = \{e\}$ then $(G/H)^{(n)} = \{e\}$.

3) Assume that $H \lhd G$, and that $H^{(m)}$, $(G/H)^{(n)}$ are trivial groups. Consider the canonical epimorphism $f\colon G \to G/H$. We have

$$f(G^{(n)}) = (G/H)^{(n)} = \{e\}$$

Therefore $G^{(n)} \subseteq \mathrm{Ker}(f) = H$. As a consequence we obtain

$$G^{(n+m)} = \left(G^{(n)}\right)^{(m)} \subseteq H^{(m)} = \{e\}$$

□

23.7 Theorem (Feit-Thompson)**.** *Every finite group of odd order is solvable.*

Proof. See: W. Feit, J.G. Thompson, *Solvability of groups of odd order*, Pacific Journal of Mathematics 13(3) (1963), 775-1029.

□

23.8 Corollary. *There are no non-abelian finite simple groups of odd order.*

Proof. Let $G \neq \{e\}$ be a simple group of odd order. By Theorem 23.7 G is solvable so $[G, G] \neq G$. Since $[G, G] \lhd G$, by simplicity of G we must have $[G, G] = \{e\}$, and so G is an abelian group. □

24 Nilpotent groups

24.1. Recall that if G is a group then

$$Z(G) = \{a \in G \mid ab = ba \text{ for all } b \in G\}$$

Note that $Z(G) \lhd G$. Take the canonical epimorphism $\pi\colon G \to G/Z(G)$. Since $Z\left(G/Z(G)\right) \lhd G/Z(G)$ we have:

$$\pi^{-1}\left(Z\left(G/Z(G)\right)\right) \lhd G$$

Define:

$$\begin{aligned} Z_1(G) :=& Z(G) \\ Z_i(G) :=& \pi_i^{-1}\left(Z\left(G/Z_{i-1}(G)\right)\right) \qquad \text{for } i > 1 \end{aligned}$$

where $\pi_i\colon G \to G/Z_{i-1}(G)$. We have $Z_i(G) \lhd G$ for all i.

24.2 Definition. The *upper central series* of a group G is a sequence of normal subgroups of G:

$$\{e\} = Z_0(G) \subseteq Z_1(G) \subseteq Z_2(G) \subseteq \dots$$

24.3 Definition. A group G is *nilpotent* if $Z_i(G) = G$ for some i.

If G is a nilpotent group then the *nilpotency class* of G is the smallest $n \geq 0$ such that $Z_n(G) = G$.

24.4 Proposition. *Every nilpotent group is solvable.*

Proof. If G is nilpotent group then the upper central series of G

$$\{e\} = Z_0(G) \subseteq Z_1(G) \subseteq \ \dots \ \subseteq Z_n(G) = G$$

is a normal series.

Moreover, for every i we have

$$Z_i(G)/Z_{i-1}(G) = Z(G/Z_{i-1}(G))$$

so all quotients of the upper central series are abelian. $\square$

24.5 Note. Not every solvable group is nilpotent. Take e.g. G_T. We have $Z(G_T) = \{I\}$, and so

$$Z_i(G_T) = \{I\}$$

for all i. Thus G_T is not nilpotent. On the other hand G_T is solvable with a composition series

$$\{I\} \subseteq \{I, R_1, R_2\} \subseteq G_T$$

24.6 Proposition.

1) *Every abelian group is nilpotent.*
2) *Every finite p-group is nilpotent.*

Proof.

1) If G is abelian then $Z_1(G) = G$.

2) If G is a p-group then so is $G/Z_i(G)$ for every i. By Theorem 16.4 if $G/Z_i(G)$ is non-trivial then its center $Z(G/Z_i(G))$ a non-trivial group. This means that if $Z_i(G) \neq G$ then $Z_i(G) \subsetneq Z_{i+1}(G)$ and $Z_i(G) \neq Z_{i+1}(G)$. Since G is finite we must have $Z_n(G) = G$ for some G. $\square$

24.7 Definition. A *central series* of a group G is a normal series

$$\{e\} = G_0 \subseteq \ldots \subseteq G_k = G$$

such $G_i \lhd G$ and $G_{i+1}/G_i \subseteq Z(G/G_i)$ for all i.

24.8 Proposition. *If $\{e\} = G_0 \subseteq \ldots \subseteq G_k = G$ is a central series of G then*

$$G_i \subseteq Z_i(G)$$

Proof. Exercise. □

24.9 Corollary. *A group G is nilpotent iff it has a central series.*

Proof. If G is nilpotent then

$$\{e\} = Z_0(G) \subseteq Z_1(G) \subseteq \ \ldots \ \subseteq Z_n(G) = G$$

is a central series of G.

Conversely, if

$$\{e\} = G_0 \subseteq \ldots \subseteq G_k = G$$

is a central series of G then by (24.9) we have $G = G_k \subseteq Z_k(G)$, so $G = Z_k(G)$, and so G is nilpotent. □

24.10 Note. Given a group G define

$$\begin{aligned}\Gamma_0(G) :=& G\\ \Gamma_i(G) :=& [G, \Gamma_{i-1}(G)] \quad \text{for } i > 0.\end{aligned}$$

We have

$$\ldots \subseteq \Gamma_1(G) \subseteq \Gamma_0(G) = G$$

24.11 Proposition. *If G is a group then*

1) *$\Gamma_i(G) \lhd G$ for all i*

2) $\Gamma_i(G)/\Gamma_{i+1}(G) \subseteq Z(G/\Gamma_{i+1}(G))$ *for all* i

Proof. Exercise. $\square$

24.12 Definition. If $\Gamma_n(G) = \{e\}$ then

$$\{e\} = \Gamma_n(G) \subseteq \ldots \subseteq \Gamma_0(G) = G$$

is a central series of G. It is called the *lower central series* of G.

24.13 Proposition. *A group* G *is nilpotent iff* $\Gamma_n(G) = \{e\}$

Proof. Exercise. $\square$

24.14 Theorem.

1) Every subgroup of a nilpotent group is nilpotent.

2) Ever quotient group of a nilpotent group is nilpotent.

Proof. Exercise. $\square$

24.15 Note. The properties of nilpotent group given in Theorem 24.14 are analogous to the first two properties of solvable groups from Theorem 23.6. The third part of that theorem (if H, G/H are solvable then so is G) is not true for nilpotent groups. For example, take $G = G_T$ and $H = \{I, R_1, R_2\}$. Both H and G/H are nilpotent, but by (24.5) G is not.

24.16 Proposition. *If $G_1, \dots, G_k$ are nilpotent groups then the direct product $G_1 \times \cdots \times G_k$ is also nilpotent.*

Proof. It will be enough to show that the statement holds for $k = 2$, then the general case will follow by induction with respect to k. Notice for any groups G_1, G_2 we have $\Gamma_i(G_1 \times G_2) = \Gamma_i(G_1) \times \Gamma_i(G_2)$. If G_1 and G_2 are nilpotent then by (24.13) there exists $n \geq 0$ such that $\Gamma_n(G_1) = \{e\}$ and $\Gamma_n(G_2) = \{e\}$. This implies that $\Gamma_n(G_1 \times G_2)$ is trivial, and so using (24.13) again we obtain that $G_1 \times G_2$ is nilpotent.

□

24.17 Corollary. *If $p_1, \dots, p_k$ are primes and P_i is a p_i-group then $P_1 \times \ldots \times P_k$ is a nilpotent group.*

Proof. Follows from (24.6) and (24.16). □

24.18 Theorem. *Let G be a finite group. The following conditions are equivalent.*

1) *G is nilpotent.*
2) *Every Sylow subgroup of G is a normal subgroup.*
3) *G isomorphic to the direct product of its Sylow subgroups.*

24.19 Lemma. *If G is a finite group and P is a Sylow p-subgroup of G then*

$$N_G(N_G(P)) = N_G(P)$$

Proof. Since $P \subseteq N_G(P) \subseteq G$ and P is a Sylow p-subgroup of G therefore P is a Sylow p-subgroup of $N_G(P)$. Moreover, $P \lhd N_G(P)$, so P is the only Sylow p-subgroup of G.

Take $a \in N_G(N_G(P))$. We will show that $a \in N_G(P)$. We have

$$aPa^{-1} \subseteq aN_G(P)a^{-1} = N_G(P)$$

As a consequence aPa^{-1} is a Sylow p-subgroup of $N_G(P)$, and thus $aPa^{-1} = P$. By the definitions of normalizer this gives $a \in N_G(P)$. $\square$

24.20 Lemma. *If H is a proper subgroup of a nilpotent group G (i.e. $H \subseteq G$, and $H \neq G$), then H is a proper subgroup of $N_G(H)$.*

Proof. Let $k \geq 0$ be the biggest integer such that $Z_k(G) \subseteq H$. Take $a \in Z_{k+1}(G)$ such that $a \notin H$. We will show that $a \in N_G(H)$.

We have

$$H/Z_k(G) \subseteq G/Z_k(G) \quad \text{and} \quad Z_{k+1}(G)/Z_k(G) = Z(G/Z_k(G))$$

If follows that for every $h \in H$ we have

$$ahZ_k(G) = (aZ_k(G))(hZ_k(G)) = (hZ_k(G))(aZ_k(G)) = haZ_k(G)$$

Therefore $ha = ahh'$ for some $h' \in Z_k(G) \subseteq H$, and so $a^{-1}ha = hh' \in H$. As a consequence $a^{-1}Ha = H$, so $a^{-1} \in N_G(H)$, and so also $a \in N_G(H)$.

$\square$

Proof of Theorem 24.18.

1) $\Rightarrow$ 2) Let P be a Sylow p-subgroup of G. It suffices to show that $N_G(P) = G$.

Assume that this is not true. Then $N_G(P)$ is a proper subgroup G, and so by Lemma 24.20 it is also a proper subgroup of $N_G(N_G(P))$. On the other hand by Lemma 24.19 we have $N_G(N_G(P)) = N_G(P)$, so we obtain a contradiction.

2) $\Rightarrow 3)$ Exercise.

3) $\Rightarrow 1)$ Follows from Corollary 24.17.

$\square$

25 Rings

25.1 Definition. A *ring* is a set R together with two binary operations: addition $(+)$ and multiplication $(\cdot)$ satisfying the following conditions:

1) R with addtion is an abelian group.
2) multiplication is associative: $(ab)c = a(bc)$
3) addition is distributive with respect to multiplication:

$$a(b+c) = ab + ac \qquad\qquad (a+b)c = ac + bc$$

The ring R is *commutative* if $ab = ba$ for all $a, b \in R$.

The ring R is a *ring with identity* if there is an element $1 \in R$ such that $a1 = 1a = a$ for all $a \in R$. (Note: if such identity element exists then it is unique)

25.2 Examples.

1) $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are commutative rings with identity.

2) $\mathbb{Z}/n\mathbb{Z}$ is a ring with multiplication given by

$$k(n\mathbb{Z}) \cdot l(n\mathbb{Z}) := kl(n\mathbb{Z})$$

3) If R is a ring then

$$R[x] = \{a_0 + a_1x + \ldots + a_nx^n \mid a_i \in R,\ n \geq 0\}$$

   is the ring of polynomials with coefficients in R and

$$R[[x]] = \{a_0 + a_1x + \ldots \mid a_i \in R\}$$

   is the ring of formal power series with coefficients in R.

   If R is a commutative ring then so are $R[x]$, $R[[x]]$. If R has identity then $R[x]$, $R[[x]]$ also have identity.

4) If R is a ring then $M_n(R)$ is the ring of $n \times n$ matrices with coefficients in R.

5) The set $2\mathbb{Z}$ of even integers with the usual addition and multiplication is a commutative ring without identity.

6) If G is an abelian group then the set $\mathrm{Hom}(G, G)$ of all homomorphisms $f\colon G \to G$ is a ring with multiplication given by composition of homomorphisms and addition defined by

$$(f+g)(a) := f(a) + g(a)$$

7) If R is a ring and G is a group then define

$$R[G] := \{ \sum_{g \in G} a_g g \mid a_g \in R,\ a_g \neq 0 \text{ for finitely many } g \text{ only } \}$$

addition in $R[G]$:

$$\sum_{g\in G} a_g g + \sum_{g \in G} b_g g = \sum_{g\in G} (a_g + b_g) g$$

multiplication in $R[G]$:

$$\left(\sum_{g\in G} a_g g\right)\left(\sum_{g\in G} b_g g\right) = \sum_{g\in G}\left(\sum_{hh'=g} a_h a_{h'}\right) g$$

The ring $R[G]$ is called the *group ring* of G with coefficients in R.

25.3 Definition. Let R be a ring. An element $0 \neq a \in R$ is a *left (resp. right) zero divisor in* R if there exists $0 \neq b \in R$ such that $ab = 0$ (resp. $ba = 0$).

An element $0 \neq a \in R$ is a *zero divisor* if it is both left and right zero divisor.

25.4 Example. In $\mathbb{Z}/6\mathbb{Z}$ we have $2 \cdot 3 = 0$, so 2 and 3 are zero divisors.

25.5 Definition. An *integral domain* is a commutative ring with identity $1 \neq 0$ that has no zero divisors.

25.6 Proposition. *Let R be an integral domain. If $a, b, c \in R$ are non-zero elements such that*

$$ac = bc$$

then $a = b$.

Proof. We have $(a - b)c = 0$. Since $c \neq 0$ and R has no zero divisors this gives $a - b = 0$, and so $a = b$. □

25.7 Definition. Let R be a ring with identity. An element a has a *left (resp. right) inverse* if there exists $b \in R$ such that $ba = 1$ (resp. there exists $c \in R$ such that $cb = 1$).

An element $a \in R$ is a *unit* if it has both a left and a right inverse.

25.8 Proposition. *If a is a unit of R then the left inverse and the right inverse of a coincide.*

Proof. If $ba = 1 = ac$ then

$$b = b \cdot 1 = b(ac) = (ba)c = 1 \cdot c = c$$

□

25.9 Note. The set of all units of a ring R forms a group R^* (with multiplication). E.g.:

$\mathbb{Z}^* = \{-1, 1\} \cong \mathbb{Z}/2\mathbb{Z}$
$\mathbb{R}^* = \mathbb{R} - \{0\}$
$(\mathbb{Z}/14\mathbb{Z})^* = \{1, 3, 5, 9, 11, 13\} \cong \mathbb{Z}/6\mathbb{Z}$

25.10 Definition. A *division ring* is a ring R with identity $1 \neq 0$ such that every non-zero element of R is a unit.

A *field* is a commutative division ring.

25.11 Examples.

1) $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{C}$ are fields.

2) $\mathbb{Z}$ is an integral domain but it is not a field.

3) The ring of *real quaternions* is defined by

$$\mathbb{H} := \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\}$$

Addition in $\mathbb{H}$ is coordinatewise. Multiplication is defined by the identities:

$$i^2 = j^2 = k^2 = -1, \quad ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j$$

The ring $\mathbb{H}$ is a (non-commutative) division ring with the identity

$$1 = 1 + 0i + 0j + 0k$$

The inverse of an element $z = a + bi + cj + dk$ is given by

$$z^{-1} = (a/\|z\|) - (b/\|z\|)i - (c/\|z\|)j - (d/\|z\|)k$$

where $\|z\| = \sqrt{a^2 + b^2 + c^2 + d^2}$

25.12 Proposition. *The following conditions are equivalent.*

1) $\mathbb{Z}/n\mathbb{Z}$ *is a field.*

2) $\mathbb{Z}/n\mathbb{Z}$ *is an integral domain.*

3) n *is a prime number.*

Proof. Exercise. $\square$

26 Ring homomorphisms and ideals

26.1 Definition. Let R, S be rings. A *ring homomorphism* is a map

$$f\colon R \to S$$

such that

1) $f(a+b) = f(a) + f(b)$
2) $f(ab) = f(a)f(b)$

26.2 Note. If R, S are rings with identity then these conditions do not guarantee that $f(1_R) = 1_S$.

Take e.g. rings with identity R_1, R_2 and define

$$R_1 \oplus R_2 = \{(r_1, r_2) \mid r_1 \in R_1, R_2\}$$

with addition and multiplication defined coordinatewise. Then $R_1 \oplus R_2$ is a ring with identity $(1_{R_1}, 1_{R_2})$. The map

$$f\colon R_1 \to R_1 \oplus R_2, \qquad f(r_1) = (r_1, 0)$$

is a ring homomorphism, but $f(1_{R_1}) \neq (1_{R_1}, 1_{R_2})$.

26.3 Note. Rings and ring homomorphisms form a category $\mathcal{R}ing$.

26.4 Proposition. *A ring homomorphism $f\colon R \to S$ is an isomorphism of rings iff f is a bijection.*

Proof. Exercise. $\square$

26.5 Definition. If $f\colon R \to S$ is a ring homomorphism then

$$\mathrm{Ker}(f) = \{a \in R \mid f(a) = 0\}$$

26.6 Proposition. *A ring homomorphism is 1-1 iff* $\mathrm{Ker}(f) = \{0\}$

Proof. The same as for groups (4.4). □

26.7 Definition. A *subring* of a ring R is a subset $S \subseteq R$ such that S is an additive subgroup of R and it is closed under the multiplication.

A *left ideal* of R is a subring $I \subseteq R$ such that for every $a \in I$ and $b \in R$ we have $ab \in I$. A *right ideal* of R is defined analogously.

A *ideal* of R is a subring $I \subseteq R$ such that I is both left and right ideal.

26.8 Notation. If I is an ideal of R then we write $I \lhd R$.

26.9 Proposition. *If* $f\colon R \to S$ *is a ring homomorphism then* $\mathrm{Ker}(f)$ *is an ideal of* R*.*

Proof. Exercise. □

26.10 Definition. If I is an ideal of a ring R then the *quotient ring* R/I is defined as follows.

$$R/I := \text{ the set of left cosets of } I \text{ in } R$$

Addition: $(a+I)+(b+I) = (a+b)+I$, multiplication: $(a+I)(b+I) = ab+I$.

26.11 Note. If $I \lhd R$ then the map

$$\pi\colon R \to R/I, \qquad \pi(a) = a + I$$

is a ring homomorphism. It is called the *canonical epimorphism* of R onto R/I.

26.12 Theorem. *If $f\colon R \to S$ is a homomorphism of rings then there is a unique homomorphism*

$$\bar{f}\colon R/\operatorname{Ker}(f) \to S$$

such that the following diagram commutes:

$$\begin{tikzcd}
R \arrow[r, "f"] \arrow[d, "\pi"'] & S \\
R/\operatorname{Ker}(f) \arrow[ur, dashed, "\bar{f}"'] &
\end{tikzcd}$$

Moreover, $\bar{f}$ is a monomorphism and $\operatorname{Im}(\bar{f}) = \operatorname{Im}(f)$.

Proof. Similar to the proof of Theorem 6.1 for groups. □

26.13 First Isomorphism Theorem. *If $f\colon R \to S$ is a homomorphism of rings that is an epimorphism then*

$$R/\operatorname{Ker}(f) \cong S$$

Proof. Take the map $\bar{f}\colon R/\operatorname{Ker}(f) \to S$. Then $\operatorname{Im}(\bar{f}) = \operatorname{Im}(f) = S$, so $\bar{f}$ is an epimorphism. Also, $\bar{f}$ is 1-1. Therefore $\bar{f}$ is a bijective homomorphism and thus it is an isomorphism. □

26.14 Note. Let $I, J \lhd R$. Check:

1) $I \cap J \lhd R$

2) $I + J \lhd R$ where $I + J = \{a + b \mid a \in I,\ b \in J\}$

26.15 Second Isomorphism Theorem. *If I, J are ideals of R then*

$$I/(I \cap J) \cong (I + J)/J$$

Proof. Exercise. $\square$

26.16 Third Isomorphism Theorem. *If I, J are ideals of R and $J \subseteq I$ then I/J is a ideal of R/J and*

$$(R/J)/(I/J) \cong R/I$$

Proof. Exercise. $\square$

Note. From now until Section 39 all rings are commutative rings with identity $1 \neq 0$ unless stated otherwise. Also, all ring homomorphisms preserve the identity.

27 Principal ideal domains and Euclidean rings

27.1 Definition. If R is a ring and S is a subset of R then denote

$$\langle S \rangle = \text{the smallest ideal of } R \text{ that contains } S$$

We say that $\langle S \rangle$ is the *ideal of R generated by the set S*.

27.2 Note. We have

$$\langle S \rangle = \{b_1 a_1 + \dots b_k a_k \mid a_i \in I,\ b_i \in R,\ k \geq 0\}$$

27.3 Definition. An ideal $I \lhd R$ is *finitely generated* if $I = \langle a_1, \dots, a_n \rangle$ for some $a_1, \dots, a_n \in R$.

An ideal $I \lhd R$ is a *principal ideal* if $I = \langle a \rangle$ for some $a \in R$.

27.4 Definition. A ring R is a *principal ideal domain* (*PID*) if it is an integral domain (25.5) such that every ideal of R is a principal ideal.

27.5 Proposition. *The ring of integers $\mathbb{Z}$ is a PID.*

Proof. Let $I \lhd \mathbb{Z}$. If $I = \{0\}$ then $I = \langle 0 \rangle$, so I is a principal ideal. If $I \neq \{0\}$ then let a be the smallest integer such that $a > 0$ and $a \in I$. We will show that $I = \langle a \rangle$.

Since $a \in I$ we have $\langle a \rangle \subseteq I$. Conversely, if $b \in I$ then we have

$$b = qa + r$$

for some $q, r \in \mathbb{Z}$, $0 \leq r \leq a - 1$. This gives $r = b - qa$, so $r \in I$. Since a is the smallest positive element of I, this implies that $r = 0$. Therefore $b = qa$, and so $b \in \langle a \rangle$. $\square$

27.6 Proposition. *If $\mathbb{F}$ is a field then $\mathbb{F}$ is a PID.*

Proof. If $I \lhd \mathbb{F}$ and $0 \neq a \in I$ then for every $b \in F$ we have

$$b = (ba^{-1})a \in I$$

And so $I = \mathbb{F}$. As a consequence the only ideals of $\mathbb{F}$ are $\{0\} = \langle 0 \rangle$ and $\mathbb{F} = \langle 1 \rangle$. $\square$

27.7 Proposition. *If $\mathbb{F}$ is a field then the ring of polynomials $\mathbb{F}[x]$ is a PID.*

Proof. Let $I \in R$. If $I = \{0\}$ then $I = \langle 0 \rangle$. Otherwise let $0 \neq p(x)$ be a polynomial such that $p(x) \in I$ and $\deg p(x) \leq \deg q(x)$ for all $q(x) \in I - \{0\}$. Check that $I = \langle p(x) \rangle$. $\square$

27.8 Note. $\mathbb{Z}[x]$ is not a PID. E.g. the ideal $\langle 2, x \rangle$ is not a principal ideal of $\mathbb{Z}[x]$ (check!).

27.9 Definition. A *Euclidean domain* is an integral domain R equipped with a function

$$N \colon R - \{0\} \longrightarrow \mathbb{N} = \{0, 1, \dots\}$$

such that

1) $N(ab) \geq N(a)$ for all $a, b \in R - \{0\}$
2) for any $a, b \in R$, $a \neq 0$ there exist $q, r \in R$ such that

$$b = qa + r$$

and either $r = 0$ or $N(r) < N(a)$.

The function N is called the *norm function* on R.

27.10 Examples.

1) $\mathbb{Z}$ is a Euclidean domain with the norm function given by the absolute value.

2) Any field $\mathbb{F}$ is a Euclidean domain with $N(a) = 0$ for all $a \in \mathbb{F} - \{0\}$.

3) If $\mathbb{F}$ is a field then $\mathbb{F}[x]$ is a Euclidean domain with $N(p(x)) = \deg p(x)$ for $p(x) \in \mathbb{F}[x]$, $p(x) \neq 0$.

   Note: $\mathbb{Z}[x]$ is not a Euclidean domain with $N(p(x)) = \deg p(x)$. E. g. there are no $q(x), r(x) \in \mathbb{Z}[x]$ such that either $r(x) = 0$ or $\deg r(x) < 0$ and that

$$x = 2q(x) + r(x)$$

4) The *ring of Gaussian integers* is the subring $\mathbb{Z}[i] \subseteq \mathbb{C}$ given by

$$\mathbb{Z}[i] := \{a + bi \in \mathbb{C} \mid a, b \in \mathbb{Z}\}$$

   $\mathbb{Z}[i]$ is a Euclidean domain with

$$N(a + bi) = a^2 + b^2 = (a + bi)(\overline{a + bi})$$

   (exercise).

5) Define

$$\mathbb{Z}[\sqrt{-5}] := \{a + b\sqrt{5}i \mid a, b \in \mathbb{Z}\}$$

   Exercise: $\mathbb{Z}[\sqrt{-5}]$ is not a Euclidean domain.

27.11 Theorem. *If R is a Euclidean domain then R is a PID.*

Proof. Let $I \lhd R$, $I \neq \{0\}$. Choose $a \in I$ such that $a \neq 0$ and that $N(a) \leq N(b)$ for all $b \in I - \{0\}$. Check that $\langle a \rangle = I$. $\square$

27.12 Note. It is not true that every PID is a Euclidean domain. Take e.g. $\alpha = \frac{1}{2} + \frac{\sqrt{19}}{2}i$ and let

$$\mathbb{Z}[\alpha] = \{a + b\alpha \mid a, b \in \mathbb{Z}\}$$

Then $\mathbb{Z}[\alpha]$ is a PID, but it is not a Euclidean domain. See J. C. Wilson, *A principal ideal ring that is not a euclidean ring*, Mathematics Magazine, 46 (1) (1973), 34-38. (Note: this link requires a JSTOR access)

28 Prime ideals and maximal ideals

28.1 Definition. Let R be a ring.

1) An ideal $I \lhd R$ is a *prime ideal* if $I \neq R$ and for any $a, b \in R$ we have

$$ab \in I \quad \text{iff} \quad \text{either } a \in I \text{ or } b \in I$$

2) An ideal $I \lhd R$ is a *maximal ideal* if $I \neq R$ and for any $J \lhd R$ such that $I \subseteq J \subseteq R$ we have either $J = I$ or $J = R$.

28.2 Examples.

1) The zero ideal $\{0\} \in R$ is a prime ideal iff R is an integral domain, and it is a maximal ideal iff R is a field.

2) Recall that if $I \lhd \mathbb{Z}$ then $I = n\mathbb{Z}$ for some $n \geq 0$. Check:

($n\mathbb{Z}$ is a prime ideal) iff ($n\mathbb{Z}$ is a maximal ideal) iff (n is a prime number)

3) $\langle x \rangle \lhd \mathbb{Z}[x]$ is prime ideal (check!) but it is not a maximal ideal:

$$\langle x \rangle \subseteq \langle 2, x \rangle \lhd \mathbb{Z}[x]$$

28.3 Proposition. *Let $I \lhd R$*

1) The ideal I is a prime ideal iff R/I is an integral domain.

2) The ideal I is a maximal ideal iff R/I is a field.

28.4 Corollary. *Every maximal ideal is a prime ideal.*

Proof. If $I \lhd R$ is a maximal ideal then R/I is a field. In particular R/I is an integral domain, and so I is a prime ideal. $\square$

Proof of Proposition 28.3.

1) Assume that $I \lhd R$ is a prime ideal. For $a+I, b+I \in R/I$ we have

$$(a+I)(b+I) = ab+I$$

Thus if $(a+I)(b+I) = 0+I$ then $ab+I = 0+I$ i.e. $ab \in I$. Since I is a prime ideal we get that either $a \in I$ (and so $a+I = 0+I$) or $b \in I$ (and so $b+I = 0+I$). Therefore R/I is an integral domain.

The other implication follows from a similar argument.

2) Assume that $I \lhd R$ is a maximal ideal. Let $a+I \in R/I$, $a+I \neq 0+I$. We want to show that there exists $b+I \in R/I$ such that

$$(a+I)(b+I) = 1+I$$

Take the ideal $J = \langle a \rangle + I$. We have

$$I \subseteq J \subseteq R$$

and $I \neq J$ (since $a \notin I$). Since I is a maximal ideal we must have $J = R$. In particular $1 \in J$, so

$$1 = ab + c$$

for some $b \in R$, $c \in I$. This gives

$$1+I = (ab+c)+I = ab+I = (a+I)(b+I)$$

Conversely, assume that R/I is a field, and let $J \lhd R$ be an ideal such that

$$I \subseteq J \subseteq R$$

We will show that either $J = I$ or $J = R$.

Take the canonical epimorphism $\pi\colon R \to R/I$. Since $\pi(J)$ is an ideal of R/I (check!) and R/I is a field we have either $\pi(J) = \{0 + I\}$ or $\pi(J) = R/I$. Also, since $I \subseteq J$, we have $\pi^{-1}(\pi(J)) = J$. If follows that either

$$J = \pi^{-1}(\{0 + I\}) = I \quad \text{or} \quad J = \pi^{-1}(R/I) = R$$

□

28.5 Examples.

1) Since an ideal $n\mathbb{Z}$ of $\mathbb{Z}$ is prime (and maximal) iff n is a prime number therefore $\mathbb{Z}/n\mathbb{Z}$ is an integral domain (and in fact a field) iff n is a prime number.

2) Take $x^2 + 1 \in \mathbb{R}[x]$. We have an epimorphism of rings

   $$f\colon \mathbb{R}[x] \longrightarrow \mathbb{C} \qquad f(p(x)) = p(i)$$

   Check: $\mathrm{Ker}(f) = \langle x^2 + 1\rangle$. By the First Isomorphism Theorem (26.13) we get $R[x]/\langle x^2 + 1\rangle \cong \mathbb{C}$. Since $\mathbb{C}$ is a field this shows that $\langle x^2 + 1\rangle$ is a maximal ideal of $\mathbb{R}[x]$.

28.6 Note. For $I, J \lhd R$ define

$$IJ := \{a_1 b_1 + \ldots + a_k b_k \mid a_i \in I,\ b_i \in J,\ k \geq 0\}$$

Check: IJ is an ideal of R.

28.7 Proposition. *Let $I \lhd R$, $I \neq R$. The ideal I is a prime ideal iff for any ideals J_1, J_2 such that*

$$J_1 J_2 \subseteq I$$

we have either $J_1 \subseteq I$ or $J_2 \subseteq I$.

Proof. Exercise. □

29 Zorn's Lemma and maximal ideals

Goal:

29.1 Theorem. *If R is a ring, $I \lhd R$, and $I \neq R$ then there exists a maximal ideal $J \lhd R$ such that $I \subseteq J$.*

29.2 Definition. A *partially ordered set* (or *poset*) is a set S equipped with a binary relation $\leq$ satisfying

1) $x \leq x$ for all $x \in S$ (reflexivity)
2) if $x \leq y$ and $y \leq z$ then $x \leq z$ (transitivity)
3) if $x \leq y$ and $y \leq x$ then $y = x$ (antisymmetry).

29.3 Example. If A is a set and S is the set of all subsets of A then S is a poset with ordering given by the inclusion of subsets.

29.4 Definition. A *linearly ordered set* is a poset $(S, \leq)$ such that for any $x, y \in S$ we have either $x \leq y$ or $y \leq x$.

29.5 Definition. If $(S, \leq)$ is a poset then an element $x \in S$ is a *maximal element* of S if we have $x \leq y$ only for $y = x$.

29.6 Example. Let S be the set of all *proper* subsets of a set A ordered with respect to the inclusion. For every $a \in A$ the set $A - \{a\}$ is a maximal element of S.

29.7 Note. If $(S, \leq)$ is a poset and $T \subseteq S$ then T is also a poset with ordering inherited from S.

29.8 Definition. Let $(S, \leq)$ is a poset and let $T \subseteq S$. An *upper bound of* T is an element $x \in S$ such that $y \leq x$ for all $y \in T$.

29.9 Definition. If $(S, \leq)$ is a poset. A *chain* in S is a subset $T \subseteq S$ such that T is linearly ordered.

29.10 Zorn's Lemma. *If $(S, \leq)$ is a non-empty poset such that every chain in S has an upper bound in S then S contains a maximal element.*

Proof of Theorem 29.1. Let $I \neq R$ be an ideal of R, and let S be the set of all ideals $J \triangleleft R$ such that $I \subseteq J$ and $J \neq R$. Notice that $S \neq \varnothing$ since $I \in S$.

The set S is a poset ordered with respect to inclusion of ideals. We will show that every chain in S has an upper bound in S. Let

$$T = \{J_i\}_{i \in A}$$

be a chain in S. Check: $J_T := \bigcup_{i \in A} J_i$ is an ideal of R. Moreover, $I \subseteq J_T$. Finally, we have $J_T \neq R$. Indeed, otherwise $1 \in J_T$, and so $1 \in J_i$ for some $i \in A$. This would give $J_i = R$, which contradicts our assumptions.

It follows that $J_T \in S$. Since $J_i \subseteq J_T$ for all $i \in A$, we obtain that J_T is an upper bound of the chain T.

By Zorn's Lemma (29.10) there is a maximal element $J \in S$. This means, in particular, that J is an ideal of R such $I \subseteq J$. Moreover, let $K \triangleleft R$ be any ideal such that $K \neq R$ and $J \subseteq K$. We have

$$I \subseteq J \subseteq K$$

which means that $K \in S$. Maximality of J is S implies then that $J = K$. This shows that J is a maximal ideal of R.

□

29.11 Corollary. *Every ring contains a maximal ideal.*

Proof. If R is a ring then by Theorem 29.1 there exists a maximal ideal $I \lhd R$ such that I contains the zero ideal $\{0\} \lhd R$. □

30 Unique factorization domains

Motivation:

30.1 Fundamental Theorem of Arithmetic.

If $n \in \mathbb{Z}$, $n > 1$ then

$$n = p_1 p_2 \cdot \ldots \cdot p_k$$

where $p_1, \ldots, p_k$ are primes. Moreover, this decomposition is unique up to re-ordering of factors.

Goal. Extend this to other rings.

30.2 Definition. Let R be an integral domain. An element $a \in R$ is *irreducible* if $a \neq 0$, a is not a unit and if $a = bc$ for some $b, c \in R$ then either b or c is a unit.

30.3 Examples.

1) $n \in \mathbb{Z}$ is irreducible iff $n = \pm p$ where p is a prime number.

2) A field has no irreducible elements.

3) Take $p(x) \in \mathbb{R}[x]$, $p(x) = x^2 + 1$. Then $p(x)$ is irreducible in $\mathbb{R}[x]$.

4) Take $p(x) \in \mathbb{C}[x]$, $p(x) = x^2 + 1$. Then $p(x)$ is not irreducible in $\mathbb{C}[x]$:

$$p(x) = (x - i)(x + i)$$

30.4 Note. If $a \in R$ is an irreducible element and $u \in R$ is a unit then ua is irreducible.

30.5 Definition. Let R be an integral domain. Elements $a, b \in R$ are *associates* if $a = ub$ for some unit $u \in R$. We write: $a \sim b$.

30.6 Examples.

1) If $m, n \in \mathbb{Z}$ then $m \sim n$ iff $m = \pm n$.

2) Check: units in $\mathbb{R}[x]$ are non-zero polynomials of degree 0. It follows that if $p(x), q(x) \in \mathbb{R}[x]$ then $p(x) \sim q(x)$ iff $p(x) = aq(x)$ for some $a \in \mathbb{R} - \{0\}$.

30.7 Definition. A *unique factorization domain* (*UFD*) is an integral domain R that satisfies the following conditions:

1) if $a \in R$ is a non-zero, non-unit element then
$$a = b_1 \cdot \ldots \cdot b_k$$
for some irreducible elements $b_1, \ldots, b_k \in R$

2) if $b_1, \ldots, b_k$, $c_1, \ldots, c_l$ are irreducible elements such that
$$b_1 \cdot \ldots \cdot b_k = c_1 \cdot \ldots \cdot c_l$$
then $k = l$ and for some permutation $\sigma\colon \{1, \ldots, k\} \to \{1, \ldots, k\}$ we have $b_1 \sim c_{\sigma(1)}, \ldots, b_k \sim c_{\sigma(k)}$.

30.8 Examples.

1) $\mathbb{Z}$ is a UFD by the Fundamental Theorem of Arithmetic (30.1).

2) If $\mathbb{F}$ is a field then $\mathbb{F}$ is a UFD since all non-zero elements of $\mathbb{F}$ are units.

Recall. $\mathbb{Z}[\sqrt{-5}]$ is the subring of $\mathbb{C}$ given by
$$\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{5}i \mid a, b \in \mathbb{Z}\}$$

30.9 Proposition. $\mathbb{Z}[\sqrt{-5}]$ *is not a UFD.*

Proof. For $a + b\sqrt{5}i \in \mathbb{Z}[\sqrt{-5}]$ define

$$N(a + b\sqrt{5}i) = (a + b\sqrt{5}i)(\overline{a + b\sqrt{5}i}) = a^2 + 5b^2 \in \mathbb{N}$$

Notice that

1) $N(\alpha) = 1$ iff $\alpha = \pm 1$
2) $N(\alpha) = 0$ iff $\alpha = 0$
3) $N(\alpha) \neq 3$ for all $\alpha \in \mathbb{Z}[\sqrt{-5}]$
4) $N(\alpha\beta) = N(\alpha)N(\beta)$

Observation 1. The only units in $\mathbb{Z}[\sqrt{-5}]$ are 1 and -1.
As a consequence for any α, β we have $\alpha \sim \beta$ iff $\alpha = \pm\beta$.

Indeed, if $\alpha \in \mathbb{Z}[\sqrt{-5}]$ is a unit then

$$N(\alpha)N(\alpha^{-1}) = N(\alpha\alpha^{-1}) = N(1) = 1$$

Therefore $N(\alpha) = 1$, and so $\alpha = \pm 1$.

Observation 2. If $\alpha \in \mathbb{Z}[\sqrt{-5}]$ is an element such that $N(\alpha) = 9$ then α is irreducible.

Indeed, if $\alpha = \beta\beta'$ then

$$N(\beta)N(\beta') = N(\alpha) = 9$$

Therefore $N(\beta)$ must be either 1 (and so β is a unit), 3 (impossible), or 9 (and then $N(\beta') = 1$, i.e. β' is a unit) .

Take $9 \in \mathbb{Z}[\sqrt{-5}]$. We have

$$3 \cdot 3 = 9 = (2 + \sqrt{5}i)(2 - \sqrt{5}i)$$

By Observation 2 the elements 3, $2 + \sqrt{5}i$, $2 - \sqrt{5}i$ are irreducible in $\mathbb{Z}[\sqrt{-5}]$. On the other hand by Observation 1 we obtain

$$3 \not\sim (2 + \sqrt{5}i), \quad 3 \not\sim (2 - \sqrt{5}i)$$

As a consequence 9 does not have a unique factorization in $\mathbb{Z}[\sqrt{-5}]$ $\square$

31 Prime elements

31.1 Definition. Let R be an integral domain, and let $a, b \in R$. We say that a *divides* b if $b = ac$ for some $c \in R$. We then write: $a \mid b$.

31.2 Proposition. *If R is an integral domain and $a, b \in R$ then $a \sim b$ iff $a \mid b$ and $b \mid a$.*

Proof. Exercise. $\square$

31.3 Definition. Let R is an integral domain. An element $a \in R$ is a *prime element* if $a \neq 0$, a is a non-unit and if $a \mid bc$ then either $a \mid b$ or $a \mid c$.

31.4 Example.

1) In $\mathbb{Z}$ we have

$$\{\text{prime elements}\} = \{\pm \text{ prime numbers}\} = \{\text{irreducible elements}\}$$

2) By the proof of Proposition 30.9 in $\mathbb{Z}[\sqrt{-5}]$ the element $\alpha = 2 + \sqrt{5}i$ is irreducible. On the other hand α is not a prime element:

$$\alpha \mid (3 \cdot 3) \quad \text{but} \quad \alpha \nmid 3$$

31.5 Proposition. *If R is an integral domain and $a \in R$ is a prime element then a is irreducible.*

Proof. Let $a \in R$ be a prime element and let $a = bc$. We want to show that either b or c must be a unit in R.

We have $a \mid (bc)$, and since a is a prime element it implies that $a \mid b$ or $a \mid c$.

We can assume that $a \mid b$. Since also $b \mid a$, thus by (31.2) we obtain that $a \sim b$, i.e. $a = bu$ for some unit $u \in R$. Therefore we have

$$bc = a = bu$$

By (25.6) this gives $u = c$, and so c is s unit. $\square$

31.6 Proposition. *If R is a UFD and $a \in R$ then a is an irreducible element iff a is a prime element.*

Proof.

($\Leftarrow$) Follows from Proposition 31.5.

($\Rightarrow$) Assume that $a \in R$ is irreducible and that $a \mid (bc)$. We want to show that either $a \mid b$ or $a \mid c$.

If $b = 0$ then $b = a \cdot 0$ so $a \mid 0$. If b is a unit then $c = b^{-1}bc$ so $a \mid c$.

As a consequence we can assume that b, c are non-zero, non-units.

Since $a \mid (bc)$ there is $d \in R$ such that $bc = ad$. Assume that d is not a unit. Since R is a UFD we have decompositions:

$$b = b_1 \cdot \ldots \cdot b_m, \quad c = c_1 \cdot \ldots \cdot c_n, \quad d = d_1 \cdot \ldots \cdot d_p$$

where b_i, c_j, d_k are irreducible. This gives

$$b_1 \cdot \ldots \cdot b_m \cdot c_1 \cdot \ldots \cdot c_n = a \cdot d_1 \cdot \ldots \cdot d_p$$

By the uniqueness of decomposition in UFDs this implies that either $a \sim b_i$ for some i or $a \sim c_j$ for some j. In the first case we get $a \mid b$, and in the second case $a \mid c$.

If d is a unit the argument is similar. $\square$

31.7 Theorem. *An integral domain R is a UFD iff*

1) *every non-zero, non-unit element of R is a product of irreducible elements*
2) *every irreducible element in R is a prime element.*

Proof.

($\Rightarrow$) Follows from the definition of UFD (30.7) and Propositon 31.6.

($\Leftarrow$) Assume that R satisfies conditions 1)-2) of the theorem. We only need to show that if $b_1, \ldots, b_k$, $c_1, \ldots, c_l$ are irreducible elements in R such that

$$b_1 \cdot \ldots \cdot b_k = c_1 \cdot \ldots \cdot c_l$$

then $k = l$, and after reordering factors we have $b_1 \sim c_1$, $\ldots$, $b_k \sim c_k$.

We argue by induction with respect to k.

If $k = 1$ then we have $b_1 = c_1 \cdot \ldots \cdot c_l$. Since b_1 is irreducible this implies that $l = 1$, and so $b_1 = c_1$.

Next, assume that the uniqueness property holds for some k and that we have

$$b_1 \cdot \ldots \cdot b_k \cdot b_{k+1} = c_1 \cdot \ldots \cdot c_l$$

where b_i, c_j are irreducible elements. This implies that $b_{k+1} \mid (c_1 \cdot \ldots \cdot c_l)$. By condition 2) of the theorem b_{k+1} is a prime element. It follows that $b_k \mid c_j$ for some $1 \leq j \leq l$. We can assume that $b_{k+1} \mid c_l$. Then $c_l = ab_{k+1}$ for some $a \in R$. Also, since c_l, b_{k+1} are irreducible a must be a unit. This shows that $b_{k+1} \sim c_l$. Furthermore, we obtain from here that

$$b_1 \cdot \ldots \cdot b_k \cdot b_{k+1} = c_1 \cdot \ldots \cdot c_{l-1} \cdot ab_{k+1}$$

Since R is an integral domain we get

$$b_1 \cdot \ldots \cdot b_k = c_1 \cdot \ldots \cdot c_{l-1}a$$

Since b_k is irreducible and a is a unit the product $c_{l-1}a$ is an irreducible element. Therefore by the inductive assumption $k = l - 1$, and after reordering of factors we have

$$b_1 \sim c_1, \quad \ldots, \quad b_{k-1} \sim c_{k-1}, \quad b_k \sim c_{l-1}a \sim c_{l-1}$$

$\square$

32 PIDs and UFDs

32.1 Theorem. *If R is a PID then it is a UFD.*

32.2 Lemma. *If R is a PID and $I_1, I_2, \dots$ are ideals of R such that*

$$I_1 \subseteq I_2 \subseteq \dots$$

then there exists $n \geq 1$ such that $I_n = I_{n+1} = \dots$.

Proof. Take $I = \bigcup_{i=1}^{\infty} I_i$. Check: I is an ideal of R. Since R is a PID we have $I = \langle a \rangle$ for some $a \in I$. Take $n \geq 1$ such that $a \in I_n$. Then we get

$$I \subseteq I_n \subseteq I_{n+1} \subseteq \ \dots \ \subseteq I$$

It follows that $I_n = I_{n+1} = \ \dots \ = I$. □

32.3 Lemma. *Let R be a PID. An element $a \in R$ is irreducible iff $\langle a \rangle$ is a maximal ideal of R.*

Proof. Exercise. □

32.4 Lemma. *If R is an integral domain then $a \in R$ is a prime element iff $\langle a \rangle$ is a non-zero prime ideal of R.*

Proof. Exercise. □

Proof of Theorem 32.1. Let R be a PID. By Theorem 31.7 it suffices to show that

1) every non-zero, non-unit element of R is a product of irreducible elements

2) every irreducible element in R is a prime element.

1) We argue by contradiction. Assume that $a_0 \in R$ is a non-zero, non-unit element that is not a product of irreducibles. This implies that

$$a_0 = a_1 b_1$$

for some non-zero, non-unit elements $a_1, b_1 \in R$.

Next, if both a_1 and b_1 were products of irreducibles, then a_0 would be also a product of irreducibles, contradicting our assumption. We can then assume that a_1 is not a product of irreducibles, and so in particular we have

$$a_1 = a_2 b_2$$

for some non-zero, non-unit elements $a_2, b_2 \in R$.

By induction we obtain that for $i = 1, 2, \ldots$ there exists non-zero, non-unit elements $a_i, b_i \in R$ such that $a_i = a_{i+1} b_{i+1}$ for all $i \geq 0$.

Consider the chain of ideals

$$\langle a_0 \rangle \subseteq \langle a_1 \rangle \subseteq \ldots$$

By Lemma 32.2 we obtain that $\langle a_n \rangle = \langle a_{n+1} \rangle$ for some $n \geq 0$. This means that $a_n = a_{n+1} u$ for some unit $u \in R$ (check!). As a consequence we obtain

$$a_{n+1} b_{n+1} = a_n = a_{n+1} u$$

and so $b_{n+1} = u$. This is a contradiction, since b_{n+1} is not a unit.

2) Let $a \in R$ be an irreducible element and let $a \mid (bc)$. We need to show that either $a \mid b$ or $a \mid c$.

Assume that $a \nmid b$. This implies that $b \notin \langle a \rangle$ and so

$$\langle a \rangle \neq \langle a \rangle + \langle b \rangle$$

Since by Lemma 32.3 the ideal $\langle a \rangle$ is a maximal ideal we obtain then that $\langle a \rangle + \langle b \rangle = R$, and so in particular $1 \in \langle a \rangle + \langle b \rangle$. Therefore

$$1 = ar + bs$$

for some $r, s \in R$, and so

$$c = a(rc) + (bc)s$$

Since $a \mid a(rc)$ and $a \mid (bc)s$ we obtain from here that $a \mid c$.

$\square$

33 Application: sums of two squares

Recall. The ring of Gaussian integers:

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$$

$\mathbb{Z}[i]$ is a Euclidean domain with the norm function $N(a + bi) = a^2 + b^2$. In particular it follows that $\mathbb{Z}[i]$ is a UFD.

33.1 Note. For $\alpha, \beta \in \mathbb{Z}[i]$ we have

$$N(\alpha\beta) = N(\alpha)N(\beta)$$

33.2 Proposition. *An element $\alpha \in \mathbb{Z}[i]$ is a unit iff $N(\alpha) = 1$.*

Proof. Exercise. $\square$

33.3 Corollary. *The only units in $\mathbb{Z}[i]$ are ± 1 and $\pm i$.*

33.4 Proposition. *If $p \in \mathbb{Z}$ is a prime number and for some $\alpha \in \mathbb{Z}[i]$ we have $N(\alpha) = p$ then α is an irreducible element in $\mathbb{Z}[i]$.*

Proof. Assume that $\alpha = \beta\gamma$. Then we have

$$p = N(\alpha) = N(\beta)N(\gamma)$$

Since p is a prime number we get that either $N(\beta) = 1$ or $N(\gamma) = 1$, and so either β or γ is a unit in $\mathbb{Z}[i]$. $\square$

33.5 Theorem. *Let p be an odd prime. The following conditions are equivalent.*

1) $p = a^2 + b^2$ *for some* $a, b \in \mathbb{Z}$
2) $p \equiv 1 \pmod 4$

33.6 Lemma. *If $p \in \mathbb{Z}$ is a prime number and $p \equiv 1 \pmod 4$ then there is $m \in \mathbb{Z}$ such that*

$$m^2 \equiv -1 \pmod p$$

33.7 Proposition. *If $p \in \mathbb{Z}$ is a prime number then the groups of units $(\mathbb{Z}/p\mathbb{Z})^*$ of the ring $\mathbb{Z}/p\mathbb{Z}$ is a cyclic group of order $(p-1)$.*

Proof. See Propositon 38.8. □

Proof of Lemma 33.6. By Proposition 33.7 we have

$$(\mathbb{Z}/p\mathbb{Z})^* \cong \mathbb{Z}/(p-1)\mathbb{Z}$$

Since $p \equiv 1 \pmod 4$, thus $4 \mid (p-1)$, and so the group $(\mathbb{Z}/p\mathbb{Z})^*$ has a cyclic subgroup of order 4. Let $a \in \mathbb{Z}/p\mathbb{Z}$ be a generator of this subgroup. Then a^2 is an element of order 2 in $(\mathbb{Z}/p\mathbb{Z})^*$, so $a^2 = -1$.

Take the canonical epimorphism $\pi\colon \mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ and let $m \in \pi^{-1}(a)$. Then

$$\pi(m^2) = a^2 = \pi(-1)$$

which gives $m^2 \equiv -1 \pmod p$ □

Proof of Theorem 33.5.

1) $\Rightarrow$ 2) Assume that $p = a^2 + b^2$. Since p is odd we can assume that a is even, $a = 2m$, and b is odd, $b = 2n + 1$. Then we have

$$p = (2m)^2 + (2n+1)^2 = 4m^2 + 4n^2 + 4n + 1$$

and so $p \equiv 1 \pmod 4$.

2) $\Rightarrow$ 1) Since $p \equiv 1 \pmod 4$ by Lemma 33.6 there exists $m \in \mathbb{Z}$ such that $p \mid (m^2 + 1)$. In $\mathbb{Z}[i]$ we have

$$m^2 + 1 = (m + i)(m - i)$$

Therefore in $\mathbb{Z}[i]$ we have $p \mid (m + i)(m - i)$. On the other hand $p \nmid (m \pm i)$ since otherwise for some $a, b \in \mathbb{Z}$ we would have

$$m \pm i = p(a + bi) = pa + pbi$$

and so $pb = \pm 1$ which is impossible.

As a consequence we obtain that p is not a prime element in $\mathbb{Z}[i]$. Since $\mathbb{Z}[i]$ is a UFD, prime elements in $\mathbb{Z}[i]$ coincide with irreducible elements and so p is not irreducible. Therefore

$$p = \alpha\beta$$

for some non-units $\alpha, \beta \in \mathbb{Z}[i]$. We have

$$p^2 = N(p) = N(\alpha)N(\beta)$$

By Lemma 33.2 we have $N(\alpha) \neq 1 \neq N(\beta)$ so $N(\alpha) = N(\beta) = p$. Therefore, if $\alpha = a + bi$ then $p = a^2 + b^2$. $\square$

33.8 Note.

1) One can use factorization in $\mathbb{Z}[i]$ to show that, in general, a positive integer n is a sum of two squares iff n is of the form

$$n = 2^k p_1^{m_i} \cdots p_r^{m_r} q_1^{2n_1} \cdots q_s^{2n_s}$$

where $k, m_i, n_j \geq 0$, and p_i, q_j are prime numbers such that $p_i \equiv 1 \pmod 4$ and $q_j \equiv 3 \pmod 4$.

2) The ring $\mathbb{Z}[i]$ can be also used to describe all Pythagorean triples, e.i. triples of positive integers (x, y, z) that satisfy the equation $x^2 + y^2 = z^2$.

34 Application: Fermat's Last Theorem

34.1 Fermat's Last Theorem.

For $n \geq 3$ there are no integers $x, y, z > 0$ such that $x^n + y^n = z^n$.

Kummer's idea of the proof.

1) It is enough to show that $x^n + y^n = z^n$ has no integral solutions for $n = 4$ and for $n = p$ where p is an odd prime. Indeed, if $n \geq 3$ is any integer then $n = mk$ where either $k = 4$ or k is an odd prime, and if $x = a, y = b, z = c$ would be a solution of

$$x^n + y^n = z^n$$

then $x = a^m, y = b^m, z = c^m$ would be a solution of

$$x^k + y^k = z^k$$

2) The case $m = 4$ was proved by Fermat.

3) Take an odd prime p. Let $\zeta = e^{2\pi i/p}$ be a p^{th} primitive root of 1, and let $\mathbb{Z}[\zeta]$ be the smallest subring of $\mathbb{C}$ that contains ζ.

For any $x, y \in \mathbb{Z}$ we have a factorization in $\mathbb{Z}[\zeta]$:

$$x^p + y^p = \prod_{i=0}^{p-1}(x + \zeta^i y)$$

Therefore, if $x^p + y^p = z^p$ then in $\mathbb{Z}[\zeta]$ we have

$$z^p = \prod_{i=0}^{p-1}(x + \zeta^i y)$$

Assuming that $\mathbb{Z}[\zeta]$ is a UFD we can compare these factorizations and try to get a contradiction.

Problem. For some values of p (e.g. $p = 23$) the ring $\mathbb{Z}[\zeta]$ is not a UFD.

35 Greatest common divisor

35.1 Definition. Let R be a ring and let $a_1, \ldots a_n \in R$. We say that $b \in R$ is a *greatest common divisor* of $a_1, \ldots, a_n$ if

1) $b \mid a_i$ for $i = 1, \ldots, n$
2) if $c \mid a_i$ for $i = 1, \ldots, n$ then $c \mid b$.

In such case we write $b \sim \gcd(a_1, \ldots, a_n)$.

35.2 Note.

1) If b is a greatest common divisor of $a_1, \ldots, a_n$ and $b' \sim b$ then b' is also a greatest common divisor of $a_1, \ldots, a_n$.

2) In general $\gcd(a_1, \ldots, a_n)$ need not exists. E.g. $\gcd(9, 3(2+\sqrt{5}i))$ does not exist in $\mathbb{Z}[\sqrt{-5}]$.

35.3 Theorem. *If R is a UFD then $\gcd(a_1, \ldots, a_n)$ exists for any $a_1, \ldots, a_n$.*

35.4 Lemma. *If R is a ring such that $\gcd(a_1, a_2)$ exists for any $a_1, a_2 \in R$ then $\gcd(a_1, \ldots, a_n)$ exists for any $a_1, \ldots, a_n \in R$.*

Proof. Check:

$$\gcd(a_1, \ldots, a_n) \sim \gcd(\gcd(a_1, \ldots, a_{n-1}), a_n)$$

$\square$

Proof of Theorem 35.3. Let $a, b \in R$. By Lemma 35.4 it is enough to show that $\gcd(a, b)$ exists.

If $a = 0$ then $\gcd(a, b) \sim b$ (check!).

If a is a unit then $\gcd(a, b) \sim a$ (check!).

Therefore we can assume that a, b are non-zero, non-unit elements of R. Since R is a UFD in such case we have

$$a = uc_1^{k_1}c_2^{k_2}\dots c_m^{k_m}, \qquad b = vc_1^{l_1}c_2^{l_2}\dots c_m^{l_m}$$

where u, v are units, $c_1, \dots, c_m$ are distinct irreducible elements, and $k_i, l_i \geq 0$. Check: we have

$$\gcd(a, b) \sim c_1^{n_1}c_2^{n_2}\dots c_m^{n_m}$$

where $n_i = \min\{k_i, l_i\}$ for $i = 1, \dots m$. $\square$

35.5 Definition. Let R be a ring. Elements $a_1, \dots, a_n \in R$ are *relatively prime* if $\gcd(a_1, \dots, a_n) \sim 1$.

35.6 Note.

1) The elements $a_1, \dots, a_n$ are relatively prime iff every common divisor of $a_1, \dots, a_n$ is a unit (check!).
2) If R is a UFD, $a, b, c \in R$, $\gcd(a, b) \sim 1$ and $a \mid bc$ then $a \mid c$ (check!).

36 Rings of fractions

Recall. If R is a PID then R is a UFD.

In particular

- $\mathbb{Z}$ is a UFD
- if $\mathbb{F}$ is a field then $\mathbb{F}[x]$ is a UFD.

Goal. If R is a UFD then so is $R[x]$.

Idea of proof.

1) Find an embedding $R \hookrightarrow \mathbb{F}$ where $\mathbb{F}$ is a field.
2) If $p(x) \in R[x]$ then $p(x) \in \mathbb{F}[x]$ and since $\mathbb{F}[x]$ is a UFD thus $p(x)$ has a unique factorization into irreducibles in $\mathbb{F}[x]$.
3) Use the factorization in $\mathbb{F}[x]$ and the fact that R is a UFD to obtain a factorization of $p(x)$ in $R[x]$.

36.1 Definition. If R is a ring then a subset $S \subseteq R$ is a *multiplicative subset* if

1) $1 \in S$
2) if $a, b \in S$ then $ab \in S$

36.2 Example.

Multiplicative subsets of $\mathbb{Z}$:

1) $S = \mathbb{Z}$

2) $S_0 = \mathbb{Z} - \{0\}$

3) $S_p = \{n \in \mathbb{Z} \mid p \nmid n\}$ where p is a prime number.

   Note: $S_p = \mathbb{Z} - \langle p \rangle$.

36.3 Proposition. *If R is a ring, and I is a prime ideal of R then $S = R - I$ is a multiplicative subset of R.*

Proof. Exercise. $\square$

36.4. Construction of a ring of fractions.

Goal. For a ring R and a multiplicative subset $S \subseteq R$ construct a ring $S^{-1}R$ such that every element of S becomes a unit in $S^{-1}R$.

Consider a relation on the set $R \times S$:

$$(a, s) \sim (a', s') \quad \text{if } \ s_0(as' - a's) = 0 \text{ for some } s_0 \in S$$

Check: $\sim$ is an equivalence relation.

Elements of $S^{-1}R \ = \ $ (equivalence classes of $R \times S$ under the relation $\sim$).

Notation. $a/s \ := \ $ the equivalence class represented by (a, s)

Note.

1) If $0 \in S$ then $(a, s) \sim (a', s')$ for all $(a, s), (a', s') \in R \times S$, and so $S^{-1}R$ consists of only one element.
2) If R is an integral domain and $0 \notin S$ then $(a, s) \sim (a', s')$ iff $as' = a's$.

Multiplication in $S^{-1}R$:

$$(a/s)(a'/s') = (aa')/ss'$$

Addition in $S^{-1}R$:

$$a/s + a'/s' = (as' + a's)/ss'$$

Check:

1) The operations of addition and multiplication in $S^{-1}R$ are well defined.
2) These operations define a commutative ring structure on $S^{-1}R$.
3) The map
$$\varphi_S\colon R \to S^{-1}R, \qquad \varphi_S(r) = r/1$$
is a ring homomorphism.

Note. If $s \in S$ then $\varphi_S(s) = s/1$ is a unit in $S^{-1}R$:

$$(s/1)(1/s) = s/s = 1/1 = 1_{S^{-1}R}$$

36.5 Definition. If S is a multiplicative subset of a ring R then $S^{-1}R$ is called the *ring of fractions of R with respect to S*.

36.6 Theorem. *If S is a multiplicative subset of a ring R and $f\colon R \to R'$ is a ring homomorphism such for ever $s \in S$ the element $f(s) \in R'$ is a unit, then there exists a unique homomorphism*

$$\bar{f}\colon S^{-1}R \to R'$$

such that the following diagram commutes:

$$\begin{tikzcd} R \arrow[r, "f"] \arrow[d, "\varphi_S"'] & R' \\ S^{-1}R \arrow[ur, dashed, "\bar{f}"'] & \end{tikzcd}$$

Proof. Define $\bar{f}(r/s) = f(r)f(s)^{-1}$. Check that

1) $\bar{f}$ is a well defined ring homomorphism
2) $\bar{f}$ is the only homomorphism such that $f = \bar{f}\varphi_S$.

□

36.7 Examples.

1) If $S \subseteq \mathbb{Z}$, $S = \mathbb{Z}$ then $S^{-1}\mathbb{Z} \cong \{0\}$.

2) If $S_0 \subseteq \mathbb{Z}$, $S_0 = \mathbb{Z} - \{0\}$ then $S_0^{-1}\mathbb{Z} \cong \mathbb{Q}$.

3) If $S_p \subseteq \mathbb{Z}$, $S_p = \mathbb{Z} - \langle p \rangle$ then $S_p^{-1}\mathbb{Z}$ is isomorphic to the subring of $\mathbb{Q}$ consisting of all fractions $\frac{m}{n}$ such that $p \nmid n$.

4) If $R = \mathbb{Z}/6\mathbb{Z}$, $S = \{1, 3\} \subseteq R$ then $S^{-1}R \cong \mathbb{Z}/2\mathbb{Z}$ (check!).

36.8 Proposition. *If S is a multiplicative subset of a ring R then*

$$\mathrm{Ker}(\varphi_S) = \{a \in R \mid sa = 0 \textit{ for some } s \in S\}$$

Proof. We have $\varphi_S(r) = 0/1$ iff $s(r \cdot 1 - 0 \cdot 1) = 0$ for some $s \in S$.

□

36.9 Corollary. *If R is an integral domain and $S \subseteq R$ is a multiplicative subset such that $0 \notin S$ then the homomorphism*

$$\varphi_S \colon R \to S^{-1}R$$

is 1-1. As a consequence in this case we can identify R with a subring of $S^{-1}R$.

36.10 Note.

1) If R is an integral domain and $S = R - \{0\}$ then the ring $S^{-1}R$ is a field. In this case $S^{-1}R$ is called the *field of fractions of* R.

2) If I is a prime ideal of a ring R then the set $S = R - I$ is a multiplicative subset of R. In this case the ring $S^{-1}R$ is called the *localization of* R *at* I and it is denoted R_I.

36.11 Definition. A ring R is a *local ring* if R has exactly one maximal ideal.

36.12 Examples.

1) If $\mathbb{F}$ is a field then it is a local ring with the maximal ideal $I = \{0\}$.

2) Check: if $\mathbb{F}$ is a field then $\mathbb{F}[[x]]$ is a local ring with the maximal ideal $\langle x \rangle$.

3) Check: if $\mathbb{F}$ is a field then $\mathbb{F}[x]$ is not a local ring since for every $a \in \mathbb{F}$ the ideal $\langle x - a \rangle$ is a maximal ideal in $\mathbb{F}[x]$.

4) $\mathbb{Z}$ is not a local ring.

36.13 Proposition. *If R is a local ring then the maximal ideal $J \lhd R$ consists of all non-units of R.*

Proof. Since $J \neq R$ thus J does not contain any units.

Conversely, if $a \in R$ is a non-unit then $1 \notin \langle a \rangle$ so $\langle a \rangle \neq R$. Therefore by Theorem 29.1 $\langle a \rangle$ is contained in some maximal ideal of R. Since J is the only maximal ideal we obtain $\langle a \rangle \subseteq J$, and so $a \in J$. $\square$

36.14 Proposition. *If R is a ring and $I \lhd R$ is a prime ideal then the ring R_I is a local ring, and the maximal ideal $J \lhd R_I$ is given by*

$$J := \{a/s \mid a \in I,\ s \notin I\}$$

Proof. Exercise. $\square$

37 Factorization in rings of polynomials

Goal:

37.1 Theorem. *If R is a UFD then so is $R[x]$.*

37.2 Lemma. *If R is an integral domain then $p(x) \in R[x]$ is a unit in $R[x]$ iff $\deg p(x) = 0$ and $p(x) = a_0$ where a_0 is a unit in R.*

Proof. Exercise. □

37.3 Lemma. *If R is an integral domain and $p(x) = a_0$ is a polynomial of degree 0 in $R[x]$ then $p(x)$ is irreducible in $R[x]$ iff a_0 is irreducible in R.*

Proof. Exercise. □

37.4 Definition. If R is a UFD and $p(x) = a_0 + a_1x + \dots a_nx^n \in R[x]$, then $p(x)$ is a *primitive polynomial* if $\gcd(a_0, \dots, a_n) \sim 1$.

37.5 Lemma. *If R is a UFD and $p(x) \in R[x]$ is an irreducible polynomial such that $\deg p(x) > 0$ then $p(x)$ is a primitive polynomial.*

Proof. If $p(x) = a_0 + a_1x + \dots a_nx^n$ is not primitive then

$$p(x) = \gcd(a_0, \dots a_n)q(x)$$

for some $q(x) \in R[x]$, $\deg q(x) > 0$. Since both $\gcd(a_0, \dots, a_n)$ and $q(x)$ are non-units in $R[x]$ the polynomial $p(x)$ is not irreducible. □

37.6 Lemma (Gauss). *If R is a UFD and $p(x), q(x) \in R[x]$ are primitive polynomials then $p(x)q(x)$ is also primitive.*

Proof. Assume that $r(x) = p(x)q(x)$ is not primitive. Then we have

$$r(x) = c \cdot \tilde{r}(x)$$

where $c \in R$ is an irreducible element, and $\tilde{r}(x) \in R[x]$. Take the canonical epimorphism

$$\pi \colon R \longrightarrow R/\langle c \rangle$$

This defines a homomorphism of rings of polynomials

$$\bar{\pi} \colon R[x] \longrightarrow (R/\langle c \rangle)[x]$$

given by

$$\bar{\pi}(a_0 + a_1 x + \ldots + a_n x^m) := \pi(a_0) + \pi_1(a)x + \ldots + \pi(a_n)x^n$$

Note:

1) Since c is irreducible element thus $\langle c \rangle$ is a prime ideal and so $R/\langle c \rangle$ is a domain. As a consequence $(R/\langle c \rangle)[x]$ is a domain.
2) We have

   $$\bar{\pi}(p(x))\bar{\pi}(q(x)) = \bar{\pi}(c \cdot \tilde{r}(x)) = 0 \cdot \bar{\pi}(\tilde{r}(x)) = 0$$

   so either $\bar{\pi}(p(x)) = 0$ or $\bar{\pi}(q(x)) = 0$.

We can assume that $\bar{\pi}(p(x)) = 0$. Then $p(x) = c \cdot \tilde{p}(x)$ for some $\tilde{p}(x) \in R[x]$. Since $p(x)$ is primitive we get a contradiction.

□

37.7 Lemma. *Let R be a UFD and K be the field of fractions of R.*

1) *For any non-zero polynomial $p(x) \in K[x]$ there is an element $c(p) \in K$ and a primitive polynomial $p^*(x) \in R[x]$ such that*

   $$p(x) = c(p)p^*(x)$$

2) If $p(x) = c(p)p^*(x)$ and $p(x) = \tilde{c}(p)\tilde{p}^*(x)$ for some $c(p), \tilde{c}(p) \in K$, and some primitive polynomials $p^*(x), \tilde{p}^*(x) \in R[x]$ then there exists a unit $u \in R$ such that

$$\tilde{c}(p) = uc(p), \qquad \tilde{p}^*(x) = u^{-1}p^*(x)$$

3) If $p(x), q(x) \in K[x]$ are non-zero polynomials then for some unit $u \in R$ we have

$$c(pq) = uc(p)c(q) \qquad (p(x)q(x))^* = u^{-1}p^*(x)q^*(x)$$

37.8 Definition. If $p(x) \in K[x]$ then the element $c(p) \in K$ is called the *content* of $p(x)$.

Proof of Lemma 37.7.

1) If $p(x) \in K[x]$ then there is there exists $0 \neq c \in R$ such that $cp(x) \in R[x]$. Let $b \in R$ be a greatest common divisor of coefficients of $cp(x)$. Take

$$p^*(x) = c/b \cdot p(x), \qquad c(p) = b/c$$

Check that $p^*(x) \in R[x]$ and $p^*(x)$ is a primitive polynomial.

2) We have

$$c(p)p^*(x) = p(x) = \tilde{c}(p)\tilde{p}^*(x)$$

where $c(p), \tilde{c}(p) \in K$ are non-zero elements and $p^*(x), \tilde{p}^*(x) \in R[x]$ are primitive polynomials. Let $p^*(x) = a_0 + a_1x + \ldots + a_nx^n$.

We have

$$\tilde{p}^*(x) = c(p)\tilde{c}(p)^{-1}p^*(x)$$

Since $c(p)\tilde{c}(p)^{-1} \in K$ there $b, d \in R$ such that $c(p)\tilde{c}(p)^{-1} = b/d$. We can assume that $\gcd(b, d) \sim 1$. We have

$$d\tilde{p}^*(x) = bp^*(x) = ba_0 + ba_1x + \ldots + ba_nx^n$$

This gives: $d \mid ba_i$ for $i = 1, \ldots, n$. By (35.6) this gives that $d \mid a_i$ for all i. Since $p^*(x)$ is a primitive polynomial it implies that d is a unit in R. By an analogous argument we obtain that b is also a unit in R.

As a consequence $u = db^{-1}$ is a unit in R and we have

$$\tilde{c}(p) = uc(p), \qquad \tilde{p}^*(x) = u^{-1}p^*(x)$$

3) For $p(x), q(x) \in K[x]$ we have

$$c(p)c(q)p^*(x)q^*(x) = p(x)q(x) = c(pq)(p(x)q(x))^*$$

We have $c(p)c(q), c(pq) \in K$, $(p(x)q(x))^* \in R[x]$ is primitive by construction and $p^*(x)q^*(x) \in R[x]$ is primitive by Lemma 37.6.

Applying part 2) we obtain that there is a unit $u \in R$ such that

$$c(pq) = uc(p)c(q) \qquad (p(x)q(x))^* = u^{-1}p^*(x)q^*(x)$$

□

37.9 Note. Check:

1) Let $p(x) \in K[x]$. Then $p(x) \in R[x]$ iff $c(p) \in R$.
2) $p(x) \in R[x]$ is primitive iff $p(x) = up^*(x)$ for some unit $u \in R$.

37.10 Proposition. *Let R be a UFD and K be the ring of fractions of R.*

1) *A polynomial $p(x) \in K[x]$ of non-zero degree is irreducible in $K[x]$ iff $p^*(x)$ is irreducible in $R[x]$.*
2) *A polynomial $p(x) \in R[x]$ of non-zero degree is irreducible in $R[x]$ iff $p(x)$ is primitive and it is irreducible in $K[x]$.*

Proof.

1) ($\Leftarrow$) If $p(x)$ is not irreducible in $K[x]$ then $p(x) = q(x)r(x)$ for some $q(x), r(x) \in K[x]$, $\deg q(x), \deg r(x) > 0$. By Lemma 37.7 we have

$$p^*(x) = uq^*(x)r^*(x)$$

for some unit $u \in R$. Therefore $p^*(x)$ is not irreducible in $R[x]$.

($\Rightarrow$) If $p^*(x)$ is not irreducible in $R[x]$ then

$$p^*(x) = q(x)r(x)$$

where $q(x), r(x)$ are non-units in $R[x]$. Since $p^*(x)$ is primitive we must have $\deg q(x), \deg r(x) > 0$. Then

$$p(x) = c(p)q(x)r(x)$$

and so $p(x)$ is not irreducible in $K[x]$.

2) ($\Rightarrow$) If $p(x) \in R[x]$ is irreducible then it must be a primitive polynomial. Therefore $p(x) = up^*(x)$ for some unit $u \in R$. Since $p(x)$ is irreducible in $R[x]$, thus $p^*(x)$ is also irreducible in $R[x]$, and so by part 1) $p(x)$ is irreducible in $K[x]$.

($\Leftarrow$) Since $p(x)$ is irreducible in $K[x]$ by part 1) we get that $p^*(x)$ is irreducible in $R[x]$. Also, since $p(x)$ is primitive we have $p(x) = up^*(x)$ for some unit $u \in R$. It follows that $p(x)$ is irreducible in $R[x]$.

$\square$

Proof of Theorem 37.1. By Theorem 31.7 we need to show that:

1) Every non-zero, non-unit $p(x) \in R[x]$ is a product of irreducible polynomials.
2) Every irreducible polynomial $p(x) \in R[x]$ is a prime element of $R[x]$.

1) Recall that irreducible polynomials of degree 0 in $R[x]$ correspond to irreducible elements of R. Since R is a UFD it follows that if $p(x) \in R[x]$ has degree 0 then $p(x)$ is a product of irreducibles in $R[x]$.

If $\deg p(x) > 0$ then $p(x)$ is a non-zero non-unit element of $K[x]$. Since $K[x]$ is a UFD we have

$$p(x) = q_1(x)q_2(x)\dots q_k(x)$$

where $q_1(x), \dots, q_k(x)$ are irreducible polynomials in $K[x]$. We have

$$p(x) = (c(q_1)c(q_2)\dots c(q_k))q_1^*(x)q_2^*(x)\dots q_k^*(x)$$

By Proposition 37.10 $q_i^*(x)$ are irreducible polynomials in $R[x]$. Moreover, by Lemma 37.7 we have

$$c(q_1)c(q_2)\dots c(q_k) = uc(p)$$

for some unit $u \in R$, and also since $p(x) \in R[x]$ we have $c(p) \in R$. Therefore $c(q_1)c(q_2)\dots c(q_k) \in R$, and since R is a UFD we have

$$c(q_1)c(q_2)\dots c(q_k) = a_1a_2\dots a_l$$

where $a_1, \dots, a_l \in R$ are irreducible elements in $R[x]$. As a consequence we obtain a factorization of $p(x)$:

$$p(x) = a_1a_2\dots a_l q_1^*(x)q_2^*(x)\dots q_k^*(x)$$

where $a_1, \dots, a_l$, $q_1^*(x), \dots, q_k^*(x)$ are irreducible elements in $R[x]$.

2) Let $p(x) \in R[x]$ be an irreducible polynomial. We need to show that if for some $q(x), r(x) \in R[x]$ we have $p(x) \mid q(x)r(x)$ then either $p(x) \mid q(x)$ or $p(x) \mid r(x)$.

Exercise: this holds if $\deg p(x) = 0$.

If $\deg p(x) > 0$, then since $p(x)$ is irreducible in $R[x]$ by Proposition 37.10 we obtain that $p(x)$ is primitive and it is irreducible in $K[x]$. Since $K[x]$ is a UFD we obtain that $p(x)$ is a prime element of $K[x]$, and so $p(x) \mid q(x)$ or $p(x) \mid r(x)$ in $K[x]$.

We can assume that $p(x) \mid q(x)$ in $K[x]$. Then there exists $h(x) \in K[x]$ such that

$$p(x)h(x) = q(x)$$

We have $c(p)c(h) = uc(q)$ for some unit $u \in R$. Also, since $q(x) \in R[x]$ we have that $c(q) \in R$ and therefore $c(p)c(h) \in R$. Finally, since $p(x)$ is primitive thus $c(p)$ is a unit in R. We obtain:

$$c(h) = c(p)^{-1}uc(q) \in R$$

Therefore $h(x) \in R[x]$, and so $p(x) \mid q(x)$ in $R[x]$. $\square$

37.11 Corollary. $\mathbb{Z}[x]$ *is a UFD*

37.12 Corollary. *If R is a UFD then the ring of polynomials of n variables $R[x_1, \ldots, x_n]$ is also a UFD.*

Proof. It is enough to notice that

$$R[x_1, \ldots, x_n] \cong R[x_1, \ldots, x_{n-1}][x_n]$$

(check!). $\square$

38 Irreducibility criteria in rings of polynomials

38.1 Theorem. *Let* $p(x), q(x) \in R[x]$ *be polynomials such that*

$$p(x) = a_0 + a_1x + \ldots + a_nx^n, \quad q(x) = b_0 + b_1x + \ldots + b_mx^m$$

and $a_n, b_m \neq 0$. *If* b_m *is a unit in* R *then there exist unique polynomials* $r(x), s(x) \in R[x]$ *such that*

$$p(x) = s(x)q(x) + r(x)$$

and either $\deg r(x) < \deg q(x)$ *or* $r(x) = 0$.

Proof. Exercise (or see Hungerford p.158). □

38.2 Definition. If R is a ring and $p(x) \in R[x]$ then $p(x)$ defines a function

$$p\colon R \longrightarrow R, \qquad a \mapsto p(a)$$

A function of this form is called a *polynomial function*.

38.3 Note. Different polynomials may define the same polynomial function.

E.g. if $p(x) = x + 1$, $q(x) = x^2 + 1$ are polynomials in $\mathbb{Z}/2\mathbb{Z}[x]$ then $p(x)$, $q(x)$ define the same function $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$:

$$p(0) = q(0) = 1, \quad p(1) = q(1) = 0$$

38.4 Definition. Let $p(x) \in R[x]$. An element $a \in R$ is a *root of* $p(x)$ if $f(a) = 0$.

38.5 Proposition. *An element $a \in R$ is a root of $p(x) \in R[x]$ iff $(x-a) \mid p(x)$.*

Proof.

($\Leftarrow$) If $(x - a) \mid p(x)$ then $p(x) = q(x)(x - a)$ for some $q(x) \in R[x]$ so $p(a) = q(a)(a - a) = 0$.

($\Rightarrow$) Assume that $p(a) = 0$. By Theorem 38.1 we have

$$p(x) = s(x)(x - a) + r(x)$$

where $\deg r(x) = 0$, so $r(x) = b$ for some $b \in R$. This gives

$$0 = p(a) = s(a)(a - a) + b$$

Thus $b = 0$, and so $p(x) = s(x)(x - a)$.

$\square$

38.6 Corollary. *If R is an integral domain and $0 \neq p(x) \in R[x]$ is a polynomial of degree n then $R[x]$ has at most n distinct roots in R.*

Proof. Let $a_1, \dots, a_k \in R$ be all distinct roots of $p(x)$. By (38.5) we have

$$p(x) = (x - a_1)q_1(x)$$

for some $q_1(x) \in R[x]$. Also we have

$$0 = p(a_2) = (a_2 - a_1)q_1(a_2)$$

Since R is an integral domain and $a_2 - a_1 \neq 0$ we obtain $q_1(a_2)$, and so

$$q_2(x) = (x - a_2)q_3(x)$$

for some $q_3(x) \in R[x]$. This gives

$$p(x) = (x - a_1)(x - a_2)q_3(x)$$

By induction we obtain

$$p(x) = (x - a_1) \cdot \ldots \cdot (x - a_k) q_k(x)$$

for some $0 \neq q_k(x) \in R[x]$. This gives

$$\deg p(x) = \deg((x - a_1) \cdot \ldots \cdot (x - a_k) q_k(x)) \geq k$$

$\square$

38.7 Note.

1) Corollary 38.6 in not true if R is not an integral domain. E.g. if $R = \mathbb{Z}/6\mathbb{Z}$ and $p(x) = x^2 + x$ then $0, 2, 3 \in \mathbb{Z}/6\mathbb{Z}$ are roots of $p(x)$.

2) Corollary 38.6 is not true is R is a non-commutative ring (even if R has no zero divisors). For example, if $R = \mathbb{H}$ and $p(x) = x^2 + 1$ then $\pm i, \pm j, \pm k \in \mathbb{H}$ are roots of $p(x)$.

38.8 Proposition. *Let R is an integral domain. If G is a finite subgroup of the multiplicative group of units of R then G is a cyclic group.*

Proof. Let $a \in G$ be an elements of a maximal order in G. We will show that $\langle a \rangle = G$. We argue by contradiction. Assume that there exists an element $b \in G - \langle a \rangle$, and let $|b| = m$. By assumption $m \leq |a|$. If $m \nmid |a|$ then $|ab| > |a|$ (check!) which is impossible by the choice of a. Therefore $|a| = km$ for some $k > 0$. This shows that there are $k + 1$ distinct elements of order m in G: $b, a^k, a^{2k}, \ldots, a^{(m-1)k}$. This is however impossible since each of these elements would be a root of the polynomial $f(x) = x^m - 1 \in R[x]$ and by Corollary 38.6 this polynomial has at most m roots in R. $\square$

38.9 Proposition. *If $\mathbb{F}$ is a field and $p(x) \in \mathbb{F}[x]$ is a polynomial such that $\deg p(x) > 1$ and $p(x)$ has a root in $\mathbb{F}$ then $p(x)$ is not irreducible in $\mathbb{F}[x]$.*

Proof. By (38.6) we have $p(x) = q(x)(x-a)$ for some $q(x) \in \mathbb{F}[x]$. Since $\deg p(x) > 1$ we have $\deg q(x) > 0$, so $q(x)$ and $(x-a)$ are not units in $\mathbb{R}[x]$. □

38.10 Corollary. *Let R be a UFD and let K be the field of fractions of R. If $p(x) \in R[x]$ is a polynomial such that $\deg p(x) > 1$ and $p(x)$ has a root in K then $p(x)$ is not irreducible in $R[x]$.*

Proof. By (38.6) $p(x)$ is not irreducible in $K[x]$, so by (37.10) it is also not irreducible in $R[x]$. □

38.11 Proposition (Integral root test)**.** *Let R be a UFD, let K be the field of fractions of R and let $p(x) \in R[x]$ be a polynomial*

$$p(x) = a_0 + a_1 x + \ldots + a_n x^n$$

where $a_n \neq 0$. If $a \in K$ is a root of $p(x)$ then a is of the form $a = b/s$ where $b, s \in R$, $\gcd(b,s) \sim 1$, $b \mid a_0$ and $s \mid a_n$.

In particular, in $a_n = 1$ then $a \in R$ and $a \mid a_0$.

Proof. Exercise. □

38.12 Theorem (Eisenstein Irreducibility Criterion)**.** *Let R be a UFD. If*

$$p(x) = a_0 + a_1 x + \ldots + a_n x^n$$

is a primitive polynomial in $R[x]$ such that $\deg p(x) > 0$, and $b \in R$ is an irreducible element $b \in R$ such that

1) $b \nmid a_n$
2) $b \mid a_i$ *for all* $i < n$

3) $b^2 \nmid a_0$
then $p(x)$ *is irreducible in* $R[x]$.

Proof. Assume that $p(x)$ is not irreducible in $R[x]$. Then we have

$$p(x) = q(x)r(x)$$

for some non-units $q(x), r(x) \in R[x]$. Since $p(x)$ is primitive we must have $\deg q(x), \deg r(x) > 0$. Let

$$q(x) = c_0 + c_1x + \ldots + c_kx^k, \quad r(x) = d_0 + d_1x + \ldots + d_lx^l$$

Notice that since b is irreducible it is a prime element of R and so by (32.4) the ideal $\langle b\rangle$ is a prime ideal of R. As a consequence $R/\langle b\rangle$ is an integral domain. Consider the canonical epimorphism $\pi\colon R \to R/\langle b\rangle$ and the induced homomorphism of rings of polynomials

$$\tilde{\pi}\colon R[x] \longrightarrow R/\langle b\rangle[x]$$

By assumption on $p(x)$ we have

$$\tilde{\pi}(p(x)) = \pi(a_n)x^n$$

On the other hand we have

$$\tilde{\pi}(p(x)) = \tilde{\pi}(q(x))\tilde{\pi}(r(x))$$

Check: since $R/\langle b\rangle$ is an integral domain we must have

$$\tilde{\pi}(q(x)) = \pi(c_k)x^k, \qquad \tilde{\pi}(r(x)) = \pi(d_l)x^l$$

In particular $\pi(c_0) = \pi(d_0) = 0$, so $b \mid c_0$ and $b \mid d_0$. On the other hand $a_0 = c_0d_0$, so $b^2 \mid a_0$ which contradicts the assumption on a_0. $\square$

38.13 Example. If $p \in \mathbb{Z}$ is a prime number then $q(x) = x^n - p$ is an irreducible polynomial in $\mathbb{Z}[x]$.

Note: by (37.10) $q(x)$ is also irreducible in $\mathbb{Q}[x]$. This shows in particular that $q(x)$ has no roots in $\mathbb{Q}$, and so that $\sqrt[n]{p}$ is an irrational number for all primes p and all $n > 1$.

38.14 Proposition. *Let R be an integral domain and let $c \in R$. A polynomial $p(x) = \sum_{i=0}^{n} a_i x^i$ is irreducible iff the polynomial $p(x-c) = \sum_{i=0}^{n}(x-c)^i$ is irreducible.*

Proof. It is enough to notice that the map

$$f\colon R[x] \to R[x], \qquad f(p(x)) = p(x-c)$$

is an isomorphism of rings. $\square$

38.15 Example. Let $p \in \mathbb{Z}$ be a prime number, and let

$$q(x) = x^{p-1} + x^{p-2} + \ldots + x + 1$$

We will show that $q(x)$ is irreducible in $\mathbb{Z}[x]$. We have

$$q(x) = \frac{x^p - 1}{x - 1}$$

This gives

$$\begin{aligned} q(x+1) &= \frac{(x+1)^p - 1}{(x+1) - 1} \\ &= \frac{(x+1)^p - 1}{x} \\ &= x^{p-1} + \binom{p}{1} x^{p-2} + \binom{p}{2} x^{p-3} + \ldots + \binom{p}{p-1} \end{aligned}$$

Since $p \mid \binom{p}{k}$ for $k = 1, \ldots, p-1$ and $p^2 \nmid \binom{p}{p-1}$ the polynomial $q(x+1)$ is irreducible in $\mathbb{Z}[x]$, and so $q(x)$ is also irreducible.

39 Modules

39.1 Definition. Let R be a (possibly non-commutative) ring. A *left R-module* is an abelian group M together with a map

$$R \times M \to M, \qquad (r, m) \mapsto rm$$

satisfying the following conditions:

1) $r(m + n) = rm + rn$
2) $(r + s)m = rm + sm$
3) $(rs)m = r(sm)$
4) If R is a ring with identity $1 \in R$ then $1m = m$ for all $m \in M$.

A *right R-module* is defined analogously.

39.2 Definition. If M, N are left R-modules then a map

$$f \colon M \longrightarrow N$$

is a *left R-modules homomorphism* if f is a homomorphism of abelian groups and $f(rm) = rf(m)$ for all $r \in R$, $m \in M$.

39.3 Note. Left R-modules and their homomorphisms form a category R-$\mathcal{M}od$. Analogously, right R-modules form a category $\mathcal{M}od$-R.

39.4 Examples.

1) If I is a left ideal of R then I is a left module of R. In particular R is a left R-module.

2) If $\mathbb{F}$ is a field then left (or right) $\mathbb{F}$-modules are vector spaces over $\mathbb{F}$, and homomorphisms of $\mathbb{F}$-modules are $\mathbb{F}$-linear maps.

3) The category of left (or right) $\mathbb{Z}$-modules is isomorphic to the category of abelian groups: if G is an abelian group then G has a natural $\mathbb{Z}$-module structure such that for $n \in \mathbb{Z}$ and $g \in G$ we have

$$ng = \begin{cases} \overbrace{g + \cdots + g}^{n \text{ times}} & \text{for } n > 0 \\ 0 & \text{for } n = 0 \\ \underbrace{(-g) + \cdots + (-g)}_{|n| \text{ times}} & \text{for } n < 0 \end{cases}$$

4) Let G be an abelian group, and let $R = \mathrm{Hom}(G, G)$ be the ring of homomorphisms of G (with the usual addition of homomorphims and multiplication given by composition of homomorphisms). We have a map

$$R \times G \to G, \quad \varphi \cdot g = \varphi(g)$$

Check: this defines a left R-module structure on G.

Note: the multiplication

$$G \times R \to G, \quad g \cdot \varphi = \varphi(g)$$

does not define a right module structure on G. Indeed, for $\varphi, \psi \in R$ we have:

$$(g \cdot \psi) \cdot \varphi = (\psi(g)) \cdot \varphi = \varphi(\psi(g))$$

One the other hand $g \cdot (\psi \cdot \varphi) = \psi(\varphi(g))$. Since in general $\psi(\varphi(g)) \neq \varphi(\psi(g))$ we get that

$$(g \cdot \psi) \cdot \varphi \neq g \cdot (\psi \cdot \varphi)$$

39.5 Note. For a ring R define a ring R^{op} as follows:

- $R^{\mathsf{op}} = R$ as abelian group
- $r \cdot_{R^{\mathsf{op}}} s := sr$

We have:

$$\text{(left } R\text{-modules)} \ = \ \text{(right } R^{\mathsf{op}}\text{-modules)}$$

(check!). In particular, if R is a commutative ring then $R = R^{\mathsf{op}}$, and so

$$\text{(left } R\text{-modules)} \ = \ \text{(right } R\text{-modules)}$$

Note. From now on by an R-module we will understand a *left* R-module.

40 Basic operations on modules

40.1 Definition. If M in an R-module then a *submodule* of M is an additive subgroup $N \subseteq M$ such that if $r \in R$ and $n \in N$ then $rn \in N$.

40.2 Note. If $f\colon M \to N$ is a homomorphism of R-modules then $\operatorname{Ker}(f) := f^{-1}(0)$ is a submodule of M and $\operatorname{Im}(f) := f(M)$ is a submodule of N.

40.3 Definition. If M is an R-module and S is a subset of M then the *module generated by* S is the submodule $\langle S \rangle \subseteq M$ that is the smallest submodule of M containing S.

If $\langle S \rangle = M$ then we say that the set S *generates* M. A module M is finitely generated if $M = \langle S \rangle$ for some finite set S.

40.4 Note. If M is an R-module and $S \subseteq M$ then

$$\langle S \rangle = \{r_1 m_1 + \ldots + r_k m_k \mid r_i \in R,\ m_i \in S\}$$

40.5 Definition. If M is an R-module and $N \subseteq M$ is a submodule then the quotient module M/N is the quotient abelian group with multiplication defined by

$$r(m + N) := rm + N$$

for $r \in R$, $m + N \in M/N$.

40.6 First Isomorphism Theorem. *If $f\colon M \to N$ is a homomorphism of R-modules that is onto then*

$$M/\operatorname{Ker}(f) \cong N$$

Proof. Similar to the proof of Theorem 6.1 for groups. □

40.7 Definition. If $\{M_i\}_{i\in I}$ is a family of R-modules then the *direct product* of $\{M_i\}_{i\in I}$ is the module

$$\prod_{i\in I} M_i = \{(m_i)_{i\in I} \mid m_i \in M_i \}$$

with addition and multiplication by R defined coordinatewise.

The *direct sum* of $\{M_i\}_{i\in I}$ is the submodule $\bigoplus_{i\in I} M_i$ of $\prod_{i\in I} M_i$ given by

$$\bigoplus_{i\in I} M_i := \{(m_i)_{i\in I} \mid m_i \neq 0 \text{ for finitely many } i \text{ only } \}$$

40.8 Note. Recall the notions of categorical products and copruducts (Section 12). Check:

$\prod_{i\in I} M_i$ is the categorical product of the family $\{M_i\}_{i\in I}$ in the category R-$\mathcal{M}od$.

$\bigoplus_{i\in I} M_i$ is the categorical coproduct of $\{M_i\}_{i\in I}$ in the category R-$\mathcal{M}od$.

41 Free modules and vector spaces

41.1 Definition. Let M be an R-module. A set $S \subseteq M$ is *linearly independent* if for any distinct $m_1, \ldots, m_k \in S$ we have

$$r_1 m_1 + \ldots + r_k m_k = 0$$

only if $r_1 = \ldots = r_k = 0$.

41.2 Definition. Let M be an R-module. A set $B \subseteq M$ is a *basis* of M if B is linearly independent and B generates M.

41.3 Definition. An R-module M is a free module if M has a basis.

41.4 Theorem. *Let R be a ring with identity $1 \neq 0$ and let F be an R-module. The following conditions are equivalent.*

1) *F is a free module.*

2) *$F \cong \bigoplus_{i \in I} R$ for some set I.*

3) *There is a non-empty subset $B \subseteq F$ satisfying the following universal property. For any R-module M and any map of sets $f\colon B \to M$ there is a unique R-module homomorphism $\bar{f}\colon F \to N$ such that the following diagram commutes:*

$$\begin{tikzcd}
B \arrow[r, "f"] \arrow[d, "i"'] & M \\
F \arrow[ur, dashed, "\bar{f}"'] &
\end{tikzcd}$$

Here $i\colon B \hookrightarrow F$ is the inclusion map.

Proof. Exercise. $\square$

41.5 Note. Let R be a ring with a an identity and let $U\colon R\text{-}Mod \to \mathcal{S}et$ be the forgetful functor. Check: U has a left adjoint functor

$$F_{RMod}\colon \mathcal{S}et \to R\text{-}Mod$$

One can show that an R-module M is free iff $M \cong F_{RMod}(S)$ for some set S (exercise).

41.6 Note. We have:

$$(\text{free } \mathbb{Z}\text{-modules}) = (\text{free abelian groups})$$

41.7 Theorem. *If R is a division ring then every R-module is free.*

Proof. Let M be an R-module. It is enough to show that M has a basis.

Let S be the set of all linearly independent subsets of M ordered with respect to inclusion of subsets.

Claim 1. S has a maximal element.

Indeed, by Zorn's Lemma (29.10) it is enough to show that every chain in S has an upper bound. Let then

$$T = \{B_i\}_{i\in I}$$

be a chain in S. Take $B := \bigcup_{i\in I} B_i$. We have $B_i \subseteq B$ for all $i \in I$, so it suffices check that B is a linearly independent. Let then $b_1, \dots, b_k \in B$, and assume that

$$r_1 b_1 + \ldots + r_k b_k = 0$$

We need to show that $r_1 = \ldots = r_k = 0$.

We have $b_1 \in B_{i_1}, \dots, b_k \in B_{i_k}$ for some $i_1, \dots, i_k \in I$. Since $\{B_i\}_{i\in I}$ is a chain we can assume that

$$B_{i_1} \subseteq B_{i_2} \subseteq \ldots \subseteq B_{i_k}$$

As a consequence $b_1, \ldots, b_k \in B_{i_k}$, and since B_{i_k} is a linearly independent set we get that $r_1 = \ldots = r_k = 0$.

Claim 2. If B is a maximal element in S then B is a basis of M.

Indeed, by the definition of S the set B is linearly independent so it is enough to show that $\langle B \rangle = M$. Assume that this is not true, and let $m \in M - \langle B \rangle$. Take the set

$$B' = B \cup \{m\}$$

Notice that the set B' is linearly independent. To see this, assume that for some $b_1, \ldots, b_k \in B$ and $r_1, \ldots r_k, s \in R$ we have

$$r_1 b_1 + \ldots + r_k b_k + sm = 0$$

If $s \neq 0$ then s is a unit (since R is a division ring) and so

$$m = (-s^{-1} r_1) b_1 + \ldots + (-s^{-1} r_k) b_k$$

This is however impossible since $m \notin \langle B \rangle$. Therefore $s = 0$, and so

$$r_1 b_1 + \ldots + r_k b_k = 0$$

Linear independence of B gives then $s = r_1 = \ldots = r_k = 0$

As a consequence we get that $B' \in S$ and $B \subsetneq B'$. This is however a contradiction since B is a maximal element of S.

□

41.8 Note. Let R be a division ring and let M be an R-module. By a similar argument as in the proof of Theorem 41.7 we can show that:

1) if $V \subseteq M$ is a linearly independent set then there is a basis B of M such that $V \subseteq B$;

2) if $V \subseteq M$ is a set generating M then then there is a basis B of M such that $B \subseteq V$.

41.9 Note.

1) For a general ring R it is not true that a linearly independent subset V of a free R-module F can be always extended to a basis. Take e.g.

$$R = \mathbb{Z}, \quad F = \mathbb{Z}, \quad V = \{2\}$$

2) It is also not true in general that if V is a set generating a free R-module then V contains a basis of F. Take e.g.

$$R = \mathbb{Z}, \quad F = \mathbb{Z}, \quad V = \{2, 3\}$$

42 Invariant basis number

42.1 Definition. A ring R has the *invariant basis number (IBN)* property if for any free R-module F and for any bases two B, B' of F we have $|B| = |B'|$.

42.2 Definition. If a ring R has IBN then for a free R-module F the *rank of* F is the cardinality of a basis of F.

42.3 Example. Since free $\mathbb{Z}$-modules correspond to free abelian groups by Proposition 13.3 the ring of integers $\mathbb{Z}$ has IBN.

42.4 Notation. For a ring R and $n > 0$ denote $R^n := \bigoplus_{i=1}^{n} R$.

42.5 Example. Let $\mathbb{F}$ be a field and let V be an $\mathbb{F}$-vector space with an infinite, countable basis. Let R be the ring of all linear maps $V \to V$:

$$R = \mathrm{Hom}_{\mathbb{F}}(V, V)$$

with pointwise addition and with multiplication given by composition. We have

$$R^n \cong R^m$$

for every $m, n \geq 0$ (exercise). Thus R does not have IBN.

42.6 Theorem. *Let R be a ring with identity and let F be a free R-module. If F has an infinite basis B then for any other basis B' of F we have $|B| = |B'|$.*

42.7 Corollary. *Let R be a ring with identity. The following conditions are equivalent.*

1) R has IBN

2) If F is a free R module with two finite bases B and B' then $|B| = |B'|$.

3) For any $m, n > 0$ if $R^m \cong R^n$ then $m = n$.

Proof. Follows directly from Theorem 42.6. □

Proof of Theorem 42.6. Let F be a free R module with an infinite basis B. Let B' be any other basis of F.

Claim 1. The basis B' is infinite.

Indeed, assume that B' is finite. Since $F = \langle B \rangle$ thus every element of B' is a linear combination of a finite number of elements of B and so $B' \subseteq \{b_1, \dots, b_n\}$ where $\{b_1, \dots, b_n\}$ is some finite subset of B. This gives

$$\langle b_1, \dots, b_n \rangle \supseteq \langle B' \rangle = F$$

so $\langle b_1, \dots, b_n \rangle = F$. Since B is an infinite set there is $b \in B$ such that $b \notin \{b_1, \dots, b_n\}$. On the other hand $b \in F = \langle b_1, \dots, b_n \rangle$. This is a contradiction since B is a linearly independent set.

Next, assume that B, B' are two infinite bases of F. We can also assume that $|B'| \leq |B|$.

Claim 2. Let $T = \{b'_1, \dots, b'_k\}$ be a finite subset of B' and let

$$B_T := \{b \in B \mid b \in \langle T \rangle\}$$

Then B_T is a finite subset of B.

Indeed, each b'_i is a linear combination of a finite number of elements of B and so $T \subseteq \langle b_1, \dots, b_n \rangle$ where $\{b_1, \dots, b_n\}$ is some finite subset of B. This gives

$$B_T \subseteq \langle T \rangle \subseteq \langle b_1, \dots, b_n \rangle$$

By linear independence of B we must then have $B_T \subseteq \{b_1, \dots, b_n\}$

Claim 3. $|B| \leq |B'|$.

Indeed, let $S_{B'}$ be the set of all finite subsets of B'. Note that since B' is an infinite set we have $|S_{B'}| = |B'|$.

We have a map of sets

$$f\colon B \to S_{B'}$$

such that $f(b) = \{b'_1, \dots, b'_k\}$ if we have

$$b = r_1 b'_1 + \cdots + r_k b'_k$$

for some non-zero elements $r_1, \dots, r_k \in R$. Since B' is a basis of F this map is well defined.

Notice that for $T \in S_{B'}$ we have

$$b \in f^{-1}(T) \quad \text{iff} \quad b \in \langle T \rangle$$

and so by Claim 2 the set $f^{-1}(T)$ is finite for all $T \in S_{B'}$. As a consequence we obtain

$$|B| = |\bigcup_{T \in S_{B'}} f^{-1}(T)| \leq |\bigcup_{T \in S_{B'}} \mathbb{N}| = |S_{B'}| \cdot \aleph_0 = |B'| \cdot \aleph_0$$

Since B' is an infinite set we have $|B'| \cdot \aleph_0 = |B'|$, and so $|B| \leq |B'|$.

Since by assumption we had $|B| \leq |B'|$ and by Claim 3 we have $|B| \leq |B'|$ we obtain that $|B| = |B'|$.

□

42.8 Theorem. *If R is a division ring then R has IBN.*

Proof. By Corollary 42.7 it is enough to show that if F is a free R-module with two finite bases $B = \{b_1, \ldots, b_n\}$ and $B' = \{b'_1, \ldots, b'_m\}$ then $n = m$.

We will argue by induction with respect to n

Assume that $n = 1$, and so $B = \{b_1\}$. If $B' = \{b'_1, \ldots, b'_m\}$ for some $m > 1$ then we have

$$b'_1 = r_1 b_1 \quad \text{and} \quad b'_2 = r_2 b_1$$

for some $r_1, r_2 \in R$, $r_1, r_2 \neq 0$. Therefore $r_1^{-1} b'_1 - r_2^{-1} b'_2 = 0$ which contradicts the assumption that B' is a inearly independent set. As a consequence we must have $m = 1$.

Next, assume that $n \geq 1$ is a number such that if a free R-module has a basis consisting of n elements then every other basis of that module also has n elements.

Let F be a free R-module with a basis $B = \{b_1, \ldots, b_{n+1}\}$ consisting of $n+1$ elements and let $B' = \{b'_1, \ldots, b'_m\}$ be another basis of F. Since $\langle B' \rangle = F$ we have

$$b_{n+1} = r_1 b'_1 + \ldots + r_m b'_m$$

for some $r_1, \ldots, r_m \in R$. Also, since $b_{n+1} \neq 0$ we have $r_i \neq 0$ for some i. We can assume that $r_m \neq 0$. Let $B'' := \{b'_1, \ldots, b'_{m-1}, b_{n+1}\}$. Check: B'' is a basis of F.

Take the canonical epimorphism

$$\pi \colon F \to F/\langle b_{n+1} \rangle$$

Check: since F is a free module with basis $B := \{b_1, \ldots, b_n, b_{n+1}\}$, thus $F/\langle b_{n+1} \rangle$ is a free module with basis $\{\pi(b_1), \ldots, \pi(b_n)\}$. On the other hand, since F has a basis $B'' := \{b'_1, \ldots, b'_{m-1}, b_{n+1}\}$ therefore $\{\pi(b'_1), \ldots, \pi(b'_{m-1})\}$ is a basis of $F/\langle b_{n+1} \rangle$.

By the inductive assumption we obtain than $n = m - 1$, and so $n + 1 = m$

□

42.9 Note. Let I be an ideal of R and let M be an R-module. Define:

$$IM := \{rm \mid r \in I,\ m \in M\}$$

Check:

1) IM a submodule of M.

2) M/IM has a structure of a R/I-module with the multiplication given by

$$(r + I)(m + IM) = rm + IM$$

42.10 Theorem. *Let R be a ring with identity and let $I \neq R$ be an ideal of R. If R/I has IBN then R also has IBN.*

Proof. Let F be a free R-module and let $B = \{b_1, \dots, b_n\}$ be a basis of F. Check: F/IF is a free R/I-module with basis $\{b_1 + IF, \dots, b_n + IF\}$.

Since R/I has IBN any basis of F/IF has n elements. As a consequence any basis of F also has n elements. □

42.11 Corollary. *If R is a commutative ring with identity $1 \neq 0$ then R has IBN.*

Proof. Let I be a maximal ideal in R. Then R/I is a field and we have the canonical homomorphism

$$\pi \colon R \to R/I$$

By Theorem 42.8 R/I has IBN, so by Theorem 42.10 R also has IBN. □

42.12 Note. Corollary 42.11 can be generalized as follows. If

$$f \colon R \to S$$

is an epimorphism of rings of identity such that S is a division ring then R has IBN.

43 Projective modules

43.1 Note. If F is a free R-module and $P \subseteq F$ is a submodule then P need not be free even if P is a direct summand of F.

Take e.g. $R = \mathbb{Z}/6\mathbb{Z}$. Notice that $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ are $\mathbb{Z}/6\mathbb{Z}$-modules and we have an isomorphism of $\mathbb{Z}/6\mathbb{Z}$-modules:

$$\mathbb{Z}/6\mathbb{Z} \;\cong\; \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$$

Thus $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ are non-free modules isomorphic to direct summands of the free module $\mathbb{Z}/6\mathbb{Z}$.

43.2 Definition. An R-module P is a *projective module* if there exists an R-module Q such that $P \oplus Q$ is a free R-module.

43.3 Examples.

1) If R is a ring with identity then every free R-module is projective.
2) $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ are non-free projective $\mathbb{Z}/6\mathbb{Z}$-modules.

43.4 Definition. Let

$$\ldots \longrightarrow M_i \xrightarrow{f_i} M_{i+1} \xrightarrow{f_{i+1}} M_{i+2} \longrightarrow \ldots$$

be a sequence of R-modules and R-module homomorphisms. This sequence is *exact* if $\mathrm{Im}(f_i) = \mathrm{Ker}(f_{i+1})$ for all i.

43.5 Definition. A *short exact sequence* is an exact sequence of R-modules the form

$$0 \longrightarrow N \xrightarrow{f} M \xrightarrow{g} K \longrightarrow 0$$

(where 0 is the trivial R-module).

43.6 Note.

1) A sequence $0 \to N \xrightarrow{f} M \xrightarrow{g} K \to 0$ is short exact iff
   - f is a monomorphism
   - g is an epimorphism
   - $\mathrm{Im}(f) = \mathrm{Ker}(g)$.

2) If M' is a submodule of M then we have a short exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M/M' \longrightarrow 0$$

Morever, up to an isomorphism, every short exact sequence is of this form:

$$\begin{tikzcd}
0 \arrow[r] & N \arrow[r, "f"] \arrow[d, "\cong"'] & M \arrow[r, "g"] \arrow[d, "="'] & K \arrow[r] \arrow[d, "\cong"] & 0 \\
0 \arrow[r] & \mathrm{Ker}(g) \arrow[r] & M \arrow[r] & M/\mathrm{Ker}(g) \arrow[r] & 0
\end{tikzcd}$$

43.7 Definition. A short exact sequence

$$0 \longrightarrow N \xrightarrow{f} M \xrightarrow{g} K \longrightarrow 0$$

is *split exact* if there is an isomorphism $\varphi\colon M \xrightarrow{\cong} N \oplus K$ such that the following diagram commutes:

$$\begin{tikzcd}
0 \arrow[r] & N \arrow[r, "f"] \arrow[d, "="'] & M \arrow[r, "g"] \arrow[d, "\varphi"', "\cong"] & K \arrow[r] \arrow[d, "="] & 0 \\
0 \arrow[r] & N \arrow[r] & N \oplus K \arrow[r] & K \arrow[r] & 0
\end{tikzcd}$$

43.8 Proposition. *Let R be a ring and let $0 \to N \xrightarrow{f} M \xrightarrow{g} K \to 0$ be a short exact sequence of R-modules. The following conditions are equivalent.*

1) The sequence is split exact.

2) *There exists a homomorphism $h\colon K \to M$ such that $gh = \mathrm{id}_K$.*

3) *There exists a homomorphism $k\colon M \to N$ such that $kf = \mathrm{id}_N$*

Proof. Exercise. □

43.9 Theorem. *Let R be a ring with identity and let P be an R-module. The following conditions are equivalent.*

1) *P is a projective module.*

2) *For any homomorphism $f\colon P \to N$ and an epimorphism $g\colon M \to N$ there is a homomorphism $h\colon P \to M$ such that the following diagram commutes:*

$$\begin{tikzcd} & P \arrow[dl, dashed, "h"'] \arrow[d, "f"] \\ M \arrow[r, "g"] & N \end{tikzcd}$$

3) *Every short exact sequence $0 \to N \xrightarrow{f} M \xrightarrow{g} P \to 0$ splits.*

Proof.

(1) $\Rightarrow$ (2) Let Q be a module such that $P \oplus Q$ is a free module, and let $B = \{b_i\}_{i\in I}$ be a basis of $P \oplus Q$. Since g is an epimorphism for every $i \in I$ we can find $m_i \in M$ such that $g(m_i) = f(b_i)$. Define

$$\tilde{h}\colon P \oplus Q \longrightarrow M$$

by

$$\tilde{h}\left(\sum_i r_i b_i\right) := \sum_i r_i m_i$$

Check: since B is a basis of $P \oplus Q$ the map $\tilde{h}$ is a well defined R-module homomorphism and $g\tilde{h} = f$. Then we can take $h = \tilde{h}|_P$.

(2) $\Rightarrow$ (3) We have a diagram

$$\begin{tikzcd} & P \arrow[d, "\mathrm{id}_P"'] \\ M \arrow[r, "g"] & P \end{tikzcd}$$

Since g is an epimorphism there is $h\colon P \to M$ such that $gh = \mathrm{id}_P$. Therefore by (43.8) the sequence $0 \to N \xrightarrow{f} M \xrightarrow{g} P \to 0$ splits. □

(3) $\Rightarrow$ (1) We have the canonical epimorphism of R-modules:

$$f\colon \bigoplus_{p\in P} R \to P$$

This gives a short exact sequence

$$0 \longrightarrow \mathrm{Ker}(f) \longrightarrow \bigoplus_{p\in P} R \xrightarrow{f} P \longrightarrow 0$$

By assumption on P this sequence splits. so we obtain

$$P \oplus \mathrm{Ker}(f) \cong \bigoplus_{p\in P} R$$

and thus P is a projective module.

43.10 Corollary. *If R is a ring with identity, P is a projective R-module and $f\colon M \to P$ is an epimorphism of R-modules then $M \cong P \oplus \mathrm{Ker}(f)$.*

Proof. We have a short exact sequence

$$0 \longrightarrow \mathrm{Ker}(f) \longrightarrow M \xrightarrow{f} P \longrightarrow 0$$

which splits by Theorem 43.9. □

44 Projective modules over PIDs

44.1 Theorem. *If R is a PID, F is a free R-module of a finite rank, and $M \subseteq F$ is a submodule then M is a free module and $\operatorname{rank} M \leq \operatorname{rank} F$.*

44.2 Corollary. *If R is a PID then every finitely generated projective R-module is free.*

Proof. If P is a finitely generated projective R-module then we have an epimorphism $f\colon R^n \to P$ for some $n > 0$. By Corollary 43.10 we have an isomorphism

$$P \oplus \operatorname{Ker}(f) \cong R^n$$

Therefore we can identify P with a submodule of R^n, and thus by Theorem 44.1 P is a free module. $\square$

44.3 Note. Theorem 44.1 is true also for infinitely generated free modules over PIDs. As a consequence Corollary 44.2 is true for all (non necessarily finitelly generated) projective modules over PIDs.

Proof of Theorem 44.1 (compare with the proof of Theorem 13.6).

We can assume that $F = R^n$.

We want to show: if $M \subseteq R^n$ then M is a free R-module and $\operatorname{rank} M \leq n$.

Induction with respect to n:

If $n = 1$ then $M \subseteq R$, so M is an ideal of R. Since R is a PID we have $M = \langle a \rangle$ for some $a \in R$. If $a = 0$ then $M = \{0\}$ is a free module of rank 0. Otherwise we have an isomorphism of R-modules

$$f\colon R \xrightarrow{\cong} M, \qquad f(r) = ra$$

and so M is a free module of rank 1.

Next, assume that for some n every submodule of R^n is a free R-module of rank $\leq n$, and let $M \subseteq R^{n+1}$. Take the homomorphism of R-modules

$$f\colon R^{n+1} \to R, \quad f(r_1, \dots, r_{n+1}) = r_{n+1}$$

We have:

$$\operatorname{Ker}(f) = \{(r_1, \dots, r_n, 0) \mid r_i \in R\} \cong R^n$$

We have an epimorphism:

$$f|_M\colon M \to \operatorname{Im}(f|_M)$$

Since $\operatorname{Im}(f|_M) \subseteq R$, thus $\operatorname{Im}(f|_M)$ is a free R-module, and so by Corollary 43.10 we have

$$M \cong \operatorname{Im}(f|_M) \oplus \operatorname{Ker}(f|_M)$$

We also have:

$$\operatorname{Ker}(f|_M) = \operatorname{Ker}(f) \cap M$$

It follows that that $\operatorname{Ker}(f|_M)$ is a submodule of $\operatorname{Ker}(f)$, and since $\operatorname{Ker}(f)$ is a free R-module of rank n by the inductive assumption we get that $\operatorname{Ker}(f|_M)$ is a free R-module of rank $\leq n$. Therefore

$$M \cong \underbrace{\operatorname{Im}(f|_M)}_{\substack{\text{free} \\ \text{rank} \leq 1}} \oplus \underbrace{\operatorname{Ker}(f|_M)}_{\substack{\text{free} \\ \text{rank} \leq n}}$$

and so M is a free R-module of rank $\leq n + 1$.

□

45 The Grothendieck group

Recall. If R is a ring with IBN and F is a free, finitely generated R-module then

$$\operatorname{rank} F = \text{number of elements of a basis of } F$$

Goal. Extend the notion of rank to finitely generated projective modules.

Idea.

1) Rank should be additive: $\operatorname{rank}(P \oplus Q) = \operatorname{rank} P + \operatorname{rank} Q$.

2) Rank of a module need not be an integer. Each ring determines a group $K_0(R)$ such that for each finitely generated projective module rank of P is an element $[P] \in K_0(R)$.

Recall. A commutative monoid is a set M together with addition

$$M \times M \to M, \quad (x, y) \mapsto x + y$$

and with a trivial element $0 \in M$ such that the addition is associative, commutative and $0 + x = x$ for all $x \in M$.

45.1 Example. Let Proj_R^{fg} be the set of isomorphism classes of finitely generated projective R-modules. For a projective finitely generated R-module P denote

$$[P] = \text{the isomorphism class of } P$$

The set Proj_R^{fg} is a commutative monoid with addition given by

$$[P] + [Q] := [P \oplus Q]$$

The identity element in Proj_R^{fg} is $[0]$, the isomorphism class of the zero module.

45.2 Theorem. *Let M be a commutative monoid. There exists an abelian group $\mathrm{Gr}(M)$ and a homomorphism of monoids*

$$\alpha_M \colon M \to \mathrm{Gr}(M)$$

that satisfies the following universal property. If G is any abelian group and $f \colon M \to G$ is a homomorphism of monoids then there exists a unique homomorphism of groups $\bar{f} \colon \mathrm{Gr}(M) \to G$ such that the following diagram commutes:

$$\begin{tikzcd}
M \arrow[r, "f"] \arrow[d, "\alpha_M"'] & G \\
\mathrm{Gr}(M) \arrow[ur, dashed, "\bar{f}"'] &
\end{tikzcd}$$

Moreover, such group $\mathrm{Gr}(M)$ is unique up to isomorphism.

45.3 Note. Let $\mathcal{CM}ono$ denote the category of commutative monoids. We have the forgetful functor

$$U \colon \mathcal{A}b \longrightarrow \mathcal{CM}ono$$

Theorem 45.2 is equivalent to the statement that this functor has a left adjoint

$$\mathrm{Gr} \colon \mathcal{CM}ono \to \mathcal{A}b, \qquad M \mapsto \mathrm{Gr}(M)$$

45.4 Definition. Let M be a commutative monoid. The group $\mathrm{Gr}(M)$ is called the *group completion* or the *Grothendieck group* of the monoid M.

Proof of Theorem 45.2.

Construction of $\mathrm{Gr}(M)$. Let M be a commutative monoid. Define

$$\mathrm{Gr}(M) := M \times M / \sim$$

where

$$(x, y) \sim (x', y') \quad \text{iff} \quad x + y' + t = x' + y + t \ \text{ for some } t \in M$$

Check: $\sim$ is an equivalence relation on $M \times M$. Notation:

$$[x, y] := \text{the equivalence class of } (x, y)$$

(Intuitively: $[x, y] = x - y$)

Note: for any $x \in M$ we have $[x, x] = [0, 0]$ since $x + 0 = 0 + x$.

Addition in $\mathrm{Gr}(M)$:

$$[x, y] + [x', y'] = [x + x', y + y']$$

Check: this operation is well defined, it is associative, and it has $[0, 0]$ as the identity element.

Additive inverses in $\mathrm{Gr}(M)$:

$$-[x, y] = [y, x]$$

Indeed: $[x, y] + [y, x] = [x + y, y + x] = [0, 0]$

Construction of the homomorphism $\alpha_M \colon M \to \mathrm{Gr}(M)$.

Define

$$\alpha_M \colon M \to \mathrm{Gr}(M), \qquad x \mapsto [x, 0]$$

The universal property of $\mathrm{Gr}(M)$.

Let G be an abelian group and let $f \colon M \to G$ be a homomorphism of commutative monoids. Define

$$\bar{f} \colon \mathrm{Gr}(M) \to G, \qquad \bar{f}([x, y]) := f(x) - f(y)$$

Check:

1) $\bar{f}$ is a well defined group homomorphism.

2) $\bar{f}\alpha_M = f$

3) $\bar{f}$ is the only homomorphism $\mathrm{Gr}(M) \to G$ satisfying condition 2).

Uniqueness of $\mathrm{Gr}(M)$ follows from the universal property.

□

45.5 Examples.

1) $\mathrm{Gr}(\mathbb{N}) \cong \mathbb{Z}$

2) Let $M = \mathbb{N} \cup \{\infty\}$ with $n + \infty = \infty$ for all $n \in M$. Then $\mathrm{Gr}(M)$ is the trivial group. Indeed, for any $m, n \in M$ we have

$$[m, n] = [\infty, \infty]$$

since $m + \infty = \infty + n$.

3) If G is an abelian group then $\mathrm{Gr}(G) \cong G$.

45.6 Definition. If R is a ring then $K_0(R) := \mathrm{Gr}(\mathrm{Proj}_R^{fg})$

45.7 Notation. For $[P], [Q] \in \mathrm{Proj}_R^{fg}$ denote

$$\begin{aligned} [P] - [Q] :=& [P, Q] \in K_0(R) \\ [P] :=& [P, 0] \\ -[Q] :=& [0, Q] \end{aligned}$$

45.8 Proposition. *Let R be a ring with identity. If P, Q are finitely generated projective R-modules then $[P] = [Q]$ in $K_0(R)$ iff there exists $n \geq 0$ such that $P \oplus R^n \cong Q \oplus R^n$.*

Proof.

($\Rightarrow$) If $P \oplus R^n \cong Q \oplus R^n$ then in $K_0(R)$ we have

$$[P] + [R^n] = [P \oplus R^n] = [Q \oplus R^n] = [Q] + [R^n]$$

and so $[P] = [Q]$

($\Rightarrow$) If $[P] = [Q]$ in $K_0(R)$ then

$$P \oplus S \cong Q \oplus S$$

for some finitely generated projective R-module S.

Exercise: If S is a finitely generated projective R-module then there is a finitely generated projective R-module T such that $S \oplus T \cong R^n$ for some $n \geq 0$.

We obtain

$$P \oplus R^n \cong P \oplus S \oplus T \cong Q \oplus S \oplus T \cong Q \oplus R^n$$

$\square$

45.9 Definition. Let R be a ring with identity. We say that R-modules M, N are *stably isomorphic* if $M \oplus R^n \cong N \oplus R^n$ for some $n \geq 0$.

45.10 Definition. Let R be a ring with identity. We say an R-module M is *stably free* if $M \oplus R^n \cong R^m$ for some $m, n \geq 0$.

45.11 Note. Let R be a ring with identity. We have a homomorphism $\varphi\colon \mathbb{Z} \to K_0(R)$ given by

$$\varphi(n) := \begin{cases} [R^n] & \text{for } n \geq 0 \\ -[R^{-n}] & \text{for } n < 0 \end{cases}$$

45.12 Proposition. *Let R be a ring with identity, and let $\varphi\colon \mathbb{Z} \to K_0(R)$ be the homomorphism as in (45.11).*

1) *φ is 1-1 iff R has IBN.*

2) *φ is an epimorphism iff every finitely generated projective R-module is stably free.*

Proof.

1) By Proposition 45.8 for $n \geq 0$ we have $n \in \operatorname{Ker}(\varphi)$ iff $R^n \oplus R^m \cong 0 \oplus R^m$ for some $m \geq 0$. If R has IBN this is possible only if $n = 0$, and so $\operatorname{Ker}(\varphi) = \{0\}$.

Conversely, assume that R does not have IBN. Then $R^n \cong R^m$ for some $n > m$. This gives $R^{n-m} \oplus R^m \cong 0 \oplus R^m$, and so $\varphi(n - m) \in \operatorname{Ker}(\varphi)$.

2) ($\Rightarrow$) Assume that φ is an epimorphism. Then for every finitely generated projective R-module P we have $[P] = [R^n]$ for some $n \geq 0$ or $[P] = -[R^n]$ for some $n \geq 0$.

If $[P] = [R^n]$ then by Proposition 45.8 we have $P \oplus R^m \cong R^n \oplus R^m$ and so P is a stably free module.

If $[P] = -[R^n]$ then

$$[0] = [P] + [R^n] = [P \oplus R^n]$$

Again by Proposition 45.8 this gives $0 \oplus R^m \cong P \oplus R^n \oplus R^m$, and again we obtain that P is stably free.

($\Leftarrow$) The group $K_0(R)$ is generated by elements $[P]$ where P is a finitely generated projective R-module, so it is enough to show that for any such P we have $[P] = \varphi(k)$ for some $k \in \mathbb{Z}$.

Since P is stably free we have $P \oplus R^n \cong R^m$ for some $n, m \geq 0$. This gives

$$[P] + [R^n] = [P \oplus R^n] = [R^m]$$

Therefore $[P] = [R^m] - [R^n] = \varphi(m) - \varphi(n) = \varphi(m - n)$.

$\square$

45.13 Example.

Here is an example of a stable free module that is not free. For details see:

R. G. Swan, *Vector bundles and projective modules*, Transactions AMS 105 (2) (1962), 264-277.

Let B be a compact, normal, topological space and let $p\colon E \to B$ be a real vector bundle over B. Define:

$$C(B) = \{f\colon B \to \mathbb{R} \mid f \text{ - continuous } \}$$

$C(B)$ is a ring (with pointwise addition and multiplication). Let $\Gamma(p)$ be the set of all continuous sections of p:

$$\Gamma(p) = \{s\colon B \to E \mid ps = \mathrm{id}_B\}$$

Note: $\Gamma(p)$ is an $C(B)$-module with poinwise addition and pointwise multiplication by elements of $C(B)$.

Fact 1. The module $\Gamma(p)$ is free iff p is a trivial vector bundle.

Fact 2. If $p\colon E \to B$ and $q\colon E' \to B$ are real vector bundles over B then we have an isomorphism of $C(B)$-modules:

$$\Gamma(p \oplus q) \cong \Gamma(p) \oplus \Gamma(q)$$

Upshot. If $p\colon E \to B$, $q\colon E' \to B$ are bundles such that p is non-trivial, but both q and $p \oplus q$ are trivial bundles then $\Gamma(p)$ is stably free $C(B)$-module that is not free.

Indeed, in such case we have:

$$\Gamma(p) \oplus \underbrace{\Gamma(q)}_{\text{free}} \cong \underbrace{\Gamma(p \oplus q)}_{\text{free}}$$

Fact 3. It is possible to find vector bundles as above. Take e.g. $p\colon TS^2 \to S^2$ to be the tangent bundle of the 2-dimensional sphere, and $q\colon S^2 \times R^1 \to S^2$ to be the 1-dimensional trivial bundle over S^2.

Note: one can also show that a $C(B)$-module M is finitely generated projective module iff $M \cong \Gamma(p)$ for some vector bundle $p\colon E \to B$.

46 Injective modules

Recall. If R is a ring with identity then an R-module P is projective iff one of the following equivalent conditions holds:

1) For any homomorphism $f\colon P \to N$ and an epimorphism $g\colon M \to N$ there is a homomorphism $h\colon P \to M$ such that the following diagram commutes:

$$\begin{tikzcd}
& P \arrow[dl, dashed, "h"'] \arrow[d, "f"'] \\
M \arrow[r, "g"] & N
\end{tikzcd}$$

2) Every short exact sequence $0 \to N \xrightarrow{f} M \xrightarrow{g} P \to 0$ splits.

46.1 Proposition. *Let R be a ring and let J be an R-module. The following conditions are equivalent.*

1) *For any homomorphism $f\colon N \to J$ and an monomorphism $g\colon M \to N$ there is a homomorphism $h\colon M \to J$ such that the following diagram commutes:*

$$\begin{tikzcd}
J & \\
M \arrow[u, "f"] \arrow[r, "g"'] & N \arrow[ul, dashed, "h"']
\end{tikzcd}$$

2) *Every short exact sequence $0 \to J \xrightarrow{f} M \xrightarrow{g} N \to 0$ splits.*

Proof. Exercise. $\square$

46.2 Definition. An R-module J is an *injective module* if J satisfies one of the equivalent conditions of Proposition 46.1.

46.3 Theorem (Baer's Criterion)**.**
Let R be a ring with identity and let J be an R-module. The following conditions are equivalent.

1) J is an injective module.

2) For every left ideal $I \lhd R$ and for every homomorphisms of R-modules $f\colon I \to J$ there is a homomorphism $\bar{f}\colon R \to J$ such $\bar{f}|_I = f$.

Proof.
1) $\Rightarrow$ 2) Given a homomorphism $f\colon I \to J$ we have a diagram

$$\begin{tikzcd} J & \\ I \arrow[u, "f"] \arrow[r, hook, "i"] & R \end{tikzcd}$$

where $i\colon I \hookrightarrow R$ is the inclusion homomorphism. By the definition of an injective module there is a homomorphism $\bar{f}\colon R \to J$ such that $f = \bar{f}i = \bar{f}|_I$

2) $\Leftarrow$ 1) Assume that J is an R-module satisfying 2). It is enough to show that if M is an R-module, N is a submodule of M, and $f\colon N \to J$ is an R-module homomorphism then there exists a homomorphism $\bar{f}\colon M \to J$ such that $\bar{f}|_N = f$

Let S be a set of all pairs (K, f_K) such that

(i) K is a submodule of M such that $N \subseteq K \subseteq M$

(ii) $f_K\colon K \to J$ is a homomorphism such that $f_K|_N = f$

Define partial ordering on S as follows:

$$(K, f_K) \leq (K', f_{K'}) \qquad \text{if} \quad K \subseteq K' \text{ and } f_{K'}|_K = f_K$$

Check: assumptions of Zorn's Lemma 29.10 are satisfied in S, and so S contains a maximal element (K_0, f_{K_0}).

It will suffice to show that $K_0 = M$. Assume, by contradiction, that $K_0 \neq M$, and let $m_0 \in M - K_0$. Define

$$I := \{r \in R \mid rm_0 \in K_0\}$$

Check: I is an ideal of R and the map

$$g\colon I \to J, \qquad g(r) = f_{K_0}(rm_0)$$

is a homomorphism of R-modules. By the assumptions on J we have a homomorphism $\bar{g}\colon R \to J$ such that $\bar{g}|_I = g$. Define

$$K_0 + Rm_0 := \{k + rm_0 \mid k \in K,\ r \in R\}$$

Check: $K_0 + Rm_0$ is a submodule of M and the map

$$f'\colon K_0 + Rm_0 \to J, \quad f'(k + rm_0) = f_{K_0}(k) + \bar{g}(r)$$

is a well defined homomorphism of R-modules such that $f'|_N = f$. This shows that $(K_0 + Rm_0, f') \in S$. We also have

$$(K_0, f_{K_0}) < (K_0 + Rm_0, f')$$

This is impossible since by assumption (K_0, f_{K_0}) is a maximal element in S. $\square$

46.4 Corollary. *Let R be an integral domain and let K the field of fractions of R. Then K is an injective R-module.*

Proof. Let I be an ideal of R and let $f\colon I \to K$ be a homomorphism of R-modules. For $0 \neq r, s \in I$ we have

$$rf(s) = f(rs) = sf(r)$$

As consequence in K we have $f(r)/r = f(s)/s$ for any $0 \neq r, s \in I$. Denote this element by a. Define

$$\bar{f}\colon R \to K, \qquad \bar{f}(r) := ra$$

Check: $\bar{f}$ is a homomorphism of R-modules and $\bar{f}|_I = f$.

By Baer's Criterion (46.3) it follows that K is an injective R-module. □

46.5 Example. $\mathbb{Q}$ is an injective $\mathbb{Z}$-module.

46.6 Definition. Let R be an integral domain. An R-module M is *divisible* if for every $r \in R - \{0\}$ and for every $m \in M$ there is $n \in M$ such that $rn = m$.

46.7 Theorem. *If R is a PID then an R-module J is injective iff J is divisible.*

Proof. Exercise. □

46.8 Example.

Since $\mathbb{Z}$ is a PID injective $\mathbb{Z}$-modules are divisible $\mathbb{Z}$-modules (i.e. divisible abelian groups).

Exercise: an abelian group G is divisible iff G is isomorphic to a direct sum of copies of $\mathbb{Q}$ and $\mathbb{Z}(p^\infty)$ for various primes p.

46.9 Corollary. *If R is a PID, J is an injective R-module and K is a submodule of J then J/K is injective.*

Proof. Since J is divisible thus so is J/K (check!). $\square$

46.10 Note. If R is not a PID then a quotient of an injective R-module need not be injective.

Note. If R is a ring with identity then for any R-module M there exists an epimorphism of R-modules:

$$f\colon P \longrightarrow M$$

where P is a projective module (take e.g. $P = \bigoplus_{m\in M} R$).

46.11 Theorem. *If R is a ring with identity then for any R-module M there exist a monomorphism*

$$j\colon M \longrightarrow J$$

where J is an injective R-module.

46.12 Lemma. *For any abelian group G there exists a monomorphism*

$$i\colon G \longrightarrow D$$

where H is a divisible abelian group.

Proof. We have an epimorphism $f\colon \bigoplus_{g\in G}\mathbb{Z} \to G$ which gives an isomorphism

$$\varphi\colon G \xrightarrow{\cong} \bigoplus_{g\in G}\mathbb{Z}/\operatorname{Ker}(f)$$

Moreover, the monomorphism $\bigoplus_{g\in G}\mathbb{Z} \to \bigoplus_{g\in G}\mathbb{Q}$ induces a monomorphism

$$\psi\colon \bigoplus_{g\in G}\mathbb{Z}/\operatorname{Ker}(f) \longrightarrow \bigoplus_{g\in G}\mathbb{Q}/\operatorname{Ker}(f)$$

We can take $D := \bigoplus_{g\in G}\mathbb{Q}/\operatorname{Ker}(f)$ and $i := \psi\varphi$. $\square$

46.13 Note. Let G is an abelian group, let R let be a ring, and let $\mathrm{Hom}_{\mathbb{Z}}(R, G)$ be the set of all homomorphisms of abelian groups $\varphi\colon R \to G$.

Check: $\mathrm{Hom}_{\mathbb{Z}}(R, G)$ is an R-module with pointwise addition and with multiplication by elements of R given by

$$(r \cdot \varphi)(s) := \varphi(sr)$$

for $r, s \in R$.

46.14 Lemma. *If D is a divisible abelian group and R is a ring with identity then $\mathrm{Hom}_{\mathbb{Z}}(R, D)$ in an injective R-module.*

Proof. Exercise. □

Proof of Theorem 46.11. Let M be an R-module. Consider M as an abelian group. By Lemma 46.12 we have a monomorphism of abelian groups

$$i\colon M \longrightarrow D$$

where D is a divisible abelian group. Consider the induced map

$$i_*\colon \mathrm{Hom}_{\mathbb{Z}}(R, M) \to \mathrm{Hom}_{\mathbb{Z}}(R, D), \qquad i_*(\varphi) = i \circ \varphi$$

Check:

1) i_* is a monomorphism.
2) i_* is a homomorphism of R-modules.

Since M is an R-module we also have a map

$$f\colon M \to \mathrm{Hom}_{\mathbb{Z}}(R, M), \qquad f(m)(r) = rm$$

Check:

1) f is a monomorphism.

2) f is a homomorphism of R-modules.

As a consequence we obtain a monomorphism of R-modules

$$i_* f\colon M \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(R, D)$$

Moreover, by Lemma 46.12 $\mathrm{Hom}_{\mathbb{Z}}(R, D)$ is an injective R-module.

$\square$

47 Exact functors

47.1 Definition. A *chain complex* of R-modules is a sequence of R-modules and R-homomorphisms

$$\dots \longrightarrow M_{i+1} \xrightarrow{d_{i+1}} M_i \xrightarrow{d_i} M_{i-1} \xrightarrow{d_{i-1}} M_{i-2} \longrightarrow \dots$$

such that $d_i d_{i+1} = 0$ for all i.

47.2 Note. If $M_* = (M_i, d_i)$ is a chain complex then $\operatorname{Im}(d_{i+1}) \subseteq \operatorname{Ker}(d_i)$.

47.3 Definition. If $M_* = (M_i, d_i)$ is a chain complex of R-modules then the *i-th homology module* of M_* is the module

$$H_i(M_*) = \operatorname{Ker}(d_i)/\operatorname{Im}(d_{i+1})$$

Recall. A sequence

$$M_* = (\dots \longrightarrow M_{i+1} \xrightarrow{d_{i+1}} M_i \xrightarrow{d_i} M_{i-1} \xrightarrow{d_{i-1}} M_{i-2} \longrightarrow \dots)$$

is exact if $\operatorname{Ker}(d_i) = \operatorname{Im}(d_{i+1})$ for all i. Therefore M_* is exact iff $H_i(M_*) = 0$ for all i.

47.4 Definition. Let R, S be rings. A functor $F\colon R\text{-}\mathcal{M}od \to S\text{-}\mathcal{M}od$ is *exact* if for every short exact sequence of R-modules

$$0 \longrightarrow N \xrightarrow{f} M \xrightarrow{g} K \longrightarrow 0$$

the sequence

$$0 \longrightarrow F(N) \xrightarrow{F(f)} F(M) \xrightarrow{F(g)} F(K) \longrightarrow 0$$

is short exact.

47.5 Note. If $F\colon R\text{-}\mathcal{M}od \to S\text{-}\mathcal{M}od$ is a functor such that $F(0) = 0$ and

$$M_* = (\ldots \longrightarrow M_{i+1} \xrightarrow{d_{i+1}} M_i \xrightarrow{d_i} M_{i-1} \longrightarrow \ldots)$$

is a chain complex of R-modules then

$$F(M_*) = (\ldots \longrightarrow F(M_{i+1}) \xrightarrow{F(d_{i+1})} F(M_i) \xrightarrow{F(d_i)} F(M_{i-1}) \longrightarrow \ldots)$$

is a chain complex of S-modules.

Moreover, the functor F is exact iff for every chain complex M_* we have isomorphisms

$$F(H_i(M_*)) \cong H_i(F(M_*))$$

for all i.

47.6 Note. For a ring R and R-modules L, M let $\mathrm{Hom}_R(L, M)$ be the set of all R-module homomorphisms $\varphi\colon L \to M$. Notice that $\mathrm{Hom}_R(L, M)$ is an abelian group (with respect to the pointwise addition of homomorphisms). Moreover, for any homomorphism of R-modules $f\colon M \to N$ the map

$$f_*\colon \mathrm{Hom}_R(L, M) \to \mathrm{Hom}_R(L, N), \qquad f_*(\varphi) = f \circ \varphi$$

is a homomorphism of abelian groups. This defines a functor

$$\mathrm{Hom}_R(L, -)\colon R\text{-}\mathcal{M}od \longrightarrow \mathcal{A}b$$

This functor is in general not exact. Take e.g. $R = \mathbb{Z}$, $L = \mathbb{Z}/2\mathbb{Z}$. We have a short exact sequence of abelian groups:

$$0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \to 0$$

On the other hand the sequence

$$0 \to \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \to 0$$

is not exact since $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) \cong 0$ and $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$.

47.7 Proposition. *Let R be a ring and let L be an R-module. If*

$$0 \longrightarrow N \xrightarrow{f} M \xrightarrow{g} K \longrightarrow 0$$

is a short exact sequence of R-modules then

$$0 \longrightarrow \mathrm{Hom}_R(L, N) \xrightarrow{f_*} \mathrm{Hom}_R(L, M) \xrightarrow{g_*} \mathrm{Hom}_R(L, K)$$

is an exact sequence of abelian groups.

Proof. Exercise. □

47.8 Theorem. *Let R be a ring with identity and let P be an R-module. The functor $\mathrm{Hom}_R(P, -)$ is exact iff P is a projective module.*

Proof. By Proposition 47.7 is suffices to show that P is a projective module iff for every epimorphism of R-modules $g\colon M \to K$ the map

$$g_*\colon \mathrm{Hom}_R(P, M) \longrightarrow \mathrm{Hom}_R(P, K)$$

is an epimorphism. This follows directly from Theorem 43.9. □

47.9 Definition. Let $\mathcal{C}, \mathcal{D}$ be categories. A *contravariant functor* $F\colon \mathcal{C} \to \mathcal{D}$ consists of

1) an assignment
$$\mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D}), \quad c \mapsto F(c)$$

2) for every $c, c' \in \mathcal{C}$ a function
$$\mathrm{Hom}_{\mathcal{C}}(c, c') \to \mathrm{Hom}_{\mathcal{D}}(F(c'), F(c)), \quad f \mapsto F(f)$$
such that $F(id_c) = \mathrm{id}_{F(c)}$ and $F(gf) = F(f)F(g)$.

47.10 Example. Let R be a ring and let L be an R-module. For any homomorphism of R-modules $f\colon M \to N$ we have a map

$$f^*\colon \mathrm{Hom}_R(N, L) \longrightarrow \mathrm{Hom}_R(M, L), \qquad f^*(\varphi) = \varphi \circ f$$

Moreover, f^* is a homomorphism of abelian groups.

This defines a contravariant functor

$$\mathrm{Hom}_R(-, L)\colon R\text{-}\mathcal{M}od \longrightarrow \mathcal{A}b$$

47.11 Note. The functor $\mathrm{Hom}_R(-, L)$ is in general not exact. Take e.g. $R = \mathbb{Z}$, $L = \mathbb{Z}$. We have a short exact sequence of abelian groups:

$$0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \to 0$$

On the other hand the sequence

$$0 \to \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}) \to 0$$

is not exact.

47.12 Proposition. *Let R be a ring and let L be an R-module. If*

$$0 \longrightarrow N \xrightarrow{f} M \xrightarrow{g} K \longrightarrow 0$$

is a short exact sequence of R-modules then

$$0 \longrightarrow \mathrm{Hom}_R(K, L) \xrightarrow{g^*} \mathrm{Hom}(M, L) \xrightarrow{f^*} \mathrm{Hom}(M, L)$$

is an exact sequence of abelian groups.

Proof. Exercise. □

47.13 Theorem. *Let R be a ring with identity and let J be an R-module. The functor $\mathrm{Hom}_R(-, J)$ is exact iff J is an injective module.*

Proof. By Proposition 47.12 is suffices to show that J is an injective module iff for every monomorphism of R-modules $f\colon N \to M$ the map

$$f^*\colon \mathrm{Hom}_R(M, J) \longrightarrow \mathrm{Hom}_R(N, J)$$

is an epimorphism. This follows directly from Proposition 46.1. $\square$

47.14 Note. Let

$$M_* = (\ldots \longrightarrow M_i \xrightarrow{d_i} M_{i-1} \longrightarrow \ldots)$$

be a chain complex of R-modules. For any R-module L we have the induced chain complex of abelian groups

$$\mathrm{Hom}_R(M_*, L) = (\ldots \longrightarrow \mathrm{Hom}_R(M_{i-1}, L) \xrightarrow{d_i^*} \mathrm{Hom}_R(M_i, L) \longrightarrow \ldots)$$

Homology groups of the complex $\mathrm{Hom}_R(M_*, L)$ are called *cohomology groups* of M_* with coefficients in L. We denote:

$$H^i(M_*, L) := H_i(\mathrm{Hom}_R(M_*, L))$$

If L is an injective module then by Theorem 47.13 the functor $\mathrm{Hom}_R(-, L)$ is exact. By (47.5) in such case we have

$$H^i(M_*, L) \cong \mathrm{Hom}_R(H_i(M_*), L)$$

48 Tensor products

Note. For now all rings are commutative rings with identity.

48.1 Definition. Let M, N, K be R-modules. A function

$$f\colon M \times N \longrightarrow K$$

is *bilinear* if

1) $f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$
   $f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$

2) $f(rm, n) = rf(m, n) = f(m, rn)$

48.2 Theorem. *Let M, N be R-modules. There exists an R-module $M \otimes_R N$ and a bilinear map*

$$\eta\colon M \times N \to M \otimes_R N$$

that satisfies the following universal property. For any R-module K and a bilinear map $f\colon M \times N \to K$ there exists a unique homomorphism of R-modules $\bar{f}\colon M \otimes_R N \to K$ such that the following diagram commutes:

$$\begin{tikzcd}
M \times N \arrow[r, "f"] \arrow[d, "\eta"'] & K \\
M \otimes_R N \arrow[ur, dashed, "\bar{f}"'] &
\end{tikzcd}$$

Moreover, such R-module $M \otimes_R N$ is unique up to isomorphism.

48.3 Definition. If M, N are R-modules then the module $M \otimes_R N$ is called the *tensor product* of M and N.

Proof of Theorem 48.2.

Construction of $M \otimes_R N$.

Let $F(M \times N)$ denote the free R-module with basis $\{(m, n) \mid m \in M, n \in N\}$
Note: every element of $F(M \times N)$ is of the form

$$\sum_i r_i(m_i, n_i)$$

for some $r_i \in R$.

Let S be the submodule of $F(M \times N)$ generated by all elements of the following forms:

(i) $(m_1 + m_2, n) - (m_1, n) - (m_2, n)$
$(m, n_1 + n_2) - (m, n_1) - (m, n_2)$

(ii) $(rm, n) - r(m, n)$
$(m, rn) - r(m, n)$

Define $M \otimes_R N := F(M \times N)/S$

Denote:

$$m \otimes n = (\text{the equvalence class of } (m, n) \text{ in } M \otimes_R N)$$

Note:

1) Every element of $M \otimes_R N$ can be written (non-uniquely!) as a linear combination
$$\sum_i r_i(m_i \otimes n_i)$$

2) In $M \otimes_R N$ we have

   (i) $(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$
   $m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$

   (ii) $(rm) \otimes n = r(m \otimes n) = m \otimes (rn)$

Construction of the map η.

Define:

$$\eta\colon M \times N \to M \otimes_R N, \qquad \eta(m,n) = m \otimes n$$

By (i)-(ii) above η is bilinear.

The universal property of $M \otimes_R N$.

If $f\colon M \times N \to K$ is a bilinear map define $\bar{f}\colon M \otimes_R N \to K$ by

$$\bar{f}\left(\sum_i r_i(m_i \otimes n_i)\right) = \sum_i r_i f(m_i, n_i)$$

Check:

1) $\bar{f}$ is a well defined R-module homomorphism
2) $\bar{f}\eta = f$
3) $\bar{f}$ is the only homomorphism $M \otimes_R N \to K$ satisfying condition 2).

Uniqueness of $M \otimes_R N$ up to isomorphism follows directly from the universal property.

$\square$

48.4 Example. Let $m, n > 0$, $\gcd(m,n) = 1$

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} = ?$$

Note: if $f\colon \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \to K$ is a bilinear map then

$$f(k,l) = f(k \cdot 1, l \cdot 1) = (kl) \cdot f(1,1)$$

Therefore the map f is determined by the value of $f(1,1)$.

Moreover, since $\gcd(m,n)=1$ we have $am+bn=1$ for some $m,n\in\mathbb{Z}$. This gives:

$$\begin{aligned}
f(1,1) &= f(1,(am+bn)\cdot 1)\\
&= f(1,(am)\cdot 1)+f(1,(bn)\cdot 1)\\
&= f((am)\cdot 1,1)+f(1,(bn)\cdot 1)\\
&= f(0,1)+f(1,0)\\
&= f(0\cdot 1,1)+f(1,0\cdot 1)\\
&= 0\cdot f(1,1)+0\cdot f(1,1)\\
&= 0
\end{aligned}$$

Therefore for any bilinear map f we have $f(1,1)=0$, and so $f=0$. It follows that

$$\mathbb{Z}/m\mathbb{Z}\otimes_{\mathbb{Z}}\mathbb{Z}/n\mathbb{Z}=0$$

since the zero module satisfies the universal property:

$$\begin{tikzcd}
\mathbb{Z}/m\mathbb{Z}\times\mathbb{Z}/n\mathbb{Z} \arrow[r, "f=0"] \arrow[d] & K \\
0 \arrow[ur, dashed, "\bar{f}=0"'] &
\end{tikzcd}$$

48.5 Proposition. *If M is an R-module then*

$$R\otimes_R M\cong M$$

Proof. We have a bilinear map

$$\eta\colon R\times M\to M,\quad \eta(r,m)=rm$$

(note: this works only if R is commutative!).

Moreover, if $f\colon R\times M\to K$ is any bilinear map then define

$$\bar{f}\colon M\to K,\quad \bar{f}(m)=f(1,m)$$

Check: $\bar{f}$ is the unique R-module homomorphism such that $\bar{f}\eta = f$.

By the universal property of tensor products this gives $M \cong R \otimes_R M$. $\square$

48.6 Proposition.

1) If M, N are R-modules then we have an isomorphism

$$\varphi\colon M \otimes_R N \to N \otimes_R M, \qquad \varphi(m \otimes n) = n \otimes m$$

2) If M, N, K are R-modules then

$$M \otimes_R (N \otimes_R K) \cong (M \otimes_R N) \otimes_R K$$

3) If M_α, N are R-modules then

$$(\bigoplus_\alpha M_\alpha) \otimes_R N \cong \bigoplus_\alpha (M_\alpha \otimes_R N)$$

Proof. Exercise. $\square$

49 Tensor products of homomorphisms

Let $f\colon M \to K$, $g\colon N \to L$ be homomorphisms of R-modules. Notice that we have a bilinear map

$$f \times g\colon N\colon M \times N \to K \otimes_R L, \quad f \times g(m, n) = f(m) \otimes g(n)$$

By the unniversal property of tensor products $f \times g$ induces a unique homomorphism of R-modules

$$f \otimes g\colon M \otimes_R N \to K \otimes_R L, \quad f \otimes g(m \otimes n) = f(m) \otimes g(n)$$

49.1 Proposition.

1) *Tensor product preserves composition of homomorphisms: if we have homomorphisms of R-modules*

$$M \xrightarrow{f} K \xrightarrow{f'} K'$$

$$N \xrightarrow{g} L \xrightarrow{g'} L'$$

then

$$(f' \circ f) \otimes (g' \circ g) = (f' \otimes g') \circ (f \otimes g)$$

2) *Tensor product preserves identity maps:*

$$\mathrm{id}_M \otimes \mathrm{id}_N = \mathrm{id}_{M \otimes_R N}$$

Proof. Exercise. □

49.2 Corollary. *If $f\colon M \to K$, $g\colon N \to L$ are isomorphisms of R-modules then*

$$f \otimes g\colon M \otimes_R N \to K \otimes_R L$$

is also an isomorphism.

Proof. Exercise. □

49.3 Proposition. *If F, F' are free R-modules then $F \otimes_R F'$ is also a free module.*

Proof. We have

$$F \cong \bigoplus_{i \in I} R, \quad F' \cong \bigoplus_{j \in J} R$$

Therefore

$$\begin{aligned} F \otimes_R F' &\cong (\bigoplus_{i \in I} R) \otimes_R (\bigoplus_{j \in J} R) \\ &\cong \bigoplus_{i \in I} \bigoplus_{j \in J} (R \oplus_R R) \\ &\cong \bigoplus_{(i,j) \in I \times J} R \end{aligned}$$

□

49.4 Note. If F, F' are free modules, $\{b_i\}_{i \in I}$ is a basis of F and $\{b'_j\}_{j \in J}$ is a basis of F' then $\{b_i \otimes b_j\}_{(i,j) \in I \times J}$ is a basis of $F \otimes_R F'$ (exercise).

49.5 Corollary. *If P, Q are projective R-modules then $P \otimes_R Q$ is also projective.*

Proof. Since P, Q are projective R-modules there exists modules P', Q' such that $P \oplus P'$, $Q \oplus Q'$ are free modules. We have

$$\underbrace{(P \oplus P') \otimes_R (Q \oplus Q')}_{\text{free}} \cong (P \otimes_R Q) \oplus (P \otimes_R Q') \oplus (P' \otimes_R Q) \oplus (P' \otimes_R Q')$$

Therefore $P \otimes_R Q$ is a direct summand of a free module, and so it is projective.

□

50 Tensor products and adjoint functors

Recall. Let R be a commutative ring. If M, N are R-modules then

$$\mathrm{Hom}_R(M, N) = (\text{the set of all } R\text{-module homomorphisms } M \to N)$$

Note.

1) $\mathrm{Hom}_R(M, N)$ is an abelian group with addition given by

$$(f + g)(m) := f(m) + g(m)$$

2) $\mathrm{Hom}_R(M, N)$ is an R-module with multiplication by elements of R given by

$$(rf)(m) := r \cdot f(m)$$

(Note: rf need not be an R-module homomorphism if R is a non-commutative ring.)

3) If $g\colon N \to N'$ is a homomorphism of R-modules then the map

$$g_*\colon \mathrm{Hom}_R(M, N) \to \mathrm{Hom}_R(M, N'), \qquad g_*(f) = g \circ f$$

is a homomorphism of R-modules.

Upshot. For any R-module M we have a functor

$$\mathrm{Hom}(M, -)\colon R\text{-}\mathcal{M}od \longrightarrow R\text{-}\mathcal{M}od$$

50.1 Proposition. *For any R-module M the functor*

$$- \otimes_R M\colon R\text{-}\mathcal{M}od \longrightarrow R\text{-}\mathcal{M}od$$

is left adjoint to the functor $\mathrm{Hom}_R(M, -)$.

50.2 Note. Proposition 50.1 says that for any R-modules N, K we have a natural bijection of sets

$$\mathrm{Hom}_R(N \otimes_R M, K) \xrightarrow{\cong} \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, K))$$

Compare this with the exponential law for in set theory: if A, B, C are sets then we have a bijection

$$\mathrm{Map}(A \times B, C) \xrightarrow{\cong} \mathrm{Map}(A, \mathrm{Map}(B, C))$$

Sketch of proof of Proposition 50.1.

Denote:

$$\mathrm{Hom}^2_R(N, M; K) = (\text{set of all bilinear maps } N \times M \to K)$$

By the universal property of tensor ptoducts we have a bijection

$$\mathrm{Hom}_R(N \otimes_R M, K) \cong \mathrm{Hom}^2_R(N, M; K)$$

Therefore we only need a bijection

$$\Phi\colon \mathrm{Hom}^2_R(N, M; K) \longrightarrow \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, K))$$

For $f \in \mathrm{Hom}^2_R(N, M; K)$ define:

$$\Phi(f)\colon N \to \mathrm{Hom}_R(M, K), \qquad \Phi(f)(n) = f(n, -)$$

$\square$

51 Tensor products for non-commutative rings

Recall. If R is a commutative ring and M, N, K are R-modules then we have a bijection

$$\begin{pmatrix}\text{bilinear maps}\\ M\times N\to K\end{pmatrix} \quad\cong\quad \begin{pmatrix}R\text{-linear maps}\\ M\otimes_R N\to K\end{pmatrix}$$

Note. If R is a non-commutative ring then there are few bilinear map since for any such map $f\colon M\times N\to K$ we have

$$(sr)\cdot f(m,n) = s\cdot f(rm,n) = f(rm,sn) = r\cdot f(m,sn) = (rs)\cdot f(m,n)$$

A more appropriate notion in this setting is a *middle linear map*.

51.1 Definition. Let R be a ring with identity. Let M be a right R-module, N be a left R-module, G be an abelian group. A map

$$f\colon M\times N\to G$$

is *middle linear* if

1) $f(m_1+m_2,n) = f(m_1,n)+f(m_2,n)$
   $f(m,n_1+n_2) = f(m,n_1)+f(m,n_2)$

2) $f(mr,n) = f(m,rn)$

51.2 Examples.

1) If R is a ring then the map

   $$\mu\colon R\times R\to R,\qquad \mu(r,s)=rs$$

   is middle linear

2) In general, if N is a left R-module then the map

$$\mu\colon R\times N\to N,\qquad \mu(r,n)=rn$$

is middle linear.

51.3 Definition. If R is a ring with identity, M is a right R-module and N is a left R-module then $M\otimes_R N$ is the abelian group given by

$$M\otimes_R N := F_{\mathsf{ab}}(M\times N)/S$$

Here $F_{\mathsf{ab}}(M\times N)$ is the free abelian group generated by the set $M\times N$ and S is the subgroup of $F_{\mathsf{ab}}(M\times N)$ generated by all elements of the following forms:

(i) $(m_1+m_2,n)-(m_1,n)-(m_2,n)$
$(m,n_1+n_2)-(m,n_1)-(m,n_2)$

(ii) $(mr,n)-(m,rn)$

51.4 Note. Let $m\otimes n$ denote the element of $M\otimes_R N$ represented by (m,n). We have a middle linear map

$$\eta\colon M\times N\to M\otimes_R N,\qquad \eta(m,n)=m\otimes n$$

51.5 Theorem. *Let R be a ring with identity, let M be a right R-module, N be a left R-module. If G be an abelian group and and $f\colon M\times N\to G$ is a middle linear map then there exists a unique homomorphism of abelian groups $\bar{f}\colon M\otimes_R N\to G$ such that the following diagram commutes:*

$$\begin{tikzcd}
M\times N \arrow[r, "f"] \arrow[d, "\eta"'] & G \\
M\otimes_R N \arrow[ur, dashed, "\bar{f}"'] &
\end{tikzcd}$$

Moreover, $M\otimes_R N$ is unique up to isomorphism abelian group satisfying this property.

Proof. Similar to the proof of Theorem 48.2. □

52 Restriction and extension of scalars

52.1 Definition. Let $f\colon R \to S$ be a homomorphism of rings, and let M be a left S-module. Let f^*M be the left R-module given by

$$\begin{aligned} f^*M :=& M && \text{as abelian group} \\ r \cdot m :=& f(r)m && \text{for } r \in R, m \in M \end{aligned}$$

We say that f^*M *is obtained from M by restriction of scalars*.

52.2 Example. Let V be an $\mathbb{R}$-vector space and let $i\colon \mathbb{Q} \to \mathbb{R}$ be the inclusion. Then i^*V is a $\mathbb{Q}$-vector space.

52.3 Note. For a ring homomorphism $f\colon R \to S$ restriction of scalars defines a functor

$$f^*\colon S\text{-}\mathcal{M}od \longrightarrow R\text{-}\mathcal{M}od$$

52.4 Note. Let $f\colon R \to S$ be a ring homomorphism. Then:

1) S is a right R-module with multiplication given by

$$s \cdot r := sf(r) \qquad \text{for } s \in S, r \in R$$

   Therefore for any left R-module N we have an abelian group $S \otimes_R N$

2) $S \otimes_R N$ has a structure of a left S-module with multiplication given by

$$s \cdot (s' \otimes n) := (ss') \otimes n$$

   (check!).

52.5 Definition. If $f\colon R \to S$ is a homomorphism of rings and N is a left R-module then we say that the left S-module $S \otimes_R N$ *is obtained from* N *by extension of scalars*. We denote:

$$f_* N := S \otimes_R N$$

52.6 Example. If V be an $\mathbb{R}$-vector space and $i\colon \mathbb{R} \to \mathbb{C}$ be the inclusion then Then $i_* V = \mathbb{C} \otimes_R V$ is a $\mathbb{C}$-vector space. We say that $i_* V$ is the *complexification* of V.

52.7 Note. For a ring homomorphism $f\colon R \to S$ extension of scalars defines a functor

$$f_*\colon R\text{-}\mathcal{M}od \longrightarrow S\text{-}\mathcal{M}od$$

52.8 Note. The functor f_* is left adjoint to the functor f^*. That is, for any left R-module N and any left S-module M we have a natural bijection of sets

$$\mathrm{Hom}_S(f_* M, N) \xrightarrow{\cong} \mathrm{Hom}_R(M, f^* N)$$

(exercise).

52.9 Proposition. *Let $f\colon R \to S$ be a homomorphism of rings.*

1) *If M, M' are left R-modules and $M \cong M'$ then $f_* M \cong f_* M'$.*

2) *$f_* R \cong S$*

3) *$f_*(\bigoplus_\alpha M_\alpha) \cong \bigoplus_\alpha f_* M_\alpha$*

Proof. Exercise. □

52.10 Corollary. *Let $f\colon R \to S$ be a homomorphism of rings. If F is a free R-module with basis $\{b_\alpha\}_{\alpha\in A}$ then f_*F is a free S-module with basis $\{1\otimes b_\alpha\}_{\alpha\in A}$.*

Sketch of proof. We have $F \cong \bigoplus_\alpha R$, so

$$f_*F \cong f_*(\bigoplus_\alpha R) \cong \bigoplus_\alpha f_*R \cong \bigoplus_\alpha S$$

□

52.11 Corollary. *Let $f\colon R \to S$ be a homomorphism of rings. If P is a projective R-module then f_*P is a projective S-module.*

Proof. Since P is projective there exists an R-module Q such that $P \oplus Q$ is a free R-module. Then we have

$$f_*(P \oplus Q) \cong f_*P \oplus f_*Q$$

By Corollary 52.10 $f_*(P \oplus Q)$ is a free S-module, so f_*P is a projective S-module. □

52.12 Note. Restriction of scalars does not preserve free and projective modules. E.g, let $i\colon \mathbb{Z} \to \mathbb{Q}$ be the inclusion. Then $\mathbb{Q}$ is a free $\mathbb{Q}$-module, but $i^*\mathbb{Q}$ is not free (or projective) $\mathbb{Z}$-module.

53 Application: rings with IBN

Recall. From Section 42:

A ring R has the *invariant basis number (IBN)* property if

$$R^m \cong R^n \quad \text{iff} \quad m = n$$

53.1 Proposition. *If $f\colon R \to S$ is a homomorphisms of rings with identity such that S has IBN then R also has IBN.*

Proof. If $R^m \cong R^n$ then $f_*R^m \cong f_*R^n$. By Corollary 52.10 we have $f_*R^m \cong S^m$, $f_*R^n \cong S^n$, and since S has IBN we have $m = n$. $\square$

Recall:

42.11 Corollary. *If R is a commutative ring with identity $1 \neq 0$ then R has IBN.*

New proof. Let $I \lhd R$ be a maximal ideal. We have a ring homomorphism

$$\pi\colon R \to R/I$$

Since R/I is a field it has IBN, and so by Proposition 53.1 the ring R also has IBN. $\square$

54 Algebras

54.1 Definition. Let R be a commutative ring with identity. A is an (associative, unital) R-*algebra* if

1) A is a ring with identity
2) A is an R-module
3) for any $r \in R$, $a, b \in A$ we have

$$r \cdot (ab) = (ra)b = a(rb)$$

A homomorphism of R-algebras is a ring homomorphism

$$f\colon A \to B$$

such that $f(ra) = rf(a)$ for all $a \in A$, $r \in R$.

54.2 Examples.

1) Every ring is a $\mathbb{Z}$-algebra.

2) If R is a commutative ring then $R[x_1, \dots, x_n]$ is an R-algebra.

3) If R is a commutative ring and $M_n(R)$ is the ring of all $n \times n$ matrices with coefficients in R then $M_n(R)$ is an R-algebra.

55 Tensor product of algebras

55.1 Note.

1) If A, B are R-algebras then they are R-modules, so $A \otimes_R B$ is also an R-module.

2) Check: $A \otimes_R B$ has a structure of an R-algebra with multiplication

$$(A \otimes_R B) \times (A \otimes_R B) \longrightarrow A \otimes_R B$$

given by:

$$\left(\sum_i r_i(a_i \otimes b_i)\right) \cdot \left(\sum_j s_j(a'_j \otimes b'_j)\right) := \sum_{i,j} r_i s_j(a_i a'_j \otimes b_i b'_j)$$

55.2 Example. Let R be a commutative ring. We have an isomorphism of R-algebras:

$$R[x] \otimes_R R[y] \cong R[x, y]$$

(exercise).

56 Fields

Recall. A field is a commutative ring K with identity $1 \neq 0$ such that all non-zero elements of K are invertible.

56.1 Examples.

1) $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{F}_p$ (p - prime number).

2) R/I where R is a commutative ring with identity $1 \neq 0$ and $I \lhd R$ is a maximal ideal.

   E.g. $R = K[x]$ where K is a field, $I = \langle p(x) \rangle$ where $p(x) \in K[x]$ is an irreducible polynomial.

3) A field of fractions: $S^{-1}R$ where R is an integral domain and $S = R - \{0\}$.

   Note. If K is a field and $R = K[x_1, \ldots, x_n]$ then

$$K(x_1, \ldots, x_n) := S^{-1}R$$

   is the field of rational functions in n variables with coefficients in K.

56.2 Proposition. *If K, L are fields and*

$$f \colon K \longrightarrow L$$

is a homomorphism then f is a monomorphism.

Proof. We have $\mathrm{Ker}(f) \lhd K$. Since K is a field we must have $\mathrm{Ker}(f) = \{0\}$ (and so f is a monomorphism), or $\mathrm{Ker}(f) = K$ (which is impossible since $f(1) = 1 \neq 0$. $\square$

Problems.

1) How do embeddings of fields $K \hookrightarrow L$ look like? (Field extensions)

2) If L is a field what are automorphisms $L \to L$? If $\varphi \colon L \to L$ is such automorphism and K is a subfield of L what is $\varphi(K)$? (Galois Theory)

57 Field extensions

57.1 Definition. If K, L are fields and $K \subseteq L$ then we say that L is a *field extension* of K.

57.2 Note. If $K \subseteq L$ then L is a vector space over K.

57.3 Notation.

- $L/K :=$ extension L of a field K.
- $[L : K] := \dim_K L$. The number $[L : K]$ is called the *degree* of the extension L/K.

57.4 Examples.

1) $[\mathbb{R} : \mathbb{Q}] = \infty$ since $|\mathbb{R}| > |\mathbb{Q}|$

   Exercise: show that the subset
   $$\{\sqrt{p} \mid p - \text{prime}\} \subseteq \mathbb{R}$$
   is linearly independent over $\mathbb{Q}$.

2) $[\mathbb{C} : \mathbb{R}] = 2$ since $\{1, i\}$ is a basis of $\mathbb{C}$ over $\mathbb{R}$.

57.5 Note. $[L : K] = 1$ iff $L = K$.

58 Prime subfield and field characteristic

58.1 Proposition. *If L is a field and $\{K_i\}_{i\in I}$ is a family of subfields of L then $\bigcap_{i\in I} K_i$ is a subfield of L.*

Proof. Exercise. $\square$

58.2 Definition. The *prime subfield* of a field L is the intersection of all subfields of L.

58.3 Definition. If L is a field then the *characteristic* of L is the smallest positive integer $\chi(L)$ such that

$$\underbrace{1_L + \cdots + 1_L}_{\chi(L)\text{ times}} = 0$$

If $\underbrace{1_L + \cdots + 1_L}_{n\text{ times}} \neq 0$ for all $n > 0$ then $\chi(L) = 0$.

58.4 Example.

$\chi(\mathbb{Q}) = \chi(\mathbb{R}) = \chi(\mathbb{C}) = 0.$

$\chi(\mathbb{F}_p) = p.$

58.5 Proposition. *If $\chi(L) \neq 0$ then it is a prime number.*

Proof. Exercise. $\square$

58.6 Theorem. *Let L be a field and let K be the prime subfield of L.*

1) If $\chi(L) = 0$ then $K \cong \mathbb{Q}$.

2) if $\chi(L) = p > 0$ then $K \cong \mathbb{F}_p$.

Proof. Exercise. □

59 Algebraic and transcendental elements

59.1 Notation. If L/K is a field extension, and S is a subset of L then $K(S)$ is the smallest subfield L such that $K \subseteq K(S)$ and $S \subseteq K(S)$.

Note.

1) If $a \in L$ then

$$K(a) = \left\{ \frac{r_0 + r_1 a + \cdots + r_n a^n}{s_0 + s_1 a + \cdots + s_m a^m} \;\middle|\; r_i, s_j \in K, \quad s_0 + \cdots + s_m a^m \neq 0 \right\}$$

2) If $a_1, \dots, a_n \in L$ then

$$K(a_1, \dots, a_n) = K(a_1, \dots, a_{n-1})(a_n)$$

59.2 Definition. Let L/K be a field extension. An element $a \in L$ is *algebraic* over K if there exists $r_0, \dots, r_n \in K$, $r_n \neq 0$ such that

$$r_0 + r_1 a + \cdots + r_n a^n = 0$$

Equivalently, $a \in L$ is algebraic over K if there exists a non-zero polynomial $p(x) = r_0 + r_1 x + \cdots + r_n x^n \in K[x]$ such that $p(a) = 0$.

An element $a \in L$ is *transcendental* over K if it is not algebraic over K.

59.3 Example.

1) $\sqrt{2} \in \mathbb{R}$ is a root of $p(x) = x^2 - 2 \in \mathbb{Q}[x]$, so $\sqrt{2}$ is algebraic over $\mathbb{Q}$.

2) e, $\pi \in \mathbb{R}$ are transcendental over $\mathbb{Q}$ (harder to show).

59.4 Proposition. *If L/K is a field extension and $a \in L$ is a transcendental element over K then*

$$K(a) \cong K(x)$$

where $K(x)$ is the field of rational functions of variable x with coefficients in K.

Proof. We have a homomorphism of rings

$$\varphi_a \colon K[x] \to L, \quad \varphi_a(f) = f(a)$$

Since a is transcendental over K this map is a monomorphism, so $\varphi_a(f)$ is an invertible element for all $f \neq 0$. Therefore, by the universal property of localizations of rings (36.6) there exists a homomorphism

$$\bar{\varphi}_a \colon K(x) \to L, \quad \bar{\varphi}_a(f/g) = f(a)/g(a)$$

We have $\mathrm{Im}(\bar{\varphi}_a) = K(a)$, so $K(a) \cong K(x)$. □

59.5 Example.

$$\mathbb{Q}(\pi) \cong \mathbb{Q}(x) \cong \mathbb{Q}(e)$$

59.6 Proposition. *Let L/K be a field extension and let $a \in L$ be an algebraic element over K.*

1) *There exists an irreducible polynomial $p(x) \in K[x]$ such that $p(a) = 0$ and if $q(a) = 0$ for some $q(x) \in K[x]$ then $p(x) \mid q(x)$.*
2) *We have an isomorphism $K(a) \cong K[x]/\langle p(x) \rangle$.*

Proof.

1) Take the homomorphism of rings

$$\varphi_a \colon K[x] \to L, \quad \varphi_a(f) = f(a)$$

Since a is algebraic over K we have $\operatorname{Ker}(\varphi_a) \neq \{0\}$. Also, since $K[x]$ is a PID thus $\operatorname{Ker}(\varphi_a) = \langle p(x) \rangle$ for some $p(x) \in K[x]$.

If $q(a) = 0$ for some $q(x) \in K[x]$ then $q(x) \in \langle p(x) \rangle$, so $p(x) \mid q(x)$.

In order to see that $p(x)$ is irreducible assume that $p(x) = g(x)h(x)$ for some $g(x), h(x) \in K[x]$. Then

$$0 = p(a) = g(a)h(a)$$

so either $g(a) = 0$ or $h(a) = 0$. We can assume that $g(a) = 0$. Then $p(x) \mid g(x)$. Since also $g(x) \mid p(x)$ we obtain that $h(x)$ must be a unit in $K[x]$.

2) We have

$$K[x]/\langle p(x) \rangle \cong \operatorname{Im}(\varphi_a)$$

It is then enough to show that $\operatorname{Im}(\varphi_a) = K(a)$.

It is clear that $\operatorname{Im}(\varphi_a) \subseteq K(a)$. On the other hand, since $p(x)$ is an irreducible element of $K[x]$ then by (32.3) $\langle p(x) \rangle$ is a maximal ideal of $K[x]$ and so $K[x]/\langle p(x) \rangle$ is a field. As a consequence $\operatorname{Im}(\varphi_a)$ is a subfield of L, and $K \cup \{a\} \subseteq \operatorname{Im}(\varphi_a)$. Since $K(a)$ is the smallest subfield containing $K \cup \{a\}$ we get that $K(a) = \operatorname{Im}(\varphi_a)$. $\square$

59.7 Notation. Let L/K be a field extension and let $a \in L$ be an algebraic element over K. By Proposition 59.6 there is a unique irreducible polynomial $p(x) \in K[x]$ of the form

$$p(x) = x^n + r_{n-1}x^{n-1} + \ldots + r_1 x + r_0$$

such that $p(a) = 0$. Denote this polynomial by $\operatorname{irr}_a^K(x)$.

Note. Polynomials of the form

$$f(x) = x^n + r_{n-1}x^{n-1} + \ldots + r_1 x + r_0$$

(with the highest degree coefficient equal to 1) are called *monic polynomials*.

59.8 Proposition. *Let L/K, M/K be extensions of a field K and let $a \in L$, $b \in M$ be algebraic elements over K. The following conditions are equivalent.*

1) $\mathrm{irr}_a^K(x) = \mathrm{irr}_b^K(x)$

2) there exists an isomorphism

$$\varphi\colon K(a) \to K(b)$$

such that $\varphi|_K = \mathrm{id}_K$ *and* $\varphi(a) = \varphi(b)$.

Proof.

(1) $\Rightarrow$ (2) If $\mathrm{irr}_a^K(x) = \mathrm{irr}_b^K(x) = p(x)$ then we have isomorphisms

$$K(a) \xleftarrow{\varphi_a} K[x]/\langle p(x)\rangle \xrightarrow{\varphi_b} K(b)$$

Take $\varphi = \varphi_b \circ \varphi_a^{-1}$.

(2) $\Rightarrow$ (1) It is enough to check that $\mathrm{irr}_a^K(b) = 0$. If $\mathrm{irr}_a^K(x) = x^n + r_{n-1}x^{n-1} + \ldots + r_1 x + r_0$ Then we have

$$\begin{aligned}
\mathrm{irr}_a^K(b) &= b^n + r_{n-1}b^{n-1} + \ldots + r_1 b + r_0 \\
&= \varphi(a)^n + r_{n-1}\varphi(a)^{n-1} + \ldots + r_1\varphi(a) + r_0
\end{aligned}$$

Since $r_i \in K$ we have $r_i = \varphi(r_i)$, so

$$\begin{aligned}
\mathrm{irr}_a^K(b) &= \varphi(a)^n + \varphi(r_{n-1})\varphi(a)^{n-1} + \ldots + \varphi(r_1)\varphi(a) + \varphi(r_0) \\
&= \varphi(a^n + r_{n-1}a^{n-1} + \ldots + r_1 a + r_0) \\
&= \varphi(0) \\
&= 0
\end{aligned}$$

$\square$

59.9 Note. If L/K is a field extension, $a, b \in L$ and $\mathrm{irr}_a^K(x) = \mathrm{irr}_b^K(x)$ then we may have $K(a) \neq K(b)$. Take e.g. $K/L = \mathbb{C}/\mathbb{Q}$, $a = \sqrt[3]{2}$, $b = \sqrt[3]{2}(\frac{1}{2} + \frac{\sqrt{3}}{2}i)$. Then

$$\mathrm{irr}_a^{\mathbb{Q}}(x) = \mathrm{irr}_b^{\mathbb{Q}}(x) = x^3 - 2$$

but $\mathbb{Q}(a) \subseteq \mathbb{R}$, $\mathbb{Q}(b) \not\subseteq \mathbb{R}$, so $\mathbb{Q}(a) \neq \mathbb{Q}(b)$.

59.10 Proposition. *Let L/K be a field extension and let $a \in L$ be an algebraic element over K, and let $p(x) = \mathrm{irr}_a^K(x)$. If $\deg p(x) = n$ then*

1) $K(a) = \{s_0 + s_1 a + \ldots + s_{n-1}a^{n-1} \mid s_i \in K\}$
2) *the set* $\{1, a, \ldots, a^{n-1}\}$ *is a basis of* $K(a)$ *over* K
3) $[K(a) : K] = n$

Proof.

1) We have the isomorphism

$$\varphi_a \colon K[x]/\langle p(x) \rangle \longrightarrow K(a), \qquad \varphi_a(f) = f(a)$$

Also, for $f(x) \in K[x]$ we have

$$f(x) = q(x)p(x) + r(x)$$

for some $q(x), r(x) \in K[x]$, $\deg r(x) < n$. This gives

$$f(a) = q(a)p(a) + r(a) = 0 + r(a) = r(a)$$

Ii follows that $K(a) = \{r(a) \mid r(x) \in K[x], \deg r(x) < n\}$.

2) By part 1) we have

$$K(a) = \mathrm{Span}_K(1, a, \ldots a^{n-1})$$

so it suffices to show that the set $\{1, a, \ldots, a^{n-1}\}$ is linearly independent over K. Assume that

$$s_0 \cdot 1 + s_1 a + \ldots + s_{n-1}a^{n-1} = 0$$

for some $s_i \in K$. Then a is a root of the polynomial $g(x) = s_0 + s_1 x + \ldots + s_{n-1}x^{n-1}$. Since $\deg g(x) < \deg p(x)$ we must have $g(x) = 0$, so $s_0 = s_1 = \ldots = s_{n-1} = 0$.

3) Obvious by part 2).

$\square$

59.11 Example.

Take $\sqrt{2} \in \mathbb{R}$. Since $\operatorname{irr}^{\mathbb{Q}}_{\sqrt{2}}(x) = x^2 - 2$, thus

$$\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} \ \mid \ a, b \in \mathbb{Q}\}$$

Notice that

$$(a + b\sqrt{2})^{-1} = \frac{1}{a^2 - 2b^2}(a - b\sqrt{2})$$

60 Algebraic extensions

60.1 Notation. If L/K is a field extension and $a \in L$ then

$$\deg_K(a) := [K(a) : K]$$

Note.

1) If a is a transcendental element over K then $K(a) \cong K(x)$, so $\deg_K(a) = \infty$.

2) If a is an algebraic element over K then $\deg_K(a) = \deg \operatorname{irr}_a^K(x) < \infty$.

Upshot. If L/K is a field extension then an element $a \in L$ is algebraic over K iff $\deg_K(a) < \infty$.

60.2 Definition. A field extension L/K is an *algebraic extension* if all elements of L are algebraic over K.

60.3 Proposition. *If $[L : K] < \infty$ then L/K is an algebraic extension.*

Note. $L = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \dots)$ is an algebraic extension of $\mathbb{Q}$ even though $[L : \mathbb{Q}] = \infty$.

60.4 Lemma. *Let $K \subseteq L \subseteq M$ be field extensions. If $\{a_i\}_{i \in I}$ is a basis of M over L and $\{b_j\}_{j \in J}$ is a basis of L over K then $\{a_i b_j\}_{(i,j) \in I \times J}$ is a basis of M over K. As a consequence*

$$[M : K] = [M : L] \cdot [L : K]$$

Proof.

Claim 1. The set $\{a_i b_j\}_{(i,j)\in I\times J}$ spans M over K.

Indeed, if $m \in M$ then

$$m = \sum_i l_i a_i$$

for some $l_i \in L$. Also, for each l_i we have

$$l_i = \sum_j k_{ij} b_j$$

for some $k_{ij} \in K$. Therefore we obtain

$$m = \sum_i \left(\sum_j k_{ij} b_j \right) a_i = \sum_{ij} k_{ij}(a_i b_j)$$

Claim 2. The set $\{a_i b_j\}_{(i,j)\in I\times J}$ is linearly independent over K.

Indeed, if $\sum_{ij} k_{ij}(a_i b_j) = 0$ for some $k_{ij} \in K$ then

$$0 = \sum_i \left(\sum_j k_{ij} a_i \right) b_j$$

Since $\sum_j k_{ij} a_i \in L$ and the set $\{b_j\}_{j\in J}$ is linearly independent over L we have $\sum_j k_{ij} a_i = 0$ for each j. Then, since the set $\{a_i\}_{i\in I}$ is linearly independent over K we have $k_{ij} = 0$ for all i, j.

$\square$

Proof of Proposition 60.3.

Let $a \in L$. We have $K \subseteq K(a) \subseteq L$, so by Lemma 60.4 we get

$$[L : K(a)] \cdot [K(a) : K] = [L : K] < \infty$$

Therefore $[K(a) : K] < \infty$, and so a is an algebraic element over K.

$\square$

60.5 Corollary. *If L/K is a field extension and $a_1, \dots, a_n \in L$ then*

$$[K(a_1, \dots, a_n) : K] \leq \deg_K(a_1) \cdot \ldots \cdot \deg_K(a_1)$$

In particular, if $a_1, \dots, a_n$ are algebraic elements over K then $[K(a_1, \dots, a_n) : K] < \infty$ and so $K(a_1, \dots, a_n)$ is an algebraic extension of K.

Proof. Exercise. □

60.6 Corollary. *If M/L, L/K are algebraic extensions then M/K is also an algebraic extension.*

Proof. Let $a \in M$. We need to show that a is algebraic over K.

Since a is algebraic over L there is a polynomial $f(x) = l_0 + l_1x + \ldots + l_nx^n$ in $L[x]$ such that $f(a) = 0$.

Let $N = K(l_0, \dots, l_n)$. Notice that $f(x) \in N[x]$, so a is an algebraic element over N. In particular we have $[N(a) : N] < \infty$. Moreover, since $l_0, \dots, l_n$ are algebraic elements over K by Corollary 60.5 we have $[N : K] < \infty$. This gives

$$[N(a) : K] = [N(a) : N] \cdot [N : K] < \infty$$

Finally, since $K(a) \subseteq N(a)$ we have

$$[K(a) : K] \leq [N(a) : K] < \infty$$

so a is an algebraic element over K. □

60.7 Proposition. *Let L/K be a field extension and let*

$$K_{\mathit{alg}}(L) = \{a \in L \mid a \textit{ is an algebraic element over } K\}$$

Then $K_{\mathit{alg}}(L)$ is a subfield of L.

Proof. We need to show that if $a, b \in L$ are algebraic elements over K, then $a \pm b$, ab, a/b are also algebraic over K. This holds since by (60.5) $K(a, b)/K$ is an algebraic extension and $a \pm b, ab, a/b \in K(a, b)$. □

61 Separable elements

61.1 Example.

Note: since $\sqrt{2}, \sqrt{3} \in \mathbb{R}$ are algebraic elements over $\mathbb{Q}$, thus $Q(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ is an algebraic extension. We have:

$$\mathbb{Q} \subseteq \mathbb{Q}(\sqrt{2} + \sqrt{2}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3})$$

Claim: $\mathbb{Q}(\sqrt{2} + \sqrt{2}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$

Indeed, it is enough to show that $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2} + \sqrt{3})] = 1$

We have $\mathrm{irr}^{\mathbb{Q}}_{\sqrt{2}}(x) = x^2 - 2$ and $\mathrm{irr}^{\mathbb{Q}}_{\sqrt{2}}(x) = x^2 - 3$, so by (60.5)

$$[Q(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] \leq \deg^{\mathbb{Q}}_{\sqrt{2}} \cdot \deg^{\mathbb{Q}}_{\sqrt{3}} = 4$$

On the other hand $\sqrt{2} + \sqrt{3}$ is a root of $p(x) = x^4 - 10x^2 + 1$, and since $p(x)$ is irreducible in $\mathbb{Q}[x]$ (check!) we have

$$[Q(\sqrt{2} + \sqrt{3}) : \mathbb{Q}] = \deg p(x) = 4$$

This gives

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2} + \sqrt{3})] \cdot \underbrace{[Q(\sqrt{2} + \sqrt{3}) : \mathbb{Q}]}_{=\,4} = \underbrace{[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}]}_{\leq\,4}$$

so $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2} + \sqrt{3})] = 1$.

Goal. If L/K is a field extension and $a_1, \ldots, a_n \in L$ are algebraic elements over K then there exists an element $c \in L$ such that

$$K(a_1, \ldots, a_n) = K(c)$$

provided that $a_1, \ldots, a_n$ are *separable* over K.

Recall. Let L/K be a field extension. An element $a \in L$ is a root of $f(x) \in K[x]$ iff

$$f(x) = (x - a)g(x)$$

for some $g(x) \in L[x]$.

61.2 Definition. If L/K is a field extension and $a \in L$ then a is a *multiple root* of $f(x) \in K[x]$ if

$$f(x) = (x - a)^k g(x)$$

for some $k > 1$ and $g(x) \in L[x]$.

61.3 Definition. A polynomial $f(x) \in K[x]$ is *separable* if it has no multiple roots in any extension of K.

61.4 Definition. Let L/K be a field extension and let $a \in L$ be an algebraic element over K. The element a is *separable* over K if $\mathrm{irr}_a^K(x) \in K[x]$ is a separable polynomial.

62 Derivatives and separable elements

62.1 Definition. If $f(x) \in K[x]$, $f(x) = a_0 + a_1x + \ldots + a_nx^n$ then the *derivative* of $f(x)$ is the polynomial

$$f'(x) = a_1 + 2a_2x + \ldots + na_nx^{n-1}$$

62.2. Properties of derivatives.

1) $(f+g)' = f' + g'$
2) $(cf)' = c \cdot f'$ for $c \in K$
3) $(fg)' = f'g + fg'$
4) $(f^n)' = nf^{n-1} \cdot f'$
5) $(f \circ g)' = (f' \circ g) \cdot g'$

62.3 Proposition. *A polynomial $f(x) \in K[x]$ is separable iff f and f' have no common roots in any extension of K.*

Proof.

$(\Rightarrow)$ Assume that there exists an extension L/K such that $f(a) = f'(a) = 0$ for some $a \in L$. We have

$$f(x) = (x-a)g(x)$$

for some $g(x) \in L[x]$. It follows that

$$f'(x) = 1 \cdot g(x) + (x-a)g'(x)$$

and so

$$g(x) = f'(x) - (x-a)g'(x)$$

This gives

$$g(a) = f'(a) - (a-a)g'(a) = 0$$

Therefore $g(x) = (x-a)h(x)$ for some $h(x) \in L[x]$, and

$$f(x) = (x-a)^2 h(x)$$

Thus a is a multiple root of $f(x)$.

($\Leftarrow$) Assume that $f(x)$ is not a separable polynomial. Then there exists an extension L/K and $a \in L$ such that

$$f(x) = (x-a)^k g(x)$$

for some $k > 1$ and $g(x) \in L[x]$. This gives

$$f'(x) = k(x-a)^{k-1}g(x) + (x-a)^k g'(x)$$

so $f'(a) = f(a) = 0$.

$\square$

62.4 Proposition. *Let L/K be a field extension, let $a \in L$ be an algebraic element over K and let $p(x) = \mathrm{irr}_a^K(x)$. The element a is separable over K iff $p'(x) \neq 0$.*

Proof. ($\Leftarrow$) Assume that $p'(x) = 0$. Then

$$p(a) = p'(a) = 0$$

and by Proposition 62.3 $p(x)$ is not a separable polynomial.

($\Rightarrow$) Assume that a is not separable over K. Then $p(x)$ is not a separable polynomial, so by Proposition 62.3 there exists an extension M/K and $b \in M$ such that

$$p(b) = p'(b) = 0$$

Since $p(x)$ is an irreducible polynomial in $K[x]$ we have $p(x) = \mathrm{irr}_b^K(x)$. It follows that $p(x) \mid p'(x)$. However, $\deg p(x) > \deg p'(x)$, so we must have $p'(x) = 0$.

□

62.5 Corollary. *Let K be a field such that $\chi(K) = 0$. If L/K is a field extension then every algebraic element $a \in L$ is separable over K.*

Proof. If $p(x) = \mathrm{irr}_a^K(x)$ then $\deg p(x) \geq 1$, so $p'(x) \neq 0$. □

62.6 Example.

Let p be a prime number, and let

$$K = \mathbb{F}_p(t)$$

be the field of rational functions of variable t with coefficients in $\mathbb{F}_p$. Check:

$$p(x) = x^p - t$$

is an irreducible polynomial in $K[x]$. Let L/K be an extension such that $p(a) = 0$ for some $a \in L$. Then $p(x) = \mathrm{irr}_a^K(x)$, and since $p'(x) = 0$ then element a is not separable over K.

62.7 Proposition. *If K is a field, $\chi(K) = p \neq 0$ and $f(x) \in K[x]$ then $f'(x) = 0$ iff $f(x) \in K[x^p]$.*

Proof.

$(\Rightarrow)$ If $f(x) \in K[x^p]$ then

$$f(x) = a_0 + a_p x^p + a_{2p} x^{2p} + \ldots + a_{np} x^{np}$$

for some $a_{kp} \in K$. Then

$$f'(x) = pa_p x^{p-1} + 2pa_{2p}x^{2p-1} + \ldots + npa_{np}x^{np-1} = 0$$

($\Leftarrow$) If $f(x) = a_0 + a_1 x + \ldots + a_n x^n$ and $f'(x) = 0$ then $ka_k = 0$ for $k = 0, \ldots, n$. Therefore either $a_k = 0$ or $p \mid k$. It follows that $f(x) \in K[x^p]$.

$\square$

62.8 Corollary. *Let L/K be a field extension, let $\chi(K) = p \neq 0$ and let $a \in L$ be an algebraic element over K. Then a is separable over K iff $\mathrm{irr}_a^K(x) \notin K[x^p]$.*

Proof. Follows from (62.4) and (62.7). $\square$

62.9 Proposition. *Let L/K be a field extension such that $\chi(K) = p \neq 0$, and let $a \in L$ be an algebraic element over K. The element a is separable over K iff $K(a) = K(a^p)$.*

Proof. ($\Leftarrow$) If a is separable over K then it is separable over $K(a^p)$ (check!). Also, a is a root of $g(x) = x^p - a^p \in K(a^p)[x]$. Let $f(x) = \mathrm{irr}_a^{K(a^p)}(x)$. Then $f(x) \mid g(x)$.

On the other hand we have $g(x) = (x - a)^p$, so $f(x) = (x - a)^k$ for some $1 \leq k \leq p$. Since $f(x)$ is a separable polynomial we must have $f(x) = x - a$, and since $f(x) \in K(a^p)[x]$ this gives that $a \in K(a^p)$. It follows that $K(a) = K(a^p)$.

($\Rightarrow$) Assume that a is not separable over K, and let $f(x) = \mathrm{irr}_a^K(x)$. By Corollary 62.8 we have $f(x) \in K[x^p]$, so

$$f(x) = r_0 + r_p x^p + \ldots + r_{np}x^{np}$$

for some $r_0, \ldots, r_{np} \in K$. Let $g(x) = r_0 + r_p x + \ldots + r_{np}x^n$. We have $g(a^p) = f(a) = 0$, so

$$\deg_K(a^p) \leq \deg g(x) < \deg f(x) = \deg_K(a)$$

This gives

$$[K(a^p) : K] = \deg_K(a^p) < \deg_K(a) = [K(a) : K]$$

so $K(a^p) \neq K(a)$. □

62.10 Corollary. *Let K be a field such that $\chi(K) = p \neq 0$. If L/K is a field extension and $a_1, \ldots, a_n \in L$ are elements separable over K then*

$$K(a_1, \ldots, a_n) = K(a_1^p, \ldots, a_n^p)$$

62.11 Notation. If $K, L \subseteq M$ are field extensions then

$$KL := \begin{pmatrix} \text{the smallest subfield of L} \\ \text{containing } K \cup L \end{pmatrix}$$

Note: if L/K is a field extension and $a_1, \ldots, a_n \in L$ then

$$K(a_1, \ldots, a_n) = K(a_1)K(a_2) \cdot \ldots \cdot K(a_n)$$

Proof of Corollary 62.10.

Using Proposition 62.9 we obtain:

$$K(a_1, \ldots, a_n) = K(a_1) \cdot \ldots \cdot K(a_n) = K(a_1^p) \cdot \ldots \cdot K(a_n^p) = K(a_1^p, \ldots, a_n^p)$$

□

63 Separable extensions

63.1 Definition. A field extension L/K is *separable* if every element $a \in L$ is separable over K.

63.2 Note. By (62.5) if $\chi(K) = 0$ then every algebraic extension of K is separable.

63.3 Notation. If L is a field and $\chi(L) = p \neq 0$ then

$$L^p := \{a^p \mid a \in L\}$$

Note.

1) L^p is a subfield of L since if $a^p, b^p \in L^p$ then $a^p b^p = (ab)^p \in L^p$ and $a^p + b^p = (a+b)^p \in L^p$.

2) The map
$$\varphi\colon L \to L^p, \quad \varphi(a) = a^p$$
is a field isomorphism.

63.4 Proposition. *Let K be a field such that $\chi(K) = p \neq 0$. If L/K is a field extension and $[L : K] < \infty$ then L/K is separable iff $KL^p = L$.*

63.5 Lemma. *If $K \subseteq M \subseteq L$ and $N \subseteq L$ are field extensions then*

$$[NM : NK] \leq [M : K]$$

Proof. Exercise. □

Proof of Proposition 63.4.

($\Leftarrow$) If L/K is separable and $a \in L$ then by Proposition 62.9 we have

$$a \in K(a) = K(a^p) \subseteq KL^p$$

This gives $L \subseteq KL^p$, and since also $KL^p \subseteq L$ we obtain $KL^p = L$.

($\Rightarrow$) Assume that $L = KL^p$, and let $b \in L$. We need to show that b is a separable element over K. By Proposition 62.9 it suffices to show that $K(b) = K(b^p)$.

Take the isomorphism

$$\varphi\colon L \to L, \qquad \varphi(a) = a^p$$

We have

$$[\varphi(L) : \varphi(K(b))] = [L : K(b)] \leq [L : K] < \infty$$

Consider the extensions $\varphi(K(b)) \subseteq \varphi(L) \subseteq L$ and $K \subseteq L$. By Lemma 63.5 we have

$$[K\varphi(L) : K\varphi(K(b))] \leq [\varphi(L) : \varphi(K(b))]$$

Notice that $K\varphi(L) = KL^p = L$ and $K\varphi(K(b)) = KK^p(b^p) = K(b^p)$, so this gives

$$[L : K(b^p)] \leq [\varphi(L) : \varphi(K(b))] = [L : K(b)]$$

One the other hand we have

$$[L : K(b^p)] = [L : K(b)] \cdot [K(b) : K(b^p)]$$

Therefore we must have $[L : K(b^p)] = [L : K(b)]$ and $[K(b) : K(b^p)] = 1$. This gives

$$K(b^p) = K(b)$$

$\square$

63.6 Theorem. *If L/K is a field extension, $\chi(K) = p \neq 0$ and $a_1, \dots, a_n \in L$ are elements separable over K then $K(a_1, \dots, a_n)$ is a separable extension of K.*

Proof. Denote $M := K(a_1, \ldots, a_n)$. Since $[M : K] < \infty$ by Proposition 63.4 it suffices to show that $KM^p = M$. We have

$$KM^p = K(K(a_1, \ldots, a_n))^p = K(K^p(a_1^p, \ldots, a_n^p)) = K(a_1^p, \ldots, a_n^p)$$

By (62.10) we also have

$$K(a_1^p, \ldots, a_n^p) = K(a_1, \ldots, a_n) = M$$

Therefore $KM^p = M$. □

63.7 Theorem. *If M/L and L/K are separable extensions then M/K is also a separable extension.*

Proof. Let $a \in M$ we need to show that a is separable over K.

Let $f(x) = \mathrm{irr}_a^L(x)$, $f(x) = x^n + r_{n-1}x^{n-1} + \ldots + r_1 x + r_0$. Define

$$N := K(r_0, \ldots, r_{n-1}) \subseteq L$$

Notice that $f(x) = \mathrm{irr}_a^N(x)$, and since $f(x)$ is a separable polynomial the element a is separable over K. Moreover, we have

1) N/K is a separable extension (since L/K is separable and $N \subseteq L$)

2) $[N : K] < \infty$

Therefore by Proposition 63.4 we have $KN^p = N$.

Similarly, since

1) $N(a)/N$ is a separable extension (by (63.6), since a is separable over N)

2) $[N(a) : N] < \infty$

thus $N(N(a))^p = N(a)$.

Finally we have:

1) $[N(a) : K] = [N(a) : N] \cdot [N : K] < \infty$

2) $K(N(a))^p = KN^p(N(a))^p = N(N(a))^p = N$

so by Proposition 63.4 $N(a)/K$ is a separable extension. Since $a \in N(a)$ it is a separable element over K. $\square$

63.8 Proposition. *Let L/K be a field extension and let*

$$K_{sep}(L) = \{a \in L \mid a \text{ is separable over } K\}$$

Then $K_{sep}(L)$ is a subfield of L.

Proof. We need to show that if $a, b \in L$ are separable elements over K, then $a \pm b$, ab, a/b are also separable over K. This holds since by (63.6) $K(a,b)/K$ is an separable extension and $a \pm b, ab, a/b \in K(a,b)$. $\square$

63.9 Definition. The *separability degree* of an extension L/K is the number

$$[L : K]_{\mathrm{sep}} := [K_{\mathrm{sep}}(L) : K]$$

64 Simple extensions

64.1 Definition. A field extension L/K is *simple* if $L = K(a)$ for some $a \in L$.

64.2 Theorem. *If K is an infinite field and L/K is a separable extension such that $[L : K] < \infty$ then L/K is a simple extension.*

64.3 Note. Theorem 64.2 is true also if K is a finite field (later, see (69.5)).

Recall. If K is a field then $K[x]$ is a Euclidean domain, so

1) for every $f(x), g(x) \in K[x]$ there exists $\gcd(f(x), g(x)) \in K[x]$

2) there exist $\lambda(x), \mu(x) \in K[x]$ such that

$$\gcd(f(x), g(x)) = \lambda(x)f(x) + \mu(x)g(x)$$

64.4 Lemma. *Let L be a field, let $f(x), g(x), h(x) \in L[x]$ be polynomials such that*

$$h(x) \sim \gcd(f(x), g(x))$$

If $K \subseteq L$ is a subfield such that $f(x), g(x) \in K[x]$ then

$$h(x) = ah_1(x)$$

for some $0 \neq a \in L$ and some $h_1(x)$. In particular is $h(x)$ is a monic polynomial then $h(x) \in K[x]$.

Proof. Exercise. □

64.5 Lemma. *Let K be an infinite field, let L/K be a field extension and let $f(x), g(x) \in L[x]$. If $f(0) \neq 0$ and $g(x) \neq 0$ then there exists $0 \neq d \in K$ such that*

$$\gcd(f(x), g(dx)) \sim 1$$

Proof. If $\deg f(x) = 0$ or $\deg g(x) = 0$ the statement is obvious. Otherwise we have

$$f(x) = a \cdot f_1(x) \cdot \ldots \cdot f_n(x)$$
$$g(x) = b \cdot g_1(x) \cdot \ldots \cdot g_n(x)$$

where $0 \neq a, b \in L$ and $f_i(x), g_j(x) \in L[x]$ are irreducible monic polynomials:

$$f_i(x) = a_{i,0} + a_{i,1}x + \ldots + x^{r_i}$$
$$g_j(x) = b_{j,0} + b_{j,1}x + \ldots + x^{s_j}$$

Note:

1) Since $f(0) \neq 0$ thus $a_{i,0} \neq 0$ for all i.

2) For any $0 \neq d \in K$ we have

$$g(dx) = b \cdot g_1(dx) \cdot \ldots \cdot g_n(dx)$$

Also, $g_1(dx), \ldots, g_n(dx)$ are irreducible polynomials in $L[x]$ since $g_j(dx) = \varphi(g_j(x))$ where φ is the ring isomorphism

$$\varphi\colon L[x] \to L[x], \quad \varphi(q(x)) := q(dx)$$

Let $d \in K$ be an element such that $\gcd(f(x), g(dx)) \not\sim 1$. Then for some i, j we have $f_i(x) \sim g_j(dx)$, i.e.

$$f_i(x) = cg_j(dx)$$

foe some $0 \neq c \in L$. This gives:

$$a_{i,0} + a_{i,1}x + \cdots + x^{r_i} = cb_{j,0} + cb_{j,1}dx + \cdots + cd^{s_j}x^{s_j}$$

As a consequence we obtain:

1) $r_i = s_j$

2) $a_{i,0} = cb_{j,0}$

3) $1 = cd^{s^j}$

Therefore
$$d^{s_j} = c^{-1} = b_{j,0}a_{i,0}^{-1}$$

Since K is an infinite field we can find $d \in K$ such that $d^{s_j} \neq b_{j,0}a_{i,0}^{-1}$ for all i, j. Then $f_i(x) \not\sim g_j(dx)$ for all i, j, and so $\gcd(f(x), g(dx)) \sim 1$.

□

64.6 Lemma. *Let K be an infinite field and let L/K be a separable extension such that $L = K(a, b)$ let $a, b \in L$. There exists $c \in L$ such that*

$$K(a, b) = K(c)$$

Proof. Let $f(x) = \mathrm{irr}_a^K(x)$, $g(x) = \mathrm{irr}_b^K(x)$. Since $f(a) = 0$ we have

$$f(x) = (x - a)f_1(x)$$

for some $f_1(x) \in L[x]$. Moreover, since a is separable over K we have $f_1(a) \neq 0$.

Denote:
$$F(x) := f_1(x + a), \qquad G(x) := g(x + b)$$

We have $F(0) = f_1(a) \neq 0$ and $G(x) \neq 0$, so by Lemma 64.5 there exists $0 \neq d \in K$ such that
$$\gcd(F(x), G(dx)) \sim 1$$

Take $c := b - da$. We will show that $K(a, b) = K(c)$.

It is enough to show that $a \in K(c)$. Indeed, we have $K(c) \subseteq K(a, b)$, and if $a \in K(c)$ then also $b = c + da \in K(c)$, so in such case $K(a, b) \subseteq K(c)$.

To see that $a \in K(x)$ consider the ring isomorphism

$$\varphi\colon L[x] \to L[x], \qquad \varphi(q(x)) = q(x - a)$$

Since $\gcd(F(x), G(dx)) \sim 1$, thus also $\gcd(\varphi(F(x)), \varphi(G(dx))) \sim 1$. Moreover we have

$$\begin{aligned}
\varphi(F(x)) &= \varphi(f_1(x+a)) = f_1((x-a)+a) = f_1(x) \\
\varphi(G(dx)) &= \varphi(g(dx+b)) = g(d(x-a)+b) = g(dx-da+b) = g(dx+c)
\end{aligned}$$

Denote $h(x) := g(dx+c) \in K(c)[x]$. We have:

$$h(a) = g(da - da + b) = 0$$

so

$$h(x) = (x-a)h_1(x)$$

for some $h_1(x) \in L[x]$. This gives:

$$\begin{aligned}
\gcd(f(x), h(x)) &= \gcd((x-a)f_1(x), (x-a)h_1(x)) \\
&= (x-a) \cdot \gcd(f_1(x), h_1(x))
\end{aligned}$$

Since $\gcd(f_1(x), h(x)) \sim 1$ and $h_1(x) \mid h(x)$ we have $\gcd(f_1(x), h_1(x)) \sim 1$. Therefore we get

$$\gcd(f(x), h(x)) \sim (x-a)$$

Finally, notice that $f(x), h(x) \in K(c)[x]$. Thus, by Lemma 64.4, $(x-a) \in K(c)[x]$, and so $a \in K(c)$.

$\square$

Proof of Theorem 64.2.

Since $[L : K] < \infty$ we have

$$L = K(a_1, \ldots, a_n)$$

for some $a_1, \ldots, a_n \in L$. We will show that L/K is a simple extension by induction with respect to n.

If $n = 1$ then $L = K(a_1)$, so L/K is simple.

Next, assume that for some n any extension $K(a_1, \ldots, a_n)/K$ is simple, and let $L = K(a_1, \ldots, a_{n+1})$. Then we have

$$L = K(a_1, \ldots, a_n)(a_{n+1})$$

By the inductive assumption $K(a_1, \ldots, a_n) = K(b)$ for some $b \in L$, so we obtain

$$L = K(b)(a) = K(b, a)$$

Finally, by Lemma 64.6 we have $K(a, b) = K(c)$ for some $c \in L$, and so L/K is simple. $\square$

65 Simple extensions and intermediate fields

65.1 Definition. If L/K is a field extension then an *intermediate field* of L/K is a field M such that

$$K \subseteq M \subseteq L$$

65.2 Theorem. *If L/K is a field extension, K is an infinite field and $[L : K] < \infty$ then L/K is a simple extension iff L/K has finitely many intermediate fields.*

65.3 Note. Theorem 65.2 is true also when K is a finite field. If K is finite and $[L : K] < \infty$ then L is also a finite field, and so it has L finitely many subfields. Also, every finite extension of a finite field is simple (see Theorem 69.5).

65.4 Corollary. *If L/K is a separable extension, K is an infinite field and $[L : K] < \infty$ then L/K has finitely many intermediate fields.*

Proof. By (64.2) L/K is a simple extension, so this follows from Theorem 65.2.

□

Proof of Theorem 65.2.

($\Rightarrow$) Let L/K be a simple extension:

$$L = K(a)$$

for some $a \in L$, and let $f(x) = \mathrm{irr}^K_a(x)$. Let M be an intermediate field of L/K and let $g(x) = \mathrm{irr}^M_a(x)$. Since $K \subseteq M$ we have $g(x) \mid f(x)$ in $L[x]$.

Claim. If $g(x) = r_0 + r_1 x + \cdots + r_{n-1}x^{n-1} + x^n$ then

$$M = K(r_0, \ldots, r_{n-1})$$

Indeed, denote $N := K(r_0, \ldots, r_{n-1})$. Notice that $g(x) = \operatorname{irr}_a^N(x)$. Since $M(a) = L = N(a)$ this gives

$$[L : M] = \deg_M(a) = \deg g(x) = \deg_N(a) = [L : N]$$

On the other hand we have

$$[L : N] = [L : M] \cdot [M : N]$$

Therefore $[M : N] = 1$, and so $M = N$.

We obtained that if $K \subseteq M \subseteq L$ then $M = K(r_0, \ldots, r_{n-1})$ where

$$g(x) = r_0 + r_1 x + \cdots + r_{n-1}x^{n-1} + x^n$$

is some monic polynomial in $L[x]$ such that $g(x) \mid f(x)$. Since there are only finitely many such polynomials there are only finitely many such fields M.

($\Leftarrow$) Assume that L/K has finitely many intermediate fields. We want to show:

$$L = K(c)$$

for some $c \in L$. Since $[L : K] < \infty$ we have

$$L = K(a_1, \ldots, a_n)$$

for some $a_1, \ldots, a_n \in L$. We can assume that $n = 2$, since then we can argue by induction as in the proof of (64.2).

Thus, assume that we have $L = K(a_1, a_2)$. For $k \in K$ define

$$c_k := a_1 + ka_2 \in L$$

We have

$$K \subseteq K(c_k) \subseteq L$$

Notice that since K is an infinite field there are infinitely many elements of the form c_k. On the other hand, by assumption L/K has finitely many intermediate fields, so there exist $k, k' \in K$, $k \neq k'$ such that

$$K(c_k) = K(c_{k'})$$

We will show $K(c_k) = K(a_1, a_2) = L$.

We have $K(c_k) \subseteq K(a_1, a_2)$. Also, since $c_k, c_{k'} \in K(c_k)$ we have

$$K(c_k) \ni c_k - c_{k'} = (a_1 + ka_2) - (a_1 + k'a_2) = (k - k')a_2$$

Since $0 \neq k - k' \in K$ this gives $a_2 \in K(c_k)$, and then also

$$a_1 \in c_k - ka_2 \in K(c_k)$$

Therefore $K(a_1, a_2) \subseteq K(c_k)$.

$\square$

66 Construction of extensions

66.1 Proposition. *Let K be a field and let $p(x) \in K[x]$ be an irreducible polynomial. There exists a filed extension L/K such that $p(x)$ has a root in L.*

Proof. Let $K[t]$ be the ring of polynomials of variable t. Since $p(t) \in K[t]$ is an irreducible polynomial the ideal $I = \langle p(x) \rangle$ is a maximal ideal of $K[t]$, so $K[t]/I$ is a field. Denote $L := K[t]/I$.

We have an embedding

$$i \colon K \to L, \qquad i(a) = a + I$$

so we can identify K with a subfield of L. Assume that

$$p(x) = a_0 + a_1 x + \ldots + a_n x^n$$

Under the above identification we have

$$p(x) = (a_0 + I) + (a_1 + I)x + \ldots + (a_n + I)x^n$$

Let $\beta := t + I \in L$. We have

$$\begin{aligned} p(\beta) &= (a_0 + I) + (a_1 + I)(t + I) + \ldots + (a_n + I)(t + I)^n \\ &= (a_0 + a_1 t + \ldots + a_n T^n) + I \\ &= p(t) + I \\ &= 0 + I \end{aligned}$$

Therefore $p(\beta) = 0$ in L.

$\square$

66.2 Definition. Let L/K be a field extension and let $p(x) \in K[x]$. We say that $p(x)$ *splits over L* if $p(x)$ decomposes into a product of polynomials of degree 1 in $L[x]$

66.3 Proposition. *Let K be a field and let $p(x) \in K[x]$ be a polynomial such that $\deg p(x) > 0$. There exists a field extension M/K such that $p(x)$ splits over M.*

Proof. We argue by induction with respect to $n = \deg p(x)$.

If $n = 1$ then $p(x) = ax + b$, and we can take $M = K$.

Next, assume that the statement holds for some n and let $\deg p(x) = n+1$. We have

$$p(x) = p_1(x) \cdot \ldots \cdot p_k(x)$$

where $p_1(x), \ldots, p_k(x) \in K[x]$ are irreducible polynomials. By Proposition 66.1 there exists an extension L/K such that $p_1(x)$ has a root $a \in L$. Then we have

$$p_1(x) = (x - a)q_1(x)$$

for some $q_1(x) \in L[x]$. Take

$$\bar{p}(x) := q_1(x)p_2(x) \cdot \ldots \cdot p_k(x) \in L[x]$$

We have $p(x) = (x - a)\bar{p}(x)$. Moreover, since $\deg \bar{p}(x) = n$ by inductive assumption there exists a field extension M/L such that $\bar{p}(x)$ splits over M. If follows that $p(x)$ also splits over M. ☐

66.4 Definition. If K is a field and $p(x) \in K[x]$ then we say that a field L is a *splitting field* of $p(x)$ if

1) $K \subseteq L$
2) $p(x)$ splits over L
3) $p(x)$ does not split over any proper subfield of L.

66.5 Proposition. *Let K be a field. For any polynomial $p(x) \in K[x]$, such that $\deg p(x) > 0$ there exists a splitting field of $p(x)$.*

Proof. By Proposition 66.3 there exists a field extension L/K such that $p(x)$ splits over L:

$$p(x) = b(x - a_1)(x - a_2) \cdot \ldots \cdot (x - a_n)$$

for some $b \in K$, $a_1, \ldots, a_n \in L$. Then $K(a_1, \ldots, a_n)$ is a splitting field of $p(x)$. $\square$

66.6 Example.

Take $p(x) = x^4 - 10x^2 + 1 \in \mathbb{Q}[x]$. The polynomial $p(x)$ splits over $\mathbb{R}[x]$:

$$p(x) = (x-(\sqrt{2}+\sqrt{3}))\cdot(x-(-\sqrt{2}-\sqrt{3}))\cdot(x-(-\sqrt{2}+\sqrt{3}))\cdot(x-(\sqrt{2}-\sqrt{3}))$$

Therefore $L = \mathbb{Q}(\pm(\sqrt{2}+\sqrt{3}),\ \pm(\sqrt{2}-\sqrt{3}))$ is a splitting field of $p(x)$.

Note: $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.

66.7 Proposition. *Let K be a field, let $p(x) \in K[x]$, $\deg p(x) > 0$ and let L, L' be splitting fields of $p(x)$. There exists an isomorphism*

$$\varphi\colon L \to L'$$

such $\varphi|_K = \mathrm{id}_K$.

Proof. Exercise. $\square$

67 Algebraically closed fields

67.1 Definition. A field is *algebraically closed* if the only irreducible polynomials in $K[x]$ are polynomials of degree 1.

67.2 Proposition. *Let K be a field. The following conditions are equivalent.*

(i) K *is algebraically closed.*

(ii) If $p(x) \in K[x]$ *and* $\deg p(x) > 0$ *then* $p(x)$ *splits over* K*.*

(iii) If $p(x) \in K[x]$ *and* $\deg p(x) > 0$ *then* $p(x)$ *has a root in* K*.*

(iv) K *is the only algebraic extension of* K*.*

Proof. Exercise. □

67.3 Proposition. *Every algebraically closed field is infinite.*

Proof. Let K be a finite field, $K = \{a_1, \dots, a_n\}$. Take the polynomial $p(x) \in K[x]$ given by

$$p(x) = (x - a_1)(x - a_2) \cdot \ldots \cdot (x - a_n) + 1$$

We have $p(a_i) = 1$ for all $a_i \in L$, so $p(x)$ has no roots in K. Therefore K is not algebraically closed. □

67.4 Definition. Let L/K be a field extension. The field L is an *algebraic closure* of K if L is an algebraically closed field and L/K is an algebraic extension.

67.5 Proposition. *Let L/K be an algebraic field extension. The following conditions are equivalent*

(i) L is an algebraic closure of K.

(ii) If $p(x) \in K[x]$ is a polynomial such that $\deg p(x) > 0$ then $p(x)$ splits over L.

Proof.

(i) $\Rightarrow$ (ii) If L is an algebraic closure of K, then L is an algebraically closed field and so by Proposition 67.2 every polynomial $p(x) \in L[x]$ such that $\deg p(x) > 0$ splits over L. Since $K[x] \subseteq L[x]$ the statement of (ii) follows.

(iI) $\Rightarrow$ (i) Since L/K is an algebraic extension we only need to show that L is an algebraically closed field. By Proposition 67.2 it will suffice to show that for any algebraic extension M/L we have $M = L$.

Let then M/L be an algebraic extension. In such case the extension M/K is also algebraic so if $a \in M$ then a is a root of some polynomial $p(x) \in K[x]$. Since by assumption $p(x)$ splits over L we get that $a \in L$. Therefore $M = L$.

□

67.6 Theorem. *For any field K there exists an extension $\bar{K}/K$ such that $\bar{K}$ is an algebraic closure of K.*

67.7 Lemma. *Let M/K be an algebraic extension.*

1) If K is an infinite field then $|K| = |M|$.

2) If K is a finite field then $|M| \leq \aleph_0$.

Proof. Exercise. □

Proof of Theorem 67.6. Let U be a set such that $K \subseteq U$ and $|U| > |K|$. If K is a finite field assume also that U is an uncountable set.

Let S be the set of all fields L such that $L \subseteq U$ and L is an algebraic extension of K. Define a partial ordering on S where $L \leq L'$ if L' is a field extension of L. Check: assumptions of Zorn's Lemma 29.10 are satisfied in S, and so S contains a maximal element $\bar{K}$.

We will show that $\bar{K}$ is an algebraic closure of K. Since the extension $\bar{K}/K$ is algebraic we only need to show that $\bar{K}$ is an algebraically closed field.

Assume, by contradiction, that $\bar{K}$ is not algebraically closed. By Proposition 67.2 there is then an algebraic extension $M/\bar{K}$ where $M \neq \bar{K}$. Then M/K is also an algebraic extension and so by Lemma 67.7 we have $|M| < |U|$. As a consequence there exists a monomorphism of sets

$$\varphi\colon M \to U$$

such that $\varphi|_{\bar{K}} = \mathrm{id}_{\bar{K}}$. We can introduce a field structure on the set $\varphi(M)$ such that φ becomes a field isomorphism. Then $\varphi(M) \in S$ and $\bar{K} < \varphi(M)$ which is impossible by maximality of $\bar{K}$.

$\square$

67.8 Proposition. *Let $\bar{K}/K$ be an algebraic closure and let L/K be any algebraic extension of K. There exists an embedding*

$$\varphi\colon L \to \bar{K}$$

such that $\varphi_K = \mathrm{id}_K$.

67.9 Lemma. *Let L/M be a field extension, let $a \in L$ be an algebraic element over M, and let*

$$\psi\colon M \to \bar{K}$$

be a field homomorphism such that $\bar{K}$ is algebraically closed. Then there exists a homomorphism $\bar{\psi}\colon M(a) \to \bar{K}$ such that $\bar{\psi}|_M = \psi$.

Proof. Exercise (compare with (59.8)). □

Proof of Proposition 67.8. Let S be a set of all pairs (M, ψ) such that

(i) M is a field such that $K \subseteq M \subseteq L$

(ii) $\psi\colon M \to \bar{K}$ is a homomorphism such that $\psi|_K = \mathrm{id}_K$

Define partial ordering on S as follows:

$$(M, \psi) \leq (M', \psi') \quad \text{if} \quad M \subseteq M' \ \text{ and } \ \psi'|_M = \psi$$

Check: assumptions of Zorn's Lemma 29.10 are satisfied in S, and so S contains a maximal element (M_0, ψ_0).

Assume that $M_0 \neq L$ and let $a \in L/M_0$. By Lemma 67.9 there exists a homomorphism

$$\bar{\psi}_0\colon M_0(a) \to \bar{K}$$

such that $\bar{\psi}_0|_{M_0} = \psi_0$. It follows that $(M_0, \psi_0) < (M_0(a), \bar{\psi}_0)$ which is impossible since (M_0, ψ_0) is a maximal element. Therefore $M_0 = L$ and we can take $\varphi = \psi_0$.

□

67.10 Note. Proposition 67.8 can be generalized as follows. Let $K \subseteq L \subseteq M$ be algebraic extensions, let $\bar{K}$ be an algebraic closure of K, and let

$$\tilde{\varphi}\colon L \to \bar{K}$$

be an embedding such that $\varphi|_K = \mathrm{id}_K$. Then there exists an embedding

$$\varphi\colon M \to \bar{K}$$

such that $\varphi_L = \tilde{\varphi}$ (exercise).

67.11 Corollary. *If L/K, L'/K are algebraic closures of K then there exists an isomorphism*

$$\varphi\colon L \to L'$$

such that $\varphi|_K = \mathrm{id}_K$.

Proof. By Proposition 67.8 there exists a homomorphism

$$\varphi\colon L \to L'$$

such that $\varphi|_K = \mathrm{id}_K$. It suffices to show that φ is an epimorphism. We have

$$K = \varphi(K) \subseteq \varphi(L) \subseteq L'$$

Since L is algebraically closed and $\varphi(L) \cong L$ we get that $\varphi(L)$ is also algebraically closed and so (by Proposition 67.2) the only algebraic extension of $\varphi(L)$ is $\varphi(L)$ itself. On the other hand, since L'/K is an algebraic extension and $K \subseteq \varphi(L)$ the extension $L'/\varphi(L)$ is algebraic. This gives that $L' = \varphi(L)$. ☐

67.12 Fundamental Theorem of Algebra. *The field $\mathbb{C}$ of complex numbers is algebraically closed.*

Proof. Later. ☐

Recall. If L/K is a field extension then

$$K_{\mathsf{alg}}(L) = \{a \in L \mid a \text{ is an algebraic element over } K\}$$

is a subfield of L.

67.13 Proposition. *If L/K is a field extension and L is an algebraically closed field then the field $K_{alg}(L)$ is also algebraically closed. As a consequence $K_{alg}(L)$ is an algebraic closure of K.*

Proof. Let $p(x) \in K_{\mathsf{alg}}(L)[x]$, $\deg p(x) > 0$. It suffices to show that $p(x)$ has a root in $K_{\mathsf{alg}}(L)$.

The field L is algebraically closed, so $p(x)$ has a root $a \in L$. Consider the extensions

$$K_{\mathsf{alg}}(L)(a)/K_{\mathsf{alg}}(L) \quad \text{and} \quad K_{\mathsf{alg}}(L)/K$$

These extensions are algebraic, so by Proposition 60.6 $K_{\mathsf{alg}}(L)(a)/K$ is also an algebraic extension. This means in particular that $a \in L$ is an algebraic element over K, so $a \in K_{\mathsf{alg}}(L)$. □

67.14 Note. Since $\mathbb{C}$ is an algebraically closed field thus $\bar{\mathbb{Q}} := \mathbb{Q}_{\mathsf{alg}}(\mathbb{C})$ is an algebraic closure of $\mathbb{Q}$.

68 Roots of unity

68.1 Definition. Let K be a field. We say that $a \in K$ is an n-th root of unity if $a^n = 1$, i.e. if a is a root of the polynomial $f_n(x) = x^n - 1 \in K[x]$.

68.2 Proposition. *Let K be a field and let*

$$\mu_n(K) := \{a \in K \mid a^n = 1\}$$

Then $\mu_n(K)$ is a subgroup of the multiplicative group $K^* = K - \{0\}$.

Proof. Exercise. □

68.3 Note.

1) If $m|n$ then $\mu_m(K)$ is a subgroup of $\mu_n(K)$.

2) If $\chi(K) = p \neq 0$ and $n = p^k m$ then $\mu_n(K) = \mu_m(K)$. Indeed, we have

$$f_n(x) = x^{p^k m} - 1^{p^k} = (x^m - 1)^{p^k} = (f_m(x))^{p^k}$$

so roots of f_n are the same as roots of f_m.

68.4 Theorem. *For any field K the group $\mu_n(K)$ is cyclic and $|\mu_n(K)| \mid n$.*

Proof. Since $\mu_n(K)$ is a finite subgroup of K^* it is cyclic by 38.8. so $\mu_n(K) = \langle a \rangle$ for some $a \in \mu_n(K)$. This gives $|\mu_n(K)| = |a|$. Also, since $a^n = 1$, we have $|a| \mid n$.

□

68.5 Corollary. *Let K be an algebraically closed field. If $\chi(K) = p \neq 0$ assume also that $p \nmid n$. Then we have*

$$\mu_n(K) \cong \mathbb{Z}/n\mathbb{Z}$$

Proof. By Theorem 68.4 it is enough to show that $|\mu_n(K)| = n$.

Let $f_n(x) = x^n - 1 \in K[x]$. We have

$$f_n'(x) = nx^{n-1}$$

Since $f_n(x)$ and $f_n'(x)$ have no common roots thus by Proposition 62.3 $f_n(x)$ is a separable polynomial. As a consequence $f_n(x)$ has n distinct roots in K, and so $|\mu_n(K)| = n$. □

68.6 Definition. An element $a \in K$ is a *primitive n-th root of unity* if $a^n = 1$ and $a^m \neq 1$ for all $0 < m < n$.

68.7 Note.

1) A primitive n-th root of unity exists in K iff $|\mu_n(K)| = n$. In such case $a \in K$ is primitive n-th roots of unity if a generates $\mu_n(K)$.

2) If $|\mu_n(K)| = m$ then $\mu_n(K) = \mu_m(K)$ and $\mu_n(K)$ is generated by a primitive m-th root of unity in K.

68.8 Examples.

1) In $\mathbb{Q}$ for any $n \geq 1$ we have

$$\mu_{2n-1}(\mathbb{Q}) = \{1\} = \mu_1(\mathbb{Q}), \qquad \mu_{2n}(\mathbb{Q}) = \{1, -1\} = \mu_2(\mathbb{Q})$$

Thus 1 is a primitive 1-st root of unity and -1 is a primitive 2-nd root of unity. For $n > 2$ there are no primitive n-roots of unity in $\mathbb{Q}$.

2) Since $\mathbb{C}$ is an algebraically closed field we have $|\mu_n(\mathbb{C})| = n$ for all $n > 0$, and so $\mathbb{C}$ contains a primitive n-th root of unity for every n. We have

$$\mu_n(\mathbb{C}) = \{\varepsilon_k := \cos(2\pi k/n) + i\sin(2\pi k/n) \mid k = 0, \ldots, n-1\}$$

and ε_k is a primitive n-th root of unity iff $\gcd(k, n) = 1$.

69 Finite fields

69.1 Proposition. *If K is a finite field and $\chi(K) = p$ then $|K| = p^n$ for some $n \geq 1$.*

Proof. Since $\chi(K)$ thus by Theorem 58.6 $\mathbb{F}_p$ can be identified with the primitive subfield of K. Since K is a finite field we have $[K : \mathbb{F}_p] = n$ for some $n \geq 1$. This means that K is an n dimensional vector space over $\mathbb{F}_p$, and so K has p^n elements. □

69.2 Proposition. *If K is a finite field, such that $\chi(K) = p$ then $|K| = p^n$ iff K is a splitting field of the polynomial*

$$f(x) = x^{p^n} - x \in \mathbb{F}_p[x]$$

69.3 Lemma. *If K be a field and $\varphi\colon K \to K$ be a field homomorphism then*

$$L := \{a \in K \mid \varphi(a) = a\}$$

is a subfield of K.

Proof. Exercise. □

Proof of Proposition 69.2.

($\Rightarrow$) We will show that if $|K| = p^n$ then every element of K is a root of $f(x)$. Since $f(x)$ can have at most p^n roots it will follow that K is a splitting field of $f(x)$.

Let $a \in K$. If $a = 0$ then a is a root of $f(x)$. Assume then that $a \neq 0$. Then a is an element of the multiplicative group $K^* = K - \{0\}$. Since $|K^*| = p^n - 1$ thus we have we have $a^{p^n-1} = 1$. This gives

$$f(a) = a^{p^n} - a = a(a^{p^n-1} - 1) = 0$$

($\Leftarrow$) Assume that K is a splitting field of $f(x)$. Consider the field homomorphism

$$\varphi\colon K \to K, \quad \varphi(a) = a^{p^n}$$

By Lemma 69.3 the set

$$L = \{a \in K \mid \varphi(a) = a\}$$

is a subfield of K. On the other hand L consists of all roots of $f(x)$ in K, and so since $f(x)$ splits over K it also splits over L. Since K is a splitting field of $f(x)$ this implies that $K = L$, i.e. every element of K is a root of $f(x)$.

It suffices to show then that $f(x)$ has p^n distinct roots, i.e. that it is a separable polynomial. This follows from Proposition 62.3, since $f'(x) = -1$, and so $f(x)$ and $f'(x)$ have no roots in common. □

69.4 Corollary. *For every prime p and $n \geq 1$ there exists a field K such that $\chi(K) = p$ and $|K| = p^n$. Moreover such field is unique up to isomorphism.*

Proof. Take the polynomial $f(x) = x^{p^n} - x \in \mathbb{F}_p[x]$. By Proposition 66.5 there exists a splitting field K of $f(x)$ and by Proposition 69.2 we have $|K| = p^n$.

Assume now that K' is another field such that $|K'| = p^n$. By Proposition 69.2 we have that K' is also a splitting field of $f(x)$. Uniqueness of splitting fields (66.7) gives then that $K' \cong K$. □

Recall:

64.2 Theorem. *If K is an infinite field and L/K is a separable extension such that $[L : K] < \infty$ then L/K is a simple extension.*

69.5 Theorem. *If K is an finite field and L/K is a field extension such that $[L : K] < \infty$ then L/K is a simple extension.*

69.6 Note. If K is a finite field then any algebraic extension L/K is a separable extension (exercise).

Proof of Theorem 69.5.

Assume that $\chi(K) = p$, and that $\mathbb{F}_p \subseteq K$. We will show that a $L = \mathbb{F}_p(c)$ for some $c \in L$. Since $\mathbb{F}_p(c) \subseteq K(c)$ this will imply that $K(c) = L$.

Since K is a finite field and $[L : K] < \infty$ the field L is finite, and so $|L| = p^n$ for some $n \geq 1$. Let $L^* = L - \{0\}$ be the multiplicative group of L. By the proof of Proposition 69.2 we have

$$L^* = \{a \mid a^{p^n - 1} = 1\}$$

i.e. L^* is the group of $(p^n - 1)$-st roots of unity in L:

$$L^* = \mu_{p^n-1}(L)$$

By Theorem 68.4 $\mu_{p^n-1}(L)$ is a cyclic group. Let c be a generator of $\mu_{p^n-1}(L)$. Then we have $L = \mathbb{F}_p(c)$.

□

70 Galois theory - motivation

70.1 Proposition. *Let L/K be a field extension and let*

$$\varphi\colon L \to L$$

be an automorphism such that $\varphi|_K = \mathrm{id}_K$. If $a \in L$ is a root of a polynomial $f(x) \in K[x]$ then $\varphi(a)$ is also a root of $f(x)$.

Proof. If $f(x) = r_0 + r_1 x + \ldots + r_n x^n$ then we have

$$f(\varphi(a)) = r_0 + r_1\varphi(a) + \ldots + r_n\varphi(a)^n$$

Also, since $r_i \in K$ we have $r_i = \varphi(r_i)$, so we obtain

$$\begin{aligned} f(\varphi(a)) &= \varphi(r_0) + \varphi(r_1)\varphi(a) + \ldots + \varphi(r_n)\varphi(a)^n \\ &= \varphi(f(a)) \\ &= \varphi(0) \\ &= 0 \end{aligned}$$

□

70.2 Corollary. *Let L/K be an algebraic extension, let $f(x) \in K[x]$, and let $S = \{a_1, \ldots, a_n\} \subseteq L$ be the set of all roots of $f(x)$ in L. If*

$$\varphi\colon L \to L$$

is an automorphism such that $\varphi|_K = \mathrm{id}_K$ then $\varphi|_S$ is a permutation of S.

Moreover, if L is a splitting field of $f(x)$ then the automorphism φ is uniquely determined by this permutation.

Proof. By Proposition 70.1 we have $\varphi(S) \subseteq S$. Also, since φ is a bijection and S is a finite set thus $\varphi|_S\colon S \to S$ is also a bijection i.e. it is a permutation of the set S.

Recall that if L is a splitting field of $f(x)$ we have

$$L = K(a_1, \ldots, a_n)$$

Thus, since $\varphi|_K = \mathrm{id}$, the automorphism φ is determined by the restriction $\varphi|_{\{a_1,\ldots,a_n\}}$. □

Note. Let L be a splitting field of a polynomial $f(x) \in K[x]$, and let $S \subseteq L$ be the set of all roots of $f(x)$. It is not true that every permutation of S can be extended to an automorphism $\varphi\colon L \to L$ such that $\varphi|_K = \mathrm{id}_K$.

70.3 Example.

Let $f(x) = x^4 - 10x^2 + 1 \in \mathbb{Q}[x]$. Roots of $f(x)$ in $\mathbb{C}$ are:

$$\sqrt{2}+\sqrt{3}, \quad \sqrt{2}-\sqrt{3}, \quad -\sqrt{2}+\sqrt{3}, \quad -\sqrt{2}-\sqrt{3}$$

It follows that the field $L := \mathbb{Q}(\sqrt{2}, \sqrt{3})$ is a splitting field of $f(x)$.

Let

$$\varphi\colon L \to L$$

be an automorphism such that $\varphi|_{\mathbb{Q}} = \mathrm{id}_{\mathbb{Q}}$. Notice that we must have $\varphi(\sqrt{2}) = \pm\sqrt{2}$ since $\pm\sqrt{2}$ are the only roots of $g(x) = x^2 - 2$. Similarly we must have $\varphi(\sqrt{3}) = \pm\sqrt{3}$.

Assume that $\varphi(\sqrt{2}+\sqrt{3}) = \sqrt{2}-\sqrt{3}$. Then we must have

$$\varphi(\sqrt{2}) = \sqrt{2}, \quad \text{and} \quad \varphi(\sqrt{3}) = -\sqrt{3}$$

This gives that

$$\begin{aligned}
\varphi(\sqrt{2}-\sqrt{3}) &= \sqrt{2}+\sqrt{3} \\
\varphi(-\sqrt{2}+\sqrt{3}) &= -\sqrt{2}-\sqrt{3} \\
\varphi(-\sqrt{2}-\sqrt{3}) &= -\sqrt{2}+\sqrt{3}
\end{aligned}$$

By this argument one gets that there are only 4 permutations of the set of roots $f(x)$ that can extend to an automorphism of L, and that each of these permutations is determined by the value of $\varphi(\sqrt{2}+\sqrt{3})$.

Note. Let L be a splitting field of $f(x) \in K[x]$, and let $S \subseteq L$ be the set of all roots of $f(x)$. Permutations of S that extend to automorphisms of L form a subgroup of the group of permutations of S (check!). We will see that properties of this group provide information about the polynomial $f(x)$ and its roots.

71 Normal extensions

71.1 Definition. Let L/K be an algebraic extension and let

$$\mathrm{Gal}(L/K) := \{\varphi\colon L \to L \mid \varphi \text{ is an automorphism and } \varphi|_K = \mathrm{id}_K\}$$

Then $\mathrm{Gal}(L/K)$ is a group (with composition of automorphism), and it is called the *Galois group* of the extension L/K.

71.2 Proposition. *Let L be a field and let $\mathrm{Aut}(L)$ be the group of all automorphisms of L. If H is a subgroup of $\mathrm{Aut}(L)$ then the set*

$$L^H = \{a \in L \mid \varphi(a) = a \text{ for all } \varphi \in H\}$$

is a subfield of L.

Proof. Exercise. □

71.3 Definition. If L is a field and $H \subseteq \mathrm{Aut}(L)$ is a subgroup then the subfield $L^H \subseteq L$ is called the *fixed field* of H.

71.4 Note. Let L be a field. We have maps

$$\Phi\colon \begin{pmatrix}\text{subfields}\\ \text{of } L\end{pmatrix} \leftrightarrows \begin{pmatrix}\text{subgroups}\\ \text{of } \mathrm{Aut}(L)\end{pmatrix} : \Psi$$

where $\Phi(K) = \mathrm{Gal}(L/K)$ and $\Psi(H) = L^H$. In general these maps are not inverses of each other. Take e.g. $L = \mathbb{Q}(\sqrt[3]{2})$ and let $K = \mathbb{Q} \subseteq L$. We have

$$\Phi(K) = \mathrm{Gal}(L/K) = \{\mathrm{id}_L\}$$

(check!). This gives

$$\Psi(\Phi(K)) = L^{\mathrm{Gal}(L/K)} = L \neq K$$

71.5 Definition. An algebraic extension L/K is called a *Galois extension* if $K = L^{\mathrm{Gal}(L/K)}$.

Goal. An extension L/K is a Galois extension iff it is a *normal* and separable extension.

71.6 Definition. An algebraic extension L/K is a *normal extension* if for every $a \in L$ the polynomial $\mathrm{irr}_a^K(x)$ splits over L.

71.7 Example.

1) If $\bar{K}$ is the algebraic closure of K then the extension $\bar{K}/K$ is normal.
2) The extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not normal. Indeed, $\sqrt[3]{2} \in \mathbb{Q}(\sqrt[3]{2})$, but the polynomial
$$\mathrm{irr}_{\sqrt[3]{2}}^{\mathbb{Q}}(x) = x^3 - 2$$
does not split over $\mathbb{Q}(\sqrt[3]{2})$.

71.8 Proposition. *Let $\bar{K}$ be an algebraic closure of a field K. An algebraic extension L/K is normal iff for any two homomorphisms*
$$\varphi, \psi \colon L \to \bar{K}$$
such that $\varphi|_K = \psi|_K = \mathrm{id}_K$ we have $\varphi(L) = \psi(L)$.

Proof.

($\Rightarrow$) Let $\varphi, \psi \colon L \to \bar{K}$ be homomorphisms satisfying $\varphi|_K = \psi|_K = \mathrm{id}_K$. It is enough to show that for any $a \in L$ we have $\varphi(a) \in \psi(L)$.

Let $a \in L$ and let $f(x) = \mathrm{irr}_a^K(x)$. Since L/K is a normal extension $f(x)$ splits over L:

$$f(x) = (x - a_1)(x - a_2) \cdot \ldots \cdot (x - a_n)$$

where $a_i \in L$ and $a_1 = a$. We have

$$\begin{aligned}\varphi(f(x)) &= (x - \varphi(a_1))(x - \varphi(a_2)) \cdot \ldots \cdot (x - \varphi(a_n)) \\ \psi(f(x)) &= (x - \psi(a_1))(x - \psi(a_2)) \cdot \ldots \cdot (x - \psi(a_n))\end{aligned}$$

On the other hand, since $f(x) \in K[x]$ and $\varphi|_K = \psi|_K = \mathrm{id}_K$ we have

$$\varphi(f(x)) = f(x) = \psi(f(x))$$

Since $\bar{K}[x]$ is a UFD and $x - \varphi(a_i)$, $x - \psi(a_i)$ are irreducible polynomials in $\bar{K}[x]$ we must have $x - \varphi(a_1) = x - \psi(a_i)$ for some i. Since $a_1 = a$ this gives

$$\varphi(a) = \psi(a_i) \in \psi(L)$$

($\Leftarrow$) Assume that L/K is an algebraic extension which is not normal. Then there exists $a \in L$ such that the polynomial $f(x) = \mathrm{irr}_a^K(x)$ does not split over L. Let $\varphi\colon L \to \bar{K}$ be a homomorphism such that $\varphi|_K = \mathrm{id}_K$. Since $L \cong \varphi(L)$ we have that $\varphi(a) \in \varphi(L)$ is a root of $f(x)$, but $f(x)$ does not split over $\varphi(L)$. On the other hand $f(x)$ splits over $\bar{K}$, so there exists $b \in \bar{K} - \varphi(L)$ such that $f(b) = 0$. By Proposition 59.8 there exists an isomorphism

$$\tilde{\psi}\colon K(a) \to K(b)$$

such that $\tilde{\psi}|_K = \mathrm{id}_K$. By (67.10) $\tilde{\psi}$ can be extended to a homomorphism

$$\psi\colon L \to \bar{K}$$

Since $b \in \psi(L)$ and $b \notin \varphi(L)$ we get that $\varphi(L) \neq \psi(L)$.

$\square$

71.9 Proposition. *Let L/K be a field extension such that $[L : K] < \infty$. The extension L/K is normal iff L is a splitting field of some polynomial $f(x) \in K[x]$.*

Proof.

($\Rightarrow$) Since $[L : K] < \infty$ we have

$$L = K(a_1, \dots, a_n)$$

for some $a_1, \dots, a_n \in L$. Let $f_i(x) = \mathrm{irr}^K_{a_i}(x)$. Since L/K is a normal extension $f_i(x)$ splits over L for each i, so the polynomial

$$f(x) = f_1(x) \cdot \ldots \cdot f_n(x)$$

also splits over L. Moreover, since $a_1, \dots, a_n$ are roots of $f(x)$ and they generate L, thus L is a splitting field of $f(x)$.

($\Leftarrow$) Let L be a splitting field of some polynomial $f(x) \in K[x]$. Then we have

$$f(x) = (x - a_1)(x - a_2) \cdot \ldots \cdot (x - a_n)$$

for some $a_1, \dots, a_n \in L$, and $L = K(a_1, \dots, a_n)$.

Let $\bar{K}$ be an algebraic closure of K and let

$$\varphi, \psi \colon L \to \bar{K}$$

be field homomorphisms such that $\varphi|_K = \psi|_K = \mathrm{id}_K$. We have

$$\begin{aligned} \varphi(f(x)) &= (x - \varphi(a_1))(x - \varphi(a_2)) \cdot \ldots \cdot (x - \varphi(a_n)) \\ \psi(f(x)) &= (x - \psi(a_1))(x - \psi(a_2)) \cdot \ldots \cdot (x - \psi(a_n)) \end{aligned}$$

On the other hand, since $f(x) \in K[x]$ e have

$$\varphi(f(x)) = f(x) = \psi(f(x))$$

It follows that for every $1 \leq i \leq n$ there exists $1 \leq j \leq n$ such that $\varphi(a_i) = \psi(a_j)$. This gives

$$\varphi(L) = K(\varphi(a_1), \dots, \varphi(a_n)) \subseteq K(\psi(a_1), \dots, \psi(a_n)) = \psi(L)$$

Similarly we get $\psi(L) \subset \varphi(L)$. Therefore $\varphi(L) = \psi(L)$, and so by Proposition 71.8 L/K is a normal extension.

$\square$

71.10 Proposition. *Let K, L, M, N be fields such that $K, L, M \subseteq N$, N/K is an algebraic extension and L/K is a normal extension. Then LM/KM is also a normal extension.*

71.11 Corollary. *If $K \subseteq L \subseteq M$ are field extensions and M/K is normal then M/L is also normal.*

Proof. We have $M = ML$, $L = KL$, and by Proposition 71.10 the extension ML/KL is normal. $\square$

Proof of Proposition 71.10. Let $\overline{KM}$ be an algebraic closure of KM, and let

$$\varphi, \psi \colon LM \to \overline{KM}$$

be two embeddings of LM such that $\varphi|_{KM} = \psi|_{KM} = \mathrm{id}|_{KM}$. By Proposition 71.8 it is enough to show that $\varphi(LM) = \psi(LM)$.

Notice that since KM/K and N/KM are algebraic extensions we have

$$\overline{KM} = \bar{N} = \bar{K}$$

Since $\varphi|_K = \psi|_K = \mathrm{id}|_K$ and L/K is a normal extension we have $\varphi(L) = \psi(L)$. Moreover, $\varphi|_M = \psi_M = \mathrm{id}_M$, so $\varphi(M) = \psi(M) = M$. This gives

$$\varphi(LM) = \varphi(L)\varphi(M) = \psi(L)\psi(M) = \psi(LM)$$

$\square$

71.12 Note. It is not true that if $K \subseteq L \subseteq M$ and M/L are L/K are normal extensions then M/K is also normal. Take e.g.

$$\mathbb{Q} \subseteq \mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt[4]{2})$$

71.13 Definition. Let L/K be an algebraic extension. An *normal closure* of L/K is an extension M/K such that

1) $L \subseteq M$
2) M/K is a normal extension
3) M does not have any proper subfields satisfying 1) and 2).

71.14 Proposition. *For any L/K is any algebraic extension there exists a normal closure M/K of L/K. Moreover, if M, M' are normal closures of L/K then there exists an isomorphism*

$$\varphi\colon M \to M'$$

such that $\varphi|_L = \mathrm{id}_L$.

Proof. Let $\bar{L}$ be an algebraic closure of L, and let $S = \{M_\alpha\}$ be the set of all subfields of $\bar{L}$ such that $L \subseteq M_\alpha$ and that M_α/K is a normal extension. The set S is non-empty since $\bar{L} \in S$. Take $M = \bigcap_\alpha M_\alpha$. Check: M/K is a normal extension, and so it is a normal closure of L/K.

Let M'/K be another normal closure of L/K. Since M'/L is an algebraic extensions by (67.8) there exists an embedding

$$\varphi\colon M' \to \bar{L}$$

such that $\varphi|_L = \mathrm{id}|_L$. Since $L \subseteq \varphi(M') \subseteq \bar{L}$ and $\varphi(M)/K$ is a normal extension, thus by the construction of M we have $M \subseteq \varphi(M')$. On the other hand $\varphi(M')$ is a normal closure of L/K. This implies that $\varphi(M) = M'$.

□

72 Galois extensions

72.1 Theorem. *An algebraic extension L/K is a Galois extension iff L/K is separable and normal.*

Proof.

($\Rightarrow$) Let L/K be a Galois extension let $a \in L$, and let $f(x) = \mathrm{irr}^K_a(x)$. It is enough to show that $f(x)$ splits over L and that it is a separable polynomial.

Let $S = \{a_1, \dots, a_n\}$ be the set of all distinct roots of $f(x)$ in L where, say, $a = a_1$. Let

$$g(x) = (x - a_1)(x - a_2) \cdot \ldots \cdot (x - a_n) \in L[x]$$

Notice that $g(x)$ is a separable polynomial that splits over L. Therefore it suffices to show that $f(x) = g(x)$.

Furthermore, it is enough to show that $f(x)|g(x)$. Indeed since $g(x)|f(x)$ and since $f(x)$ and $g(x)$ are monic polynomials this will imply that $f(x) = g(x)$.

Let $\varphi \in \mathrm{Gal}(L/K)$. By Corollary 70.2 φ permutes the set $\{a_1, \dots, a_n\}$. This gives

$$\begin{aligned}\varphi(g(x)) &= (x - \varphi(a_1))(x - \varphi(a_2)) \cdot \ldots \cdot (x - \varphi(a_n)) \\ &= (x - a_1)(x - a_2) \cdot \ldots \cdot (x - a_n) \\ &= g(x)\end{aligned}$$

Therefore, if $g(x) = r_0 + r_1 x + \dots + r_n x^n$ then we have $\varphi(r_i) = r_i$, i.e. $r_i \in L^{\mathrm{Gal}(L/K)}$ for $i = 0, \dots, n$. Since L/K is a Galois extension we have $L^{\mathrm{Gal}(L/K)} = K$, and so we obtain that $g(x) \in K[x]$.

Since $g(a) = g(a_1) = 0$ and $f(x) = \mathrm{irr}^K_a(x)$ this gives $f(x)|g(x)$.

($\Leftarrow$) Let L/K be a separable and normal extension and let $a \in L - K$. We need to show that there exists $\varphi \in \mathrm{Gal}(L/K)$ such that $\varphi(a) \neq a$.

Let $\bar{K}$ be an algebraic closure of K such that $L \subseteq \bar{K}$. If $f(x) = \mathrm{irr}_a^K(x) \in K[x]$ then $\deg f(x) > 1$ (since $a \notin K$), and $f(x)$ has no multiple roots (since a is separable over K). It follows that there exists $b \in \bar{K}$ such that $b \neq a$ and $f(b) = 0$. By Proposition 59.8 there exists an isomorphism

$$\tilde{\varphi}\colon K(a) \to K(b) \subseteq \bar{K}$$

such that $\tilde{\varphi}|_K = \mathrm{id}_K$. By (67.10) we can extend $\tilde{\varphi}$ to an embedding $\varphi\colon L \to \bar{K}$.

Let $i\colon L \hookrightarrow \bar{K}$ be the inclusion homomorphism. Since L/K is a normal extension thus by (71.8) we have $\varphi(L) = i(L) = L$. It follows that φ gives an automorphism $\varphi\colon L \to L$. Since $\varphi|_K = \mathrm{id}_K$ we have $\varphi \in \mathrm{Gal}(L/K)$ and $\varphi(a) = b \neq a$.

□

72.2 Corollary. *If $K \subseteq L \subseteq M$ are fields and M/K is a Galois extension then M/L is also a Galois extension.*

Proof. The extension M/K is normal and separable, so M/L is also normal (by (71.11)) and separable. □

72.3 Theorem. *Let L be a field, let $\mathrm{Aut}(L)$ be the group of automorphisms of L, and let H be a finite subgroup of $\mathrm{Aut}(L)$. Then:*

1) L/L^H *is a Galois extension*

2) $[L : L^H] = |H|$

3) $\mathrm{Gal}(L/L^H) = H$

72.4 Note. Recall that for a field L we have maps of sets:

$$\Phi\colon \begin{pmatrix}\text{subfields}\\ \text{of } L\end{pmatrix} \leftrightarrows \begin{pmatrix}\text{subgroups}\\ \text{of } \mathrm{Aut}(L)\end{pmatrix} : \Psi$$

where $\Phi(K) = \mathrm{Gal}(L/K)$ and $\Psi(H) = L^H$. By part 3) of Theorem 72.3 we obtain that if $H \subseteq \mathrm{Aut}(L)$ is a finite subgroup then

$$\Phi(\Psi(H)) = H$$

Proof of Theorem 72.3.

1) We need to show that L/L^H is an algebraic extension and that

$$L^H = L^{\mathrm{Gal}(L/L^H)}$$

Notice that if G, H are subgroups of $\mathrm{Aut}(L)$ and $G \subseteq H$ then $L^H \subseteq L^G$. Since $H \subseteq \mathrm{Gal}(L/L^H)$ this gives $L^{\mathrm{Gal}(L/L^H)} \subseteq L^H$. On the other hand, by definition of $\mathrm{Gal}(L/L^H)$, if $\varphi \in \mathrm{Gal}(L/L^H)$ then $\varphi|_{L^H} = \mathrm{id}_{L^H}$, so $L^H \subseteq L^{\mathrm{Gal}(L/L^H)}$. Thus we obtain $L^H = L^{\mathrm{Gal}(L/L^H)}$.

It remains to show that L/L^H is an algebraic extension. Let $a \in L$ and let

$$Ha := \{\varphi(a) \mid \varphi \in H\}$$

Since H is a finite group the set Ha is finite, say $Ha = \{a_1, \dots, a_n\}$ where $a_1 = a$. Let

$$g(x) = (x - a_1)(x - a_2) \cdot \ldots \cdot (x - a_n) \in L[x]$$

Notice that if $\varphi \in H$ then $\varphi|_{Ha}$ is a bijection:

$$\varphi|_{Ha}\colon Ha \to Ha$$

As a consequence we have $\varphi(g(x)) = g(x)$ for all $\varphi \in H$. This means that $g(x) \in L^H[x]$. Since $g(a) = g(a_1) = 0$ we get that a is an algebraic element over L^H.

2) Let $a \in L$. The argument in part 1) shows that a is a root of a polynomial $g(x) \in L^H[x]$ such that $\deg g(x) \leq |H|$. It follows that for any $a \in L$ we have

$$[L^H(a) : L^H] \leq |H| \tag{$*$}$$

Claim. $[L : L^H] \leq |H|$.

Indeed, if $[L : L^H] > |H|$ then we can find an intermediate field

$$L^H \subseteq K \subseteq L$$

such that $|H| < [K : L^H] < \infty$. By part 1) L/L^H is a Galois extension, so it is separable. This implies that K/L^H is also a a separable extension. By (64.2) and (69.5) we obtain then that K/L^H is a simple extension i.e.

$$K = L^H(a)$$

for some $a \in K$. This gives

$$[L^H(a) : L^H] = [K : L^H] > |H|$$

which contradicts the inequality $(*)$.

We obtain that L/L^H is a separable extension, and $[L : L^H] \leq |H| < \infty$. It follows that L/L^H is a simple extention, $L = L^H(a)$ for some $a \in L$. Let $f(x) = \mathrm{irr}_a^K(x)$. We have

$$\deg f(x) = [L : L^H] \leq |H|$$

On the other hand each automorphism $\varphi \in \mathrm{Gal}(L/L^H)$ is uniquely determined by the value of $\varphi(a)$. Since $\varphi(a)$ is a root of $f(x)$ we obtain that

$$|\,\mathrm{Gal}(L/L^H)| \leq \deg(f(x))$$

Since $H \subseteq \mathrm{Gal}(L/L^H)$ this gives

$$|H| \leq |\,\mathrm{Gal}(L/L^H)| \leq \deg f(x) \leq |H|$$

Therefore $|H| = |\,\mathrm{Gal}(L/L^H)| = [L : L^H]$.

3) We have $H \subseteq \mathrm{Gal}(L/L^H)$, and by the proof of part 2), $|H| = |\mathrm{Gal}(L/L^H)|$. Since H is a finite group this implies that $H = \mathrm{Gal}(L/L^H)$.

$\square$

72.5 Corollary. *Let L/K be a field extension such that $[L : K] < \infty$. The extension L/K is a Galois extension iff $[L : K] = |\mathrm{Gal}(L/K)|$.*

Proof.

($\Leftarrow$) If L/K is a Galois extension then $K = L^{\mathrm{Gal}(L/K)}$. Since $[L : K] < \infty$ by part 2) of Theorem 72.3 we obtain $[L : K] = |\mathrm{Gal}(L/K)|$.

($\Rightarrow$) Assume that $[L : K] = |\mathrm{Gal}(L/K)|$. We want to show that $K = L^{\mathrm{Gal}(L/K)}$.

We have $K \subseteq L^{\mathrm{Gal}(L/K)} \subseteq L$, so

$$|\mathrm{Gal}(L/K)| = [L : K] = [L : L^{\mathrm{Gal}(L/K)}] \cdot [L^{\mathrm{Gal}(L/K)} : K]$$

By part 2) of Theorem 72.3 we also have $[L : L^{\mathrm{Gal}(L/K)}] = |\mathrm{Gal}(L/K)|$. It follows that $[L^{\mathrm{Gal}(L/K)} : K] = 1$, and so $K = L^{\mathrm{Gal}(L/K)}$. $\square$

73 Application: rational symmetric functions

Let $L = K(t_1, \ldots, t_n)$ be the field of rational functions of variables $t_1, \ldots, t_n$ with coefficients in a field K, and let S_n be the symmetric group on n letters.

The group S_n can be identified with a subgroup of the group $\mathrm{Aut}(L)$ of automorphims of L as follows. If $\sigma \in S_n$ then σ induces an automorphism

$$\sigma \colon L \longrightarrow L$$

such that $\sigma|_K = \mathrm{id}|_K$ and $\sigma(t_i) = t_{\sigma(i)}$.

73.1 Definition. If $\frac{f(t_1,\ldots,t_n)}{g(t_1,\ldots,t_n)} \in L^{S_n}$ then we say that $\frac{f(t_1,\ldots,t_n)}{g(t_1,\ldots,t_n)}$ is *symmetric function.*

73.2 Example. Let

$$f(x) = (x - t_1)(x - t_2) \cdot \ldots \cdot (x - t_n) \in L[x]$$

For $\sigma \in S_n$ we have

$$\sigma(f(x)) = (x - t_{\sigma(1)})(x - t_{\sigma(2)}) \cdot \ldots \cdot (x - t_{\sigma(n)}) = f(x)$$

Therefore $f(x) \in L^{S_n}[x]$.

Notice that we have

$$f(x) = x^n - \sigma_1 x^{n-1} + \sigma_2 x^{n-2} - \cdots + (-1)^{n-1}\sigma_{n-1}x + (-1)^n \sigma_n$$

where

$$\begin{aligned}
\sigma_1 &= \sum_i t_i \\
\sigma_2 &= \sum_{i<j} t_i t_j \\
\sigma_3 &= \sum_{i<j<k} t_i t_j t_k \\
&\vdots \\
\sigma_n &= t_1 t_2 \cdot \ldots \cdot t_n
\end{aligned}$$

If follows that $\sigma_1, \ldots, \sigma_n \in L^{S_n}$.

73.3 Definition. The polynomials $\sigma_1, \dots, \sigma_n$ are called *elementary symmetric polynomials* in variables $t_1, \dots, t_n$.

73.4 Theorem.

$$K(t_1, \dots, t_n)^{S_n} = K(\sigma_1, \dots, \sigma_n) \subseteq K(t_1, \dots, t_n)$$

Proof. Denote

$$L := K(t_1, \dots, t_n), \qquad M := K(\sigma_1, \dots, \sigma_n)$$

We have $M \subseteq L^{S_n}$ so it suffices to show that $[L^{S_n} : M] = 1$

Notice that L is a splitting field of the separable polynomial

$$f(x) = x^n - \sigma_1 x^{n-1} + \dots + (-1)^{n-1}\sigma_{n-1} + (-1)^n \sigma_n$$

where $f(x) \in M[x]$, and so L/M is a Galois extension. By Corollary 70.2 we obtain that $\mathrm{Gal}(L/M)$ is a subgroup of the group of permutations of the set $\{t_1, \dots, t_n\}$ of roots of $f(x)$. From Corollary 72.3 it follows that

$$[L : M] = |\,\mathrm{Gal}(L/M)| \leq |S_n|$$

On the other hand by Theorem 72.3 we have

$$[L : L^{S_n}] = |S_n|$$

Since $[L : M] = [L : L^{S_n}] \cdot [L^{S_n} : M]$ this implies that $[L : M] = |S_n|$ and that $[L^{S_n} : M] = 1$.

□

74 The fundamental theorem of Galois theory

Let L/K be an algebraic extension. We have maps of sets

$$\Phi\colon \begin{pmatrix} \text{intermediate fields} \\ K \subseteq M \subseteq L \end{pmatrix} \leftrightarrows \begin{pmatrix} \text{subgroups} \\ H \subseteq \mathrm{Gal}(L/K) \end{pmatrix} : \Psi$$

where $\Phi(M) = \mathrm{Gal}(L/M)$ and $\Psi(H) = L^H$.

74.1 Theorem. *Let L/K be a Galois extension such that $[L : K] < \infty$.*

1) *The maps Φ and Ψ are bijections and $\Psi = \Phi^{-1}$.*
2) *If M is an intermediate field of L/K then M/K is a Galois extension iff $\Phi(M) = \mathrm{Gal}(L/M)$ is a normal subgroup of $\mathrm{Gal}(L/K)$. Moreover in such case we have*

$$\mathrm{Gal}(M/K) \cong \mathrm{Gal}(L/K)/\Phi(M)$$

Proof.

1) We will show that $\Psi(\Phi(M)) = M$ and $\Phi(\Psi(H)) = H$.

Let $K \subseteq M \subseteq L$. We have

$$\Psi(\Phi(M)) = \Psi(\mathrm{Gal}(L/M)) = L^{\mathrm{Gal}(L/M)}$$

Since L/K is a Galois extension thus L/M is also a Galois extension (72.2), so $L^{\mathrm{Gal}(L/M)} = M$.

Next, if H is a subgroup of $\mathrm{Gal}(L/K)$ then

$$\Psi(\Phi(H)) = \Psi(L^H) = \mathrm{Gal}(L/L^H)$$

and by Theorem 72.3 we have $\mathrm{Gal}(L/L^H) = H$.

2) ($\Rightarrow$) Assume that M is an intermediate field of L/K such that M/K is a Galois extension.

Claim. If $\varphi \in \mathrm{Gal}(L/K)$ then $\varphi(M) = M$.

Indeed, let $\bar{K}$ be an algebraic closure of K such that $L \subseteq \bar{K}$. We have inclusions

$$K \subseteq M \subseteq L \subseteq \bar{K}$$

Let $i\colon M \hookrightarrow \bar{K}$ be the inclusion map. For $\varphi \in \mathrm{Gal}(L/K)$ consider the embedding

$$\varphi\colon M \to L \hookrightarrow \bar{K}$$

Since M/K is a normal extension by Proposition 71.8 we have

$$\varphi(M) = i(M) = M$$

As a consequence we obtain a group homomorphism

$$f\colon \mathrm{Gal}(L/K) \to \mathrm{Gal}(M/K)$$

given by $f(\varphi) = \varphi|_M$ Notice that

$$\mathrm{Ker}(f) = \{\varphi \in \mathrm{Gal}(L/K) \mid \varphi|_M = \mathrm{id}_M\} = \mathrm{Gal}(L/M) = \Phi(M)$$

Therefore $\Phi(M)$ is a normal subgroup of $\mathrm{Gal}(L/K)$.

($\Leftarrow$) Assume that M is an intermediate field of L/K such that $\Phi(M) = \mathrm{Gal}(L/M)$ is a normal subgroup of $\mathrm{Gal}(L/K)$.

Claim. If $\varphi \in \mathrm{Gal}(L/K)$ then $\varphi(M) = M$.

Indeed, let $\varphi \in \mathrm{Gal}(L/K)$ and let $a \in M$. For any $\psi \in \mathrm{Gal}(L/M)$ we have $\varphi^{-1}\psi\varphi \in \mathrm{Gal}(L/M)$ since $\mathrm{Gal}(L/M)$ is a normal subgroup of $\mathrm{Gal}(L/K)$. This gives $\varphi^{-1}\psi\varphi(a) = a$, or equivalently:

$$\psi(\varphi(a)) = \varphi(a)$$

for all $\psi \in \mathrm{Gal}(L/M)$. Thus, $\varphi(a) \in L^{\mathrm{Gal}(L/M)}$. Since L/M is a Galois extension we have $L^{\mathrm{Gal}(L/M)} = M$. It follows that $\varphi(a) \in M$ for any $a \in M$ and so $\varphi(M) \subseteq M$. Substituting φ^{-1} in place of φ we also obtain from here $\varphi^{-1}(M) \subseteq M$. This gives

$$M = \varphi(\varphi^{-1}(M)) \subseteq \varphi(M)$$

and so we obtain that $\varphi(M) = M$ for all $\varphi \in \mathrm{Gal}(L/M)$.

As before we obtain then that there exists a group homomorphism

$$f\colon \mathrm{Gal}(L/K) \to \mathrm{Gal}(M/K)$$

where $f(\varphi) = \varphi|_M$ and $\mathrm{Ker}(f) = \mathrm{Gal}(L/M)$. Moreover we have $M^{\mathrm{Im}(f)} = K$ (since $L^{\mathrm{Gal}(L/K)} = K$), and so by Theorem 72.3 M/K is a Galois extension with $\mathrm{Gal}(M/K) = \mathrm{Im}(f)$. By the first isomorphism theorem for groups we also get

$$\mathrm{Im}(f) \cong \mathrm{Gal}(L/K)/\,\mathrm{Ker}(f) = \mathrm{Gal}(L/K)/\,\mathrm{Gal}(L/M) = \mathrm{Gal}(L/K)/\Phi(M)$$

□

74.2 Theorem. *Let L/K be a Galois extension. Let H_1, H_2 be subgroups of $\mathrm{Gal}(L/K)$ and let $\Phi(H_1) = M_1$, $\Phi(H_2) = M_2$. Then:*

1) $\Phi(H_1 \cap H_2) = M_1 M_2$

2) $\Phi(\langle H_1, H_2\rangle) = M_1 \cap M_2$ *where $\langle H_1, H_2\rangle$ is the subgroup of $\mathrm{Gal}(L/K)$ generated by H_1, H_2.*

Proof. Exercise. □

74.3 Example.

Take the extension $L := \mathbb{Q}(\sqrt{2}, \sqrt{3})$ of $\mathbb{Q}$. This is a Galois extension since $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is a splitting field of a polynomial $f(x) = (x^2 - 2)(x^2 - 3) \in \mathbb{Q}[x]$. We have $[L : \mathbb{Q}] = 4$ so $|\,\mathrm{Gal}(L/\mathbb{Q})| = 4$.

If $\sigma \in \mathrm{Gal}(L/\mathbb{Q})$ then σ is determined by the values of $\sigma(\sqrt{2})$ and $\sigma(\sqrt{3})$. We also have $\sigma(\sqrt{2}) = \pm\sqrt{2}$ and $\sigma(\sqrt{3}) = \pm\sqrt{3}$. This gives 4 possible combinations for the values of $\sigma(\sqrt{2})$, $\sigma(\sqrt{3})$, and since $\mathrm{Gal}(L/\mathbb{Q})$ has 4 elements each of these possibilities defines an element of $\mathrm{Gal}(L/\mathbb{Q})$.

There are 2 non-isomorphic groups of order 4: $\mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$. Notice that for any $\sigma \in \mathrm{Gal}(L/\mathbb{Q})$ we have $\sigma^2 = \mathrm{id}_L$. Thus $\mathrm{Gal}(L/\mathbb{Q})$ does not contain any elements of order 4, and so $\mathrm{Gal}(L/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.

Let $\varphi, \psi \in \mathrm{Gal}(L/\mathbb{Q})$ be the automorphism given by

$$\begin{aligned} \varphi(\sqrt{2}) &= -\sqrt{2}, \qquad & \varphi(\sqrt{3}) &= \sqrt{3} \\ \psi(\sqrt{2}) &= \sqrt{2}, \qquad & \psi(\sqrt{3}) &= -\sqrt{3} \end{aligned}$$

Notice that $\mathrm{Gal}(L/\mathbb{Q}) = \{\mathrm{id}_L, \varphi, \psi, \psi\varphi\}$ The lattice of subgroups of $\mathrm{Gal}(L/\mathbb{Q})$ looks as follows:

$$\begin{tikzcd}
& \mathrm{Gal}(L/\mathbb{Q}) \arrow[dl] \arrow[d] \arrow[dr] & \\
\langle\varphi\rangle \arrow[dr] & \langle\psi\rangle \arrow[d] & \langle\psi\varphi\rangle \arrow[dl] \\
& \{\mathrm{id}_L\} &
\end{tikzcd}$$

We have:

$$L^{\langle\varphi\rangle} = \mathbb{Q}(\sqrt{3}), \qquad L^{\langle\psi\rangle} = \mathbb{Q}(\sqrt{2}), \qquad L^{\langle\psi\varphi\rangle} = \mathbb{Q}(\sqrt{2}\sqrt{3}) = \mathbb{Q}(\sqrt{6})$$

Therefore the corresponding lattice of intermediate fields of the extension $L/\mathbb{Q}$ looks as follows:

$$\begin{tikzcd}
& \mathbb{Q} & \\
\mathbb{Q}(\sqrt{3}) \arrow[ur] & \mathbb{Q}(\sqrt{2}) \arrow[u] & \mathbb{Q}(\sqrt{6}) \arrow[ul] \\
& L \arrow[ul] \arrow[u] \arrow[ur] &
\end{tikzcd}$$

Notice that since $\mathrm{Gal}(L/\mathbb{Q})$ is abelian all of its subgroups are normal and so all extensions $M/\mathbb{Q}$ where $M = \mathbb{Q}(\sqrt{2}),\ \mathbb{Q}(\sqrt{3}),\ \mathbb{Q}(\sqrt{6})$ are Galois extensions.

74.4 Example.

Take the polynomial $p(x) = x^4 - 2 \in \mathbb{Q}[x]$, and let $L \subseteq \mathbb{C}$ be a splitting field of $f(x)$. The roots of $p(x)$ in $\mathbb{C}$ are $\pm\sqrt[4]{2}$ and $\pm i\sqrt[4]{2}$, so we have

$$L = \mathbb{Q}(\sqrt[4]{2}, i)$$

Since $p(x)$ is irreducible in $\mathbb{Q}[x]$ we have $[\mathbb{Q}(\sqrt[4]{2}) : \mathbb{Q}] = 4$. Also, $[L : \mathbb{Q}(\sqrt[4]{2})] = 2$, so we obtain

$$[L : \mathbb{Q}] = [L : \mathbb{Q}(\sqrt[4]{2})] \cdot [\mathbb{Q}(\sqrt[4]{2}) : \mathbb{Q}] = 8$$

Thus $|\operatorname{Gal}(L/\mathbb{Q})| = 8$.

If $\sigma \in \operatorname{Gal}(L/\mathbb{Q})$ then σ is determined by the values of $\sigma(\sqrt[4]{2})$ and $\sigma(i)$. The possible values of $\sigma(\sqrt[4]{2})$ are $\pm\sqrt[4]{2}$ and $\pm i\sqrt[4]{2}$, while the possible values of $\sigma(i)$ are $\pm i$. This gives 8 possible combinations, and since $\operatorname{Gal}(L/\mathbb{Q})$ has 8 elements each of these combinations defines an element of $\operatorname{Gal}(L/\mathbb{Q})$.

Let $\varphi, \psi \in \operatorname{Gal}(L/\mathbb{Q})$ be the automorphism given by

$$\begin{aligned} \varphi(\sqrt[4]{2}) &= i\sqrt[4]{2}, \qquad & \varphi(i) &= i \\ \psi(\sqrt[4]{2}) &= \sqrt[4]{2}, \qquad & \psi(i) &= -i \end{aligned}$$

Notice that

$$\psi\varphi(\sqrt[4]{2}) = -i\sqrt[4]{2} \neq i\sqrt[4]{2} = \varphi\psi(\sqrt[4]{2})$$

so $\operatorname{Gal}(L/\mathbb{Q})$ is a non-abelian group. We have a group homomorphism

$$f\colon \operatorname{Gal}(L/\mathbb{Q}) \to \mathbb{Z}/2$$

where for $\sigma \in \operatorname{Gal}(L/\mathbb{Q})$ we set $f(\sigma) = 0$ if $\sigma(i) = i$ and $f(\sigma) = 1$ if $\sigma(i) = -i$. Notice that $\operatorname{Ker}(f)$ is the subgroup generated by φ, and since $|\varphi| = 4$ we have $\operatorname{Ker}(f) \cong \mathbb{Z}/4\mathbb{Z}$. We also have a group homomorphism

$$g\colon \mathbb{Z}/2\mathbb{Z} \to \operatorname{Gal}(L/\mathbb{Q})$$

such that $g(1) = \psi$. Since $fg = \mathrm{id}_{\mathbb{Z}/2\mathbb{Z}}$ by Proposition 18.6 we obtain that $\operatorname{Gal}(L/\mathbb{Q})$ is isomorphic to the semidirect product:

$$\operatorname{Gal}(L/\mathbb{Q}) \cong \mathbb{Z}/4\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z} \cong \langle\varphi\rangle \rtimes \langle\psi\rangle$$

which means that $\mathrm{Gal}(L/\mathbb{Q})$ is isomorphic to the dihedral group D_4 (18.5). In particular every element of $\mathrm{Gal}(L/\mathbb{Q})$ is of the form $\varphi^i\psi^j$ for $0 \leq i \leq 3$, $0 \leq j \leq 1$.

The lattice of subgroups of $\mathrm{Gal}(L/\mathbb{Q})$ looks as follows:

$$
\begin{tikzcd}
 & & \mathrm{Gal}(L/\mathbb{Q}) \arrow[dl] \arrow[d] \arrow[dr] & & \\
 & \langle \varphi^2, \psi \rangle \arrow[dl] \arrow[d] \arrow[dr] & \langle \varphi \rangle \arrow[d] & \langle \varphi^2, \varphi\psi \rangle \arrow[dl] \arrow[d] \arrow[dr] & \\
\langle \psi \rangle \arrow[drr] & \langle \varphi^2\psi \rangle \arrow[dr] & \langle \varphi^2 \rangle \arrow[d] & \langle \varphi^3\psi \rangle \arrow[dl] & \langle \varphi\psi \rangle \arrow[dll] \\
 & & \{\mathrm{id}_L\} & &
\end{tikzcd}
$$

This corresponds to the following lattice of intermediate fields of $L/\mathbb{Q}$:

$$
\begin{tikzcd}
 & & \mathbb{Q} & & \\
 & \mathbb{Q}(\sqrt{2}) \arrow[ur] & \mathbb{Q}(i) \arrow[u] & \mathbb{Q}(i\sqrt{2}) \arrow[ul] & \\
\mathbb{Q}(\sqrt[4]{2}) \arrow[ur] & \mathbb{Q}(i\sqrt[4]{2}) \arrow[u] & \mathbb{Q}(\sqrt{2}, i) \arrow[ul] \arrow[u] \arrow[ur] & \mathbb{Q}((1+i)\sqrt[4]{2}) \arrow[u] & \mathbb{Q}((1-i)\sqrt[4]{2}) \arrow[ul] \\
 & & L \arrow[ull] \arrow[ul] \arrow[u] \arrow[ur] \arrow[urr] & &
\end{tikzcd}
$$

75 The fundamental theorem of algebra

Recall:

67.12 Fundamental Theorem of Algebra. *The field $\mathbb{C}$ of complex numbers is algebraically closed.*

75.1 Lemma. *1) If $f(x) \in \mathbb{R}[x]$ is a polynomial of an odd degree then $f(x)$ has a root in $\mathbb{R}$. In particular if $f(x) \in \mathbb{R}[x]$ is a polynomial of an odd degree then $f(x)$ is irreducible iff $\deg f(x) = 1$.*

2) If $f(x) \in \mathbb{C}[x]$ is a polynomial of degree 2 then $f(x)$ splits over $\mathbb{C}$. In particular there are no extensions $L/\mathbb{C}$ such that $[L : \mathbb{C}] = 2$.

Proof. Exercise. □

75.2 Lemma. *If $L/\mathbb{R}$ is an algebraic extension, $[L : \mathbb{R}] < \infty$, and $[L : \mathbb{R}]$ is an odd number then $L = \mathbb{R}$.*

Proof. Since $L/\mathbb{R}$ is a separable extension and $[L : \mathbb{R}] < \infty$ thus by (64.2) we have $L = \mathbb{R}(a)$ for some $a \in L$. Let $f(x) = \mathrm{irr}^{\mathbb{R}}_{a}(x)$. Then $\deg f(x) = [L : \mathbb{R}]$ is an odd number. Since $f(x)$ is an irreducible polynomial by Lemma 75.1 we obtain $[L : \mathbb{R}] = \deg f(x) = 1$. Thus $L = \mathbb{R}$. □

75.3 Lemma. *If $L/\mathbb{R}$ is an algebraic extension such that $[L : \mathbb{R}] < \infty$ then $[L : \mathbb{R}] = 2^n$ for some $n \geq 0$.*

Proof. Assume that there exists an odd prime p such that $p \mid [L : \mathbb{R}]$. Let M be a normal closure of the extension L/K. We have

$$[M : \mathbb{R}] = [M : L] \cdot [L : \mathbb{R}]$$

so $p \mid [M : \mathbb{R}]$.

Let P be the Sylow 2-subgroup of $\mathrm{Gal}(M/\mathbb{R})$. Since $M/\mathbb{R}$ is a Galois extension we have $|\mathrm{Gal}(M/\mathbb{R})| = [M : \mathbb{R}]$, and so $p \mid |\mathrm{Gal}(M/\mathbb{R})|$. On the other hand $p \nmid |P|$ so $P \neq \mathrm{Gal}(M/\mathbb{R})$. It follows that $M^P \neq M^{\mathrm{Gal}(M/\mathbb{R})} = \mathbb{R}$. Moreover we have

$$[M : \mathbb{R}] = [M : M^P] \cdot [M^P : \mathbb{R}]$$

so:

$$[M^P : \mathbb{R}] = [M : \mathbb{R}]/[M : M^P] = |\mathrm{Gal}(M/\mathbb{R})|/|P|$$

and so $[M^P : \mathbb{R}]$ is an odd number.

As a consequence we obtain that $M^P/\mathbb{R}$ is an extension of an odd degree and that $M^P \neq \mathbb{R}$. This is however impossible by Lemma 75.2.

□

Proof of Theorem 67.12.

Let $L/\mathbb{C}$ be an algebraic extension, $[L : \mathbb{C}] < \infty$, and let M be a normal closure of $L/\mathbb{C}$. Then $M/\mathbb{C}$ is a Galois extension. We have

$$[M : \mathbb{R}] = [M : \mathbb{C}] \cdot [\mathbb{C} : \mathbb{R}] = 2[M : \mathbb{C}]$$

By Lemma 75.3 we have $[M : \mathbb{R}] = 2^n$ for some n, and thus $[M : \mathbb{C}] = 2^{n-1}$.

Assume that $n - 1 > 0$. Then $\mathrm{Gal}(M/\mathbb{C})$ is a non-trivial 2-group, and so it has a normal subgroup H such that

$$\mathrm{Gal}(M/\mathbb{C})/H \cong \mathbb{Z}/2\mathbb{Z}$$

By Theorem 74.1 we obtain that $M^H/\mathbb{C}$ is a Galois extension and

$$[M^H : \mathbb{C}] = |\mathrm{Gal}(M^H/\mathbb{C})| = |\mathbb{Z}/2\mathbb{Z}| = 2$$

which is impossible by Lemma 75.1.

Therefore $[M : \mathbb{C}] = 1$, i.e. $M = \mathbb{C}$. Since $\mathbb{C} \subseteq L \subseteq M$ this also gives that $L = \mathbb{C}$. As a consequence we obtain that there are no non-trivial algebraic extensions of $\mathbb{C}$. By Proposition 67.2 this shows that $\mathbb{C}$ is an algebraically closed field. □

76 Infinite Galois extensions

76.1 Note. If L/K is a Galois extension such that $[L : K] = \infty$ then the maps

$$\Phi\colon \begin{pmatrix} \text{intermediate fields} \\ K \subseteq M \subseteq L \end{pmatrix} \leftrightarrows \begin{pmatrix} \text{subgroups} \\ H \subseteq \mathrm{Gal}(L/K) \end{pmatrix} : \Psi$$

need not be bijections since not every subgroup of of $\mathrm{Gal}(L/K)$ will be of the form $\mathrm{Gal}(L/M)$ for some intermediate field M.

76.2 Example.

Take

$$L = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \dots)$$

Then $L/\mathbb{Q}$ is a Galois extension, $[L : \mathbb{Q}] = \infty$.

If $\sigma \in \mathrm{Gal}(L/\mathbb{Q})$ then $\sigma(\sqrt{p}) = \pm\sqrt{p}$ for each prime p. This gives $\sigma^2 = \mathrm{id}_L$, so all elements of $\mathrm{Gal}(L/\mathbb{Q})$ are of order 2, and so $\mathrm{Gal}(L/\mathbb{Q})$ is an infinite dimensions vector space over $\mathbb{F}_2$. If follows that $\mathrm{Gal}(L/\mathbb{Q})$ has uncountably many subgroups H such that $[\mathrm{Gal}(L/\mathbb{Q}) : H] = 2$ (check).

Let H be such subgroup and assume that we have an intermediate field $K \subseteq M \subseteq L$ such that $\mathrm{Gal}(L/M) = H$. Since H is a normal subgroup of $\mathrm{Gal}(L/\mathbb{Q})$ we obtain that $M/\mathbb{Q}$ is a Galois extension and

$$\mathrm{Gal}(M/\mathbb{Q}) = \mathrm{Gal}(L/\mathbb{Q})/H \cong \mathbb{Z}/2\mathbb{Z}$$

In particular $[M/\mathbb{Q}] = 2 < \infty$, so by (64.2) we have $M = K(a)$ for some $a \in \mathbb{R}$, $\deg_{\mathbb{Q}}(a) = 2$. The set of such elements a is countable, so there are only countably many such intermediate fields M. As a consequence there is a subgroup $H \subseteq \mathrm{Gal}(L/\mathbb{Q})$ such that $H \neq \mathrm{Gal}(L/M)$ for any intermediate field M.

In order to fix it we need to describe, given a Galois extension L/K, which subgroups of $\mathrm{Gal}(L/K)$ are of the form $\mathrm{Gal}(L/M)$ for some intermediate field $K \subseteq M \subseteq L$.

Let L/K be an algebraic field extension and let $A = \{a_1, \dots, a_n\}$, $B = \{b_1, \dots, b_n\}$ be finite subsets of L. Denote

$$U(A, B) := \{\sigma \in \mathrm{Gal}(L/K) \mid \sigma(a_i) = b_i\}$$

76.3 Definition/Proposition. For any algebraic field extension L/K the collection of subsets

$$\mathcal{U} := \{U(A, B) \subseteq \mathrm{Gal}(L/K) \mid A, B \subseteq L,\ |A| = |B| < \infty\}$$

is a basis of some topology on $\mathrm{Gal}(L/K)$. This topology is called the *Krull topology*.

76.4 Proposition. *Let L/K be a algebraic field extension. If $K \subseteq M \subseteq L$ is an intermediate field. then $\mathrm{Gal}(L/M)$ is a closed subset of $\mathrm{Gal}(L/K)$.*

Proof. Let $\overline{\mathrm{Gal}(L/M)}$ denote the closure of $\mathrm{Gal}(L/M)$ in the Krull topology on $\mathrm{Gal}(L/K)$. We will show that $\overline{\mathrm{Gal}(L/M)} = \mathrm{Gal}(L/M)$.

Let $\psi \in \overline{\mathrm{Gal}(L/M)}$. Take an element $a \in M$ and let

$$A := \{a\}, \quad B := \{\psi(a)\}$$

Then $U(A, B)$ is an open neighborhood of ψ in $\mathrm{Gal}(L/K)$. Since $\psi \in \overline{\mathrm{Gal}(L/M)}$ there exists $\varphi \in U(A, B)$ such that $\varphi \in \mathrm{Gal}(L/M)$. We have

$$\varphi(a) = \psi(a)$$

On the other hand, since $a \in M$ we have $\varphi(a) = a$. Therefore we obtain $\psi(a) = a$. Since this holds for an arbitrary element $a \in M$ we get that $\psi \in \mathrm{Gal}(L/M)$.

$\square$

76.5 Proposition. *Let L/K be a Galois extension and let $H \subset \mathrm{Gal}(L/K)$ be a closed subgroup. Then $H = \mathrm{Gal}(L/L^H)$.*

Proof. We have $H \subseteq \mathrm{Gal}(L/L^H)$. Conversely, let $\varphi \in \mathrm{Gal}(L/L^H)$. We will show that $\varphi \in H$.

We argue by contradiction. Assume that $\varphi \notin H$. Since H is a closed subgroup there exists an open neighborhood $U(A, B) \subseteq \mathrm{Gal}(L/K)$ such that $\varphi \in U(A, B)$ and $H \cap \mathrm{Gal}(L/L^H) = \varnothing$.

Let $A = \{a_1, \dots, a_n\}$ and let $N \subseteq L$ be the normal closure of $L^H(a_1, \dots, a_n)$. Since N/L^H is a normal extension for any $\psi \in \mathrm{Gal}(L/L^H)$ we have $\psi(N) = N$. Define

$$H' := \{\psi|_N \colon N \to N \mid \psi \in H\} \subseteq \mathrm{Gal}(N/L^H)$$

Notice that $\varphi|_N \notin H'$, so $H' \neq \mathrm{Gal}(N/L^H)$. Since N/L^H is a Galois extension and $[N : L^H] < \infty$ by the Fundamental Theorem of Galois Theory (74.1) we obtain

$$N^{H'} \neq N^{\mathrm{Gal}(N/L^H)} = L^H$$

On the other hand if $a \in N^{H'}$ then $\psi(a) = a$ for all $\psi \in H$, so $a \in L^H$. Thus $N^{H'} \subseteq L^H$. Since obviously we also have $L^H \subseteq N^{H'}$ we get $N^{H'} = L^H$ which is a contradiction.

$\square$

76.6 Corollary. *If L/K is a Galois extension then the maps*

$$\Phi\colon \begin{pmatrix} \textit{intermediate fields} \\ K \subseteq M \subseteq L \end{pmatrix} \leftrightarrows \begin{pmatrix} \textit{closed subgroups} \\ H \subseteq \mathrm{Gal}(L/K) \end{pmatrix} : \Psi$$

given by $\Phi(M) = \mathrm{Gal}(L/M)$, $\Psi(H) = L^H$ are bijections and $\Psi = \Phi^{-1}$.

76.7 Note. If L/K is a Galois extension such that $[L : K] < \infty$ then Krull topology on $\mathrm{Gal}(L/K)$ is the discrete topology (check!), and so any subgroup of $\mathrm{Gal}(L/K)$ is a closed subgroup.

77 Abelian and cyclic extensions

77.1 Definition. Let L/K be a Galois extension, $[L : K] < \infty$. The extension L/K is *abelian* (resp. *cyclic*) if the group $\mathrm{Gal}(L/K)$ is abelian (resp. cyclic).

77.2 Proposition. *If L/K is an abelian (resp. cyclic) extension and M is an intermediate field then L/M and M/K are also abelian (resp. cyclic) extensions.*

Proof. Exercise. $\square$

77.3 Definition. An extension L/K is a *cyclotomic extension of degree n* if L is a splitting field of the polynomial $f(x) = x^n - 1 \in K[x]$.

77.4 Note. If $\chi(K) = p \neq 0$ and $n = mp^k$ then $x^n - 1 = (x^m - 1)^{p^k}$, so cyclotomic extensions of orders n and m coincide. Thus we can always assume that $p \nmid n$.

77.5 Theorem. *Let L/K be a cyclotomic extension of degree n. If $\chi(K) = p \neq 0$ assume that $p \nmid n$. Then*

1) $L = K(\zeta)$ *where ζ is a primitive n-th root of unity;*

2) $\mathrm{Gal}(L/K)$ *is isomorphic to a subgroup of $(\mathbb{Z}/n\mathbb{Z})^*$, the multiplicative of units of the ring $\mathbb{Z}/n\mathbb{Z}$.*

Proof.

1) Since L is a splitting field of $f(x) = x^n - 1$ thus L contains a primitive n-th root of unity. Moreover, every other root of $f(x) = x^n - 1$ in L is of the form ζ^k for some $1 \leq k \leq n$, so $L = K(\zeta)$.

2) If $\varphi \in \mathrm{Gal}(L/K)$ then φ is uniquely determined determined by the value of $\varphi(\zeta) \in L$. Since ζ is a primitive n-th root of unity $\varphi(\zeta)$ also must be a primitive n-th root of unity. This means that $\varphi(\zeta) = \zeta^k$ where $1 \leq k \leq n$, $\gcd(n, k) = 1$.

Define a map

$$\Phi\colon \mathrm{Gal}(L/K) \to (\mathbb{Z}/n\mathbb{Z})^*$$

where $\Phi(\varphi) = k$ if $\varphi(\zeta) = \zeta^k$. Then Φ is a monomorphism and it is a homomorphism of groups (check!). It follows that $\mathrm{Gal}(L/K)$ is isomorphic to the subgroup $\Phi(\mathrm{Gal}(L/K)) \subseteq (\mathbb{Z}/n\mathbb{Z})^*$.

□

77.6 Note. If L is the cyclotomic extension of $\mathbb{Q}$ of degree n then

$$\mathrm{Gal}(L/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^*$$

(see Hungerford Ch. V, Proposition 8.3).

77.7 Corollary. *If L/K is a cyclotomic extension of degree n then L/K is an abelian extension. Moreover, if n is a prime number and $\chi(K) \nmid n$ then L/K is a cyclic extension.*

Proof. By Theorem 77.5 the group $\mathrm{Gal}(L/K)$ can be identified with a subgroup of $(\mathbb{Z}/n\mathbb{Z})^*$. Since the group $(\mathbb{Z}/n\mathbb{Z})^*$ is abelian thus so is $\mathrm{Gal}(L/K)$.

If n is prime then $(\mathbb{Z}/n\mathbb{Z})^*$ is a cyclic group (see Proposition 38.8), so $\mathrm{Gal}(L/K)$ is cyclic. □

77.8 Theorem (Kronecker-Weber)**.**
If $L/\mathbb{Q}$ is an abelian extension then $L \subseteq M$ where $M/\mathbb{Q}$ is some cyclotomic extension of $\mathbb{Q}$.

Proof. See e.g.:

M.J. Greenberg, *An elementary proof of the Kronecker-Weber theorem*, The American Mathematical Monthly 81(6) (1974), 601-607.

(with erratum in The American Mathematical Monthly 82(8) (1975), 803)

$\square$

77.9 Theorem. *Let K be a field such that K contains a primitive n-th root of unity ζ and let L/K be a field extension.*

1) If L/K is a cyclic extension and $[L : K] = n$ then $L = K(b)$ for some $b \in L$ such that $b^n \in K$

2) If $L = K(b)$ for some $b \in L$ such that $b^n \in K$ then L/K is a cyclic extension.

77.10 Lemma. *Let L be a field and let $\varphi_1, \dots, \varphi_n$ be distinct automorphisms of L. For $c_1, \dots, c_n \in L$ consider the map*

$$\psi \colon L \to L, \qquad \psi(a) = \sum_{i=1}^{n} c_i \varphi_i(a)$$

If $\varphi(a) = 0$ for all $a \in L$ then $c_1 = \cdots = c_n = 0$

Proof. We argue by induction with respect to n.

If $n = 1$ the statement is obvious.

Assume that the statement of the lemma holds for some n, and let $\varphi_1, \dots, \varphi_{n+1} \in \mathrm{Aut}(L)$, $c_1, \dots, c_{n+1} \in L$ be such that

$$\sum_{i=1}^{n+1} c_i \varphi_i(a) = 0$$

for all $a \in L$. For any $a, b \in L$, $a \neq 0$ we have

$$\begin{aligned} 0 &= \sum_{i=1}^{n+1} c_i \varphi_i(ab) \\ &= \sum_{i=1}^{n+1} c_i \varphi_i(a)\varphi_i(b) \end{aligned}$$

Since $a \neq 0$ we have $\varphi_{n+1}(a) \neq 0$ so we have

$$\begin{aligned} 0 &= \frac{1}{\varphi_{n+1}(a)} \sum_{i=1}^{n+1} c_i \varphi_i(a)\varphi_i(b) \\ &= \sum_{i=1}^{n+1} c_i \frac{\varphi_i(a)}{\varphi_{n+1}(a)} \varphi_i(b) \end{aligned}$$

Since also $\sum_i c_i \varphi_i(b) = 0$ we obtain

$$\begin{aligned} 0 &= \sum_{i=1}^{n+1} c_i \frac{\varphi_i(a)}{\varphi_{n+1}(a)} \varphi_i(b) - \sum_i c_i \varphi_i(b) \\ &= \sum_{i=1}^{n+1} c_i \left(\frac{\varphi_i(a)}{\varphi_{n+1}(a)} - 1 \right) \varphi_i(b) \\ &= \sum_{i=1}^{n} c_i \left(\frac{\varphi_i(a)}{\varphi_{n+1}(a)} - 1 \right) \varphi_i(b) \end{aligned}$$

By the inductive assumption this gives

$$c_i \left(\frac{\varphi_i(a)}{\varphi_{n+1}(a)} - 1 \right) = 0$$

for all $i = 1, \dots, n$ and all $0 \neq a \in L$. If $c_i \neq 0$ for some $1 \leq i \leq n$ this would give

$$\varphi_i(a) = \varphi_{n+1}(a)$$

for all $a \in L$ which is impossible since $\varphi_1, \dots, \varphi_{n+1}$ are distinct automorphisms. Thus we obtain $c_1 = \dots = c_n = 0$, and so also $c_{n+1} = 0$. □

Proof of Theorem 77.9.

1) Assume that L/K is a cyclic extension, $[L : K] = n$, and let φ be a generator of $\mathrm{Gal}(L/K)$. Let $\psi\colon L \to L$ be the map given by

$$\psi(a) = \sum_{i=1}^{n} \zeta^i \varphi^i(a)$$

By Lemma 77.10 there exists $a \in L$ such that $\psi(a) \neq 0$.

Notice that we have

$$\begin{aligned}\varphi(\psi(a)) &= \sum_{i=1}^{n} \zeta^i \varphi^{i+1}(a) \\ &= \zeta^{-1} \sum_{i=1}^{n} \zeta^{i+1} \varphi^{i+1}(a) \\ &= \zeta^{-1}\psi(a)\end{aligned}$$

By induction we obtain

$$\varphi^k(\psi(a)) = \zeta^{-k}\psi(a)$$

for all k.

Claim 1. $L = K(\psi(a))$.

Indeed, it is enough to show that $\deg_K(\psi(a)) = [L : K] = n$.

Let $f(x) = \mathrm{irr}^K_{\psi(a)}(x)$. By (70.1) for any $1 \leq k \leq n$ the element $\varphi^k(\psi(a))$ is a root of $f(x)$. Moreover, if $1 \leq k, l \leq n$ and $k \neq l$ then we have

$$\varphi^k(\psi(a)) = \zeta^{-k}\psi(a) \neq \zeta^{-l}\psi(a) = \varphi^l(\psi(a))$$

so $f(x)$ has n distinct roots in L. It follows that $\deg f(x) \geq n$, and so $\deg_K(\psi(a)) \geq n$. Also, since $\psi(a) \in L$ we have $\deg_K(\psi(a)) \leq [L : K] = n$. Therefore $\deg_K(\psi(a)) = n$.

Claim 2. $\psi(a)^n \in K$.

Indeed, notice that since L/K is a Galois extension we have $K = L^{\mathrm{Gal}(L/K)}$. Also, since φ generates $\mathrm{Gal}(L/K)$ we have

$$K = L^{\mathrm{Gal}(L/K)} = \{c \in L \mid \varphi(c) = c\}$$

Thus it suffices to show that $\varphi(\psi(a)^n) = \psi(a)^n$. We have

$$\varphi(\psi(a)^n) = \varphi(\psi(a))^n = (\zeta^{-1}\psi(a))^n = \zeta^{-n}\psi(a)^n = \psi(a)$$

From Claims 1 and 2 it follows that in the statement of the theorem we can take $b = \psi(a)$.

2) Assume that $L = K(b)$ for some $b \in L$ such that $b^n \in K$. We need to show that L/K is a Galois extension and that $\mathrm{Gal}(L/K)$ is a cyclic group.

Consider the polynomial $f(x) = x^n - b^n \in K[x]$. Notice that $f(x)$ has n distinct roots $b, \zeta b, \dots, \zeta^{n-1}b \in L$. It follows that L is a splitting field of $f(x)$ and that $f(x)$ is a separable polynomial. As a consequence L/K is a Galois extension.

Next, notice that any automorphism $\varphi \in \mathrm{Gal}(L/K)$ is uniquely determined by the value of $\varphi(b)$, and that $\varphi(b) = \zeta^k b$ for some $0 \leq k \leq n-1$.

Let $\mu_n(K)$ denote the multiplicative group of n-th roots of unity in K (68.2). Define a map

$$\Phi\colon\ \mathrm{Gal}(L/K) \to \mu_n(K)$$

by $\Phi(\varphi) = \zeta^k$ if $\varphi(b) = \zeta^k b$. Then Φ is a monomorphism, and it is a homomorphism of groups (check!). It follows that $\mathrm{Gal}(L/K)$ is isomorphic to a subgroup of $\mu_n(K)$. By (68.4) the group $\mu_n(K)$ is cyclic, and so $\mathrm{Gal}(L/K)$ is also a cyclic group.

$\square$

78 Radical extensions

78.1 Definition. A field extension L/K is a *radical extension* if there exist elements $a_1, \dots, a_n \in L$ such that $L = K(a_1, \dots, a_n)$ and for every $i = 1, \dots, n$ there exists $m_i \geq 1$ such that

$$a_i^{m_i} \in K(a_1, \dots, a_{i-1})$$

78.2 Definition. Let $\bar{K}$ be an algebraic closure of a field K. An element $a \in \bar{K}$ *can be expressed by radicals* over K if $a \in L$ for some radical extension L/K.

78.3 Note. An element $a \in \bar{K}$ can expressed by radicals over K iff a can be obtained from elements of K using arithmetic operators $+$, $-$, $\cdot$, $/$ and root extraction.

78.4 Definition. A polynomial $f(x) \in K[x]$ *can be solved by radicals* if all roots of $f(x)$ can be expressed by radicals over K.

78.5 Proposition. *If $f(x) \in K[x]$ and L is a splitting field of $f(x)$ then $f(x)$ can be solved by radicals iff $L \subseteq M$ where M/K is a radical extension.*

78.6 Lemma. *If $\bar{K}$ is an algebraic closure of K and L/K, M/K are radical extensions such that $L, M \subseteq \bar{K}$ then LM/K is also a radical extension.*

Proof. Exercise. $\square$

Proof of Proposition 78.5.

($\Rightarrow$) Let $L = K(a_1, \dots, a_n)$ where $a_1, \dots, a_n$ are roots of $f(x)$. Let $\bar{K}$ be an algebraic closure of K such that $L \subseteq \bar{K}$. By assumption for every a_i there exists $L_i \subseteq \bar{K}$ such that $a_i \in L_i$ and L_i/K is a radical extension. Then we have

$$L = K(a_1, \dots, a_n) \subseteq L_1 L_2 \cdots L_n$$

and by Lemma 78.6 $L_1 L_2 \cdots L_n/K$ is a radical extension.

($\Leftarrow$) Obvious.

$\square$

78.7 Proposition. *Let L/K be a radical extension. If N is a normal closure of L/K then N/K is also a radical extension.*

Proof. Let N be a normal closure of of L/K. We have:

$$N = \prod_{\varphi \in \mathrm{Gal}(N/K)} \varphi(L)$$

(exercise). Since L/K is a radical extension thus for every $\varphi \in \mathrm{Gal}(L/K)$ the extension $\varphi(L)/K$ is also a radical extension (check!). By Lemma 78.6 it follows that that $\left(\prod_{\varphi \in \mathrm{Gal}(N/K)} \varphi(L)\right)/K$ is also a radical extension. $\square$

79 Solvable extensions

Recall.

- A group G is *solvable* if it has a normal series
$$\{e\} = G_0 \subseteq \ldots \subseteq G_k = G$$
such that for every i the group G_i/G_{i-1} is abelian.

- Abelian groups and finite p-groups are solvable.

- The symmetric group S_n is solvable iff $n \leq 4$.

- Subgroups and quotients groups of a solvable group are solvable.

- If $H \lhd G$ and H, G/H are solvable groups then G is solvable.

79.1 Definition. A Galois extension L/K is *solvable* if $[L : K] < \infty$ and $\mathrm{Gal}(L/K)$ is a solvable group.

79.2 Theorem. *Let L/K be a Galois extension such that $[L : K] < \infty$ and $\chi(K) = 0$. The following conditions are equivalent:*

(i) There exists a radical extension M/K such that $L \subseteq M$.

(ii) L/K is a solvable extension.

79.3 Note. Theorem 79.2 holds also if $\chi(K) = p \neq 0$ and $p \nmid [L : K]$.

79.4 Corollary. *If K is a field, $\chi(K) = 0$, $f(x) \in K[x]$ and L is a splitting field of $f(x)$ then $f(x)$ can be solved by radicals iff L/K is a solvable extension.*

Proof. Follows from Proposition 78.5 and Theorem 79.2. ☐

Proof of Theorem 79.2.

(i) $\Rightarrow$ (ii) It is enough to show that for any radical Galois extension N/K the group $\mathrm{Gal}(N/K)$ is solvable.

Indeed, if $L \subseteq M$ where M/K is a radical extension then let N be a normal closure of M/K. By Proposition 78.7 N/K is a radical Galois extension. We have

$$K \subseteq L \subseteq N$$

Since L/K is a Galois extension the group $\mathrm{Gal}(N/L)$ is a normal subgroup of $\mathrm{Gal}(N/K)$, and

$$\mathrm{Gal}(L/K) \cong \mathrm{Gal}(N/K)/\,\mathrm{Gal}(N/L)$$

Therefore if $\mathrm{Gal}(N/K)$ is solvable then so is $\mathrm{Gal}(L/K)$.

Assume then that N/K is a radical Galois extension. Then we have $a_1, \ldots, a_n \in N$ such that

$$N = K(a_1, \ldots, a_n)$$

and $a_i^{m_i} \in K(a_1, \ldots, a_{i-1})$ for some $m_i \geq 1$. Denote $L_i := K(a_1, \ldots, a_i)$. We have

$$K = L_0 \subseteq L_1 \subseteq \ldots \subseteq L_n = N$$

This gives

$$\mathrm{Gal}(N/K) = \mathrm{Gal}(N/L_0) \supseteq \mathrm{Gal}(N/L_1) \supseteq \ldots \supseteq \mathrm{Gal}(N/L_n) = \{\mathrm{id}_N\} \qquad (*)$$

Let $\zeta \in \bar{K}$ be a primitive root of unity of order $m_1 m_2 \cdots m_n$. Assume first that $\zeta \in K$.

Claim. For every $i = 1, \ldots, n$ the group $\mathrm{Gal}(N/L_i)$ is a normal subgroup of $\mathrm{Gal}(N/L_{i-1})$ and $\mathrm{Gal}(N/L_{i-1})/\,\mathrm{Gal}(N/L_i)$ is a cyclic group.

Indeed, $L_i = L_{i-1}(a_i)$ and $a_i^{m_i} \in L_{i-1}$, and L_{i-1} contains a primitive root of unity of degree m_i, so by Theorem 77.9 L_i/L_{i-1} is a cyclic Galois extension. We have

$$L_{i-1} \subseteq L_i \subseteq N$$

Since L_i/L_{i-1} is a Galois extension thus $\mathrm{Gal}(N/L_i)$ is a normal subgroup of $\mathrm{Gal}(N/L_{i-1})$ and

$$\mathrm{Gal}(N/L_{i-1})/\mathrm{Gal}(N/L_i) \cong \mathrm{Gal}(L_i/L_{i-1})$$

As a consequence we obtain that the sequence $(*)$ is a normal series of $\mathrm{Gal}(N/K)$ with cyclic quotients, and so $\mathrm{Gal}(N/K)$ is a solvable group.

Next, assume that K does not contain the primitive root ζ. In such case consider the extensions

$$K \subseteq K(\zeta) \subseteq N(\zeta)$$

The extensions $N(\zeta)/K$, $K(\zeta)/K$ are Galois extensions (check!), so we have

$$\mathrm{Gal}(N(\zeta)/K(\zeta)) \lhd \mathrm{Gal}(N(\zeta)/K)$$

and

$$\mathrm{Gal}(K(\zeta)/K) \cong \mathrm{Gal}(N(\zeta)/K)/\mathrm{Gal}(N(\zeta)/K(\zeta))$$

By Theorem 77.5 the group $\mathrm{Gal}(K(\zeta)/K)$ is abelian. Also, since $N(\zeta)/K(\zeta)$ is a radical extension thus by the Claim above the group $\mathrm{Gal}(N(\zeta)/K(\zeta))$ is solvable. As a consequence $\mathrm{Gal}(N(\zeta)/K)$ is also a solvable group. Finally we have

$$K \subseteq N \subseteq N(\zeta)$$

and since N/K is a Galois extension we have

$$\mathrm{Gal}(N/K) \cong \mathrm{Gal}(N(\zeta)/K)/\mathrm{Gal}(N(\zeta)/N)$$

so $\mathrm{Gal}(N/K)$ is also a solvable group.

(ii) $\Rightarrow$ (i) Let L/K be a solvable extension. We want to show that $L \subseteq M$ where M/K is a radical extension.

Let $[L : K] = m$ and let $\zeta \in \bar{L}$ be a primitive root of unity of degree m. Assume first that $\zeta \in K$.

Since $\mathrm{Gal}(L/K)$ is a solvable group we have a normal series

$$\mathrm{Gal}(L/K) = G_0 \supseteq G_1 \supseteq \ldots \supseteq G_n = \{\mathrm{id}_L\}$$

such that G_i/G_{i+1} is a cyclic group for all i. Take $L_i := L^{G_i}$. This gives a sequence of field extensions:

$$K = L_0 \subseteq L_1 \subseteq \ldots \subseteq L_n = L$$

such that $G_i = \mathrm{Gal}(L/L_i)$.

Consider the extensions $L_i \subseteq L_{i+1} \subseteq L$. Since $G_{i+1} = \mathrm{Gal}(L/L_{i+1})$ is a normal subgroup of $G_i = \mathrm{Gal}(L/L_i)$ thus L_{i+1}/L_i is a Galois extension, and $\mathrm{Gal}(L_{i+1}/L_i) \cong G_i/G_{i+1}$ is a cyclic group. If $[L_{i+1} : L_i] = m_{i+1}$ then $m_{i+1} \mid m$, and so L_i contains a primitive root of unity of degree m_{i+1}. By Theorem 77.9 there exists $b_{i+1} \in L_{i+1}$ such that $L_{i+1} = L_i(b_{i+1})$ and $b_{i+1}^{m_{i+1}} \in L_i$. As a consequence L/K is a radical extension.

If $\zeta \notin K$ then consider the extension $L(\zeta)/K(\zeta)$. This extension is solvable (exercise) and $[L(\zeta) : K(\zeta)] \mid [L : K]$. By the argument above we obtain that $L(\zeta)/K(\zeta)$ is a radical extension. Since, $K(\zeta)/K$ is also radical extension we obtain that $L(\zeta)/K$ is a radical extension and $L \subseteq L(\zeta)$. Therefore we can take $M = L(\zeta)$.

$\square$

79.5 Note. Theorem 79.2 can be modified to hold when $\chi(K) = p \neq 0$ as follows. Let L/K be a Galois extension such that $[L : K] < \infty$.

1) If L/K is a radical extension then it is a solvable extension.

2) If L/K is a solvable extension then there exist elements $a_1, \ldots, a_n \in L$ such that $L = K(a_1, \ldots, a_n)$ and for each $i = 1, \ldots, n$ we have either

$$a_i^{m_i} \in K(a_1, \ldots, a_{i-1})$$

for some $m_i \geq 1$, or

$$a_i^p - a_i \in K(a_1, \ldots, a_{i-1})$$

(exercise).

80 Solvability of polynomials by radicals

80.1 Theorem. *If K is a field, $\chi(K) = 0$ and $f(x) \in K[x]$ is a polynomial such that $\deg f(x) \leq 4$ then $f(x)$ is solvable by radicals.*

80.2 Lemma. *If G is a group such that $|G| < 60$ then G is a solvable group.*

Proof. By the Jordan – Hölder Theorem (19.10) we have a composition series

$$\{e\} \subseteq G_1 \subseteq \ldots \subseteq G_k = G$$

The groups G_i/G_{i-1} are simple and $|G_i/G_{i-1}| < 60$. By Theorem 20.4 it follows that the groups G_i/G_{i-1} are abelian and so G is a solvable group. □

Proof of Theorem 80.1. Let L be a splitting field of $f(x)$. Since $\deg f(x) = 4$ we have

$$[L : K] \leq 4! = 24$$

so by Lemma 80.2 $\mathrm{Gal}(L/K)$ is a solvable group. Therefore by Corollary 79.4 $f(x)$ is solvable by radicals.

□

Note. See e.g. Section 4.1 of

J. J. Rotman *Advanced modern algebra*, Prentice Hall, 2002

for explicit formulas for roots of polynomials $f(x) \in \mathbb{Q}[x]$, $\deg f(x) \leq 4$.

Next goal. There are polynomials $f(x) \in \mathbb{Q}[x]$, $\deg f(x) > 4$ that are not solvable by radicals.

80.3 Lemma. *Let p be a prime number and let G be a subgroup of the symmetric group S_p. If $p \mid |G|$ and G contains any transposition then $G = S_p$.*

Proof. Exercise. □

80.4 Proposition. *Let p be a prime number and let K be a subfield of $\mathbb{R}$. Let $f(x) \in K[x]$ be an irreducible polynomial such that $\deg f(x) = p$, and let $L \subseteq \mathbb{C}$ be a splitting field of $f(x)$. If $f(x)$ has exactly $p-2$ roots in $\mathbb{R}$ then*

$$\mathrm{Gal}(L/K) \cong S_p$$

Proof. Let $a_1, \dots, a_{p-2} \in L$ be the real roots of $f(x)$, and let $a_{p-1}, a_p \in L$ be the complex roots. Notice that a_p is the complex conjugate of a_{p-1}, i.e. $a_p = \bar{a}_{p-1}$. We have

$$L = K(a_1, \dots, a_p)$$

and $\mathrm{Gal}(L/K)$ can be identified with a subgroup of the group of permutations of the set $\{a_1, \dots, a_p\}$. Since $f(x)$ is an irreducible polynomial of degree p we get that p divides $[L : K] = |\mathrm{Gal}(L/K)|$ (check!). By Lemma 80.3 it will suffice to show that $\mathrm{Gal}(L/K)$ contains a transposition of the set $\{a_1, \dots, a_p\}$.

Take the map

$$\varphi\colon \mathbb{C} \to \mathbb{C}, \quad \varphi(z) = \bar{z}$$

Then φ is a field homomorphism. We have $\varphi(a_i) = a_i$ for $1 \leq i \leq p-2$, $\varphi(a_{p-1}) = a_p$, and $\varphi(a_p) = a_{p-1}$. If follows that $\varphi|_L$ is an automorphism of L. Since $K \subseteq \mathbb{R}$ we also have $\varphi|_K = \mathrm{id}_K$. Therefore $\varphi|_L \in \mathrm{Gal}(L/K)$ and $\varphi|_L$ corresponds to the transposition (a_{p-1}, a_p) of the set $\{a_1, \dots, a_p\}$. □

80.5 Lemma. *Let $f(x) \in \mathbb{R}[x]$ be a polynomial of the form*

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

If $a_{n-1} = a_{n-2} = 0$ and all roots of $f(x)$ are real numbers then $f(x) = x^n$.

Proof. We have

$$f(x) = (x - b_1)(x - b_2) \cdot \ldots (x - b_n)$$

for some $b_i \in \mathbb{R}$. Then

$$\sum_i b_i = a_{n-1} = 0, \qquad \sum_{i<j} b_i b_j = a_{n-2} = 0$$

This gives

$$0 = \left(\sum_i b_i\right)^2 = \sum_i b_i^2 \ + \ 2\sum_{i<j} b_i b_j = \sum_i b_i^2$$

Therefore $b_i = 0$ for $i = 1, \ldots, n$ and so $f(x) = x^n$. □

80.6 Proposition. *Let q be a prime number and let L be a splitting field of the polynomial*

$$f(x) = x^5 - 2qx - q \in \mathbb{Q}[x]$$

Then $\mathrm{Gal}(L/\mathbb{Q}) = S_5$. As a consequence $f(x)$ cannot be solved by radicals over $\mathbb{Q}$.

Proof. By the Eisenstein Irreducibility Criterion (38.12) $f(x)$ is irreducible in $\mathbb{Q}[x]$. By Proposition 80.4 it is enough to show that $f(x)$ has exactly 3 roots in $\mathbb{R}$.

By Lemma 80.5 $f(x)$ has at least 2 roots in $\mathbb{C} - \mathbb{R}$, and so at most 3 roots in $\mathbb{R}$. Moreover we have

$$\begin{aligned}
&f(-q) = -q(q^4 - 2q + 1) < 0 \\
&f(-1) = q - 1 > 0 \\
&f(0) = -1 < 0 \\
&f(q) = q(q^4 - 2q + 1) > 0
\end{aligned}$$

Thus $f(x)$ has a real root in each of the intervals $(-q, -1)$, $(-1, 0)$, and $(0, q)$. It follows that $f(x)$ has exactly 3 real roots. □

Propostion 80.6 can be generalized as follows:

80.7 Proposition. *Let p, q be prime numbers such that $p \geq 5$ and let L be a splitting field of the polynomial*

$$f(x) = x^p - 2qx - q \in \mathbb{Q}[x]$$

Then $\mathrm{Gal}(L/\mathbb{Q})$ is not a solvable group and as a consequence $f(x)$ cannot be solved by radicals over $\mathbb{Q}$.

80.8 Lemma. *Let G be a solvable subgroup of S_p that acts transitively on the set $\{1, \ldots, p\}$. If $1 \neq \sigma \in G$ then $\sigma(i) = i$ for at most one $i \in \{1, \ldots, p\}$.*

Proof. Exercise. □

80.9 Lemma. *Let K be a field, let $f(x) \in K[x]$ be an irreducible, separable polynomial, and let L be a splitting field of $f(x)$. If $\deg f(x)$ is a prime number and $\mathrm{Gal}(L/K)$ is a solvable group then $L = K(a_1, a_2)$ where $a_1, a_2 \in L$ are any two distinct roots of $f(x)$.*

Proof. Let $\deg f(x) = p$ and let $\{a_1, \ldots, a_p\} \subseteq L$ be the set of all roots of $f(x)$. Then $\mathrm{Gal}(L/K)$ can be identified with a subgroup of the group of permutations of $\{a_1, \ldots, a_p\}$. Also, since $f(x)$ is an irreducible polynomial $\mathrm{Gal}(L/K)$ acts transitively on the set $\{a_1, \ldots, a_p\}$ (check!).

Take $\varphi \in \mathrm{Gal}(L/K(a_1, a_2))$. We have $\varphi(a_i) = a_i$ for $i = 1, 2$. By Lemma 80.8 we get $\varphi = \mathrm{id}_L$. Therefore $\mathrm{Gal}(L/K(a_1, a_2)) = \{\mathrm{id}_L\}$. Since $L/K(a_1, a_2)$ is a Galois extension we obtain

$$K(a_1, a_2) = L^{\mathrm{Gal}(L/K(a_1, a_2))} = L$$

□

80.10 Lemma. *Let K be a field, let $f(x) \in K[x]$ be an irreducible, separable polynomial, and let L be a splitting field of $f(x)$. If $p = \deg f(x)$ is a prime number and $\mathrm{Gal}(L/K)$ is a solvable group then any intermediate field*

$$K \subseteq M \subseteq L$$

contains either 0, 1, or p roots of $f(x)$.

Proof. By Lemma 80.9 if $f(x)$ contains two roots of $f(x)$ then $M = L$, and so M contains all roots of $f(x)$. $\square$

Proof of Proposition 80.7.

The polynomial $f(x)$ is irreducible in $\mathbb{Q}[x]$ by the Eisenstein Irreducibility Criterion 38.12). Let $L \subseteq \mathbb{C}$ be a splitting field of $f(x)$ and let $M = L \cap \mathbb{R}$.

If $\mathrm{Gal}(L/K)$ were a solvable group then by Lemma 80.10 $f(x)$ would have either 0, 1, or p roots in M. By Lemma 80.5 $f(x)$ can't have all real roots thus the number of roots in M would have to be either 0 or 1. On the other hand we have

$$\begin{aligned} f(-q) &= -q^p + 2q^2 - q < 0 \\ f(-1) &= q - 1 > 0 \\ f(0) &= -1 < 0 \end{aligned}$$

so $f(x)$ has a real root in each of the intervals $(-q, -1)$ and $(-1, 0)$. It follows that $\mathrm{Gal}(L/K)$ cannot be a solvable group. $\square$

81 Straightedge and compass constructions

81.1. Motivation. Three great problems of antiquity.

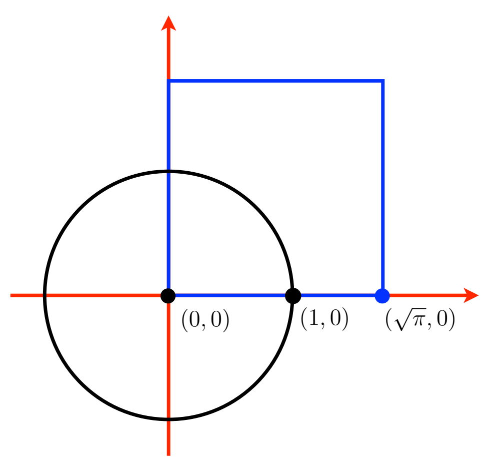
1) Squaring of a circle. Using a straightedge and a compass construct a square whose area is equal to the area of a given circle.

2) Doubling of a cube. Using a straightedge and a compass construct a cube whose volume is double the volume of a given cube.

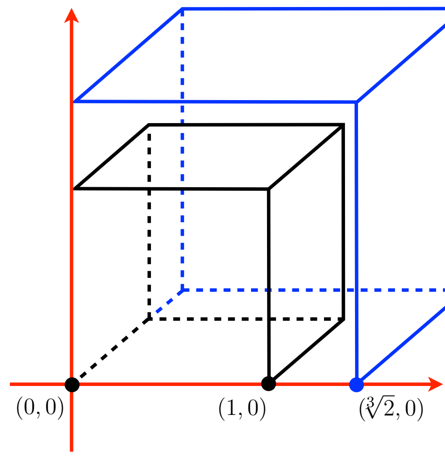
3) Trisection of an angle Using a straightedge and a compass construct an angle whose measure is one third of the measure of a given angle.

Cartesian reformulation.

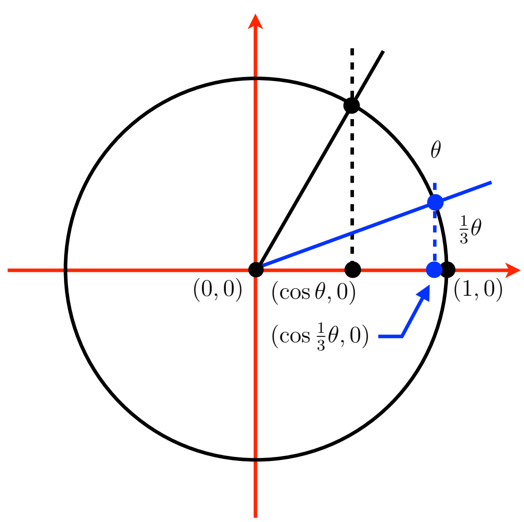
1) Squaring of a circle. We can assume that the given circle has its center at the point $(0,0)$ and that it passes through the point $(1,0)$. Squaring of this circle amounts to constructing the point with coordinates $(\sqrt{\pi}, 0)$



2) Doubling of a cube. We can assume that two vertices of the front face of the cube have coordinates $(0,0)$ and $(1,0)$. Doubling of this cube amounts to constructing the point with coordinates $(\sqrt[3]{2}, 0)$



3) Trisection of an angle. We can assume that one of the arms of the given angle θ coincides with the x-axis. Being given such an angle is equivalent to being given the point with coordinates $(\cos\theta, 0)$. Trisection of this angle is equivalent to constructing the point with coordinates $(\cos\frac{1}{3}\theta, 0)$



81.2. General constructibility problem. Assume that on the coordinate plane $\mathbb{R}^2$ we are given points a set of points

$$S_0 = \{P_1, \ldots, P_n\}$$

where $P_i = (a_i, b_i)$. Assume also that $(0,0), (1,0) \in S_0$. We can perform two operations:

- draw a line through any two points in S_0
- draw a circle with the center in one point of S_0 and passing through another point of S_0.

Let S_1 be the set of all intersection points of these lines and circles. Recursively, for each $k \geq 1$ let S_k be the set of all points obtained in this way from the set S_{k-1}. Denote

$$S = \bigcup_{k=0}^{\infty} S_k$$

81.3 Definition. A point $Q \in \mathbb{R}^2$ is *constructible from the set* S_0 if $Q \in S$. A point $Q \in \mathbb{R}^2$ is *constructible* if it is constructible from the set $\{(0,0), (1,0)\}$.

Problem. Given a set $S_0 \in \mathbb{R}^2$ determine which points of $\mathbb{R}^2$ are constructible from S_0.

81.4 Note.

1) Squaring of the circle can be performed if the point $(\sqrt{\pi}, 0)$ is constructible.

2) Doubling of a cube can be performed if the point $(\sqrt[3]{2}, 0)$ is constructible.

3) Trisection of the angle θ can be performed if the point $(\cos\frac{1}{3}\theta, 0)$ is constructible from the set $\{(0,0), (1,0), (\cos\theta, 0)\}$.

Constructible points and radical extensions.

81.5 Definition. A field extension L/K is a *degree* 2 *radical extension* if there exist elements $a_1, \dots, a_n \in L$ such that $L = K(a_1, \dots, a_n)$ and for every $i = 1, \dots, n$ we have
$$a_i^2 \in K(a_1, \dots, a_{i-1})$$

81.6 Theorem. *Let $S_0 = \{P_1, \dots, P_n\}$ be set of points in $\mathbb{R}^2$ where $P_i = (a_i, b_i)$. Let*
$$K = \mathbb{Q}(a_1, b_1, \dots, a_n, b_n)$$
A point $Q = (c, d)$ is constructible from the set S_0 iff there exists a degree 2 radical extension L/K, $L \subseteq \mathbb{R}$ such that $c, d \in L$.

In particular a point $Q = (c, d)$ is constructible iff there exists a degree 2 radical extension $L/\mathbb{Q}$, $L \subseteq \mathbb{C}$ such that $c, d \in L$.

Proof. Exercise. □

81.7 Corollary. *Let $S_0 = \{P_1, \dots, P_n\}$ be a set of points in $\mathbb{R}^2$ where $P_i = (a_i, b_i)$. and let*
$$K = \mathbb{Q}(a_1, b_1, \dots, a_n, b_n)$$

If a point $Q = (c, d)$ is constructible then $[K(c, d) : K] = 2^k$ for some $k \geq 1$.

Proof. By Theorem 81.6 we have $K(c, d) \subseteq L$ where L is some degree 2 radical extension of K. We have
$$[L : K] = 2^m$$
for some $m \geq 1$, so $[K(c, d) : K] = 2^k$ for some $1 \geq k \geq m$.

□

81.8 Example.

Since $\sqrt{\pi}$ is a transcendental number thus $\sqrt{\pi} \notin L$ for any degree 2 radical extension of $\mathbb{Q}$. It follows that the point $(\sqrt{\pi}, 0)$ is not constructible and it is impossible to square a circle using straightedge and compass.

81.9 Example.

We have $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$. By Corollary 81.7 we obtain that the point $(\sqrt[3]{2}, 0)$ is not constructible. As a consequence it is impossible to double a cube using straightedge and compass.

81.10 Example.

Recall that we can perform a trisection of an angle θ iff the point $(0, \cos \frac{1}{3}\theta)$ can be constructed from the set $S_0 = \{(0,0), (1,0), (\cos\theta, 0)\}$. By Theorem 81.6 this is possible iff $\cos\frac{1}{3}\theta \in L$ for some degree 2 radical extension L of the field $K := \mathbb{Q}(\cos\theta)$.

Claim. $\cos\frac{1}{3}\theta \in L$ where L is some degree 2 radical extension of K iff the polynomial

$$f(x) = x^3 - \frac{3}{4}x - \frac{1}{4}\cos\theta$$

is not irreducible in $K[x]$.

Indeed, by de Moivre formula for any angle α we have:

$$\cos 3\alpha = 4\cos^3\alpha - 3\cos\alpha$$

It follows from here that $\cos\frac{1}{3}\theta$ is a root of $f(x)$.

If $f(x)$ is irreducible over K then $[K(\cos\frac{1}{3}\theta) : K] = 3$, and so by Corollary 81.7 the point $(\cos\frac{1}{3}\theta, 0)$ is not constructible from S_0.

Conversely, if $f(x)$ is not irreducible then we have

$$f(x) = (x - a)g(x)$$

where $a \in K$, $g(x) \in K[x]$, $\deg g(x) = 2$. Since $\cos \frac{1}{3}\theta$ is a root $f(x)$ we obtain that either $\cos \frac{1}{3}\theta = a$ or $\cos \frac{1}{3}$ is a root of $g(x)$, and so it can be expressed by elements of K and square roots of such elements. In both cases $\cos \frac{1}{3}\theta$ belongs to some degree 2 radical extension of K.

Some special cases:

- $\theta = \dfrac{\pi}{4}$

We have $\cos\theta = \frac{\sqrt{2}}{2}$, and so $\mathbb{Q}(\cos\theta) = \mathbb{Q}(\sqrt{2})$. Moreover

$$f(x) = x^3 - \frac{3}{4}x - \frac{\sqrt{2}}{8}$$

Check: $f(x)$ has a root $-\frac{\sqrt{2}}{2} \in \mathbb{Q}(\sqrt{2})$ so it is not irreducible. Thus the angle $\theta = \frac{\pi}{4}$ can be trisected using straightedge and compass.

- $\theta = \dfrac{\pi}{3}$

We have $\cos\theta = \frac{1}{2}$, so $\mathbb{Q}(\cos\theta) = \mathbb{Q}$ and

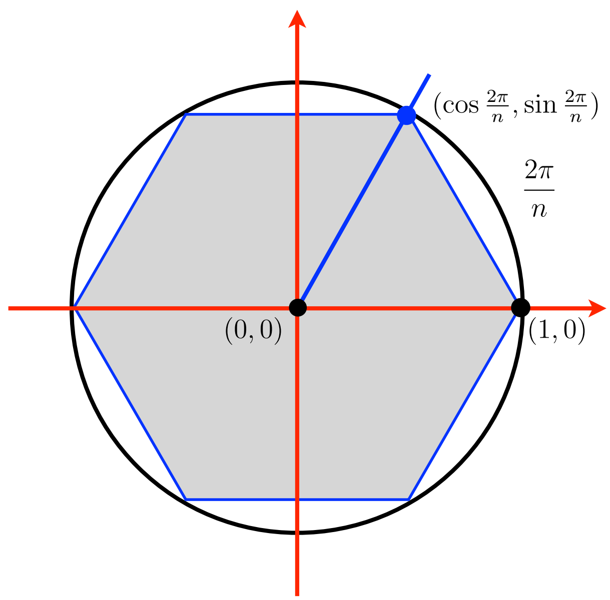
$$f(x) = x^3 - \frac{3}{4}x - \frac{1}{8}$$

Check: $f(x)$ has no roots in $\mathbb{Q}$, and so it is irreducible in $\mathbb{Q}[x]$. As a consequence the angle $\theta = \frac{\pi}{3}$ cannot be trisected using straightedge and compass.

82 Construction of regular polygons

Problem. For which n it is possible to construct a regular polygon with n sides using a straightedge and a compass?

Note. Construction of a regular polygon with n sides is equivalent to the construction of the point with coordinates $(\cos\frac{2\pi}{n}, \sin\frac{2\pi}{n})$.



82.1 Lemma. *The numbers $\cos\frac{2\pi}{n}$, $\sin\frac{2\pi}{n}$ belong to some degree 2 radical extension L of $\mathbb{Q}$ iff $\zeta_n = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$ ome degree 2 radical extension of $\mathbb{Q}$.*

Proof.

($\Rightarrow$) If $\cos\frac{2\pi}{n}$, $\sin\frac{2\pi}{n} \in L$ where L is a degree 2 radical extension of $\mathbb{Q}$ then $\zeta_n \in L(i)$, and $L(i)$ is a degree 2 radical extension of $\mathbb{Q}$.

($\Leftarrow$) Assume that $\zeta_n \in L$ for some degree 2 radical extension L of $\mathbb{Q}$. We can assume that $i \in L$. We have $\zeta^{-1} = \cos\frac{2\pi}{n} - i\sin\frac{2\pi}{n}$, which gives

$$\cos\frac{2\pi}{n} = \frac{1}{2}\left(\zeta_n + \frac{1}{\zeta_n}\right), \qquad \sin\frac{2\pi}{n} = \frac{1}{2i}\left(\zeta_n - \frac{1}{\zeta_n}\right)$$

Therefore $\cos\frac{2\pi}{n},\ \sin\frac{2\pi}{n} \in L$.

$\square$

82.2 Note. The element ζ_n is a primitive root of unity of degree n, so $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is the cyclotomic Galois extension of degree n (77.3). By (77.6) we have

$$\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^*$$

82.3 Lemma. *If $L/\mathbb{Q}$ is a Galois extension and $[L:\mathbb{Q}] < \infty$ then $L \subseteq M$ for some degree 2 radical extension M iff $\mathrm{Gal}(L/\mathbb{Q}) = 2^m$ for some $m \geq 0$*

Proof. Exercise. $\square$

82.4 Corollary. *A regular polygon with n sides can be constructed using a straightedge and a compass iff $|(\mathbb{Z}/n\mathbb{Z})^*| = 2^m$ for some m.*

82.5 Note.

1) We have $|(\mathbb{Z}/n\mathbb{Z})^*| = \varphi(n)$ where $\varphi(n)$ is the Euler function:

$$\varphi(n) = |\{m \mid 1 \geq m \geq n, \quad \gcd(m,n) = 1\}|$$

2) If $n = p_1^{k_1} \cdot \ldots \cdot p_r^{k_r}$ where $p_1, \ldots, p_r$ are distinct primes then

$$\varphi(n) = \varphi(p_1^{k_1}) \cdot \ldots \cdot \varphi(p_r^{k_r})$$

and $\varphi(p_i^{k_i}) = (p_i - 1) \cdot p_i^{k_i - 1}$.

Upshot. The number $\varphi(n)$ is a power of two iff

$$n = 2^k p_1 \cdot \ldots \cdot p_r$$

where $p_1, \ldots, p_r$ are distinct primes such that $p_i = 2^{m_i} + 1$ for some $m_i \geq 1$.

Note. If $2^m + 1$ is a prime then m must be of the form 2^s for some $s \geq 0$. Prime numbers of the form

$$F_s = 2^{2^s} + 1$$

are called Fermat primes.

We obtain:

82.6 Theorem. *A regular polygon with n sides can be constructed using a straightedge and a compass iff*

$$n = 2^k p_1 \cdot \ldots \cdot p_r$$

where $p_1, \ldots, p_r$ are distinct Fermat primes.

82.7 Note. The only known Fermat primes are

$$F_0 = 3, \quad F_1 = 5, \quad F_2 = 17, \quad F_3 = 257, \quad F_4 = 65537$$

It is not known whether there are any other prime numbers of this form.

83 Transcendental extensions

83.1 Definition. Let L/K be a field extension. A set $S \subseteq L$ is *algebraically independent* over K if for every finite sequence $(s_1, \dots, s_n)$ of distinct elements of S and for every non-zero polynomial $f(x_1, \dots, x_n) \in K[x_1, \dots, x_n]$ we have

$$f(s_1, \dots, s_n) \neq 0$$

Otherwise we say that the set S is *algebraically dependent* over K.

83.2 Note. If L/K is a field extension then an element $s \in L$ is transcendental over K iff the set $S = \{s\}$ is algebraically independent over K.

83.3 Proposition. *Let L/K be a field extension. A set $S \subseteq L$ is algebraically independent over K iff every element $s_0 \in S$ is transcendental over the field $K(S - \{s_0\})$.*

Proof.

($\Rightarrow$) Assume that there exists an element $s_0 \in S$ which is algebraic over the field $K(S - \{s_0\})$. Then there exists a finite subset $\{s_1, \dots, s_n\}$ such that s_0 is algebraic over $K(s_1, \dots, s_n)$. Let $0 \neq f(x) \in K(s_1, \dots, s_n)[x]$ be a polynomial such that $f(s_0) = 0$. Notice that $K(s_1, \dots, s_n)$ is the field of fractions of the ring $K[s_1, \dots, s_n]$. As a consequence we can assume that $f(x) \in K[s_1, \dots, s_n][x]$:

$$f(x) = f_r(s_1, \dots, s_n)x^r + \dots + f_1(s_1, \dots, s_n)x + f_0(s_1, \dots, s_n)$$

for some $f_r, \dots, f_0 \in K[s_1, \dots, s_n]$. Take $F \in K[x_1, \dots, x_n, x_{n+1}]$ given by

$$F(x_1, \dots, x_{n+1}) = f_r(x_1, \dots, x_n)x_{n+1}^r + \dots + f_1(x_1, \dots, x_n)x_{n+1} + f_0(x_1, \dots, x_n)$$

Then $F \neq 0$ and $F(s_1, \dots, s_n, s_0) = 0$. This shows that the set S is not algebraically independent over K.

($\Leftarrow$) Assume that the set is not algebraically independent over K, and let n be the minimal number such that there exists a polynomial $0 \neq F \in K[x_1, \dots, x_n]$ satisfying $F(s_1, \dots, s_n) = 0$ for some distinct elements $s_1, \dots, s_n \in S$. We have

$$F(x_1, \dots, x_n) = \\ f_r(x_1, \dots, x_{n-1})x_n^r + \ldots + f_1(x_1, \dots, x_{n-1})x_n + f_0(x_1, \dots, x_{n-1})$$

for some $f_r, \dots, f_0 \in K[x_1, \dots, x_{n-1}]$. Since $F \neq 0$ we must have $f_i \neq 0$ for some $0 \leq i \leq r$. Then by the minimality of n we have $f_i(s_1, \dots, s_{n-1}) \neq 0$. Take $G(x) \in K(s_1, \dots, s_{n-1})[x]$ defined by

$$G(x) = f_r(s_1, \dots, s_{n-1})x^r + \ldots + f_1(s_1, \dots, s_{n-1})x + f_0(s_1, \dots, s_{n-1})$$

We have $G(x) \neq 0$ and $G(s_n) = F(s_1, \dots, s_n) = 0$. Therefore s_n is an algebraic element over $K(s_1, \dots, s_{n-1})$, and so it is also algebraic over $K(S - \{s_n\})$.

□

83.4 Note. If $L = K(x_1, \dots, x_n)$ is the field of rational functions in variables $x_1, \dots, x_n$ with coefficients in K then the set $S = \{x_1, \dots, x_n\}$ is algebraically independent over K.

83.5 Proposition. *If L/K is a field extension and $S = \{s_1, \dots, s_n\}$ is an algebraically independent set over K then we have an isomorphism*

$$K(x_1, \dots, x_n) \cong K(s_1, \dots, s_n)$$

Proof. Define $\varphi\colon K(x_1, \dots, x_n) \to K(s_1, \dots, s_n)$ by

$$\varphi\left(\frac{f(x_1, \dots, x_n)}{g(x_1, \dots, x_n)}\right) = \frac{f(s_1, \dots, s_n)}{g(s_1, \dots, s_n)}$$

Check that this an isomorphism of fields. □

83.6 Definition. Let L/K be a field extension. A transcendence basis of the extension L/K is a set $S \subseteq L$ such that

1) S is algebraically independent over K
2) if $S' \subseteq L$ is an algebraically independent set such that $S \subseteq S'$ then $S = S'$.

83.7 Proposition. *For any field extension L/K there exists a transcendence basis of L/K.*

Proof. Exercise. $\square$

83.8 Theorem. *Let L/K be a field extension. A subset $S \subseteq L$ is a transcendence basis of L/K iff S a set algebraically independent over K and $L/K(S)$ is an algebraic extension.*

83.9 Lemma. *Let L/K be a field extension. If $S, T \subseteq L$ are sets such that S is algebraically independent over K and T is algebraically independent over $K(S)$ then $S \cup T$ is algebraically independent over K.*

Proof. Let $s_1, \dots, s_m \in S$, $t_1, \dots, t_n \in T$ be distinct elements, and let

$$0 \neq F \in K[x_1, \dots, x_m, y_1, \dots, y_n]$$

We need to show that $F(s_1, \dots, s_m, t_1, \dots, t_n) \neq 0$ We have

$$F = \sum_{i_1, \dots, i_n} f_{i_1, \dots, i_n} y_1^{i_1} \cdot \dots \cdot y_n^{i_n}$$

for some $f_{i_1, \dots, i_n} \in K[x_1, \dots, x_m]$. Since $F \neq 0$ we have $f_{i_1, \dots, i_n} \neq 0$ for some $i_1, \dots i_n$, and so by the algebraic independence of the set S over K we get that $f_{i_1, \dots, i_n}(s_1, \dots, s_m) \neq 0$. As a consequence the polynomial

$$G(y_1, \dots, y_m) = \sum_{i_1, \dots, i_n} f_{i_1, \dots, i_n}(s_1, \dots, s_m) y_1^{i_1} \cdot \dots \cdot y_n^{i_n}$$

is a non-zero polynomial in $K(S)[y_1, \dots, y_n]$. By the algebraic independence of the set T over $K(S)$ we obtain

$$F(s_1, \dots, s_m, t_1, \dots, t_n) = G(t_1, \dots, t_n) \neq 0$$

$\square$

Proof of Theorem 83.8.

($\Rightarrow$) Let S be a transcendence basis of L/K. It is enough to show that any element $a \in L - K(S)$ is algebraic over $K(S)$.

Assume by contradiction that a is transcendental over $K(S)$. Then the set $\{a\}$ is algebraically independent over $K(S)$, and so, by Lemma 83.9 the set $S \cup \{a\}$ is algebraically independent over K. This is impossible since S is a proper subset of $S \cup \{a\}$.

($\Leftarrow$) Assume that S is a set algebraically independent over K and that $L/K(S)$ is an algebraic extension. It is enough to show that for any $a \in L - S$ the set $S \cup \{a\}$ is not algebraically independent over K.

Assume, by contradiction, that for some $a \in L - S$ the set $S \cup \{a\}$ is algebraically independent over K. By Proposition 83.3 we would have then that a is a transcendental element over $K(S)$. This contradicts the assumption that all elements of L are algebraic over $K(S)$. $\square$

83.10 Definition. A field extension L/K is *purely transcendental* if $L = K(S)$ where $S \subseteq L$ is a set algebraically independent over K.

83.11 Corollary. *For any field extension L/K there exists an intermediate field $K \subseteq M \subseteq L$ such that M/K is a purely transcendental extension and L/M is an algebraic extension.*

Proof. By Theorem 83.8 is it enough to take $M = K(S)$ where $S \subseteq L$ is a transcendence basis of L/K. $\square$

83.12 Proposition. *If L/K is a field extension and $L = K(A)$ for some set $A \subseteq L$ then there exists $S \subseteq A$ such that S is a transcendence basis of L/K.*

Proof. By Zorn's Lemma 29.10 we can find a maximal subset $S \subseteq A$ that is algebraically independent over K. We will show S is a transcendence basis of L/K.

By Theorem 83.8 it suffices to show that $L/K(S)$ is an algebraic extension. Since $L = K(A) = K(S)(A - S)$ this amounts to showing that every element $a \in A - S$ is algebraic over $K(S)$.

Assume, by contradiction, that there exists an element $a \in A - S$ that is transcendental over $K(S)$. Then the set $\{a\}$ is algebraically independent over $K(S)$, and so by Lemma 83.9 the set $S \cup \{a\} \subseteq A$ is algebraically independent over K. This however is impossible by the maximality of S. $\square$

83.13 Proposition. *If If $K \subseteq L \subseteq M$ are field extensions, $S \subseteq L$ be a transcendence basis of L/K and $T \subseteq M$ be a transcendence basis of M/L then $S \cup T$ is a transcendence basis of M/K.*

Proof. The set T is algebraically independent over L, so it is also algebraically independent over $K(S) \subseteq L$. By Lemma 83.9 it follows that the set $S \cup T$ is algebraically independent over K.

By Theorem 83.8 the field L is an algebraic extension of $K(S)$. It follows that $L(T)$ is an algebraic extension of $K(S)(T) = K(S \cup T)$ (check!). Moreover, by Theorem 83.8 again, M is an algebraic extension of $L(T)$. As a consequence M is an algebraic extension of $K(S \cup T)$. Applying Theorem 83.8 one more time we get from here that $S \cup T$ is a transcendence basis of M over K. $\square$

83.14 Corollary. *If L/K is a field extension and $S \subseteq L$ is a set algebraically independent over K then there exists a transcendence basis S' of L/K such that $S \subseteq S'$.*

Proof. The set S is a transcendence basis of the extension $K(S)/K$. Let T be a transcendence basis of the extension $L/K(S)$. By Proposition 83.13 $S \cup T$ is a transcendence basis of L/K. Therefore we can take $S' = S \cup T$. □

83.15 Theorem. *If L/K is a field extension then any two transcendence bases of L over K have the same cardinality.*

Proof. See Hungerford Theorem 1.8 and Theorem 1.9. □

83.16 Definition. The *transcedence degree* of a field extension L/K is the cardinality of a transcendence basis of L over K. It is denoted by $\mathrm{tr}(L/K)$.

83.17 Proposition. *If $K \subseteq L \subseteq M$ are field extensions then*

$$\mathrm{tr}(M/K) = \mathrm{tr}(L/K) + \mathrm{tr}(M/L)$$

Proof. Let $S \subseteq L$ be a transcendence basis of L/K and let $T \subseteq M$ be a transcendence basis of M/L over L. By Proposition 83.13 the set $S \cup T$ is a transcendence basis of M/K. Since $S \cap T = \varnothing$ we obtain

$$\mathrm{tr}(M/K) = |S \cup T| = |S| + |T| = \mathrm{tr}(L/K) + \mathrm{tr}(M/L)$$

□

84 Algebraic sets

84.1 Definition. Let K be an algebraically closed field, and let

$$K^n = \underbrace{K \times \ldots \times K}_{n \text{ times}}$$

A subset $A \subseteq K^n$ is an *algebraic set* if A is the set of solution of some system of polynomial equations

$$\begin{cases} f_1(x_1, \ldots, x_n) = 0 \\ \vdots \\ f_r(x_1, \ldots, x_n) = 0 \end{cases}$$

for some $f_1, \ldots, f_r \in K[x_1, \ldots, x_n]$. In such case we write:

$$A = V(f_1, \ldots, f_r)$$

84.2 Note. Every finite subset $A = \{a_1, \ldots, a_m\} \subseteq K$ is an algebraic set since this is the set of solutions of the equation

$$(x - a_1) \cdot \ldots \cdot (x - a_m) = 0$$

84.3 Note. Different sets of equations may give the same algebraic set. Take e.g. $f_1, f_2, g_1, g_2 \in K[x_1, x_2]$ given by:

$$f_1(x_1, x_2) = x_1, \quad f_2(x_1, x_2) = x_2, \quad g_1(x_1, x_2) = x_1 + x_2, \quad g_2(x_1, x_2) = x_1 - x_2$$

Then

$$V(f_1, f_2) = V(g_1, g_2) = \{(0, 0)\} \subseteq K^2$$

84.4 Definition. Let K be a field and let I be an ideal of $K[x_1, \ldots, x_n]$. Denote:

$$V(I) := \{(a_1, \ldots, a_n) \mid f(a_1, \ldots, a_n) = 0 \text{ for all } f \in I\}$$

84.5 Proposition. *Let K be a field, let $f_1, \dots, f_r \in K[x_1, \dots, x_n]$, and let $I = \langle f_1, \dots, f_r \rangle$ be the ideal generated by $f_1, \dots f_r$. Then*

$$V(I) = V(f_1, \dots, f_r)$$

Proof. Since $\{f_1, \dots, f_r\} \subseteq I$ we have $V(I) \subseteq V(f_1, \dots, f_r)$.

Conversely, assume that $(a_1, \dots, a_n) \in V(f_1, \dots, f_r)$. If $g \in I$ then we have

$$g = \sum_{i=1}^{r} h_i f_i$$

for some $g_i \in K[x_1, \dots, x_n]$. Therefore

$$g(a_1, \dots, a_n) = \sum_{i=1}^{r} g_i(a_1, \dots, a_n) f_i(a_1, \dots, a_n) = 0$$

Therefore $(a_1, \dots, a_n) \in V(I)$, and as a consequence $V(f_1, \dots, f_r) \subseteq V(I)$. $\square$

84.6 Corollary. *If K is a field, $f_1, \dots, f_r, g_1, \dots, g_s \in K[x_1, \dots, x_n]$ and*

$$\langle f_1, \dots, f_r \rangle = \langle g_1, \dots, g_s \rangle$$

then $V(f_1, \dots, f_r) = V(g_1, \dots, g_s)$.

85 Hilbert basis theorem

Goal:

85.1 Theorem. *If K is a field and $I \lhd K[x_1, \dots, x_n]$ then the ideal I is finitely generated:*

$$I = \langle f_1, \dots, f_r \rangle$$

for some $f_1, \dots, f_r \in K[x_1, \dots, x_n]$

85.2 Note. Let K be an algebraically closed set. If $I \lhd K[x_1, \dots, x_n]$ and $I = \langle f_1, \dots, f_r \rangle$ then by Proposition 84.5 we have

$$V(I) = V(f_1, \dots, f_r)$$

i.e. $V(I)$ is an algebraic set. Thus by Theorem 85.1 we obtain an epimorphism:

$$V\colon \begin{pmatrix} \text{ideals} \\ I \lhd K[x_1, \dots, x_n] \end{pmatrix} \longrightarrow \begin{pmatrix} \text{algebraic sets} \\ A \subseteq K^n \end{pmatrix}$$

which sends an ideal I to the algebraic set $V(I)$.

85.3 Definition. Let R be a commutative ring with identity. The ring R is a *Noetherian ring* if every ideal $I \lhd R$ is finitely generated:

$$I = \langle r_1, \dots, r_n \rangle$$

for some $r_1, \dots, r_n \in R$.

85.4 Example. If R is a PID then R is a Noetherian ring. In particular

1) $\mathbb{Z}$ is Noetherian
2) If K is a field then $K[x]$ is Noetherian.

85.5 Theorem. *Let R be a commutative ring with identity. The following conditions are equivalent.*

1) *R is a Noetherian ring.*
2) *If*
$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$$
is any increasing sequence of ideals of R then there exists $n \geq 1$ such that $I_n = I_m$ for all $m \geq n$.
3) *Every family of ideals of R contains a maximal element.*

85.6 Note. The condition 2) is called the *ascending chain condition* on ideals of R.

Proof of Theorem 85.5.

1) $\Rightarrow$ 2) Let $I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$ be a chain of ideals in R. Take
$$I := \bigcup_{k=1}^{\infty} I_k$$
Then I is an ideal of R, and since R is a Noetherian ring this $I = \langle r_1, \dots, r_n \rangle$ for some $r_1, \dots, r_n \in R$. For $j = 1, \dots, n$ there exists k_j such that $r_j \in I_{k_j}$. Take
$$n = \max\{k_1, \dots, k_n\}$$
For any $m \geq n$ we have $r_1, \dots, r_n \in I_n$ and so
$$I \subseteq I_m \subseteq I$$
As a consequence for any $m \geq n$ we get $I_n = I = I_m$.

2) $\Rightarrow$ 3) Let $\mathcal{J}$ be a family of ideals in R. Assume that $\mathcal{J}$ does not contain a maximal element. Take any $I_1 \in \mathcal{J}$. Since I_1 is not a maximal element of $\mathcal{J}$

therefore there exists $I_2 \in \mathcal{J}$ such that $I_1 \subset I_2$ and $I_1 \neq I_2$. By induction we can find in $\mathcal{J}$ an infinite sequence of ideals

$$I_1 \subset I_2 \subset I_3 \subset \dots$$

such that $I_k \neq I_{k+1}$ for all $k \geq 1$. This however contradicts the assumption that ideals of R satisfy the ascending chain condition.

3) $\Rightarrow$ 2) Let I be an ideal of R. We need to show that I is finitely generated.

Let $\mathcal{J}$ be the family of all finitely generated ideals J such that $J \subseteq I$. By assumption $\mathcal{J}$ contains a maximal element $J_0 = \langle r_1, \dots, r_n \rangle$. Assume the $J_0 \neq I$ and let $s \in I - J_0$. Then $J_1 := \langle r_1, \dots, r_n, s \rangle$ is an ideal such that $J_1 \in \mathcal{J}$, $J_0 \subseteq J_1$, and $J_0 \neq J_1$. This is impossible since J_0 is a maximal element of $\mathcal{J}$. As a consequence we get that $I = J_0$, and so I is fintely generated.

$\square$

85.7 Hilbert Basis Theorem. *Let R be a commutative ring with identity. If R is a Noetherian ring then $R[x]$ is also Noetherian.*

Proof. Let $I \lhd R[x]$. We will show that I is generated by a finite set.

Let $I_n \subseteq R$ be the set such that $a \in I_n$ if either $a = 0$ or if there exists $f(x) = a_n x^n + \dots + a_1 x + a_0$ such that $f(x) \in I$ and $a = a_n$. Check: $I_n \lhd R$. Moreover we have

$$I_0 \subseteq I_1 \subseteq \dots$$

Since R is a Noetherian ring we obtain:

1) for any $n \geq 0$ there exists $a_{n,1}, \dots, a_{n,m_n} \in R$ such that

$$I_n = \langle a_{n,1}, \dots, a_{n,m_n} \rangle$$

2) there exists $N \geq 0$ such that $I_N = I_{N+1} = \dots$.

By the definition of I_n for any $0 \leq n \leq N$ and $1 \leq i \leq m_n$ there exists a polynomial $f_{n,i}(x) \in I$ such that $f_{n,i}(x) = a_{n,i}x^n + \ldots$. Define

$$S := \{f_{n,i}(x) \mid 0 \leq n \leq N, \;\; 1 \leq i \leq m_n\}$$

Notice that S is a finite set and $S \subseteq I$. We will show that $I = \langle S \rangle$.

It suffices to show that if $g(x) \in I$ then $g(x) \in \langle S \rangle$. We will argue by induction with respect to the degree of $g(x)$.

If $\deg g(x) = 0$ then $g(x) = b_0$ for some $b_0 \in I_0$. Then

$$b_0 = \sum_{i=1}^{m_0} r_i a_{0,i}$$

for some $r_i \in R$. Since $f_{0,i}(x) = a_{0,i}$ this gives

$$g(x) = \sum_{i=1}^{m_0} r_i f_{0,i}(x)$$

and so $g(x) \in \langle S \rangle$.

Next, assume that for some $n > 0$ if $h(x) \in I$ and $\deg h(x) \leq n-1$ then $h(x) \in \langle S \rangle$. Let $g(x) \in I$ be a polynomial of degree n:

$$g(x) = b_n x^n + \ldots b_1 x + b_0$$

Assume first that $n \leq N$ By the definition of I_n we have $b_n \in I_n$, and so

$$b_n = \sum_{i=1}^{m_n} r_i a_{n,i}$$

for some $r_i \in R$. Define

$$f(x) := \sum_{i=1}^{m_n} r_i f_{n,i}(x)$$

Notice that $f(x) = b_n x^n + \ldots$, and so $\deg(g(x) - f(x)) < n$. Also, since $g(x), f(x) \in I$ we have $g(x) - f(x) \in I$ and thus by the inductive assumption $g(x) - f(x) \in \langle S \rangle$. Therefore

$$g(x) = \underbrace{f(x)}_{\in \langle S \rangle} + \underbrace{(g(x) - f(x))}_{\in \langle S \rangle}$$

is an element of $\langle S \rangle$.

If $n > N$ then $I_n = I_N$ so we have

$$b_n = \sum_{i=1}^{m_N} r_i a_{N,i}$$

Then define

$$f(x) := x^{n-N} \left(\sum_{i=1}^{m_N} r_i f_{N,i}(x) \right) = b_n x^n + \dots$$

Then $f(x) \in \langle S \rangle$ and $\deg(g(x) - f(x)) < n$. Similarly as before we obtain from here that $g(x) \in \langle S \rangle$.

85.8 Corollary. *If R is a Noetherian ring then the ring $R[x_1, \dots, x_n]$ is also Noetherian for any $n \geq 0$.*

Proof. This follows from Theorem 85.7 by induction with respect to n. □

Proof of Theorem 85.1.

Since K is a field it is a Noetherian ring. Thus we can apply Corollary 85.8 □

□

86 Radical ideals

86.1 Note. Let K be an algebraically closed field. For any subset $X \subseteq K^n$ define

$$J(X) := \{f \in K[x_1, \dots, x_n] \mid f(a_1, \dots, a_n) = 0 \text{ for all } (a_1, \dots, a_n) \in X\}$$

Check: $J(X)$ is an ideal of $K[x_1, \dots, x_n]$.

We get maps of sets:

$$V\colon \begin{pmatrix} \text{ideals} \\ I \lhd K[x_1, \dots, x_n] \end{pmatrix} \rightleftarrows \begin{pmatrix} \text{algebraic sets} \\ A \subseteq K^n \end{pmatrix} : J$$

Notice that

1) if $I, I' \lhd K[x_1, \dots, x_n]$ and $I \subseteq I'$ then $V(I) \supseteq V(I')$
2) if $A, A' \subseteq K^n$ are algebraic sets and $A \subseteq A'$ then $J(A) \supseteq J(A')$.

86.2 Proposition. *If $A \subseteq K^n$ is an algebraic set then $V(J(A)) = A$.*

Proof. If $(a_1, \dots, a_n) \in A$ then $f(a_1, \dots, a_n) = 0$ for all $f \in J(A)$. Therefore $(a_1, \dots, a_n) \in V(J(A))$. This show that $A \subseteq V(J(A))$.

On the other hand, A is an algebraic set, so $A = V(I)$ for some $I \lhd K[x_1, \dots, x_n]$. We have $I \subseteq J(A)$ which gives

$$A = V(I) \supseteq V(J(A))$$

□

86.3 Note. It is not true that for any ideal $I \in K[x_1, \dots, x_n]$ we have $J(V(I)) = I$. Take e.g. $I = \langle x^2 \rangle \lhd K[x]$. We have

$$V(I) = \{a \in K \mid a^2 = 0\} = \{0\}$$

Then

$$J(V(I)) = \{g(x) \in K[x] \mid g(0) = 0\} = \langle x \rangle$$

Therefore $J(V(I)) \neq I$.

86.4 Definition. Let R be a commutative ring with identity. An ideal $I \lhd R$ is a *radical ideal* if for every $a \in R$ such that $a^n \in I$ for some $n \geq 1$ we have $a \in I$.

86.5 Proposition. *If K is an algebraically closed field and $A \subseteq K^n$ is an algebraic set then $J(A)$ is a radical ideal of $K[x_1, \dots, x_n]$.*

Proof. Exercise. $\square$

Goal. The following maps are inverse bijections of sets:

$$V\colon \begin{pmatrix} \text{radical ideals} \\ I \lhd K[x_1, \dots, x_n] \end{pmatrix} \rightleftarrows \begin{pmatrix} \text{algebraic sets} \\ A \subseteq K^n \end{pmatrix} : J$$

Note. In Sections 87-88 all rings are commutative rings with identity $1 \neq 0$.

87 Integral extensions of rings

87.1 Definition. Let R, S be rings. We say that S is a ring extension of R if R is a subring of S.

87.2 Notation. If $R \subseteq S$ is a ring extension and $A \subseteq S$ is a subset of S then $R[A]$ is the smallest subring of S such that $R \cup A \subseteq R[A]$. If A is a finite set, $A = \{a_1, \ldots, a_n\}$ then we write

$$R[A] = R[a_1, \ldots, a_n]$$

87.3 Note. We have

$$R[a] = \{b \in S \mid b = r_0 + r_1 a + \ldots r_k a^k \text{ for some } r_0, \ldots, r_k \in R\}$$

and $R[a_1, \ldots, a_n] = R[a_1, \ldots, a_{n-1}][a_n]$.

87.4 Definition. Let $R \subseteq S$ be a ring extension. An element $a \in S$ is an *integral element* over R if a is a root of a monic polynomial $f(x) \in R[x]$.

87.5 Example.

1) If K, L are fields and $K \subseteq L$ then an element $a \in L$ is an integral element over L iff a is an algebraic element over K.

2) $\sqrt{2}$ is integral over $\mathbb{Z}$ since it is a root of the monic polynomial $f(x) = x^2 - 2 \in \mathbb{Z}[x]$.

3) $\frac{1}{2}$ is not integral over $\mathbb{Z}$. Indeed, if $f(x) \in \mathbb{Z}[x]$ is a monic polynomial

$$f(x) = x^n + a_{n-1}x^{n-1} + \ldots + a_0$$

then

$$2^n f(\tfrac{1}{2}) = \underbrace{1 + 2a_{n-1} + \cdots + 2^n a_0}_{\text{odd}} \neq 0$$

so $f(\frac{1}{2}) = 0$.

87.6 Definition. Let R be a ring. An R-module M is *faithful* if for every $0 \neq r \in R$ there exists $m \in M$ such that $rm \neq 0$.

87.7 Theorem. *Let $R \subseteq S$ be a ring extension and let $a \in S$. The following conditions are equivalent.*

1) *The element a is integral over R.*
2) *$R[a]$ is a finitely generated R-module.*
3) *There exists $M \subseteq S$ such that M is a a faithful $R[a]$-module and it is finitely generated as an R-module.*

87.8 Lemma. *Let S be a ring and let $A = (a_{ij})$ be an $n \times n$ matrix with coefficients in S. If $s_1, \ldots, s_n \in S$ are elements such that*

$$A \begin{bmatrix} s_1 \\ \vdots \\ s_n \end{bmatrix} = 0$$

then $(\det A) \cdot s_i = 0$ for $i = 1, \ldots, n$.

Proof. See Hungerford p.354 Exercise 8. $\square$

Proof of Theorem 87.7.

1) $\Rightarrow$ 2) Since the element a is integral over R there exists a monic polynomial $f(x) \in R[x]$ such that $f(0)$. Assume that $\deg f(x) = n$.

If $b \in R[a]$ then $b = g(a)$ for some $g(x) \in R[x]$. By (38.1) there exists $q(x), r(x) \in R[x]$ such that $\deg r(x) \leq n-1$ and

$$g(x) = q(x)f(x) + r(x)$$

This gives $g(a) = r(a)$. As a consequence we obtain

$$\begin{aligned} R[a] &= \{r(a) \ \mid \ r(x) \in R[x], \ \ \deg r(x) \leq n-1\} \\ &= \{b_0 + b_1 a + \ldots + b_{n-1}a^{n-1} \ \mid \ b_1, \ldots, b_{n-1} \in R\} \end{aligned}$$

Therefore the elements $1, a, \ldots, a^{n-1}$ generate $R[a]$ as an R-module.

2) $\Rightarrow$ 3) $R[a]$ is a faithful $R[a]$-module and by assumption it is finitely generated as an R-module. Thus we can take $M = R[a]$.

3) $\Rightarrow$ 1) Assume that the set $\{s_1, \ldots s_n\}$ generates M as an R-module. For every $i = 1, \ldots n$ there exists $a_{i,j} \in R$ such that

$$as_i = a_{i1}s_1 + \ldots + a_{in}s_n$$

This gives a system of equations:

$$\begin{aligned} &(a_{11} - a)s_1 + a_{12}s_2 + \ldots a_{1n}s_n = 0 \\ &a_{21}s_1 + (a_{22} - a)s_2 + \ldots a_{2n}s_n = 0 \\ &\vdots \ \vdots \\ &a_{n1}s_1 + a_{n2}s_2 + \ldots (a_{nn} - a)s_n = 0 \end{aligned}$$

In the matrix notation this gives:

$$((a_{ij}) - aI)\begin{bmatrix} s_1 \\ \vdots \\ s_n \end{bmatrix} = 0$$

By Lemma 87.8 we get

$$\det((a_{ij}) - aI)s_i = 0$$

for $i = 1, \dots, n$. Since the elements $s_1, \dots, s_n$ generate M this shows that $\det((a_{ij}) - aI)m = 0$ for all $m \in M$. Furthermore, since $\det((a_{ij}) - aI) \in R[a]$ and M is a faithful $R[a]$-module we obtain that $\det((a_{ij}) - aI) = 0$.

Take $f(x) = \det((a_{ij}) - xI) \in R[x]$. This is a monic polynomial and $f(a) = 0$. This shows that a is an integral element over R. □

87.9 Definition. A ring extension $R \subseteq S$ is an *integral extension* if every element of S is integral over R.

87.10 Lemma. *Let $R \subseteq S \subseteq T$ be ring extensions. If T is finitely generated as an S-module and S is finitely generated as an R-module then T is finitely generated as an R-module.*

Proof. Exercise. □

87.11 Proposition. *Let $R \subseteq S$ be a ring extension and let $a_1, \dots, a_n \in S$ be elements integral over R. Then $R[a_1, \dots, a_n]$ is finitely generated R-module and it is an integral extension of R.*

Proof. We argue by induction with respect to n.

If $n = 1$ then $R[a_1]$ is finitely generated R-module by Theorem 87.7. Also, if $b \in R[a_1]$ then $R[a_1]$ is a faithful $R[b]$-module. Using Theorem 87.7 again we obtain that b is integral over R.

Assume the statement of the proposition holds for some n, and let $a_1, \dots, a_{n+1} \in S$ be elements integral over R. Then a_{n+1} is integral over $R[a_1, \dots, a_n]$, so

$R[a_1, \ldots, a_{n+1}]$ is a finitely generated $R[a_1, \ldots, a_n]$-module. By Lemma 87.10 we obtain that $R[a_1, \ldots, a_{n+1}]$ is then a finitely generated R-module.

If $b \in R[a_1, \ldots, a_{n+1}]$ then $R[a_1, \ldots, a_{n+1}]$ is a faithful $R[b]$ module, so by Theorem 87.7 b is integral over R. $\square$

87.12 Corollary. *Let $R \subseteq S$ be a ring extension and let*

$$R_{\mathit{int}} := \{a \in S \mid a \text{ is integral over } R\}$$

then R_{int} is a subring of S.

Proof. Let $a, b \in R_{\text{int}}$. It is enough to show that $a \pm b$, ab are integral elements over R. This holds since $a \pm b$, $ab \in R[a, b]$, and by Proposition 87.11 $R[a, b]$ is an integral extension of R. $\square$

87.13 Proposition. *Let $R \subseteq S \subseteq T$ be ring extensions. If T is an integral extension of S and S is an integral extension of R then T is an integral extension of R.*

Proof. It is enough to show that if $b \in T$ then b is integral over R.

Since b is integral over S there exists a monic polynomial $f(x) \in S[x]$

$$f(x) = x^n + a_{n-1}x^{n-1} + \ldots + a_0$$

such that $f(b) = 0$. Since $f(x) \in R[a_{n-1}, \ldots, a_0]$ it follows that b is integral over $R[a_{n-1}, \ldots, a_0]$. By Theorem 87.7 $R[a_{n-1}, \ldots, a_0, b]$ is then a finitely generated $R[a_{n-1}, \ldots, a_0]$-module. Also, since $a_{n-1}, \ldots, a_0$ are integral over R by Proposition 87.11 $R[a_1, \ldots, a_{n-1}]$ is a finitely generated R-module. Therefore, by Lemma 87.10 $R[a_{n-1}, \ldots, a_0, b]$ is a finitely generated R-module. Since $R[a_{n-1}, \ldots, a_0, b]$ is a faithful $R[b]$ module applying Theorem 87.7 we get that b is integral over R.

$\square$

88 Noether Normalization Lemma

88.1 Noether Normalization Lemma. *Let $K \subseteq S$ be a ring extension such that K is a field and S is finitely generated over K:*

$$S = K[a_1, \dots, a_n]$$

for some $a_1, \dots, a_n \in S$. There exists a ring $K \subseteq R \subseteq S$ such that

$$R \cong K[x_1, \dots, x_m]$$

for some $m \leq n$ and that $R \subseteq S$ is an integral extension

88.2 Note. Compare this with the decomposition of field extensions into a purely transcendental extension and an algebraic extension (Corollary 83.11).

Proof of Theorem 88.1.

We argue by induction with respect to n.

For $n = 1$ we have $S = K[a_1]$. Consider the homomorphism

$$\varphi\colon K[x] \to K[a_1]$$

given by $\varphi(f(x)) = f(a)$. This is an epimorphism, so if $\mathrm{Ker}(\varphi) = \{0\}$ we get that $K[x] \cong K[a_1]$, and then we can take $R = S$.

If $\mathrm{Ker}(\varphi) \neq \{0\}$ then $f(a) = 0$ for some $0 \neq f(x) \in K[x]$. Since K is a field we can assume that $f(x)$ is a monic polynomial. It follows that a is an integral element over K, and so by (87.11) S is an integral extension of K. In this case take $R = K$.

Next, assume that the statement of the theorem holds for some $n - 1$ and let $S = K[a_1, \dots, a_n]$. We have a homomorphism

$$\varphi\colon K[x_1, \dots, x_n] \to K[a_1, \dots, a_n]$$

given by $\varphi(f(x_1, \ldots, x_n)) = f(a_1, \ldots, a_n)$. If $\mathrm{Ker}(\varphi) = \{0\}$ then as before we obtain $K[x_1, \ldots, x_n] \cong K[a_1, \ldots, a_n]$, so we can take $R = S$.

Assume then that there exists $0 \neq f \in \mathrm{Ker}(\varphi)$. We have

$$f(x_1, \ldots, x_n) = \sum_{(i_1, \ldots, i_n) \in I} b_{i_1, \ldots, i_n} x_1^{i_1} \ldots x_n^{i_n}$$

for some $0 \neq b_{i_1, \ldots, i_n} \in K$. Let $(j_1, \ldots, j_n)$ be the element of I that is the biggest with respect to the lexicographic order. Since K is a field we can assume that $a_{j_1, \ldots, j_n} = 1$.

Take

$$d = \max \{ \ i_k \ \mid \ 1 \leq k \leq n, \ (i_1, \ldots, i_n) \in I \ \}$$

and let $h \in K[x_1, \ldots, x_n]$ be a polynomial given by

$$h(x_1, \ldots, x_n) = f(x_1 + x_n^{d^n}, \ x_2 + x_n^{d^{n-1}}, \ldots, x_{n-1} + x_n^d, x_n)$$

Claim. Consider h as a polynomial of the variable x_n with coefficients in the ring $K[x_1, \ldots, x_{n-1}]$. Then h is a monic polynomial.

Indeed, notice that

$$\begin{aligned} h &= \sum_{(i_1, \ldots, i_n) \in I} b_{i_1, \ldots, i_n} (x_1 + x_n^{d^n})^{i_1} (x_2 + x_n^{d^{n-1}})^{i_2} \ldots x_n^{i_n} \\ &= \sum_{(i_1, \ldots, i_n) \in I} b_{i_1, \ldots, i_n} (x_n^{i_1 d^n + i_2 d^{n-1} + \ldots i_n} \ + \ \text{lower powers of } x_n) \end{aligned}$$

By the choice of $(j_1, \ldots, j_n)$ the highest degree monomial of h is then

$$b_{j_1, \ldots, j_n} (x_n^{j_1 d^n + j_2 d^{n-1} + \ldots j_n})$$

and by assumption $b_{j_1, \ldots, j_n} = 1$

Next, let $g(x)$ be the polynomial given by

$$g(x) = h(a_1 - a_n^{d^n}, a_2 - a_n^{d^{n-1}}, \ldots, a_{n-1} - a_n^d, \ x)$$

Notice that

(i) $g(x) \in K[a_1 - a_n^{d_n}, \dots, a_{n-1} - a_n^d][x]$

(ii) $g(x)$ is a monic polynomial (since h is monic in x_n)

(iii) $g(a_n) = f(a_1, \dots, a_n) = 0$ (since $f \in \mathrm{Ker}(\varphi)$)

Denote $S' := K[a_1 - a_n^{d_n}, \dots, a_{n-1} - a_n^d]$. By (i)-(iii) above we obtain that a_n is an integral element over S', and so $S'[a_n] = K[a_1, \dots, a_n]$ is an integral extension of S'. On the other hand S' is generated over K by $n-1$ elements, so by the inductive assumption there is a ring

$$K \subseteq R \subseteq S'$$

such that $R \cong K[x_1, \dots x_m]$ for some $m \leq n-1$ and that S' is an integral extension of R. Consider the extensions

$$K \subseteq R \subseteq S$$

To finish the proof it is enough to notice that since $R \subseteq S'$ and $S' \subseteq S$ are integral extensions thus, by Proposition 87.13, the extension $R \subseteq S$ is also integral. $\square$

88.3 Corollary. *Let K be a field and let L be a finitely generated ring extension of K:*

$$L = K[a_1, \dots, a_n]$$

If L is a field then it is an algebraic extension of K.

Proof. By the Noether Normalization Lemma 88.1 there exists a ring R such that $K \subseteq R \subseteq L$, $R \cong K[x_1, \dots, x_m]$ for some $0 \leq m \leq n$, and that $R \subseteq L$ is an integral extension.

It is enough to show that R is a field. Indeed, this will imply that $m = 0$, (i.e. $K = R$), and that as a consequence $K \subseteq L$ is an integral (and thus algebraic) extension.

Take $b \in R$. Since L is a field we have $b^{-1} \in L$. We need to show that $b^{-1} \in R$.

Since L is an integral extension of R there exists a monic polynomial $f(x) = x^k + r_{k-1}x^{k-1} + \ldots + r_1 x + r_0 \in R[x]$ such that $f(b^{-1}) = 0$. This gives

$$b^{-k} = -r_{k-1}b^{-(k-1)} - r_{k-2}b^{-(k-2)} \ldots - r_1 b^{-1} - r_0$$

Multiplying both sides by b^{k-1} we obtain

$$b^{-1} = -r_{k-1} - r_{k-2}b - \ldots - r_1 b^{k-2} - r_0 b^{k-1}$$

Since $r_i, b \in R$ therefore $b^{-1} \in R$. $\square$

89 Hilbert Nullstellensatz

Recall.

1) Let K be an algebraically closed field. We have maps

$$V\colon \begin{pmatrix} \text{ideals} \\ I \lhd K[x_1, \dots, x_n] \end{pmatrix} \rightleftarrows \begin{pmatrix} \text{algebraic sets} \\ A \subseteq K^n \end{pmatrix} : J$$

   where

$$\begin{aligned} V(I) &= \{(a_1, \dots, a_n) \in K^n \mid f(a_1, \dots, a_n) = 0 \ \ \forall f \in I\} \\ J(A) &= \{f \in K[x_1, \dots, x_n] \mid f(a_1, \dots, a_n) = 0 \ \ \forall (a_1, \dots a_n) \in A\ \} \end{aligned}$$

2) For any algebraic set $A \subseteq K^n$ we have $V(J(A)) = A$ (86.2).

3) If $A \subseteq K^n$ is an algebraic set then $J(A)$ is a radical ideal (86.5).

Goal. If $I \lhd K[x_1, \dots, x_n]$ is a radical ideal then $J(V(I)) = I$. As a consequence we have inverse bijections of sets

$$V\colon \begin{pmatrix} \text{radical ideals} \\ I \lhd K[x_1, \dots, x_n] \end{pmatrix} \rightleftarrows \begin{pmatrix} \text{algebraic sets} \\ A \subseteq K^n \end{pmatrix} : J$$

89.1 Theorem (Weak Nullstellensatz)**.** *Let K be an algebraically closed field. If $I \in K[x_1, \dots, x_n]$ then $V(I) = \varnothing$ iff $I = K[x_1, \dots, x_n]$*

89.2 Note. If $I \lhd K[x]$ then $I = \langle f(x) \rangle$ for some $f(x) \in K[x]$ since $K[x]$ is a PID. This gives

$$V(I) = \{a \in K \mid f(a) = 0\}$$

Therefore in the case of a single variable Theorem 89.1 says that a polynomial $f(x) \in K[x]$ has no roots iff $\langle f(x)\rangle = K[x]$, i.e. iff $f(x)$ is a non-zero constant polynomial. This property defines algebraically closed fields, so in this case Theorem 89.1 is tautologically true.

Proof of Theorem 89.1.

If $I = K[x_1, \dots, x_n]$ then $1 \in I$, and so $V(I) = \varnothing$.

Assume then that $I \neq K[x_1, \dots, x_n]$. By (29.1) there exists a maximal ideal P such that $I \subseteq P$. We have $V(P) \subseteq V(I)$, so it will suffice to show that $V(P) \neq \varnothing$.

Let

$$\varphi\colon K[x_1, \dots, x_n] \to K$$

be a ring homomorphism and let $a_i = \varphi(x_i)$ for $i = 1, \dots, n$. Notice that if $\varphi|_K = \mathrm{id}_K$ then φ is given by

$$\varphi(f(x_1, \dots, x_n)) = f(a_1, \dots, a_n)$$

Moreover, $(a_1, \dots, a_n) \in V(P)$ iff $P \subseteq \operatorname{Ker}\varphi$. This shows that we have a bijection

$$\Phi\colon \left(\begin{array}{c}\text{ring homomorphisms}\\ \varphi\colon K[x_1, \dots, x_n] \to K\\ \varphi|_K = \mathrm{id}_K,\ P \subseteq \operatorname{Ker}(\varphi)\end{array}\right) \longrightarrow \left(\begin{array}{c}\text{points}\\ (a_1, \dots, a_n) \in V(P)\\ \end{array}\right)$$

where $\Phi(\varphi) = (\varphi(x_1), \dots, \varphi(x_n))$. In order to show that $V(P) \neq \varnothing$ is suffices then to prove that there exists a homomorphism $\varphi\colon K[x_1, \dots, x_n] \to K$ satisfying the above conditions.

Take the quotient ring $K[x_1, \dots, x_n]/P$ and let

$$\pi\colon K[x_1, \dots, x_n] \to K[x_1, \dots, x_n]/P$$

be the canonical epimorphism. Identifying $a \in K$ with $\pi(a) = a + P$ we get that $\pi|_K = \mathrm{id}_K$ and we can consider $K[x_1, \dots, x_n]/P$ as a ring extension of K. This extension is finitely generated:

$$K[x_1, \dots, x_n]/P = K[\pi(x_1), \dots, \pi(x_n)]$$

Since P is a maximal ideal $K[x_1, \dots, x_n]/P$ is a field. By Corollary 88.3 we obtain that $K[x_1, \dots, x_n]/P$ is an algebraic extension of K. Since K is an algebraically closed field this implies that $K[x_1, \dots, x_n]/P = K$. Therefore we can take $\varphi := \pi$.

□

89.3 Corollary. *If K is an algebraically closed field and $P \lhd K[x_1, \dots, x_n]$ is a maximal ideal then*

$$P = \langle x_1 - a_1, \dots, x_n - a_n \rangle$$

for some $a_1, \dots, a_n \in K$.

Proof. Since $P \neq K[x_1, \dots, x_n]$ by Theorem 89.1 we get $V(P) \neq \varnothing$. Let $(a_1, \dots, a_n) \in V(P)$. Denote

$$I := \langle x_1 - a_1, \dots, x_n - a_n \rangle$$

Notice that the set $V(I) = \{(a_1, \dots, a_n)\}$. Since $V(I) \neq \varnothing$ the ideal I is a proper ideal of $K[x_1, \dots, x_n]$. Also, since $V(I) \subseteq V(P)$ we have $P \subseteq I$. By maximality of P we obtain $P = I$. □

89.4 Definition. Let R be a commutative ring and let $I \lhd R$. The *radical of I* is the ideal

$$\sqrt{I} = \{a \in R \mid a^n \in I \quad \text{for some} \quad n \geq 1\}$$

89.5 Note. If $I \lhd R$ then $\sqrt{I} = I$ iff I is a radical ideal.

89.6 Theorem (Hilbert Nullstellensatz). *Let K be an algebraically closed field. If $I \lhd K[x_1, \dots, x_n]$ then*

$$J(V(I)) = \sqrt{I}$$

In particular if I is a radical ideal then $J(V(I)) = I$.

Proof. Since $J(V(I))$ is a radical ideal and $I \subseteq J(V(I))$ we have $\sqrt{I} \subseteq J(V(I))$. It remains to show then that $J(V(I)) \subseteq \sqrt{I}$.

Let $0 \neq F \in J(V(I))$. It will be enough to show that $F^N \in I$ for some $N \geq 1$.

Consider the inclusion homomorphism

$$K[x_1, \dots, x_n] \hookrightarrow K[x_1, \dots, x_n, y]$$

Let $g := (1 - yF) \in K[x_1, \dots, x_n, y]$ and let $J \lhd K[x_1, \dots, x_n, y]$ be the ideal given by the set $I \cup \{g\}$.

Claim. $V(J) = \varnothing$.

Indeed, assume that $(a_1, \dots, a_n, b) \in V(J)$. Since $F \in J(V(I))$ we have $F(a_1, \dots, a_n) = 0$. Therefore

$$g(a_1, \dots, a_n, b) = 1 - bF(a_1, \dots, a_n) = 1$$

which is a contradiction since $g \in J$.

By the Weak Nullstellensatz (89.1) we obtain that $J = K[x_1, \dots, x_n, y]$. This means that there exists $f_1, \dots, f_k \in I$ and $h_0, h_1, \dots h_k \in K[x_1, \dots, x_n, y]$ such that

$$gh_0 + \sum_{i=1}^{k} f_i h_i = 1 \tag{$*$}$$

Consider the ring homomorphism

$$\varphi\colon K[x_1, \dots, x_n, y] \to K(x_1, \dots, x_n)$$

such that $\varphi|_K = \mathrm{id}_K$, $\varphi(x_i) = x_i$, $\varphi(y) = \frac{1}{F}$. Notice that $\varphi(g) = 1 - \frac{1}{F}F = 0$. Therefore from the equation $(*)$ we obtain

$$1 = \varphi(1) = \sum_{i=1}^{k} f_i \varphi(h_i)$$

For $i = 1, \dots k$ we have $h_i = \sum_{j=0}^{n_i} h_{ij} y^j$ for some $h_{ij} \in K[x_1, \dots, x_n]$. This gives:

$$1 = \sum_{i=1}^{k} f_i \varphi(h_i) = \sum_{i=1}^{k} f_i \left(\sum_{j=1}^{n_i} h_{ij} \frac{1}{F^j} \right)$$

Take $N = \max\{n_1, \dots, n_k\}$. We obtain

$$F^N = F^N \left(\sum_{i=1}^{k} f_i \left(\sum_{j=1}^{n_i} h_{ij} \frac{1}{F^j} \right) \right) = \sum_{i=1}^{k} f_i \left(\sum_{j=1}^{n_i} h_{ij} F^{N-j} \right)$$

where $N - j \geq 0$. Since $f_i \in I$ and $h_{ij} F^{N-j} \in K[x_1, \dots, x_n]$ for all i, j this shows that $F^N \in I$. $\square$

89.7 Corollary. *Let K be an algebraically closed field. We inverse bijections of sets:*

$$V\colon \begin{pmatrix} \text{radical ideals} \\ I \lhd K[x_1, \dots, x_n] \end{pmatrix} \rightleftarrows \begin{pmatrix} \text{algebraic sets} \\ A \subseteq K^n \end{pmatrix} : J$$

Proof. This follows directly from Proposition 86.2 and Theorem 89.6. $\square$

89.8 Corollary. *If $I_1, I_2 \lhd K[x_1, \dots, x_n]$ then $V(I_1) = V(I_2)$ iff $\sqrt{I_1} = \sqrt{I_2}$.*

Proof. If $I \lhd K[x_1, \dots x_n]$ then by (86.2) and (89.6) we have

$$V(I) = V(J(V(I))) = V(\sqrt{I})$$

Therefore if $\sqrt{I_1} = \sqrt{I_2}$ then $V(I_1) = V(I_2)$.

Conversely, if $V(I_1) = V(I_2)$ then

$$\sqrt{I_1} = J(V(I_1)) = J(V(I_2)) = \sqrt{I_2}$$

$\square$

90 Zariski topology

Recall. If R is a commutative ring and $I, I \lhd R$ then

$$I + J := \{a + b \mid r \in I, s \in J\}$$

In general, if $\{I_i\}_{i \in S}$ then

$$\sum_{i \in S} I_i := \left\{ \sum_{i \in S} a_i \;\middle|\; a_i \in I_i,\ a_i \neq 0 \text{ for finitely many } i \text{ only} \right\}$$

$\sum_{i \in S} I_i$ is the smallest ideal of R containing I_i for all $i \in S$.

90.1 Proposition. *Let K be an algebraically closed set.*

1) If $\{I_1, \dots, I_k\}$ is a finite family of ideals of $K[x_1, \dots, x_n]$ then

$$V(I_1) \cup \dots \cup V(I_k) = V(I_1 \cap \dots \cap I_k)$$

2) If $\{I_i\}_{i \in S}$ is an arbitrary family of ideals of $K[x_1, \dots, x_n]$ then

$$\bigcup_{i \in S} V(I_i) = V(\textstyle\sum_{i \in S} I_i)$$

Proof.

1) Denote $J := I_1 \cap \dots \cap I_k$. Since $J \subseteq I_i$ for $i = 1, \dots, k$ thus $V(I_i) \subseteq V(J)$ and so

$$\bigcup_{i=1}^{n} V(I_i) \subseteq V(J)$$

Conversely, assume that $a \notin \bigcup_{i=1}^{n} V(I_i)$. Then for $i = 1, \dots, k$ there exists $f_i \in I_i$ such that $f_i(a) = 0$. Take $f = f_1 \cdot \ldots \cdot f_n$. We have $f \in J$, and $f(a) \neq 0$. Thus $a \notin V(J)$. Therefore we obtain

$$V(J) \subseteq \bigcup_{i=1}^{n} V(I_i)$$

2) Denote $J := \sum_{i\in S} I_i$. Since $I_i \subseteq J$ for all $i \in S$ thus $V(J) \subseteq V(I_i)$ and so

$$V(J) \subseteq \bigcap_{i\in S} V(I_i)$$

Conversely, if $a \in \bigcap_{i\in S} V(I_i)$ then $f(a) = 0$ for all $f \in I_i$ Since the set $\bigcup_{i\in S} I_i$ generates J we get that $f(a) = 0$ for all $f \in J$, and so $a \in V(J)$.

□

90.2 Corollary. *Let K be an algebraically closed field.*

1) $\varnothing$ and K^n are algebraic sets in K^n.

2) If $\{A_1, \dots A_n\}$ is a finite family of algebraic sets in K^n then $\bigcup_{i=1}^{n} A_i$ is also an algebraic set.

3) If $\{A_i\}_{i\in S}$ is an arbitrary family of algebraic sets in K^n then $\bigcap_{i\in S} A_i$ is also an algebraic set.

Proof.

1) We have $\varnothing = V(\langle 1\rangle)$ and $K^n = V(\{0\})$.

2) – 3) This follows directly from Proposition 90.1.

□

90.3 Definition/Proposition. Let K be an algebraically closed field. There exists a topology on K^n such that closed sets in this topology are algebraic sets $A \subseteq K^n$. This topology is called the *Zariski topology* on K^n.

Proof. This follows directly from Corollary 90.2. □

90.4 Note. A set $A \subseteq K^1$ is algebraic iff A is the set of zeros of some polynomial $f(x) \in K[x]$. If $f(x) = 0$ then $A = K^1$, if $f(x) = 1$, then $A = \varnothing$, and if $\deg f(x) > 0$ then A is a finite set. Thus the only closed sets in the Zariski topology on K^1 are K^1, $\varnothing$ and all finite subsets $A \subseteq K^1$.

91 Algebraic varieties

91.1 Definition. Let K be an algebraically closed field. An algebraic set $A \subseteq K^n$ is *irreducible* if for any algebraic sets $A_1, A_2 \subseteq K^n$ satisfying $A = A_1 \cup A_2$ we have either $A = A_1$ or $A = A_2$.

An irreducible algebraic set is called an *algebraic variety*.

91.2 Theorem. *Let K be an algebraically closed field. Every algebraic set in K^n is a union of a finite number of algebraic varieties.*

Proof. We argue by contradiction. Let $\mathcal{J}$ be the family of all radical ideals $I \lhd K[x_1, \dots, x_n]$ such that $V(I)$ is not a union of a finite number of algebraic varieties. Assume that $\mathcal{J} \neq \varnothing$.

Since the ring $K[x_1, \dots, x_n]$ is Noetherian, by Theorem 85.5 there exists an ideal J that is a maximal element in $\mathcal{J}$. The set $V(J)$ is not an algebraic variety (since by the definition of $\mathcal{J}$ it is not a union of a finite number of algebraic varieties), so

$$V(J) = V(I_1) \cup V(I_1)$$

for some radical ideals $I_1, I_2 \lhd K[x_1, \dots, x_n]$. such that $V(I_1) \neq V(J) \neq V(I_2)$. Since $V(I_i) \subseteq V(J)$ for $i = 1, 2$ we have $J \subseteq I_i$. Also, since $V(J) \neq V(I_i)$ and J, I_i are radical ideals, thus by Corollary 89.7 we have $J \neq I_i$ for $i = 1, 2$. Since J is a maximal element in $\mathcal{J}$ this implies that $I_1, I_2 \notin \mathcal{J}$. As a consequence the sets $V(I_1)$ and $V(I_2)$ are finite unions of algebraic varieties. Since $V(J) = V(I_1) \cup V(I_2)$ this implies that $V(J)$ is also a union of a finite number of algebraic varieties which is a contradiction.

□

91.3 Definition. Let K be an algebraically closed field, let $A \subseteq K^n$ be an algebraic set and let

$$A = A_1 \cup \dots \cup A_r \qquad (*)$$

where $A_1, \dots, A_r$ are algebraic varieties. The decomposition $(*)$ is *irredundant* if

$$A \neq A_1 \cup \dots \cup A_{i-1} \cup A_{i+1} \cup \dots \cup A_r$$

for all $i = 1, \dots, r$.

91.4 Theorem. *Let K be an algebraically closed field and let $A \subseteq K^n$ be an algebraic set. Assume we have two irredundant decompositions of A into algebraic varieties:*

$$B_1 \cup \dots \cup B_r \;=\; A \;=\; C_1 \cup \dots \cup C_s$$

Then $s = r$ and there exists a permutation $\sigma\colon \{1, \dots, r\} \to \{1, \dots, r\}$ such that $B_i = C_{\sigma(i)}$ for $i = 1, \dots, r$.

Proof. Let $1 \leq i \leq r$. We have

$$B_i = (B_i \cap C_1) \cup \dots \cup (B_i \cap C_s)$$

and since B_i is an algebraic variety thus $B_i = B_i \cap C_{\sigma(i)}$ for some $1 \leq \sigma(i) \leq s$. Therefore $B_i \subseteq C_{\sigma(i)}$. By the same argument we obtain that $C_{\sigma(i)} \subseteq B_k$ for some $1 \leq k \leq r$. This gives

$$B_i \subseteq C_{\sigma(i)} \subseteq B_k$$

Since the decomposition $A = B_1 \cup \dots \cup B_r$ is irredundant we have $i = k$, and so $B_i = C_{\sigma(i)}$. Since the decomposition $A = C_1 \cup \dots \cup C_s$ is irredundant the map

$$\sigma\colon \{1, \dots, r\} \to \{1, \dots, s\}$$

is onto, and in particular $s \leq r$. On the other hand, since the decomposition $A = B_1 \cup \dots \cup B_r$ is irredundant σ must be 1-1, and so $r \leq s$. As a consequence $r = s$ and σ is a bijection. $\square$

91.5 Theorem. *Let K be an algebraically closed field. An algebraic set $V \subseteq K^n$ is an algebraic variety iff $J(V) \lhd K[x_1, \dots, x_n]$ is a prime ideal.*

91.6 Lemma. *Let K be an algebraically closed field. If $I_1, I_2 \lhd K[x_1, \dots, x_n]$ then*

$$\sqrt{I_1 I_2} = \sqrt{I_1 \cap I_2}$$

As a consequence we have

$$V(I_1 I_2) = V(I_1 \cap I_2) = V(I_1) \cup V(I_2)$$

Proof. Exercise. $\square$

Proof of Theorem 91.5.

($\Rightarrow$) Assume that V is an algebraic variety, and let $I_1, I_2 \lhd K[x_1, \dots, x_n]$ be ideals such that $I_1 I_2 \subseteq J(V)$. By Proposition 28.7 it suffices to show that either $I_1 \subseteq J(V)$ or $I_2 \subseteq J(V)$.

By Lemma 91.6 we have

$$V \subseteq V(I_1 I_2) = V(I_1 \cap I_2) = V(I_1) \cup V(I_2)$$

As a consequence

$$V = (V \cap V(I_1)) \cup (V \cap V(I_1))$$

Since V is a variety we obtain that either $V = V \cap V(I_1)$ or $V = V \cap V(I_2)$. We can assume that $V = V \cap V(I_1)$. This gives $V \subseteq V(I_1)$, and so $I_1 \subseteq J(V)$.

($\Leftarrow$) Assume that $J(V)$ is a prime ideal and let

$$V = V_1 \cup V_2$$

where V_1, V_2 are algebraic sets. For $i = 1, 2$ let $J_i = J(V_i)$. Since J_1, J_2 are radical ideals the ideal $J_1 \cap J_2$ is also radical (check). By Proposition 90.1 we have

$$V(J_1 \cap J_2) = V(J_1) \cup V(J_2) = V_1 \cup V_2 = V$$

so the bijective correspondence (89.7) implies that $J(V) = J_1 \cap J_2$. This in turn gives

$$J_1 J_2 \subseteq J(V)$$

Since $J(V)$ is a prime ideal by (28.7) we have that either $J_1 \subseteq J(V)$ or $J_2 \subseteq J(V)$. This gives that either $V \subseteq V(J_1) = V_1$ or $V \subseteq V(J_2) = V_2$. Therefore either $V = V_1$ or $V = V_2$.

□

91.7 Corollary. *Let K be an algebraically closed set. We have inverse bijections of sets:*

$$V\colon \begin{pmatrix} \textit{prime ideals} \\ I \lhd K[x_1, \dots, x_n] \end{pmatrix} \rightleftarrows \begin{pmatrix} \textit{algebraic varieties} \\ A \subseteq K^n \end{pmatrix} : J$$

Proof. This follows from Corollary 89.7 and Theorem 91.5. □

91.8 Corollary. *Let K be an algebraically closed field and let $I \lhd K[x_1, \dots, x_n]$ be a radical ideal. Then there exist prime ideals $P_1, \dots, P_m \lhd K[x_1, \dots, x_n]$ such that*

$$I = P_1 \cap \dots \cap P_n$$

Proof. By Theorem 91.2 we have

$$V(I) = V_1 \cup \dots \cup V_k$$

where $V_1, \dots, V_k$ are algebraic varieties. Let $P_i = J(V_i)$. By Theorem 91.5 $P_1, \dots, P_n$ are prime ideals. Since $V_i = V(P_i)$ we obtain

$$V(I) = V(P_1) \cup \dots \cup V(P_k) = V(P_1 \cap \dots \cap P_k)$$

Both I and $P_1 \cap \dots \cap P_k$ are radicals ideals so by (89.7) we have

$$I = P_1 \cap \dots \cap P_k$$

□

91.9 Corollary. *Let K be an algebraically closed field and let $I \lhd K[x_1, \dots, x_n]$. Then*

$$\sqrt{I} = \bigcap_{i \in S} P_i$$

where $\{P_i\}_{i \in S}$ is the family of all prime ideals of $K[x_1, \dots, x_n]$ such that $I \subseteq P_i$.

Proof. Since prime ideals are radical ideals, for any prime ideal P such that $I \subseteq P$ we have $\sqrt{I} \subseteq P$. Therefore

$$\sqrt{I} \subseteq \bigcap_{i \in S} P_i$$

Conversely, by Corollary 91.8 there exist ideals $P_{i_1}, \dots, P_{i_k} \in \{P_i\}_{i \in S}$ such that $\sqrt{I} = P_{i_1} \cap \dots \cap P_{i_k}$. This gives

$$\bigcap_{i \in S} P_i \subseteq P_{i_1} \cap \dots \cap P_{i_k} = \sqrt{I}$$

As a consequence $\sqrt{I} = \bigcap_{i \in S} P_i$. □

91.10 Note.

1) The statement of Corollary 91.9 holds if we replace $K[x_1, \dots, x_n]$ by an arbitrary commutative ring with identity (exercise).

2) As a consequence for the zero ideal $\{0\} \lhd R$ we obtain that $\sqrt{\{0\}}$ is the intersection of all prime ideals of R. Notice that we have

$$\sqrt{\{0\}} = \{a \in R \mid a^k = 0 \text{ for some } k \geq 0\}$$

The ideal $\sqrt{\{0\}}$ is called the *nilradical* of R. We denote:

$$\mathrm{nil}(R) := \sqrt{\{0\}}$$

92 Regular functions

92.1 Definition. Let K be an algebraically closed field and let $V \subseteq K^n$ be an algebraic set. A *regular function* is a function $V \to K$ given by

$$(a_1, \ldots, a_n) \mapsto f(a_1, \ldots, a_n)$$

where f is some polynomial in $K[x_1, \ldots, x_n]$.

92.2 Note.

1) Different polynomials may define the same regular function. E.g. if $V = \{0\} \subseteq K^1$ then $f(x) = x$ and $g(x) = x^2 + x$ define the same regular function since $f(0) = g(0) = 0$.

2) Polynomials $f, g \in K[x_1, \ldots, x_n]$ define the same regular function on V iff for every $a \in V$ we have

$$f(a) - g(a) = 0$$

Equivalently, f, g define the same regular function on V iff $f - g \in J(V)$. This gives a bijection of sets:

$$\begin{pmatrix} \text{regular functions} \\ V \to K \end{pmatrix} \rightleftarrows \begin{pmatrix} \text{elements of the rings} \\ K[x_1, \ldots, x_n]/J(V) \end{pmatrix}$$

92.3 Definition. Let K be an algebraically closed field and let $V \subseteq$ be an algebraic set. The coordinate ring of V is the ring

$$K[V] := K[x_1, \ldots, x_n]/J(V)$$

92.4 Note.

1) The ring $K[V]$ is a finitely generated K-algebra.

2) Check: since $J(V)$ is a radical ideal we have $\mathrm{nil}(K[V]) = \{0\}$ (see 91.10).

3) If V is an algebraic variety then $J(V)$ is a prime ideal and so $K[V]$ is an integral domain.

92.5 Proposition. *Let K be an algebraically closed field. If A is a finitely generated K-algebra such that $\mathrm{nil}(A) = \{0\}$ then there exists an algebraic set V such that $A = K[V]$. Moreover, if A is in integral domain then V is an algebraic variety.*

Proof. Since A is finitely generated we have $A = K[a_1, \dots, a_n]$ for some elements $a_1, \dots, a_n \in A$. We have a ring epimorphism

$$\varphi\colon K[x_1, \dots, x_n] \to A$$

given by $\varphi(x_i) = a_i$ for $i = 1, \dots, n$ and $\varphi|_K = \mathrm{id}_K$. Take $V := V(\operatorname{Ker}\varphi)$. Since $\mathrm{nil}(A) = \{0\}$ thus $\operatorname{Ker}(\varphi)$ is a radical ideal (check!) and so by Theorem 89.6 we have $J(V) = \operatorname{Ker}(\varphi)$. Therefore we have

$$K[V] = \operatorname{Ker}[x_1, \dots, x_n] / \operatorname{Ker}\varphi \cong A$$

If A is an integral domain then $\operatorname{Ker}(\varphi)$ is a prime ideal and so V is an algebraic variety.

□

92.6 Definition. Let K be an algebraically closed field and let $V \subseteq K^n$, $W \subseteq K^m$ be algebraic sets. Let $p_i\colon W \to K$ be the map given by

$$p_i(a_1, \dots, a_m) = a_i$$

A map

$$\varphi\colon V \to W$$

is a *morphism of algebraic sets* if $p_i\varphi\colon V \to K$ is a regular function for $i = 1, \dots, m$.

92.7 Note. If V is an algebraic set over K then morphisms $V \to K$ coincide with regular functions on V (check).

92.8 Proposition. *If K is an algebraically closed field and $\varphi\colon V \to W$, $\psi\colon W \to Z$ are morphism of algebraic sets over K then $\psi\varphi\colon V \to Z$ is also a morphism of algebraic sets.*

Proof. Exercise. $\square$

92.9 Definition. For an algebraically closed field K denote by $\mathcal{A}lg\mathcal{S}et(K)$ the category of algebraic sets over K and morphisms of algebraic sets. Let $\mathcal{V}ar(K)$ denote the subcategory of algebraic varieties over K.

92.10 Note. Let K be an algebraically closed field and let $\varphi\colon V \to W$ be a morphism of algebraic sets over K. For any regular function $f\colon W \to K$ the map $f\varphi\colon V \to K$ is a regular function on V. Using the bijective correspondence between regular functions on W and elements of the coefficient ring $K[W]$ we obtain a map

$$\varphi^{\sharp}\colon K[W] \to K[V]$$

Check: $\varphi^{\sharp}$ is a homomorphism of K-algebras.

92.11 Proposition. *Let K be an algebraically closed field and let $\mathcal{FD}_K$ be the category of finitely generated K-algebras that are integral domains. The functor*

$$F\colon \mathcal{V}ar(K) \to \mathcal{FD}_K$$

given by $F(V) = K[V]$ and $F(\varphi) = \varphi^{\sharp}$ is an equivalence of categories.

92.12 Corollary. *If K is an algebraically closed field and V, W are algebraic varieties over K then $V \cong W$ in $\mathcal{V}ar(K)$ iff $K[V] \cong K[W]$ as K-algebras.*

93 Suggested further reading

1) M. Aschbacher, *Finite group theory*, 2000.
2) J.P. Serre, *Linear representations of finite groups*, GTM 42, 1977
3) J.P. Serre, *Trees*, 2003
4) K. S. Brown, *Cohomology of groups*, GTM 87, 1982.
5) Ch. Weibel, *Homological algebra*, 1995.
6) M.I. Atiyah, I.G. MacDonald, *Introduction to commutative algebra*, 1969.
7) D. Eisenbud, *Commutative algebra with a view toward algebraic geometry*, GMT 150, 1995.
8) I.N. Herstein, *Noncommutative rings*, 1968.
9) J.W. Milnor, *Introduction to algebraic K-theory*, 1972.
10) S. Mac Lane, *Categories for the working mathematician*, GTM 5, 1998.